ALGEBRAIC LOCI OF WEIL CLASSES ON ABELIAN VARIETIES

DEEP BHATTACHARJEE[†] AND USHASHI BHATTACHARYA

Abstract. We prove that in every family of abelian varieties of Weil type, for every CM field, dimension and discriminant, the algebraic locus of the Weil classes contains an explicit point of complex multiplication, is dense, is a countable union of closed algebraic strata of the Shimura variety of the family, and is either the whole family or meagre, so that the Weil classes are algebraic on every member exactly when that locus is closed, and the Hodge conjecture for the very general member is that one topological statement.

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This paper and its companions [II, III] together replace the earlier paper *Algebraic loci of Weil classes on abelian varieties* by the authors and Priyabrata Mandal (archive [BMB26]); every result of that paper appears in one of the three with its proof. The preprint *Relative secant cycles and Hodge classes* [BB26] contains errors and incomplete proofs and is replaced by them.

Every proof is given in the text. Programs that repeat much of the finite arithmetic, and a Lean 4 certificate of part of it, are archived as described in the statement on data availability at the end of the paper; no statement rests on them.

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1. Introduction

Let X be a smooth complex projective variety. The Hodge conjecture asserts that the cycle class map

$$\mathrm{cl}^p_X\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X) \coloneqq H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$$

is surjective for every p. For $p = 1$ this is the theorem of Lefschetz on $(1,1)$ classes.[1] Beyond degree two the best understood Hodge classes are those of abelian varieties, and the first of them that divisors do not produce are the Weil classes. Weil exhibited them on abelian varieties with an action of an imaginary quadratic field whose two eigenspaces on the tangent space have the same dimension, showed that they are Hodge classes outside the subring generated by divisor classes, and proposed them as test cases for the Hodge conjecture [Wei77]. The construction works for every CM field F in place of the quadratic one. On a general abelian variety of Weil type the Hodge ring is generated by divisor classes together with the Weil classes, a flat F-line of classes of middle degree, so for these varieties the Hodge conjecture is the algebraicity of the Weil classes.

Markman proved that the Weil classes are algebraic on the abelian sixfolds of split Weil type for every imaginary quadratic field, and, through a product argument of Schoen [Sch98, Proposition 10], on every abelian fourfold of Weil type [Mar25a, Mar25b].[2] With the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [MZ99], this gives the Hodge conjecture for abelian fourfolds and fivefolds. His sheaves are built from secant sheaves on an abelian variety of half the dimension, and they extend along the family because they satisfy an equivariant form of semiregularity. This paper is about the set of members of a family on which a flat Hodge class is algebraic, which we call its algebraic locus. For the Weil classes we determine it for every CM field and every dimension at once. The companion paper [II] determines what closing that locus requires of a single object at a single point and examines the objects that the known constructions supply, and the companion paper [III] places the Weil classes inside the whole conjecture.

1.1. The main theorem.

Fix a CM field F of degree $2m$, an integer $n \geq 2$ and a discriminant δ. The abelian varieties of (F,n,δ)-Weil type are the members of one family over a connected period domain $\mathcal{D}_F$ of dimension mn^2, a product of m copies of the

[1] The rational coefficients are essential. The theorem of Lefschetz holds integrally in degree two, but the integral analogue fails in higher degree: Atiyah and Hirzebruch found torsion classes of type (p,p) that are not classes of cycles, and Kollár found non-torsion ones. Every statement in this paper concerns $\mathbb{Q}$-coefficients.

[2] In the normalisation of Definition 15.22 the discriminant is the class of $(-1)^n \det H$ in $\mathbb{Q}^\times_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^\times)$. Markman uses the class of $\det H$ itself, in which the same sixfolds have discriminant $(-1)^3 = -1$.

bounded symmetric domain of $\mathrm{SU}(n,n)$, and their Weil classes form a flat local system of rank $2m$ over it (Setup 17.1, Proposition 17.3). We call the set

$$\Sigma_F = \{\, s \in \mathcal{D}_F : \text{the Weil classes of } A_s \text{ are algebraic} \,\}$$

the *algebraic locus* of the family. A property holds for the *very general member* if it holds outside a countable union of proper closed analytic subsets of $\mathcal{D}_F$.[3]

Theorem 1.1 (Main theorem)**.** *For every CM field F, every $n \geq 2$ and every discriminant δ, the algebraic locus Σ_F of the Weil classes has the following properties.*

(i) *It contains an explicit point s_0 at which A_{s_0} is isogenous to a product of abelian varieties with complex multiplication by F, and at which the Weil classes are polynomials with rational coefficients in divisor classes.*

(ii) *It is dense in $\mathcal{D}_F$.*[4]

(iii) *It is the preimage of a countable union of Zariski closed subsets of the Shimura variety of the family, and either $\Sigma_F = \mathcal{D}_F$ or Σ_F is meagre.*

(iv) *The following are equivalent: Σ_F is closed; Σ_F has an interior point; the Weil classes are algebraic on every member of the family; the Hodge conjecture holds for the very general member of the family.*

For $n = 1$ the Weil classes have degree two and are algebraic by the theorem of Lefschetz, and all four statements hold trivially. The proof of Theorem 1.1 is completed in Section 17, after the imaginary quadratic case has been treated in full in Sections 14 to 16. Figure 1 draws the picture it describes: a connected domain, a dense set built from one explicit point, and a decomposition of that set into countably many closed algebraic strata.

The theorem carries the Hodge conjecture for the very general abelian variety of Weil type, for every CM field, every dimension and every discriminant, to a single topological statement about one explicit set: the dense set Σ_F, which is a countable union of closed algebraic subsets of a connected Shimura variety, has to be closed. The paper is organised around that statement. Sections 2 to 4 fix the conventions, the families and the explicit form of the Weil classes; Sections 5 to 13 show why the classical constructions do not produce Weil classes and what any construction must look like; Sections 14 to 16 prove the theorem for imaginary quadratic fields, where everything can be written in coordinates; and Section 17 completes the proof for every CM field. The companion papers take the statement further: [II] shows that one semiregular object at one base point would close Σ_F, computes the semiregularity condition exactly and excludes the objects of the classical constructions, and [III] shows that the whole Hodge conjecture is two statements about such loci.

1.2. **The proof.** The first step is a base point. In Theorem 17.5 the member A_{s_0} is isogenous to $B_\Phi^n \times B_{\overline{\Phi}}^n$, where B_Φ is an abelian m-fold with complex multiplication by F of a CM type determined by the polarisation, and the Weil classes there are

[3] A subset of $\mathcal{D}_F$ is *meagre* if it is contained in a countable union of closed subsets with empty interior. Since $\mathcal{D}_F$ is a complex manifold, the Baire category theorem shows that a meagre set has dense complement, so a statement that holds off a meagre set holds on a dense set of members; the very general members are those off a particular meagre set, a countable union of proper analytic subsets.

[4] Density comes from the Hecke orbit of s_0: the group $G_F(\mathbb{Q})$ of rational points of the unitary group of the hermitian form acts on $\mathcal{D}_F$ through isogenies that respect the F-action and the polarisation up to a scalar, so it preserves the algebraicity of the Weil classes, and the closure of $G_F(\mathbb{Q})$ contains the identity component of $G_F(\mathbb{R})$, which acts transitively on $\mathcal{D}_F$; the proof uses the rational isotropic vectors of the hermitian form, which exist by the Hasse principle.

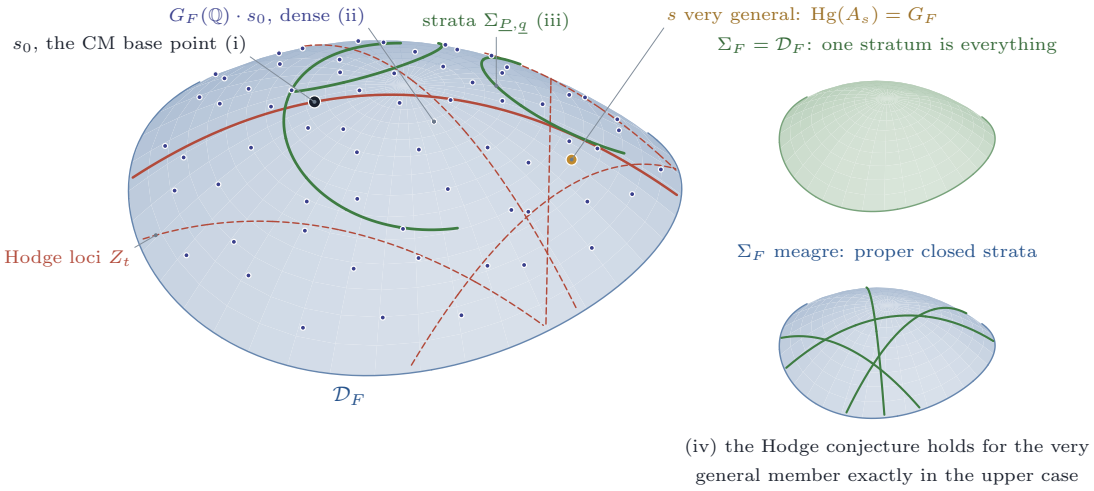


FIGURE 1. The algebraic locus of Theorem 1.1, drawn over a real slice of the period domain $\mathcal{D}_F$. Left: the explicit CM base point s_0 of part (i); its Hecke orbit $G_F(\mathbb{Q}) \cdot s_0$, which is dense by part (ii); three of the strata $\Sigma_{\underline{P},\underline{q}}$ of part (iii), each closed and algebraic, through points of the orbit; the Hodge loci Z_t, dashed, one of them through s_0, off which the Hodge group of the member is all of G_F (Lemma 17.4); and a very general point s. Right: the two cases of the dichotomy of part (iii). By part (iv) the Hodge conjecture holds for the very general member exactly in the upper case, where one stratum is the whole domain.

written out as polynomials with rational coefficients in F-balanced divisor classes. For an imaginary quadratic field K there is a second base point in every family, with quaternionic multiplication: whenever an algebra $(-d, b)_{\mathbb{Q}}$ with $b > 0$ acts compatibly with K, the Weil line is spanned by the real and imaginary parts of Θ^n for an explicit $(1,1)$ class Θ, and the discriminant of the family is the class of b^n, which can be any discriminant (Theorem 15.16, Theorem 15.25, Theorem 15.27). Products of abelian surfaces of Weil type give a third, since the Weil class is multiplicative: $\omega(A_1)\,\omega(A_2) = \omega(A_1 \times A_2)$ for the K-valued Weil class ω (Theorem 15.7, Theorem 15.10). None of these base points uses anything beyond the theorem of Lefschetz on $(1,1)$ classes.

The second step moves the base point. An element of $G_F(\mathbb{Q})$, where $G_F = \mathrm{Res}_{F_0/\mathbb{Q}}\,\mathrm{SU}(V, H)$ and F_0 is the totally real subfield, induces after clearing denominators an isogeny between two members and fixes the Weil classes, so Σ_F is a union of $G_F(\mathbb{Q})$-orbits (Lemma 16.6). The hermitian form is isotropic by the Hasse principle, the rational points of the unipotent radicals of the parabolic subgroups are dense in their real points, and these real points generate the identity component of $G_F(\mathbb{R})$; so every orbit is dense (Proposition 16.7, Theorem 17.7(ii)).

The third step is properness. The relative Hilbert scheme of the universal family, for one tuple of Hilbert polynomials, is proper over the Shimura variety,[5] and the class of a flat family of cycles is locally constant, so the set of points at which a Weil class is represented

[5]By Grothendieck's construction the Hilbert scheme of a projective morphism, for a fixed Hilbert polynomial, is projective over the base; here the base is the Shimura variety of the family, after a level structure that makes the universal family exist, and the morphism is the universal abelian scheme.

by cycles with prescribed Hilbert data is Zariski closed; Σ_F is the countable union of these strata (Theorem 16.4). The Baire category theorem then gives the dichotomy of part (iii) (Corollary 16.5). Part (iv) needs one more fact, that the very general member has Hodge group G_F; we prove it directly, from the simplicity of the real factors $\mathrm{SU}(n,n)$ (Lemma 17.4, Corollary 17.8).

For an imaginary quadratic field everything can be written in coordinates. On the standard model of Section 4, built from a K-vector space of dimension $2n$ with a hermitian form of signature (n,n), the Weil line has the integral basis

$$\omega_1 = \sum_{|T| \text{ even}} (-1)^{|T|/2} d^{\,n-|T|/2}\, w^{(T)}, \qquad \omega_2 = \sum_{|T| \text{ odd}} (-1)^{(|T|-1)/2} d^{\,n-(|T|+1)/2}\, w^{(T)},$$

where T runs over the subsets of $\{1,\dots,2n\}$ and $w^{(T)}$ is the monomial taking y_j for $j \in T$ and x_j otherwise (Theorem 4.9). A member with Hodge group $\mathrm{SU}(V,H)$ has exactly three Hodge classes in degree $2n$ and one in every other degree (Theorem 4.14), so the Hodge conjecture for it, in every degree at once, is the algebraicity of the one class ω_1, with 2^{2n-1} integer coefficients (Corollary 4.15). By Deligne's theorem [Del82] that class is absolutely Hodge at every point,[6] and the Hodge conjecture for the family is equivalent to its algebraicity at every point of one connected domain (Corollary 15.35).

1.3. **The classical constructions.** Sections 5 to 13 explain why two classical lines of attack do not produce Weil classes, and the explanations are different in kind. The first imports an object from a projective space, restricts it to the dual abelian variety and transports it back by the Fourier–Mukai transform. Its Chern character stays in the subring generated by the polarisation (Theorem 7.3), because the cohomological transform carries powers of the dual polarisation to rational multiples of powers of the polarisation in closed form (Theorem 6.5); and a construction equivariant for the field lands in the isotypic component of the norm character, which the Weil line misses, since it sits at the two ends of the grading (Theorem 5.6). The second is induction on dimension through hyperplane sections, and the restriction map kills every primitive class of middle degree (Theorem 10.3); below the middle degree the return trip is the Lefschetz standard conjecture (Theorem 10.8). The Kuga–Satake construction, which changes the variety rather than cutting it, reaches the Weil line only in dimension two (Proposition 11.11).[7] A rational class of the Weil line has a $K^\times$-orbit spanning a plane, so any construction reaching it must be bilinear in a pair of inputs exchanged by $\mathrm{Gal}(K/\mathbb{Q})$ (Proposition 13.1), and Markman's construction is of that kind.

1.4. **The companion papers.** By Theorem 1.1(iv) the Hodge conjecture for the very general member of a Weil family is the closedness of Σ_F, and by (iii) it is enough to show that one stratum has an interior point. Two statements would do this. The first bounds the Hilbert data of the cycles along the orbit of the base point (Theorem 16.9); it is equivalent to its conclusion (Theorem 16.17), and it is needed only along one sequence of CM points that leaves every proper special subvariety, which exists in the orbit (Theorem 17.14).

[6] A class is absolutely Hodge if its image under every automorphism σ of $\mathbb{C}$, through the comparison of Betti cohomology with algebraic de Rham cohomology of the conjugate variety X^σ, is again a Hodge class. Algebraic classes have this property, so it is a necessary condition for algebraicity; Deligne proved it for every Hodge class on an abelian variety.

[7] The construction attaches to a polarised Hodge structure T of weight two with $h^{2,0} = 1$ an abelian variety A with $H^1(A,\mathbb{Q}) = C^+(T)$, the even Clifford algebra, and an embedding of T into $\mathrm{End}(H^1(A,\mathbb{Q}))$ by Hodge structures, which is a Hodge class on the product of a variety carrying T with $A \times A$; the Kuga–Satake Hodge conjecture asks that this class be algebraic.

The second is semiregularity: a perfect complex at one base point whose Chern character is a nonzero multiple of a Weil class plus a polynomial in the polarisation, and whose semiregularity map is injective, extends along the whole family. The companion paper [II] proves this ([II, Theorem 2.3]), computes the injectivity condition as one number, shows that it is never met when the Chern character is exactly a Weil class, for any CM field, computes it exactly for quartic CM fields, examines the secant objects of the split member, the convolutions of line bundles and the supports of a complex meeting the criterion. The companion paper [III] shows that the Hodge conjecture is equivalent to two statements about algebraic loci, one on abelian varieties and one on the classes reached from them by correspondences, proves each where it can be proved, and proves the conjecture itself for the very general complete intersection of diagonal quadrics of Vandermonde type. The annihilator computation of Section 9, which shows that no object on which $H^2(\mathcal{O}_A)$ acts faithfully is semiregular, is the first step of that analysis and is proved here because it uses only the explicit form of the classes.

Every proof is written to be checked by hand. Where an argument rests on a finite calculation, the calculation is given where it is used, as an explicit identity, a determinant or a case analysis. The archive [BMB26] holds programs that repeat much of this arithmetic and a Lean 4 certificate of part of it; no statement of the paper rests on them.

2. Conventions

All varieties are over $\mathbb{C}$ and all cohomology is singular cohomology with the indicated coefficients. For a smooth projective variety X we write $H^{p,q}(X)$ for the Hodge components of $H^{p+q}(X,\mathbb{C})$, and

$$\mathrm{Hdg}^p(X) = H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$$

for the space of rational Hodge classes of type (p,p). A class is *algebraic* if it lies in the image of the cycle class map $\mathrm{cl}^p_X\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \to \mathrm{Hdg}^p(X)$. The algebraic classes form a subring of $H^*(X,\mathbb{Q})$, because the cycle class map carries intersection products to cup products.[8]

An abelian variety A of dimension g has $H^1(A,\mathbb{Q})$ of dimension $2g$ and a canonical isomorphism

$$H^k(A,\mathbb{Q}) \;\cong\; \bigwedge^k H^1(A,\mathbb{Q}) \tag{1}$$

of Hodge structures, with the cup product on the left corresponding to the wedge product on the right. We use (1) without further comment; it is what makes every computation below finite.[9] The dual abelian variety is $A^\vee = \mathrm{Pic}^0(A)$, and $H^1(A^\vee,\mathbb{Q}) \cong H^1(A,\mathbb{Q})^\vee$.

A polarisation of A is a class $\theta \in \mathrm{NS}(A)\otimes\mathbb{Q}$ that is ample; equivalently, an alternating form E on $H_1(A,\mathbb{Q})$ satisfying the Riemann relations. The polarisation is principal when E is unimodular on $H_1(A,\mathbb{Z})$, and then $\theta^g/g!$ is the class of a point. We write θ' for the polarisation on $A^\vee$ induced by θ through the isogeny $\phi_\theta\colon A \to A^\vee$; when θ is principal, ϕ_θ is an isomorphism and $\phi_\theta^*\theta' = \theta$.

For an endomorphism f of A we write f^* for the induced map on cohomology. When a field K acts on A through $\iota\colon K \hookrightarrow \mathrm{End}^0(A) = \mathrm{End}(A)\otimes\mathbb{Q}$, each $\tau \in K$ acts on $H^*(A,\mathbb{Q})$

[8] The intersection product on $\mathrm{CH}^*(X)_{\mathbb{Q}}$ is defined because X is smooth, and its compatibility with the cycle class map is proved in [Ful98, Chapter 19]. The integral form of the question, with $\mathbb{Z}$ in place of $\mathbb{Q}$, is a different problem and is not considered in this paper.

[9] Topologically $A = V/\Lambda$ is a real torus of dimension $2g$, so $H^*(A,\mathbb{Z}) = \bigwedge^* \mathrm{Hom}(\Lambda,\mathbb{Z})$ by the Künneth formula for $(S^1)^{2g}$, with the cup product corresponding to the wedge product. The Hodge decomposition of H^k is the k-th exterior power of $H^1 = H^{1,0}\oplus H^{0,1}$, because the translation-invariant forms $dz_I \wedge d\overline{z}_J$ are harmonic for the flat metric and represent every class.

through $\iota(\tau)$, and we speak of the action of τ when ι is understood. For nonzero τ in the order $\iota^{-1}(\mathrm{End}(A))$ the element $\iota(\tau)$ is an isogeny; every element of K is a rational multiple of an element of this order, so the general case follows by $\mathbb{Q}$-linearity, and no statement below depends on the distinction.

The Chern character of a bounded complex $\mathcal{E}^\bullet$ of locally free sheaves is $\mathrm{ch}(\mathcal{E}^\bullet) = \sum_i (-1)^i \mathrm{ch}(\mathcal{E}^i)$. On a smooth projective variety every bounded complex of coherent sheaves is quasi-isomorphic to such a complex, and the resulting class depends only on the object of D^b. We write ch_k for the component in H^{2k}. On an abelian variety the tangent bundle is trivial, and so is the Todd class. Grothendieck–Riemann–Roch therefore reads $\mathrm{ch}(f_! x) = f_* \mathrm{ch}(x)$ for a morphism f between abelian varieties, and $\mathrm{ch}\colon K_0(A)_{\mathbb{Q}} \to \mathrm{CH}^*(A)_{\mathbb{Q}}$ is an isomorphism of rings [Ful98, Chapter 15]; both facts are used without further mention. For the inclusion of a smooth subvariety, whose normal bundle need not be trivial, the Todd class of the normal bundle enters, and it is written out where it matters.

For the Fourier–Mukai transform we follow [Muk81]. The Poincaré bundle $\mathcal{P}$ on $A \times A^\vee$ is normalised so that $\mathcal{P}|_{A\times\{0\}}$ and $\mathcal{P}|_{\{0\}\times A^\vee}$ are trivial. With $p\colon A \times A^\vee \to A$ and $q\colon A \times A^\vee \to A^\vee$ the projections, the transform is

$$\Phi_{\mathcal{P}}\colon D^b(A^\vee) \to D^b(A), \qquad \Phi_{\mathcal{P}}(\mathcal{G}) = Rp_*(Lq^*\mathcal{G} \otimes^L \mathcal{P}).$$

On cohomology we write the induced map with the same symbol,

$$\Phi_{\mathcal{P}}\colon H^*(A^\vee, \mathbb{Q}) \longrightarrow H^*(A, \mathbb{Q}), \qquad \Phi_{\mathcal{P}}(x) = p_*(q^*x \cdot \mathrm{ch}(\mathcal{P})),$$

and the Grothendieck–Riemann–Roch theorem gives $\mathrm{ch}(\Phi_{\mathcal{P}}(\mathcal{G})) = \Phi_{\mathcal{P}}(\mathrm{ch}(\mathcal{G}))$ because the Todd class of an abelian variety is trivial; see Proposition 6.1.

For an imaginary quadratic field K we write $\mathrm{N} = \mathrm{N}_{K/\mathbb{Q}}$ for the norm and $\overline{\tau}$ for the conjugate of $\tau \in K$. Characters of $K^\times$ are written multiplicatively, and N^n denotes $\tau \mapsto \mathrm{N}(\tau)^n$.

Finally, we say that a construction is *of ambient type* if it has the shape of the hypothesis of Theorem 7.3: a morphism $i\colon A^\vee \to \mathbb{P}^{N-1}$ defined by a linear system, an object $\mathcal{F}$ of $D^b(\mathbb{P}^{N-1})$, and the class $\mathrm{ch}_k(\Phi_{\mathcal{P}}(Li^*\mathcal{F}))$. Nothing is assumed about $\mathcal{F}$: it may be the structure sheaf of a secant variety, of a join, of a determinantal locus or of any other subscheme, and it may be twisted by any line bundle from the ambient space.

3. Abelian varieties of Weil type

We fix the notion of Weil type, construct the Weil line, prove that it consists of Hodge classes, and record the two facts about it that the obstruction theorems use: its dimension, and its position relative to the subring generated by divisor classes. The standard references are [BL04] for abelian varieties and [Voi02] for Hodge theory. Apart from the Riemann relations, the Lefschetz theorem on $(1,1)$ classes and the description of the Néron–Severi group by symmetric endomorphisms, every result of the section is proved here or in Section 5.

3.1. **Weil type.** Let $K = \mathbb{Q}(\sqrt{-d})$ be an imaginary quadratic field, with d a squarefree positive integer, and let $^-$ denote the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$.

Definition 3.1 (Weil type)**.** Let A be a complex abelian variety, let $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$ be an embedding, and let η be a polarisation class on A. The triple (A, ι, η) is of $(K, -1, n)$-*Weil type* if

(i) $\dim A = 2n$;

(ii) the endomorphism $\iota(\sqrt{-d})$ acts on the tangent space $H^0(A,\Omega^1_A)^\vee \cong H^{1,0}(A)^\vee$ with the two eigenvalues $\pm\sqrt{-d}$ each occurring with multiplicity n;
(iii) the Rosati involution attached to η restricts on $\iota(K) \subset \mathrm{End}^0(A)$ to complex conjugation, that is $\iota(\tau)^\dagger = \iota(\overline{\tau})$ for all $\tau \in K$.

Condition (ii) is the balanced signature condition. It separates Weil type from the unbalanced case, in which the eigenvalues $\pm\sqrt{-d}$ occur with different multiplicities, and it is the condition under which the Weil line consists of Hodge classes (Remark 4.10). Condition (iii) says that the polarisation is compatible with the K-action, and the middle entry -1 of $(K,-1,n)$ records the sign in this compatibility: the Rosati involution restricts to conjugation on $\iota(K)$, not to the identity. By Lemma C.4 the sign is forced once $\iota(K)$ is stable under the Rosati involution. Some authors suppress the sign and speak of K-Weil type of dimension $2n$.

Lemma 3.2 (H^1 is a K-space)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type and put $V = H^1(A,\mathbb{Q})$. Then ι makes V a K-vector space of dimension $2n$, and $\dim_{\mathbb{Q}} V = 4n$.*

Proof. The algebra $\mathrm{End}^0(A)$ acts $\mathbb{Q}$-linearly on V by pullback. Pullback reverses the order of composition, but K is commutative, so $\tau \mapsto \iota(\tau)^*$ is a unital ring homomorphism $K \to \mathrm{End}_{\mathbb{Q}}(V)$. Since K is a field, V becomes a K-vector space, and since $\dim_{\mathbb{Q}} V = 2\dim A = 4n$, we get $\dim_K V = 4n/[K:\mathbb{Q}] = 2n$. ∎

Notation 3.3. Write $V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C}$. Since $\iota(\sqrt{-d})^2 = -d$ on V, the operator $\iota(\sqrt{-d}) \otimes 1$ on $V_{\mathbb{C}}$ has the two eigenvalues $\pm\sqrt{-d}$, and

$$V_{\mathbb{C}} = V_+ \oplus V_-, \qquad V_\pm = \ker(\iota(\sqrt{-d}) \mp \sqrt{-d}), \tag{2}$$

with $\dim_{\mathbb{C}} V_\pm = 2n$ each, because V is free of rank $2n$ over K and $K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$. Complex conjugation on $V_{\mathbb{C}}$, coming from the real structure $V \otimes \mathbb{R}$, interchanges V_+ and V_-. For general $\tau = a + b\sqrt{-d} \in K$ the operator $\iota(\tau)$ acts on V_+ by the scalar τ and on V_- by $\overline{\tau}$.

Remark 3.4. The two scalars in Notation 3.3 are different functions of τ. For $\tau \notin \mathbb{Q}$ the operator $\iota(\tau)$ is $\mathbb{Q}$-linear on V but not a scalar; it becomes a scalar on each piece only after extension to $\mathbb{C}$ and splitting. This is the source of the characters studied in Section 5. A class defined over $\mathbb{Q}$ can be an eigenvector of every $\iota(\tau)$ only with an eigenvalue in $\mathbb{Q}$; the function $\mathrm{N}(\tau) = \tau\overline{\tau}$ is $\mathbb{Q}$-valued, and τ^{2n} is not (Remark 5.8).

3.2. **The Hodge structure.** Write the Hodge decomposition as $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, with $V^{1,0} = H^{1,0}(A)$ and $V^{0,1} = H^{0,1}(A)$. The K-action preserves it, because $\iota(\tau)$ is induced by an endomorphism of A and is therefore a morphism of Hodge structures. The eigenspaces of an operator that preserves both summands are the direct sums of its eigenspaces on the two summands, so each of V_+ and V_- splits: with $V_\pm^{p,q} = V_\pm \cap V^{p,q}$,

$$V_\pm = V_\pm^{1,0} \oplus V_\pm^{0,1}. \tag{3}$$

Lemma 3.5 (Balanced signature)**.** *For (A,ι,η) of $(K,-1,n)$-Weil type,*

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_-^{1,0} = n, \qquad \dim_{\mathbb{C}} V_+^{0,1} = \dim_{\mathbb{C}} V_-^{0,1} = n.$$

Proof. By Definition 3.1(ii), $\iota(\sqrt{-d})$ acts on the tangent space with each eigenvalue of multiplicity n. The tangent space is dual to $H^{1,0}(A) = V^{1,0}$, and the action of $\iota(\sqrt{-d})$ on $V^{1,0}$ by pullback is the transpose of its action on the tangent space, so it has the same eigenvalues with the same multiplicities; that is, $\dim V_+^{1,0} = \dim V_-^{1,0} = n$. Since

$\dim_{\mathbb{C}} V_{\pm} = 2n$ by Notation 3.3 and $V_{\pm} = V_{\pm}^{1,0} \oplus V_{\pm}^{0,1}$, the remaining two dimensions are n each. ∎

3.3. **The Weil line.**

Definition 3.6 (Weil line)**.** Let (A, ι, η) be of $(K, -1, n)$-Weil type and $V = H^1(A, \mathbb{Q})$. The *Weil line* is

$$W(A) \;=\; \Big(\textstyle\bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_-\Big) \cap H^{2n}(A, \mathbb{Q}),$$

the intersection being taken inside $H^{2n}(A, \mathbb{C}) = \bigwedge_{\mathbb{C}}^{2n} V_{\mathbb{C}}$.

Proposition 3.7 (Structure of the Weil line)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type. Then*

(i) *$W(A)$ is a $\mathbb{Q}$-subspace of $H^{2n}(A, \mathbb{Q})$ of dimension 2, stable under $\iota(K)$, and free of rank one as a K-module;*
(ii) *$W(A) \subset H^{n,n}(A)$, so every element of $W(A)$ is a Hodge class;*
(iii) *$W(A)$ is canonically isomorphic to $\bigwedge_K^{2n} V$ as a K-module.*

Proof. (i) Write $W_{\mathbb{C}} = \bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$. Each summand is one-dimensional over $\mathbb{C}$ because $\dim_{\mathbb{C}} V_{\pm} = 2n$, so $\dim_{\mathbb{C}} W_{\mathbb{C}} = 2$. The splitting (2) is the eigenspace decomposition of the operator $\iota(\sqrt{-d})$, which is defined over $\mathbb{Q}$, so every automorphism of $\mathbb{C}$, acting on $V_{\mathbb{C}}$ through the second factor, either preserves V_+ and V_- or interchanges them, according to its restriction to K; complex conjugation interchanges them. Hence $W_{\mathbb{C}}$ is stable under $\mathrm{Aut}(\mathbb{C})$, and by Galois descent it is defined over $\mathbb{Q}$: the space $W(A) = W_{\mathbb{C}} \cap H^{2n}(A, \mathbb{Q})$ has $\dim_{\mathbb{Q}} W(A) = 2$ and $W(A) \otimes \mathbb{C} = W_{\mathbb{C}}$. It is stable under $\iota(K)$ because $\iota(\tau)$ preserves each of $V_{\pm}$, hence each summand. A two-dimensional $\mathbb{Q}$-space with a faithful K-action is free of rank one over K.

(ii) By (3) and Lemma 3.5, the line $\bigwedge^{2n} V_+$ equals $\bigwedge^n V_+^{1,0} \otimes \bigwedge^n V_+^{0,1}$, which is of Hodge bidegree (n, n). The same computation applies to $\bigwedge^{2n} V_-$. Hence $W_{\mathbb{C}} \subset H^{n,n}(A)$ and so $W(A) \subset H^{2n}(A, \mathbb{Q}) \cap H^{n,n}(A) = \mathrm{Hdg}^n(A)$.

(iii) Since $K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$, with the two factors corresponding to the two embeddings of K, we have $V \otimes_{\mathbb{Q}} \mathbb{C} = V \otimes_K (K \otimes_{\mathbb{Q}} \mathbb{C}) = V_+ \oplus V_-$, and exterior powers over K commute with this base change:

$$\Big(\textstyle\bigwedge_K^{2n} V\Big) \otimes_{\mathbb{Q}} \mathbb{C} \;\cong\; \bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_- \;=\; W_{\mathbb{C}} \;\subset\; H^{2n}(A, \mathbb{C}).$$

Every map in this chain is compatible with the action of $\mathrm{Aut}(\mathbb{C})$ through the second factor, which on the right permutes V_+ and V_- as in (i). By Galois descent the composite is induced by an injective $\mathbb{Q}$-linear map $\bigwedge_K^{2n} V \to H^{2n}(A, \mathbb{Q})$, which commutes with $\iota(K)$ because each map in the chain does. Its image is a $\mathbb{Q}$-subspace of $W_{\mathbb{C}} \cap H^{2n}(A, \mathbb{Q}) = W(A)$ of dimension two, hence all of $W(A)$. ∎

Remark 3.8. Some authors call a generator of $W(A)$ *the* Weil class and speak of a one-dimensional space; the difference is the usual one between a K-line and its underlying $\mathbb{Q}$-plane. Nothing below depends on the convention, since the obstruction theorems concern the whole of $W(A)$.

3.4. **The Weil line and the divisor subring.** The Weil line is not accounted for by divisor classes. We record the statement in the form in which it is used and derive it from the character computation of Section 5.

Definition 3.9. The *divisor subring* $D^*(A) \subset H^*(A, \mathbb{Q})$ is the $\mathbb{Q}$-subalgebra generated by $\mathrm{NS}(A) \otimes \mathbb{Q} \subset H^2(A, \mathbb{Q})$.

Every element of $D^*(A)$ is algebraic, by the Lefschetz theorem on $(1,1)$ classes and the fact that the algebraic classes form a subring. The Hodge conjecture for A in degree n is therefore equivalent to the algebraicity of a complement of $D^{2n}(A)$ in $\mathrm{Hdg}^n(A)$. For a very general member of a Weil family the Weil line is such a complement, by Theorem 4.14; at special members the divisor subring can grow and meet the Weil line ([III, Remark 2.5]).

Proposition 3.10. *Let (A,ι,η) be of $(K,-1,n)$-Weil type with $\mathrm{End}^0(A)=\iota(K)$, and suppose $n\geq 1$. Then*

$$W(A)\cap D^{2n}(A)=0.$$

Proof. By Proposition 3.7(i) and Theorem 5.6, the space $W(A)\otimes\mathbb{C}$ is the sum of the $K^\times$-eigenspaces for the characters $\tau\mapsto\tau^{2n}$ and $\tau\mapsto\overline{\tau}^{2n}$. The space $\mathrm{NS}(A)\otimes\mathbb{Q}$ is isomorphic to the space of elements of $\mathrm{End}^0(A)$ fixed by the Rosati involution [BL04, Section 5.2]. Since $\mathrm{End}^0(A)=\iota(K)$ and the Rosati involution restricts to conjugation on $\iota(K)$, the fixed elements are those of $\iota(\mathbb{Q})$, so $\mathrm{NS}(A)\otimes\mathbb{Q}=\mathbb{Q}\,\eta$. By Lemma 5.2 the class η is a $K^\times$-eigenclass for the norm character N, so every element of $D^{2n}(A)=\mathbb{Q}\,\eta^n$ is an eigenclass for N^n. By Lemma 5.4 the characters $\tau\mapsto\tau^{2n}$ and $\tau\mapsto\overline{\tau}^{2n}$ both differ from $\tau\mapsto\mathrm{N}(\tau)^n$. Eigenvectors for pairwise distinct characters are linearly independent, so $W(A)\otimes\mathbb{C}$ meets the N^n-eigenspace, which contains $D^{2n}(A)$, only in zero. ∎

The mechanism behind the proof is that of Remark 5.8: $K^\times$ can act on a rational class through a character only if the character is $\mathbb{Q}$-valued, and the $\mathbb{Q}$-valued characters occurring in the cohomology are the powers of the norm. The Weil line has rank one over K, not over $\mathbb{Q}$, and as a K-line of exterior degree $2n$ it carries the $2n$-th power of τ itself. Theorem 5.6 and Lemmas 5.2 and 5.4 are proved in Section 5, and nothing there uses Proposition 3.10, so the forward reference is not circular.

4. Weil classes in coordinates

Section 3 defines the Weil line as an intersection inside $H^{2n}(A,\mathbb{C})$. That description is convenient for proofs but exhibits no single class, and in this section we write the Weil classes out. We build an abelian variety of $(K,-1,n)$-Weil type from a $2n$-dimensional K-vector space with a hermitian form, write two integral generators ω_1,ω_2 of the Weil line as sums of 2^{2n-1} monomials each, and display them in full for $n=2$ and in a table for $n=3$. The coordinates then give two things that the abstract description does not: an elementary second proof that the Weil line misses the powers of the polarisation, and a complete count of the Hodge classes in every degree, which reduces the Hodge conjecture for a general member to the algebraicity of the single class ω_1.

4.1. The standard model.

Setup 4.1 (The model)**.** Fix a squarefree positive integer d, let $K=\mathbb{Q}(\sqrt{-d})$, and fix the embedding $K\hookrightarrow\mathbb{C}$ carrying $\sqrt{-d}$ to $\delta:=i\sqrt{d}$. Fix $n\geq 1$ and let

$$V=K^{2n},\qquad u_1,\dots,u_{2n}\text{ the standard }K\text{-basis},$$

regarded as a $\mathbb{Q}$-vector space of dimension $4n$ with the basis

$$x_j:=u_j,\qquad y_j:=\sqrt{-d}\,u_j,\qquad 1\leq j\leq 2n.$$

Let $S\colon V\to V$ be multiplication by $\sqrt{-d}$, so that in this basis

$$Sx_j=y_j,\qquad Sy_j=-d\,x_j. \tag{4}$$

Let $\epsilon_j = -1$ for $1 \le j \le n$ and $\epsilon_j = +1$ for $n < j \le 2n$, and let Q be the alternating $\mathbb{Q}$-bilinear form on V with

$$Q(x_j, y_j) = \epsilon_j, \qquad Q(x_j, x_k) = Q(y_j, y_k) = 0, \qquad Q(x_j, y_k) = 0 \ \ (j \neq k). \tag{5}$$

Finally let $\eta \in \bigwedge^2 V$ be the class with the same coefficients,

$$\eta = \sum_{j=1}^{2n} \epsilon_j \, x_j \wedge y_j. \tag{6}$$

Over $\mathbb{C}$ the operator S has the two eigenvalues $\pm\delta$, and its eigenvectors can be written down directly.

Lemma 4.2 (Eigenbasis)**.** *For $1 \le j \le 2n$ and $s \in \{+, -\}$ put*

$$u_j^s \ = \ \tfrac{1}{2}\Big(x_j - s\,\tfrac{\delta}{d}\,y_j\Big) \ \in V_{\mathbb{C}}. \tag{7}$$

Then $Su_j^s = s\,\delta\,u_j^s$, the vectors $u_1^+, \dots, u_{2n}^+$ form a basis of V_+ and $u_1^-, \dots, u_{2n}^-$ a basis of V_-, and complex conjugation on $V_{\mathbb{C}}$ interchanges u_j^+ and u_j^-.

Proof. Using (4) and $\delta^2 = -d$,

$$Su_j^s = \tfrac{1}{2}(y_j + s\,\tfrac{\delta}{d}\,d\,x_j) = \tfrac{s\delta}{2}(x_j + \tfrac{1}{s\delta}y_j) = \tfrac{s\delta}{2}(x_j - s\tfrac{\delta}{d}y_j) = s\,\delta\,u_j^s,$$

where the third equality uses $1/\delta = -\delta/d$. The change of basis $(x_j, y_j) \mapsto (u_j^+, u_j^-)$ is block diagonal with 2×2 blocks of determinant $\delta/(2d) \neq 0$, so the $4n$ vectors $u_j^\pm$ are linearly independent. Since V_+ and V_- have dimension $2n$ each by Notation 3.3, the u_j^+ form a basis of V_+ and the u_j^- a basis of V_-. Conjugation fixes x_j and y_j and sends δ to $-\delta$, which interchanges the two formulas. ∎

Definition 4.3 (The complex structure)**.** Define a Hodge structure of weight one on V by

$$V^{1,0} = \langle u_1^+, \dots, u_n^+,\ u_{n+1}^-, \dots, u_{2n}^- \rangle_{\mathbb{C}}, \qquad V^{0,1} = \overline{V^{1,0}}. \tag{8}$$

The first n basis directions contribute to $V^{1,0}$ through their $+$-eigenvector and the last n through their $-$-eigenvector, so each of $V_\pm$ meets $V^{1,0}$ in a subspace of dimension n. The signs ϵ_j of (5) are matched to this choice and are not free: the proof of Theorem 4.4 shows that the second Riemann relation holds for them and fails as soon as one of them is reversed (see also Remark 4.6). Reversing all of them replaces Q by $-Q$, and the construction then goes through with $V^{1,0}$ and $V^{0,1}$ interchanged.

Theorem 4.4 (The model is of Weil type)**.** *With the data of Setup 4.1 and Definition 4.3, the pair $(V, V^{1,0})$ is a polarised rational Hodge structure of weight one with polarising form Q, so $V = H^1(A, \mathbb{Q})$ for an abelian variety A of dimension $2n$; the multiplication S makes K act on A up to isogeny; η is the class of the polarisation; and (A, ι, η) is of $(K, -1, n)$-Weil type in the sense of Definition 3.1.*

Proof. We verify the two Riemann bilinear relations, then the three conditions of Definition 3.1.

The form vanishes on $V^{1,0}$. By (5) the form Q is block diagonal for the decomposition of V into the planes $\langle x_j, y_j \rangle$, so $Q(u_j^s, u_k^t) = 0$ whenever $j \neq k$. Within one block, $Q(u_j^s, u_j^s) = 0$ because Q is alternating, and

$$Q(u_j^+, u_j^-) = \tfrac{1}{4}\,Q(x_j - \tfrac{\delta}{d}y_j,\ x_j + \tfrac{\delta}{d}y_j) = \tfrac{1}{4} \cdot \tfrac{2\delta}{d}\,Q(x_j, y_j) = \frac{\delta\,\epsilon_j}{2d}. \tag{9}$$

The basis (8) of $V^{1,0}$ contains, for each j, exactly one of u_j^+, u_j^-. Hence any two of its members either lie in different blocks or are equal, and in both cases Q vanishes on them. So $Q(V^{1,0}, V^{1,0}) = 0$.

Positivity. Let $0 \neq v = \sum_j a_j v_j \in V^{1,0}$, where v_j is u_j^+ for $j \leq n$ and u_j^- for $j > n$. Then $\overline{v} = \sum_j \overline{a_j}\,\overline{v_j}$ with $\overline{v_j}$ the other eigenvector in the same block, so again only the diagonal terms survive:

$$i\,Q(v,\overline{v}) = \sum_{j\leq n} i\,|a_j|^2\,Q(u_j^+, u_j^-) + \sum_{j>n} i\,|a_j|^2\,Q(u_j^-, u_j^+).$$

By (9) and $\delta = i\sqrt{d}$ the first kind of term is $i\cdot\delta\epsilon_j/(2d) = -\epsilon_j/(2\sqrt{d})$ and the second is $+\epsilon_j/(2\sqrt{d})$. With $\epsilon_j = -1$ for $j \leq n$ and $+1$ for $j > n$ both are $+1/(2\sqrt{d})$, so $i\,Q(v,\overline{v}) = \|a\|^2/(2\sqrt{d}) > 0$. This is the second Riemann relation. For the complex structure (8) it forces the signature of ϵ to be (n,n), since reversing any single ϵ_j makes the corresponding term negative.

Weil type. Condition (i) is $\dim A = \frac{1}{2}\dim_{\mathbb{Q}} V = 2n$. For (ii), $V^{1,0}$ is spanned by the u_j^+ with $j \leq n$ and the u_j^- with $j > n$, so $\dim V_+^{1,0} = \dim V_-^{1,0} = n$; by duality, as in the proof of Lemma 3.5, this is the balanced eigenvalue condition on the tangent space. For (iii), write $\tau = a + b\sqrt{-d}$, so that $\iota(\tau)$ acts by

$$x_j \longmapsto a\,x_j + b\,y_j, \qquad y_j \longmapsto -bd\,x_j + a\,y_j,$$

and $\iota(\overline{\tau})$ by the same formulas with b replaced by $-b$. Since Q is block diagonal and $\iota(\tau)$ preserves each block, it suffices to compare $Q(\iota(\tau)v, w)$ with $Q(v, \iota(\overline{\tau})w)$ for basis vectors v, w of one block, where

$$\begin{aligned} Q(\iota(\tau)x_j, y_j) &= a\epsilon_j = Q(x_j, \iota(\overline{\tau})y_j),\\ Q(\iota(\tau)x_j, x_j) &= -b\epsilon_j = Q(x_j, \iota(\overline{\tau})x_j),\\ Q(\iota(\tau)y_j, y_j) &= -bd\,\epsilon_j = Q(y_j, \iota(\overline{\tau})y_j), \end{aligned}$$

and the remaining pair (y_j, x_j) follows from the first line with τ replaced by $\overline{\tau}$ and the antisymmetry of Q. So $\iota(\tau)^\dagger = \iota(\overline{\tau})$. Equivalently, in the form used later, $\iota(\tau)^*\eta = \mathrm{N}(\tau)\,\eta$: by (7), $x_j \wedge y_j = \frac{2d}{\delta}\,u_j^+ \wedge u_j^-$, and $\iota(\tau)$ multiplies $u_j^+ \wedge u_j^-$ by $\tau\overline{\tau}$.

Finally, η is of type $(1,1)$, because each $x_j \wedge y_j$ is a multiple of $u_j^+ \wedge u_j^-$ and in each block one of u_j^+, u_j^- lies in $V^{1,0}$ and the other in $V^{0,1}$. Since in addition $\eta^{2n} \neq 0$, the class η is a polarisation class. ∎

Lemma 4.5 (The hermitian form)**.** *On the model of Setup 4.1 there is a unique K-hermitian form H on V, linear in the first variable, with*

$$Q(v,w) = \tfrac{1}{2}\,\mathrm{Tr}_{K/\mathbb{Q}}(\delta^{-1}H(v,w)), \qquad \textit{namely} \qquad H(v,w) = Q(\sqrt{-d}\,v,\ w) + \sqrt{-d}\,Q(v,w), \tag{10}$$

and in the basis $u_1, \ldots, u_{2n}$ it is $H = \mathrm{diag}(-\epsilon_1, \ldots, -\epsilon_{2n})$, of signature (n,n).

Proof. K-linearity in the first variable is the computation

$$H(\sqrt{-d}v, w) = Q(-dv, w) + \sqrt{-d}\,Q(\sqrt{-d}v, w) = \sqrt{-d}\,H(v,w),$$

and the hermitian symmetry $\overline{H(w,v)} = H(v,w)$ follows from $Q(v, \sqrt{-d}\,w) = -Q(\sqrt{-d}\,v, w)$, which is condition (iii) of Definition 3.1 for $\tau = \sqrt{-d}$. Multiplying by δ^{-1} and taking the trace kills the first summand of (10), since Q is $\mathbb{Q}$-valued and $\mathrm{Tr}_{K/\mathbb{Q}}(\delta^{-1}) = 0$, and doubles the second; this gives Q. For uniqueness, if H' is a second such form and $z = H(v,w) - H'(v,w)$, then $\mathrm{Tr}_{K/\mathbb{Q}}(\delta^{-1}z) = 0$, and replacing v by $\sqrt{-d}\,v$

gives $\mathrm{Tr}_{K/\mathbb{Q}}(z)=0$; together these force $z=0$. Finally $H(u_j,u_j)=Q(y_j,x_j)=-\epsilon_j$ and $H(u_j,u_k)=0$ for $j\neq k$, so the signature is the number of indices with $\epsilon_j=-1$ against the number with $\epsilon_j=+1$, that is (n,n). ∎

Remark 4.6. The two signatures that have appeared express the same condition. The hermitian form H has signature (n,n) by Lemma 4.5, and the K-action on the tangent space has eigenvalue multiplicities (n,n) by Definition 3.1(ii). The positivity computation in the proof of Theorem 4.4 passes from one to the other: the sign ϵ_j decides both which eigenvector of the j-th block is holomorphic and the sign of H on that block.

Remark 4.7. Taking the lattice $\mathcal{O}^{2n}$ for an order $\mathcal{O}\subset K$ and scaling Q to be integral on it produces an abelian variety and not only a Hodge structure. Nothing below depends on the lattice, so we work with the rational Hodge structure throughout. Varying (8) over the period domain of the hermitian form gives the Weil family, of dimension n^2 inside the $n(2n+1)$-dimensional moduli of principally polarised abelian $2n$-folds (Proposition B.8).

4.2. **The two generators.**

Notation 4.8. For $T\subseteq\{1,\dots,2n\}$ write

$$w^{(T)}=w_1^{(T)}\wedge w_2^{(T)}\wedge\cdots\wedge w_{2n}^{(T)},\qquad w_j^{(T)}=\begin{cases}y_j, & j\in T,\\ x_j, & j\notin T.\end{cases}$$

The 2^{2n} elements $w^{(T)}$ are part of the monomial basis of $\bigwedge^{2n}V$, which has dimension $\binom{4n}{2n}$; they are the monomials that contain exactly one of x_j,y_j for each j.

Theorem 4.9 (The Weil line in coordinates)**.** *Let (A,ι,η) be the model of Theorem 4.4 and let*

$$\omega_1=\sum_{\substack{T\subseteq\{1,\dots,2n\}\\ |T|\text{ even}}}(-1)^{|T|/2}\,d^{\,n-|T|/2}\,w^{(T)},\qquad \omega_2=\sum_{\substack{T\subseteq\{1,\dots,2n\}\\ |T|\text{ odd}}}(-1)^{(|T|-1)/2}\,d^{\,n-(|T|+1)/2}\,w^{(T)}. \tag{11}$$

Then

(i) *ω_1 and ω_2 have integer coefficients and each has exactly 2^{2n-1} nonzero terms;*

(ii) *ω_1,ω_2 is a $\mathbb{Q}$-basis of the Weil line $W(A)$ of Definition 3.6;*

(iii) *writing $\omega_\pm=u_1^\pm\wedge\cdots\wedge u_{2n}^\pm$ for the generators of $\bigwedge^{2n}V_\pm$,*

$$\omega_++\omega_-=\frac{\omega_1}{2^{2n-1}d^{\,n}},\qquad \frac{\omega_+-\omega_-}{\delta}=-\frac{\omega_2}{2^{2n-1}d^{\,n}};$$

(iv) *$\iota(\tau)^*\omega_+=\tau^{2n}\omega_+$ and $\iota(\tau)^*\omega_-=\overline{\tau}^{\,2n}\omega_-$ for all $\tau\in K^\times$, so on the plane $\langle\omega_1,\omega_2\rangle$ the element $\iota(\tau)$ acts by the $\mathbb{Q}$-rational matrix of multiplication by τ^{2n} on K;*

(v) *ω_1 and ω_2 are of Hodge type (n,n), hence are Hodge classes.*

Proof. (iii) and (i). Expanding (7) in the product defining ω_s and collecting the monomials $w^{(T)}$,

$$\omega_s=\frac{1}{2^{2n}}\sum_T\Big(-s\,\frac{\delta}{d}\Big)^{|T|}w^{(T)}, \tag{12}$$

because the factor from index j contributes x_j with coefficient $1/2$ and y_j with coefficient $-s\delta/(2d)$, and the monomials $w^{(T)}$ appear in the order $j=1,\dots,2n$ with no reordering sign. Therefore

$$\omega_++\omega_-=\frac{1}{2^{2n-1}}\sum_{|T|\text{ even}}\Big(\frac{\delta}{d}\Big)^{|T|}w^{(T)},\qquad \frac{\omega_+-\omega_-}{\delta}=-\frac{1}{2^{2n-1}}\sum_{|T|\text{ odd}}\frac{\delta^{|T|-1}}{d^{|T|}}\,w^{(T)},$$

since the terms with $|T|$ odd cancel in the first sum and those with $|T|$ even in the second. Now $\delta^{2m} = (-d)^m$, so for $|T| = 2m$ the coefficient in the first sum is $(-1)^m d^{-m}$, and for $|T| = 2m+1$ the coefficient in the second is $-(-1)^m d^{-m-1}$. Multiplying by $2^{2n-1} d^n$ clears denominators, since $m \le n$ in the first case and $m+1 \le n$ in the second, and gives (11). The term counts are $\sum_m \binom{2n}{2m} = \sum_m \binom{2n}{2m+1} = 2^{2n-1}$.

(ii). By (iii), ω_1 and ω_2 lie in $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$. They are fixed by complex conjugation, which interchanges ω_+ and ω_- by Lemma 4.2 and sends δ to $-\delta$, and by (i) they have rational coefficients. Hence they lie in $(\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-) \cap H^{2n}(A, \mathbb{Q})$, which is $W(A)$ by Definition 3.6 and has dimension two by Proposition 3.7(i). They are linearly independent over $\mathbb{C}$ since ω_+, ω_- are, so they form a basis of $W(A)$.

(iv). The first two identities are immediate, since $\iota(\tau)$ multiplies each of the $2n$ factors u_j^+ by τ and each u_j^- by $\overline{\tau}$. For the last clause, by (iii) the space $W(A) \otimes \mathbb{C}$ is $\mathbb{C}\omega_+ \oplus \mathbb{C}\omega_-$, on which $\iota(\tau)$ acts by τ^{2n} and $\overline{\tau}^{\,2n}$; so under the isomorphism $W(A) \cong K$ of Proposition 3.7(iii) the action of $\iota(\tau)$ is multiplication by τ^{2n}, whose matrix in the basis ω_1, ω_2 has rational entries.

(v). By (8) the monomial $\omega_+ = u_1^+ \wedge \cdots \wedge u_{2n}^+$ has n factors in $V^{1,0}$, namely those with $j \le n$, and n factors in $V^{0,1}$; so ω_+ is of type (n,n), and likewise ω_-. By (iii), so are ω_1 and ω_2. ∎

Remark 4.10 (Balanced signature). The balanced condition enters the proof of Theorem 4.9 only in part (v). If (8) is replaced by a complex structure with $\dim V_+^{1,0} = p$ and $\dim V_-^{1,0} = 2n - p$, everything else being kept, then ω_+ has p holomorphic factors and is of type $(p, 2n-p)$, so $W(A)$ is a rational sub-Hodge structure of $H^{2n}(A, \mathbb{Q})$ of type $\{(p, 2n-p), (2n-p, p)\}$, and it contains a nonzero Hodge class if and only if $p = n$. The balanced signature in Definition 3.1(ii) is therefore the condition under which the Weil line contains Hodge classes at all. Proposition C.7 proves this for every abelian variety with a Rosati-stable action of K.

4.3. **The case $n = 2$.** Here $\dim A = 4$, $\dim_{\mathbb{Q}} V = 8$, $H^4(A, \mathbb{Q})$ has dimension $\binom{8}{4} = 70$, and each generator has $2^3 = 8$ terms. Writing the monomials in the order $j = 1, 2, 3, 4$ as in Notation 4.8, (11) reads

$$\omega_1 = d^2\, x_1x_2x_3x_4 \;-\; d \sum_{|T|=2} w^{(T)} \;+\; y_1y_2y_3y_4, \tag{13}$$

$$\omega_2 = d \sum_{|T|=1} w^{(T)} \;-\; \sum_{|T|=3} w^{(T)}, \tag{14}$$

with the wedge signs omitted. For $d = 1$, written out in full,

$$\begin{aligned}
\omega_1 &= x_1x_2x_3x_4 - y_1y_2x_3x_4 - y_1x_2y_3x_4 - y_1x_2x_3y_4 - x_1y_2y_3x_4 - x_1y_2x_3y_4 - x_1x_2y_3y_4 \\
&\qquad + y_1y_2y_3y_4, \\
\omega_2 &= y_1x_2x_3x_4 + x_1y_2x_3x_4 + x_1x_2y_3x_4 + x_1x_2x_3y_4 \\
&\qquad - y_1y_2y_3x_4 - y_1y_2x_3y_4 - y_1x_2y_3y_4 - x_1y_2y_3y_4.
\end{aligned}$$

The square of the polarisation is $\eta^2 = 2 \sum_{j<k} \epsilon_j \epsilon_k\, x_j y_j x_k y_k$, a sum of six monomials, none of which is among the sixteen above; Proposition 4.11 below shows that this separation holds in every dimension. Figure 2 arranges the sixteen monomials $w^{(T)}$ on the vertices of the 4-cube.

4.4. **The case $n = 3$.** Here $\dim A = 6$, $\dim_{\mathbb{Q}} V = 12$, $H^6(A, \mathbb{Q})$ has dimension $\binom{12}{6} = 924$, and each generator has $2^5 = 32$ terms. The coefficients depend only on $|T|$, so the two classes are recorded completely by Table 1.

ω_1				ω_2			
$\lvert T\rvert$	terms	coefficient	at $d = 2$	$\lvert T\rvert$	terms	coefficient	at $d = 2$
0	1	$+d^3$	$+8$	1	6	$+d^2$	$+4$
2	15	$-d^2$	-4	3	20	$-d$	-2
4	15	$+d$	$+2$	5	6	$+1$	$+1$
6	1	-1	-1				
	32				32		

TABLE 1. The Weil classes at $n = 3$. Each row gives the number of index sets T of that size and the common coefficient of the corresponding monomials $w^{(T)}$ in (11). The two classes together involve all 64 monomials containing exactly one of x_j, y_j for each j, out of the 924 monomials of H^6.

The pattern has a closed description. By Theorem 4.9(iii) and (7),

$$\omega_1 + \delta\,\omega_2 = 2^{2n} d^n\,\omega_- = d^{-n} \prod_{j=1}^{2n} (d\,x_j + \delta\,y_j), \tag{15}$$

the product being the wedge product taken in the order $j = 1, \dots, 2n$. The proof of Theorem 4.9 uses of the basis only that the u_j form a K-basis of V, so (11) and (15) define generators of the Weil line, and hold, for every K-basis $u_1, \dots, u_{2n}$ of $H^1(A, \mathbb{Q})$ of an abelian variety of $(K, -1, n)$-Weil type, with $x_j = u_j$ and $y_j = \sqrt{-d}\,u_j$. For $n = 3$ and $d = 1$ the coefficients in Table 1 are $+1, -1, +1, -1$ for ω_1 and $+1, -1, +1$ for ω_2, and the two classes are the real and imaginary parts of $\prod_{j=1}^{6}(x_j + \sqrt{-1}\,y_j)$.

4.5. **Disjoint supports.** Proposition 3.10 is deduced from the character computation of Section 5. In the model the Weil line can be separated from the powers of the polarisation without any representation theory: the Weil classes and the powers of η are supported on disjoint sets of monomials.

Proposition 4.11 (Disjoint supports)**.** *In the model of Theorem 4.4, every monomial occurring in ω_1 or in ω_2 contains exactly one of x_j, y_j for every $j \in \{1, \dots, 2n\}$, and every monomial occurring in η^k contains either both or neither, for every j and every $k \geq 0$. Consequently*

$$W(A) \cap \mathbb{Q}\,\eta^n = 0,$$

and more generally $W(A)$ meets the span of the monomials occurring in powers of η in zero.

Proof. The first assertion holds by (11) and Notation 4.8. For the second, η is the combination (6) of the $2n$ two-forms $x_j \wedge y_j$, which pairwise commute and square to zero, so

$$\eta^k = k! \sum_{\substack{S \subseteq \{1,\dots,2n\} \\ |S| = k}} \Big(\prod_{j \in S} \epsilon_j\Big) \bigwedge_{j \in S} (x_j \wedge y_j),$$

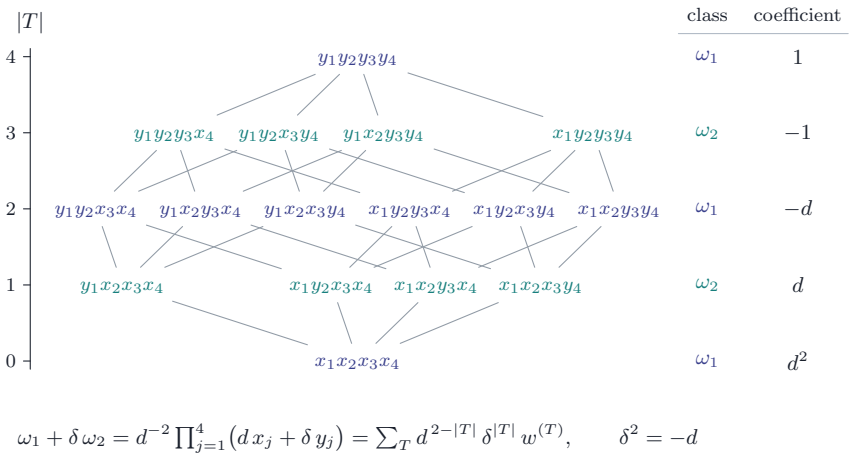


FIGURE 2. The $2^{2n} = 16$ monomials $w^{(T)}$ of Notation 4.8 at $n = 2$, as the vertices of the 4-cube, projected so that the height of $w^{(T)}$ is $|T|$. An edge joins two monomials whose index sets differ in one element, that is, two monomials related by a single substitution $x_j \mapsto y_j$. The two colours are the bipartition by the parity of $|T|$: the eight even vertices carry ω_1 and the eight odd ones carry ω_2. The coefficient of $w^{(T)}$ depends only on $|T|$ and is given by (13) and (14).

and the monomial indexed by S contains both x_j and y_j for $j \in S$ and neither for $j \notin S$. A monomial cannot satisfy both patterns for a given j, and since $2n \geq 1$ there is at least one index at which to test, so the two families of monomials are disjoint. Since ω_1, ω_2 span $W(A)$ by Theorem 4.9(ii), a nonzero element of $W(A)$ has a nonzero coefficient on some $w^{(T)}$. As the monomials form a basis of $\bigwedge^{2n} V$ and $w^{(T)}$ occurs in no power of η, such an element does not lie in the span of the monomials occurring in powers of η, and in particular not in $\mathbb{Q}\,\eta^n$. ∎

Remark 4.12. The two proofs that the Weil line misses the powers of the polarisation differ in scope. Proposition 4.11 concerns one model and one basis, and it is short because the basis makes the K-action block diagonal. The character proof is basis free, applies to every abelian variety of Weil type, and identifies the obstruction as a difference of characters. The first can be checked by hand; the second is the argument that Theorem 5.6 extends.

4.6. **The Hodge ring of a general member.** It remains to count the Hodge classes, which measures how much of the Hodge conjecture for A is at stake. The Hodge classes of an abelian variety are the invariants of its Hodge group, the special Mumford–Tate group $\mathrm{Hg}(A) \subseteq \mathrm{SL}(V)$ [Del82]: $\mathrm{Hdg}^k(A) = H^{2k}(A, \mathbb{Q})^{\mathrm{Hg}(A)}$.

Lemma 4.13 (The Hodge group)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type and let H be the K-hermitian form of Lemma 4.5. Then*

$$\mathrm{Hg}(A) \subseteq \mathrm{SU}(V, H).$$

Proof. The group $\mathrm{Hg}(A)$ is a $\mathbb{Q}$-algebraic subgroup of $\mathrm{SL}(V)$ which fixes every Hodge class in every tensor construction on V. It fixes $\eta \in \bigwedge^2 V$, hence preserves Q; and it commutes with $\iota(K)$, because each $\iota(\tau)$ is a Hodge class in $V \otimes V^\vee$. A $\mathbb{Q}$-linear automorphism of V commuting with the K-action and preserving Q preserves H, by the formula (10) expressing

H through Q and the K-action. Hence $\mathrm{Hg}(A) \subseteq \mathrm{U}(V,H)$. Now $W(A)$ consists of Hodge classes by Proposition 3.7(ii), so $\mathrm{Hg}(A)$ fixes $W(A)\otimes\mathbb{C}$ pointwise, and in particular it fixes the line $\bigwedge^{2n} V_+$ pointwise. An element $g \in \mathrm{U}(V,H)$ acts on $\bigwedge^{2n} V_+$ by $\det_K(g)$, so every $g \in \mathrm{Hg}(A)$ has $\det_K(g) = 1$, which is the definition of $\mathrm{SU}(V,H)$. ∎

For a very general member of the Weil family, that is, for every member whose point in the period domain $\mathcal{D}_{n,n}$ of Proposition B.8 lies outside a certain countable union of proper closed analytic subsets, the containment of Lemma 4.13 is an equality; this is Weil's theorem [Wei77], in the form given by Moonen and Zarhin [MZ99]. The restriction to very general members cannot be replaced by a Zariski open condition, since the locus where the endomorphism algebra jumps is a countable union of such subsets, by Theorem B.6, and is dense in the classical topology. For such a member the Hodge classes can be counted in every degree.

Theorem 4.14 (The Hodge ring of a general member)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$. Then for every k with $0 \le k \le 2n$,*

$$\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = \begin{cases} 3, & k = n, \\ 1, & k \neq n, \end{cases} \qquad \text{and} \qquad \mathrm{Hdg}^n(A) = \mathbb{Q}\,\eta^n \oplus W(A).$$

Proof. Complexify. The group $\mathrm{SU}(V,H)\otimes\mathbb{C}$ is $\mathrm{SL}(V_+) \cong \mathrm{SL}_{2n}(\mathbb{C})$, acting on V_+ tautologically and on V_- through the isomorphism $V_- \cong V_+^\vee$ induced by H. By Proposition 5.1,

$$H^{2k}(A,\mathbb{C}) = \bigoplus_{r+s=2k} \textstyle\bigwedge^r V_+ \otimes \bigwedge^s V_+^\vee,$$

so it suffices to find the invariants in each summand. An invariant is a weight-zero vector annihilated by every raising operator. Fix a basis of V_+, take the diagonal torus of $\mathrm{SL}(V_+)$ as maximal torus, and for $T,U \subseteq \{1,\dots,2n\}$ let $v_T \in \bigwedge^{|T|} V_+$ and $v_U^\vee \in \bigwedge^{|U|} V_+^\vee$ be the corresponding monomials in the basis and its dual. In the weight lattice of SL_{2n} the weight of $v_T \otimes v_U^\vee$ is $\mathbf{1}_T - \mathbf{1}_U$ modulo $(1,1,\dots,1)$; since the entries of $\mathbf{1}_T - \mathbf{1}_U$ lie in $\{-1,0,1\}$, the only multiples of $(1,\dots,1)$ it can equal are 0 and $\pm(1,\dots,1)$, giving

$$T = U \quad (\text{so } r = s), \qquad (T,U) = (\{1,\dots,2n\},\emptyset), \qquad (T,U) = (\emptyset,\{1,\dots,2n\}).$$

The last two occur only for $(r,s) = (2n,0)$ and $(0,2n)$, each with a one-dimensional weight-zero space consisting of the invariant lines $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$, which are SL-invariant because SL has trivial determinant. For $r = s$ the weight-zero space has dimension $\binom{2n}{r}$ and its invariant subspace is the line spanned by the image of the identity under $\mathrm{End}(\bigwedge^r V_+) \cong \bigwedge^r V_+ \otimes \bigwedge^r V_+^\vee$, which is one dimensional because $\bigwedge^r V_+$ is an irreducible SL_{2n}-module and Schur's lemma applies.

Collecting the contributions: for $k \neq n$ the only one is $(r,s) = (k,k)$, of dimension one and spanned by η^k, which is a nonzero invariant lying in that summand by Lemma 5.2. For $k = n$ there are three contributions, (n,n), $(2n,0)$ and $(0,2n)$, of total dimension three; the first is spanned by η^n and the last two span $W(A)\otimes\mathbb{C}$ by Lemma 5.3. Taking invariants commutes with extension of scalars from $\mathbb{Q}$ to $\mathbb{C}$, so the dimensions over $\mathbb{Q}$ are the same. ∎

Figure 3 displays these dimensions for $n = 2,3,4$.

Corollary 4.15 (Reduction to one class)**.** *Let (A,ι,η) be as in Theorem 4.14. The Hodge conjecture holds for A in every degree if and only if the single class ω_1 of (11) is algebraic.*

Proof. For $k \neq n$ the Hodge classes are the multiples of η^k, which are algebraic, being powers of a divisor class. For $k = n$ they are spanned by η^n, ω_1 and ω_2 by Theorem 4.14. Finally, ω_2 is algebraic as soon as ω_1 is. By Theorem 4.9(iv) the plane $W(A)$ is a free K-module of rank one generated by ω_1. For $\tau \in K^\times$ some positive integer multiple $m\,\iota(\tau)$ lies in $\mathrm{End}(A)$, and on $H^{2n}(A, \mathbb{Q})$ the operator $\iota(\tau)^*$ is m^{-2n} times the pullback by $m\,\iota(\tau)$, so it carries algebraic classes to algebraic classes. Any τ with $\tau^{2n} \notin \mathbb{Q}$, which exists by the argument of Lemma 5.4, moves ω_1 to a class with a nonzero ω_2 component, and so ω_2 is a rational combination of the algebraic classes ω_1 and $\iota(\tau)^*\omega_1$. ∎

Remark 4.16. Corollary 4.15 explains why the Weil line is the standard test case. On a general abelian variety of Weil type the Hodge conjecture concerns one explicitly written class with 2^{2n-1} integer coefficients, and no space of classes of unknown structure. The obstruction theorems of this paper constrain the ways in which that class can be produced, and Section 15 exhibits in every Weil family a member on which the whole Weil line consists of polynomials in divisor classes.

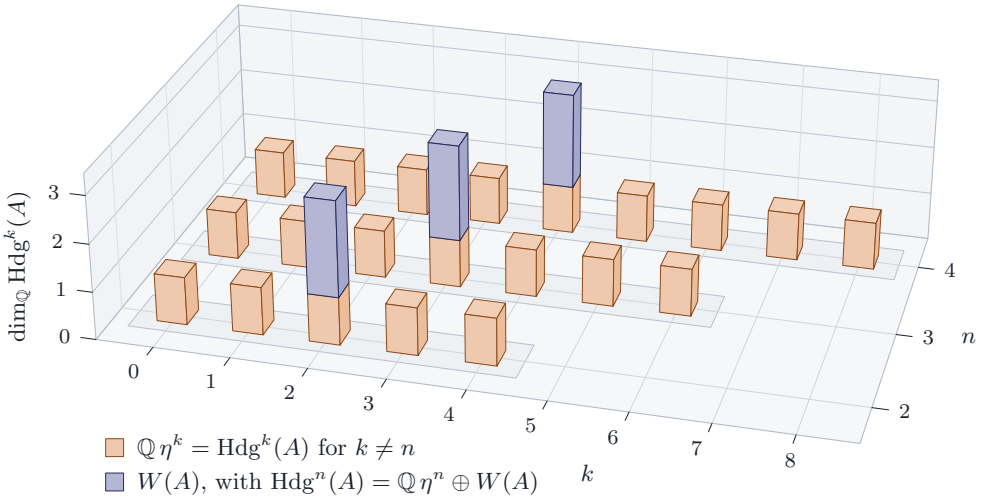


FIGURE 3. The dimensions of $\mathrm{Hdg}^k(A)$ for a general abelian variety of $(K, -1, n)$-Weil type, $0 \leq k \leq 2n$, at $n = 2, 3, 4$, as computed in Theorem 4.14. Every degree carries the line $\mathbb{Q}\,\eta^k$; the middle degree $k = n$ carries in addition the two dimensions of the Weil line $W(A)$, and nothing else occurs. The profile is the same for every n.

5. The character grading and the position of the Weil line

This section proves the character obstruction, Theorem 5.6, which is used in the proof of Proposition 3.10. The group $K^\times$ acts on $H^{2n}(A, \mathbb{Q})$ through the endomorphisms $\iota(\tau)$. This action grades the cohomology, and the Weil line and the power η^n of the polarisation lie in different graded pieces; once the grading is written down, the obstruction follows in a few lines. The one point that needs care is the character through which $K^\times$ acts on the Weil line. It is $\tau \mapsto \tau^{2n}$, together with its conjugate, and not a power of the norm, even though the Weil line is a K-line and multiplication by τ on a K-line has determinant $\mathrm{N}(\tau)$ (Remark 5.9).

5.1. **The grading.** Throughout this section (A, ι, η) is of $(K, -1, n)$-Weil type, $V = H^1(A, \mathbb{Q})$, and $V_{\mathbb{C}} = V_+ \oplus V_-$ is the splitting of Notation 3.3.

Proposition 5.1 (Character grading)**.** *For every $k \geq 0$,*

$$H^k(A, \mathbb{C}) \;=\; \bigoplus_{r+s=k} \Big(\textstyle\bigwedge^r V_+ \otimes \bigwedge^s V_- \Big), \tag{16}$$

and $\iota(\tau)^*$ acts on the summand indexed by (r, s) by the scalar $\tau^r \overline{\tau}^{\,s}$. The summands are the isotypic components for the action of $K^\times$, and the characters $\tau \mapsto \tau^r \overline{\tau}^{\,s}$ for distinct pairs (r, s) with $r + s = k$ are pairwise distinct.

Proof. By (1), $H^k(A, \mathbb{C}) = \bigwedge^k V_{\mathbb{C}}$, and the exterior power of a direct sum decomposes as $\bigwedge^k (V_+ \oplus V_-) = \bigoplus_{r+s=k} \bigwedge^r V_+ \otimes \bigwedge^s V_-$, which is (16). By Notation 3.3, $\iota(\tau)^*$ acts on V_+ by τ and on V_- by $\overline{\tau}$. Since it is an algebra homomorphism for the wedge product, it acts on $\bigwedge^r V_+ \otimes \bigwedge^s V_-$ by $\tau^r \overline{\tau}^{\,s}$.

For the last statement, suppose that $\tau^r \overline{\tau}^{\,s} = \tau^{r'} \overline{\tau}^{\,s'}$ for all $\tau \in K^\times$, where $r + s = r' + s' = k$. Then $s - s' = r' - r$, and dividing gives $(\tau/\overline{\tau})^{r-r'} = 1$ for every $\tau \in K^\times$. If $r \neq r'$, the image of the map $c(\tau) = \tau/\overline{\tau}$ therefore lies in the group of $(r - r')$-th roots of unity of K, which is finite: the group of all roots of unity of the imaginary quadratic field K has order 2, 4 or 6. But c has infinite image. Two elements have the same image exactly when their ratio lies in $\mathbb{Q}^\times$, and the elements $m + \sqrt{-d}$ with $m \in \mathbb{Z}$ are pairwise non-proportional over $\mathbb{Q}$. Hence $r = r'$, and so $s = s'$. Since the characters of the summands are pairwise distinct, the summands are the isotypic components.

The distinctness is that of characters of the torus $\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m$, not of their values at a given τ: the values at τ for (r, s) and (r', s') agree if and only if $\tau/\overline{\tau}$ is a root of unity of order dividing $r - r'$, and this can happen (Remark 5.5). ∎

Lemma 5.2 (The polarisation sits in the middle)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type. Then*

$$\iota(\tau)^* \eta = \mathrm{N}_{K/\mathbb{Q}}(\tau)\, \eta \qquad \text{for all } \tau \in K^\times,$$

so η lies in the summand $(r, s) = (1, 1)$ of (16) and η^n lies in the summand $(r, s) = (n, n)$, on which $K^\times$ acts by N^n.

Proof. Let E be the alternating form on $H_1(A, \mathbb{Q})$ attached to η. By condition (iii) of Definition 3.1, the Rosati involution $\dagger$ of η satisfies $\iota(\tau)^\dagger = \iota(\overline{\tau})$, so by the defining property of the Rosati involution

$$E(\iota(\tau)x,\, y) = E(x,\, \iota(\tau)^\dagger y) = E(x,\, \iota(\overline{\tau})y) \qquad \text{for all } x, y.$$

Therefore

$$\begin{aligned} (\iota(\tau)^* E)(x, y) &= E(\iota(\tau)x, \iota(\tau)y) = E(x, \iota(\overline{\tau})\iota(\tau)y) \\ &= E(x, \iota(\mathrm{N}(\tau))y) = \mathrm{N}(\tau)\, E(x, y), \end{aligned}$$

using $\overline{\tau}\tau = \mathrm{N}(\tau) \in \mathbb{Q}$ and the $\mathbb{Q}$-linearity of ι. Under the identification of $H^2(A, \mathbb{Q}) = \bigwedge^2 H^1(A, \mathbb{Q})$ with the alternating forms on $H_1(A, \mathbb{Q})$, which is compatible with pullback, η corresponds to E, so this is the displayed identity. Since $\mathrm{N}(\tau) = \tau\overline{\tau}$, Proposition 5.1 identifies the isotypic component of N in $H^2(A, \mathbb{C})$ with the summand $(r, s) = (1, 1)$, and η lies in it. Because $\iota(\tau)^*$ is a ring homomorphism, $\iota(\tau)^* \eta^n = \mathrm{N}(\tau)^n \eta^n$, and in the same way this places η^n in the summand (n, n). ∎

Lemma 5.3 (The Weil line sits at the ends)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type. Then*

$$W(A) \otimes_{\mathbb{Q}} \mathbb{C} = \left(\bigwedge^{2n} V_+\right) \oplus \left(\bigwedge^{2n} V_-\right),$$

the summands $(r, s) = (2n, 0)$ and $(0, 2n)$ of (16) in degree $2n$, on which $K^\times$ acts by $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \overline{\tau}^{2n}$.

Proof. This is Definition 3.6 together with Proposition 3.7(i), which says that $W(A)$ is the $\mathbb{Q}$-structure on that pair of lines; the characters are read off from Proposition 5.1 with $(r, s) = (2n, 0)$ and $(0, 2n)$. ∎

Lemma 5.4 (The two characters differ)**.** *For every $n \geq 1$ the characters $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \mathrm{N}_{K/\mathbb{Q}}(\tau)^n = (\tau\overline{\tau})^n$ of $K^\times$ are distinct, as are $\tau \mapsto \overline{\tau}^{2n}$ and N^n.*

Proof. This is the case $(r, s) = (2n, 0)$, $(r', s') = (n, n)$ of the last assertion of Proposition 5.1; we write the argument out in this case. For $\tau \neq 0$ the identity $\tau^{2n} = (\tau\overline{\tau})^n$ is equivalent to $(\tau/\overline{\tau})^n = 1$, so the two characters agree if and only if $(\tau/\overline{\tau})^n = 1$ for every $\tau \in K^\times$.

Consider the map $c\colon K^\times \to K^\times$, $c(\tau) = \tau/\overline{\tau}$. Two elements have the same image if and only if their ratio is fixed by conjugation, that is, lies in $\mathbb{Q}^\times$. The elements $m + \sqrt{-d}$ with $m \in \mathbb{Z}$ are pairwise non-proportional over $\mathbb{Q}$, so c takes infinitely many values.

On the other hand, $\{\zeta \in K : \zeta^n = 1\}$ is contained in the group $\mu(K)$ of roots of unity of K, and for an imaginary quadratic field $\mu(K)$ is finite: it has order 4 for $d = 1$, order 6 for $d = 3$, and order 2 otherwise. Since the image of c is infinite, it is not contained in this finite set, so some $\tau \in K^\times$ has $(\tau/\overline{\tau})^n \neq 1$, and the characters differ. The second pair is obtained from the first by conjugation. ∎

Remark 5.5 (Explicit witnesses)*.* The proof does not exhibit a τ at which the two characters differ, and the natural candidate $\tau = 1 + \sqrt{-d}$ is not always one. For $d = 1$ its ratio $\tau/\overline{\tau}$ is $\sqrt{-1}$, a primitive fourth root of unity, so the two characters agree at this τ whenever $4 \mid n$; for $d = 3$ the ratio is $(-1 + \sqrt{-3})/2$, a primitive cube root of unity, and they agree whenever $3 \mid n$. The candidate $\tau = 3 + \sqrt{-d}$ fails as well, at $d = 3$ with $6 \mid n$, where the ratio is $(1 + \sqrt{-3})/2$, a primitive sixth root of unity. In each case the characters agree at τ exactly when the order of the ratio divides n. The element $\tau = 2 + \sqrt{-d}$ is a witness for every d and every n. Its ratio is $(4 - d + 4\sqrt{-d})/(4 + d)$, which is not ± 1, and for $d \notin \{1, 3\}$ these are the only roots of unity in K; for $d = 1$ and $d = 3$ the ratios $(3 + 4\sqrt{-1})/5$ and $(1 + 4\sqrt{-3})/7$ are not roots of unity either, since their traces $6/5$ and $2/7$ are not integers. The same holds for $4 + \sqrt{-d}$ and $3 + 2\sqrt{-d}$. The finiteness argument in the proof covers all fields at once and needs no witness. Figure 4 shows the powers of these ratios for $d = 1$ and $d = 3$.

5.2. **The obstruction.**

Theorem 5.6 (Character obstruction)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type and write $H^1(A, \mathbb{Q}) \otimes \mathbb{C} = V_+ \oplus V_-$ for the decomposition into eigenspaces of $\iota(\sqrt{-d})$. Then*

$$H^{2n}(A, \mathbb{Q}) \otimes \mathbb{C} = \bigoplus_{r+s=2n} \bigwedge^r V_+ \otimes \bigwedge^s V_-,$$

with $\iota(\tau)$ acting on the (r, s) summand by $\tau^r \overline{\tau}^s$. The Weil line $W(A)$ spans the two extreme summands $(r, s) = (2n, 0)$ and $(0, 2n)$, on which $K^\times$ acts by τ^{2n} and $\overline{\tau}^{2n}$, while η^n lies in the middle summand $(r, s) = (n, n)$, on which $K^\times$ acts by the norm character $\mathrm{N}_{K/\mathbb{Q}}(\tau)^n$. In particular, for every $n \geq 1$,

$$W(A) \cap H^{2n}(A, \mathbb{Q})^{\mathrm{N}^n} = 0,$$

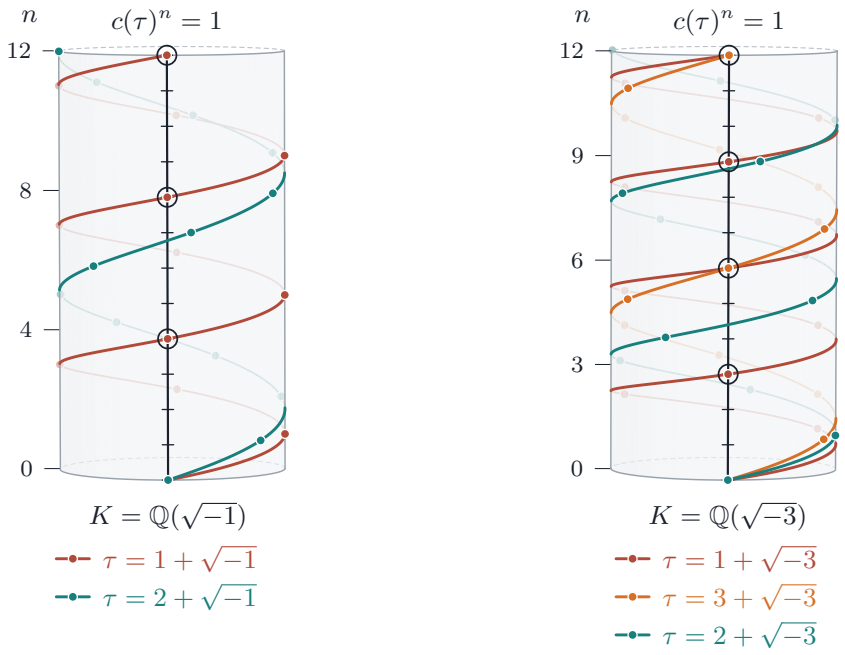


FIGURE 4. Remark 5.5. The values $c(\tau)^n = (\tau/\overline{\tau})^n$, $0 \le n \le 12$, on the unit circle, stacked over n as a helix; the characters τ^{2n} and N^n agree at τ exactly where the helix meets the vertical line $c^n = 1$ (circled). Left, $K = \mathbb{Q}(\sqrt{-1})$: the candidate $1+\sqrt{-1}$ returns every fourth step, and $2+\sqrt{-1}$ never. Right, $K = \mathbb{Q}(\sqrt{-3})$: $1+\sqrt{-3}$ returns every third step, $3+\sqrt{-3}$ every sixth, and $2+\sqrt{-3}$ never.

so a class that is a $K^\times$-eigenclass for N^n has zero component along the Weil line.

The theorem shows that the obstruction is structural. A construction that is equivariant for the K-action returns an eigenclass, so it must choose a character in advance, and the norm character is the one that the polarisation already occupies. The Weil line sits at the two ends of the $K^\times$-grading, as far from the middle as the grading allows; Figure 5 draws the grading and marks the two positions.

Proof of Theorem 5.6. The grading (16) and the action on its summands are given by Proposition 5.1, the position of $W(A)$ by Lemma 5.3, and that of η^n by Lemma 5.2. For the last assertion, write $H^{2n}(A,\mathbb{Q})^{\mathrm{N}^n}$ for the space of classes x with $\iota(\tau)^*x = \mathrm{N}(\tau)^n x$ for all $\tau \in K^\times$. After extension of scalars to $\mathbb{C}$ this is the isotypic component of the character N^n, which by Proposition 5.1 is the single summand $(r,s) = (n,n)$. By Lemma 5.4 the characters τ^{2n} and $\overline{\tau}^{2n}$ differ from N^n, so by Lemma 5.3 the space $W(A) \otimes \mathbb{C}$ meets the summand (n,n) in zero. Intersecting with $H^{2n}(A,\mathbb{Q})$ gives $W(A) \cap H^{2n}(A,\mathbb{Q})^{\mathrm{N}^n} = 0$. ∎

Corollary 5.7 (Equivariant classes)**.** *Let $x \in H^{2n}(A,\mathbb{Q})$ satisfy $\iota(\tau)^*x = \mathrm{N}(\tau)^n x$ for all $\tau \in K^\times$. Then the component of x along $W(A)$, with respect to any $K^\times$-stable complement of $W(A)$, is zero. In particular, x is a Weil class only if $x = 0$.*

Proof. Let C be a $K^\times$-stable complement of $W(A)$ in $H^{2n}(A,\mathbb{Q})$ and let p be the projection onto $W(A)$ along C. Since both summands are $K^\times$-stable, p commutes with every $\iota(\tau)^*$,

so $\iota(\tau)^* p(x) = \mathrm{N}(\tau)^n p(x)$ for all $\tau \in K^\times$. Hence $p(x)$ lies in $W(A) \cap H^{2n}(A,\mathbb{Q})^{\mathrm{N}^n}$, which is zero by Theorem 5.6. If x is a Weil class, then $x = p(x) = 0$. ∎

Corollary 5.7 is the form of the obstruction that applies to constructions. A construction that is natural in the K-action produces an eigenclass, whose character is determined by how the input transforms. The only character available to a rational eigenclass in $H^{2n}(A,\mathbb{Q})$ is N^n (Remark 5.8). By Lemma 5.2 this character is already occupied by the polarisation, and by Lemma 5.4 it is not a character of the Weil line.

Remark 5.8 (Rational eigenclasses). To land on the Weil line, a construction must produce classes on which $K^\times$ acts through $\tau \mapsto \tau^{2n}$ or $\tau \mapsto \overline{\tau}^{2n}$, and neither character is $\mathbb{Q}$-valued. This rules out rational eigenclasses. Indeed, let $0 \neq x \in H^k(A,\mathbb{Q})$ satisfy $\iota(\tau)^* x = \chi(\tau)\, x$ for all $\tau \in K^\times$. Then $\chi(\tau) \in \mathbb{Q}$, because $\iota(\tau)^*$ preserves $H^k(A,\mathbb{Q})$. By Proposition 5.1, x lies in a single summand (r,s) and $\chi(\tau) = \tau^r \overline{\tau}^{\,s}$. Since χ is $\mathbb{Q}$-valued, it equals its conjugate $\tau \mapsto \tau^s \overline{\tau}^{\,r}$, and the distinctness of the characters in Proposition 5.1 gives $r = s$. So every rational $K^\times$-eigenclass in $H^{2n}(A,\mathbb{Q})$ is an eigenclass for N^n, and Corollary 5.7 applies to it. A construction that reaches the Weil line must therefore produce a class that is not a $K^\times$-eigenclass. A nonzero element of $W(A)$ has nonzero components in both summands $(2n,0)$ and $(0,2n)$, since complex conjugation interchanges them, and its $K^\times$-orbit spans the $\mathbb{Q}$-plane $W(A)$. No rational eigenvalue condition isolates the Weil line: such conditions cut out only the diagonal summands (r,r), and the Weil line lies at the two ends of the grading.

Remark 5.9 (Determinant and character). A natural chain of statements leads to the opposite conclusion, that $K^\times$ acts on the Weil line through a power of the norm. The Weil line is $\bigwedge_K^{2n} V$, a free K-module of rank one; such a module is two-dimensional over $\mathbb{Q}$; and multiplication by τ on it is a $\mathbb{Q}$-linear map of determinant $\mathrm{N}(\tau)$. Each statement is true, but the last concerns the K-module structure and not the action of $\iota(\tau)$. By Lemma 5.3, $\iota(\tau)$ acts on the Weil line as multiplication by τ^{2n}, with determinant $\mathrm{N}(\tau)^{2n}$, and a space on which $K^\times$ acts through multiplication by τ^{2n} is not one on which it acts by the scalar $\mathrm{N}(\tau)^n$. The first is a K-line and the second a $\mathbb{Q}$-eigenspace, and they would agree only if $\tau^n = \overline{\tau}^{\,n}$ for all $\tau \in K^\times$, which Lemma 5.4 excludes. Figure 5 draws the distinction.

6. The cohomological Fourier–Mukai transform

With the character obstruction proved, we turn to the divisor obstruction of Theorem 7.3. Its one nontrivial ingredient is a computation: the cohomological Fourier–Mukai transform sends each power of the polarisation class to a rational multiple of a power of the polarisation class on the other side. We prove this in closed form. The computation takes place in a cohomology ring which, together with the classes involved, depends up to isomorphism only on the dimension, so one calculation covers every principally polarised abelian variety at once. Since the Todd class of an abelian variety is trivial, the passage from sheaves to cohomology is a formal step.

6.1. Grothendieck–Riemann–Roch on an abelian variety.

Proposition 6.1. *Let A be an abelian variety of dimension g and $\mathcal{G}$ an object of $D^b(A^\vee)$. Then*

$$\mathrm{ch}(\Phi_{\mathcal{P}}(\mathcal{G})) = \Phi_{\mathcal{P}}(\mathrm{ch}(\mathcal{G})),$$

where the transform on the right is the cohomological one, $\Phi_{\mathcal{P}}(x) = p_*(q^* x \cdot \mathrm{ch}(\mathcal{P}))$.

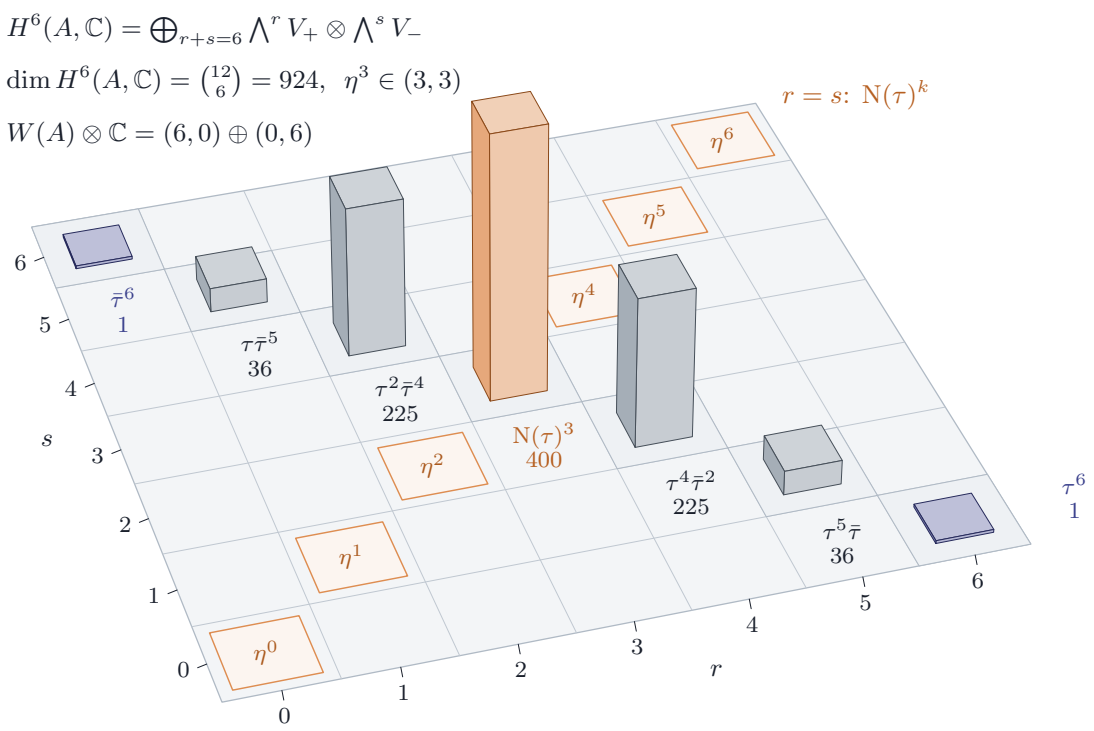


FIGURE 5. The $K^\times$-grading at $n = 3$ on the grid of pairs (r, s), $0 \le r, s \le 6$. The summands $\bigwedge^r V_+ \otimes \bigwedge^s V_-$ of $H^6(A,\mathbb{C})$ lie on the antidiagonal $r + s = 6$ and are drawn at their dimensions $\binom{6}{r}\binom{6}{s}$, which sum to $\binom{12}{6} = 924$; on the summand (r, s) the group $K^\times$ acts by $\tau^r\bar\tau^s$. The two end cells $(6, 0)$ and $(0, 6)$, with characters τ^6 and $\bar\tau^6$, span the Weil line; the middle cell $(3, 3)$, with character $\mathrm{N}(\tau)^3$, contains η^3. The outlined cells on the diagonal $r = s$ are the summands on which $K^\times$ acts through a power of the norm, and they carry the powers η^k. That the end cells and the middle cell differ is the content of Theorem 5.6.

Proof. Grothendieck–Riemann–Roch for the projection $p\colon A\times A^\vee \to A$, a proper morphism of smooth varieties, reads

$$\mathrm{ch}\big(Rp_*\mathcal{K}\big)\cdot \mathrm{td}(T_A) = p_*\big(\mathrm{ch}(\mathcal{K})\cdot \mathrm{td}(T_{A\times A^\vee})\big)$$

for $\mathcal{K} \in D^b(A \times A^\vee)$. The tangent bundle of an abelian variety is trivial, so $\mathrm{td}(T_A) = 1$ and $\mathrm{td}(T_{A\times A^\vee}) = 1$. Take $\mathcal{K} = Lq^*\mathcal{G} \otimes^L \mathcal{P}$, so that $Rp_*\mathcal{K} = \Phi_{\mathcal{P}}(\mathcal{G})$; the multiplicativity of the Chern character and $\mathrm{ch}(Lq^*\mathcal{G}) = q^*\,\mathrm{ch}(\mathcal{G})$ give the statement. ∎

6.2. **The universal model.**

Notation 6.2. Let A be a principally polarised abelian variety of dimension g with polarisation class θ, and put $V = H^1(A,\mathbb{Q})$, a $\mathbb{Q}$-vector space of dimension $2g$ carrying the symplectic form attached to θ. Choose a symplectic basis $e_1, \dots, e_{2g}$ of V for this form, so that

$$\theta = \sum_{j=1}^{g} e_j \wedge e_{g+j}. \tag{17}$$

The principal polarisation identifies $A^\vee$ with A and $H^1(A^\vee,\mathbb{Q})$ with the dual $V^\vee$; let $f_1,\dots,f_{2g}$ be the basis dual to $e_1,\dots,e_{2g}$, so that

$$\theta' = \sum_{j=1}^{g} f_j \wedge f_{g+j}. \tag{18}$$

Then $H^*(A\times A^\vee,\mathbb{Q}) = \bigwedge^*(V\oplus V^\vee)$.

Lemma 6.3 (The Poincaré class)**.** *In the model of Notation 6.2,*

$$\ell := c_1(\mathcal{P}) = \sum_{i=1}^{2g} e_i \wedge f_i \;\in\; V\otimes V^\vee \subset \bigwedge^2(V\oplus V^\vee), \tag{19}$$

the canonical element of $V\otimes V^\vee = \mathrm{Hom}(V,V)$ corresponding to the identity, and $\mathrm{ch}(\mathcal{P}) = e^{\ell}$.

Proof. The Poincaré bundle is normalised to be trivial on $A\times\{0\}$ and on $\{0\}\times A^\vee$. The components of $c_1(\mathcal{P})$ in $\bigwedge^2 V$ and in $\bigwedge^2 V^\vee$ are its restrictions to these two subvarieties, so they vanish, and $c_1(\mathcal{P})$ lies in the middle summand $V\otimes V^\vee$ of $\bigwedge^2(V\oplus V^\vee) = \bigwedge^2 V \oplus (V\otimes V^\vee)\oplus \bigwedge^2 V^\vee$. The defining property of $\mathcal{P}$ is that its restriction to $A\times\{\lambda\}$ is the line bundle λ; on cohomology this says that the induced map $H^1(A^\vee,\mathbb{Q})\to H^1(A,\mathbb{Q})^\vee$ obtained by contracting $c_1(\mathcal{P})$ is the canonical identification. Equivalently, the element of $\mathrm{Hom}(V,V)$ determined by $c_1(\mathcal{P})$ is the identity, which is (19). Since $\mathcal{P}$ is a line bundle, $\mathrm{ch}(\mathcal{P}) = e^{c_1(\mathcal{P})}$. ∎

Remark 6.4. Every principally polarised abelian variety of dimension g gives, up to isomorphism, the same data in Notation 6.2: a symplectic space of dimension $2g$ over $\mathbb{Q}$, its dual, the symplectic form, and the canonical element. The constants computed below therefore depend only on g and k, and not on A.

6.3. **The closed form.**

Theorem 6.5 (Transform of the polarisation powers)**.** *Let A be a principally polarised abelian variety of dimension g. Then for every $0\le k\le g$,*

$$\Phi_{\mathcal{P}}\big(\theta'^{\,k}\big) \;=\; (-1)^{\,g(g+1)/2\,+\,k}\,\frac{k!}{(g-k)!}\,\theta^{\,g-k}, \tag{20}$$

and $\Phi_{\mathcal{P}}(\theta'^{\,k}) = 0$ for $k>g$. In particular $\Phi_{\mathcal{P}}(\mathbb{Q}[\theta']) = \mathbb{Q}[\theta]$.

Proof. Work in the model of Notation 6.2. By definition $\Phi_{\mathcal{P}}(x) = p_*(q^*x\cdot e^{\ell})$, and p_* extracts the coefficient of the top exterior form $f_1\wedge\cdots\wedge f_{2g}$ in the $V^\vee$ variables: it sends $y\wedge f_1\wedge\cdots\wedge f_{2g}$ to y for $y\in\bigwedge^* V$, and kills every class of lower degree in the $V^\vee$ variables.

The classes $e_i\wedge f_i$ have even degree, so they commute, and they have square zero. Hence for every m,

$$\frac{\ell^m}{m!} = \sum_{\substack{S\subseteq\{1,\dots,2g\}\\ |S|=m}} \prod_{i\in S} e_i\wedge f_i, \tag{21}$$

the product being taken in increasing order of i. Likewise, from (18),

$$\frac{\theta'^{\,k}}{k!} = \sum_{\substack{T\subseteq\{1,\dots,g\}\\ |T|=k}} \prod_{j\in T} f_j\wedge f_{g+j}, \qquad \frac{\theta^{\,g-k}}{(g-k)!} = \sum_{\substack{U\subseteq\{1,\dots,g\}\\ |U|=g-k}} \prod_{j\in U} e_j\wedge e_{g+j}. \tag{22}$$

Fix k. In the product $\theta'^{\,k}\cdot e^{\ell}$, the terms that survive p_* are those of total degree $2g$ in the f variables. A term of (22) indexed by T carries the $2k$ variables $\{f_j, f_{g+j} : j\in T\}$,

and a term of (21) indexed by S carries the $|S|$ variables $\{f_i : i \in S\}$. For the product to be nonzero the two index sets must be disjoint, and for the total f-degree to be $2g$ we need

$$S = \{1, \dots, 2g\} \setminus (T \cup (g + T)), \qquad |S| = 2g - 2k.$$

So S is determined by T, only the term $\ell^{2g-2k}/(2g-2k)!$ of e^ℓ contributes, and the surviving terms are in bijection with the k-subsets T of $\{1, \dots, g\}$. Writing $U = \{1, \dots, g\} \setminus T$ we have $S = U \cup (g + U)$, so the V-part of the term indexed by T is $\prod_{i \in U \cup (g+U)} e_i$, which up to sign is the term of (22) indexed by U. The bijection $T \leftrightarrow U$ is the one between the terms of ${\theta'}^k/k!$ and those of $\theta^{g-k}/(g-k)!$.

It remains to compute the sign σ with which each term contributes, and to check that it is the same for all T. The sign is that of the rearrangement of the ordered word

$$\Big(\prod_{j \in T} f_j f_{g+j}\Big) \cdot \Big(\prod_{i \in S} e_i f_i\Big)$$

into the target word $\big(\prod_{j \in U} e_j e_{g+j}\big) \cdot (f_1 \cdots f_{2g})$, which uses the same generators. Relabelling the g symplectic pairs by a permutation π of $\{1, \dots, g\}$ is an automorphism of the exterior algebra. It carries the word for T to the word for $\pi(T)$, because the blocks $f_j f_{g+j}$ and $e_i f_i$ have degree two and may be reordered freely, and it carries the target word for U to the target word for $\pi(U)$, because reordering $f_{\pi(1)} \cdots f_{\pi(g)} f_{g+\pi(1)} \cdots f_{g+\pi(g)}$ costs the square of the sign of π. Since the permutations of $\{1, \dots, g\}$ act transitively on the k-subsets, $\sigma = \sigma(g, k)$, and we evaluate it at $T = \{1, \dots, k\}$. Put $r = g - k$. Moving the $2r$ variables e_i of the second factor to its front costs $(-1)^{r(2r-1)} = (-1)^r$, moving them further past the $2k$ variables of the first factor costs nothing, and arranging them as $\prod_{j \in U} e_j e_{g+j}$ costs $(-1)^{r(r-1)/2}$. The f variables are left in the order $f_1 f_{g+1} \cdots f_k f_{g+k}\, f_{k+1} \cdots f_g\, f_{g+k+1} \cdots f_{2g}$, and bringing them to $f_1 \cdots f_{2g}$ costs $(-1)^{k(k-1)/2 + rk}$. As $r + r(r-1)/2 + k(k-1)/2 + rk = g(g+1)/2 - k$, this gives

$$\sigma(g, k) = (-1)^{\,g(g+1)/2 \,+\, k}. \tag{23}$$

Assembling, and accounting for the factorials in (22),

$$\Phi_{\mathcal{P}}({\theta'}^k) = k! \cdot \sigma(g, k) \cdot \frac{\theta^{\,g-k}}{(g-k)!},$$

which is (20). For $k > g$ we have ${\theta'}^k = 0$, because ${\theta'}^k$ lies in $\bigwedge^{2k} V^\vee$, which vanishes for $2k > 2g$. The last assertion follows because the constants in (20) are nonzero for every $0 \le k \le g$. ∎

Remark 6.6 (The sign). The sign (23) depends only on g and k, by the transitivity argument in the proof, and the count at $T = \{1, \dots, k\}$ fixes it. For even $g = 2n$, the case of Weil type, $g(g+1)/2 = n(2n+1)$ has the parity of n, and (23) reads $\sigma(2n, k) = (-1)^{n+k}$.

Corollary 6.7 (Stability of the polarisation subring)**.** *Let A be a principally polarised abelian variety and let $x \in H^*(A^\vee, \mathbb{Q})$ lie in the subring $\mathbb{Q}[\theta']$ generated by θ'. Then $\Phi_{\mathcal{P}}(x) \in \mathbb{Q}[\theta]$.*

Proof. The ring $\mathbb{Q}[\theta']$ is spanned over $\mathbb{Q}$ by the powers ${\theta'}^k$, Theorem 6.5 sends each of them to a rational multiple of a power of θ, and $\Phi_{\mathcal{P}}$ is linear. ∎

Remark 6.8 (Non-principal polarisations). If θ is a polarisation of degree D^2 rather than a principal one, the isogeny $\phi_\theta \colon A \to A^\vee$ has degree D^2, and $\phi_\theta^* \theta' = D\theta$ for a suitable normalisation of θ'. The argument of Theorem 6.5 goes through with θ replaced by $D^{-1}\phi_\theta^* \theta'$, and the constants change by powers of D, all of which are nonzero rationals. The

statements below concern only membership in $\mathbb{Q}[\theta]$, not the values of the constants, so we state them for principal polarisations and use them in general.

7. The divisor obstruction

The first way of reaching a Weil class is to import an object from a projective space. One restricts an object $\mathcal{F}$ of $D^b(\mathbb{P}^{N-1})$ to the dual abelian variety through a morphism $i\colon A^\vee \to \mathbb{P}^{N-1}$ and carries the result back to A by the Fourier–Mukai transform $\Phi_{\mathcal{P}}$. Secant varieties of $A^\vee$ are the usual choice of $\mathcal{F}$, because a dimension count makes the codimension come out right, and this is the construction of the preprint cited on the title page. This section shows that no construction of this shape produces a Weil class. The proof needs, besides the closed form of Section 6, two facts about a morphism to projective space: the Chern character of a restricted object is a polynomial in the pulled-back hyperplane class, and this class is the first Chern class of the line bundle defining the morphism. When the Néron–Severi group of $A^\vee$ has rank one, the transform therefore lands in $\mathbb{Q}[\theta]$. We then give two proofs of Corollary 7.6, one through Proposition 3.10 and one through the characters of Section 5, and isolate the equivariance hypothesis on which the second depends.

7.1. Statement and proof.

Lemma 7.1 (Restriction from an ambient projective space)**.** *Let Y be a smooth projective variety, let $i\colon Y \to \mathbb{P}^{N-1}$ be a morphism defined by a linear system, let $h \in H^2(\mathbb{P}^{N-1}, \mathbb{Q})$ be the hyperplane class, and let $\mathcal{F} \in D^b(\mathbb{P}^{N-1})$. Then*

$$\operatorname{ch}(Li^*\mathcal{F}) \in \mathbb{Q}\big[\, i^*h \,\big] \subset H^*(Y, \mathbb{Q}).$$

Proof. The Chern character commutes with derived pullback: $\operatorname{ch}(Li^*\mathcal{F}) = i^* \operatorname{ch}(\mathcal{F})$. For a locally free sheaf this follows from the naturality of Chern classes. It extends to $D^b(\mathbb{P}^{N-1})$ because the Chern character is additive on distinguished triangles and, by Beilinson's theorem, every object of $D^b(\mathbb{P}^{N-1})$ is quasi-isomorphic to a finite complex whose terms are direct sums of line bundles $\mathcal{O}(-m)$ with $0 \le m \le N-1$. All that is needed is that the classes of line bundles generate $K_0(\mathbb{P}^{N-1})$, which also follows from the Koszul resolutions of the structure sheaves of linear subspaces. The cohomology ring of $\mathbb{P}^{N-1}$ is $\mathbb{Q}[h]/(h^N)$, so $\operatorname{ch}(\mathcal{F})$ is a polynomial in h with rational coefficients, and applying the ring homomorphism i^* gives a polynomial in i^*h. ∎

[III, Remark 2.3] records which other ambient varieties behave in the same way.

Lemma 7.2 (The hyperplane class of a linear system)**.** *Let A be an abelian variety and $i\colon A^\vee \to \mathbb{P}^{N-1}$ a morphism defined by a linear system $|M|$ with $M \in \operatorname{Pic}(A^\vee)$. Then $i^*h = c_1(M) \in \operatorname{NS}(A^\vee) \otimes \mathbb{Q}$.*

Proof. A morphism to $\mathbb{P}^{N-1}$ defined by the linear system $|M|$, or by a subsystem of it, satisfies $i^*\mathcal{O}(1) \cong M$ by construction. Hence $i^*h = c_1(i^*\mathcal{O}(1)) = c_1(M)$. ∎

Theorem 7.3 (Divisor obstruction)**.** *Let A be a principally polarised complex abelian variety of dimension g with $\operatorname{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q} \cdot \theta'$, let $i\colon A^\vee \to \mathbb{P}^{N-1}$ be any morphism defined by a linear system, and let $\mathcal{F}$ be any object of $D^b(\mathbb{P}^{N-1})$. Put $\mathcal{E} = \Phi_{\mathcal{P}}(Li^*\mathcal{F})$. Then*

$$\operatorname{ch}_k(\mathcal{E}) \in \mathbb{Q} \cdot \theta^k \qquad \text{for every } k \ge 0.$$

Proof. By Lemma 7.1, $\operatorname{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[i^*h]$, and by Lemma 7.2, $i^*h = c_1(M)$ for the line bundle M defining i. Since $\operatorname{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q}\theta'$, we have $c_1(M) = m\theta'$ for some $m \in \mathbb{Q}$, and therefore $\operatorname{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[\theta']$. By Proposition 6.1, $\operatorname{ch}(\mathcal{E})$ is the image of $\operatorname{ch}(Li^*\mathcal{F})$ under

the cohomological transform $\Phi_{\mathcal{P}}$, which carries $\mathbb{Q}[\theta']$ into $\mathbb{Q}[\theta]$ by Corollary 6.7. Hence $\mathrm{ch}(\mathcal{E}) \in \mathbb{Q}[\theta]$, and since $\mathbb{Q}[\theta] \cap H^{2k}(A,\mathbb{Q}) = \mathbb{Q}\theta^k$, taking the component in $H^{2k}(A,\mathbb{Q})$ gives the statement. ∎

Remark 7.4 (The hypothesis on the Néron–Severi group). The hypothesis $\mathrm{NS}(A^\vee)\otimes\mathbb{Q} = \mathbb{Q}\theta'$ holds at the very general point of any positive-dimensional family of abelian varieties of Weil type with $\mathrm{End}^0(A) = \iota(K)$, which is the case in which the Weil classes lie outside the divisor subring (Proposition 3.10); see Lemma 7.5. Without it, the conclusion is only that $\mathrm{ch}_k(\mathcal{E})$ lies in the image under $\Phi_{\mathcal{P}}$ of the divisor subring of $A^\vee$. Theorem 5.6 makes no hypothesis on NS, which is one reason for giving two proofs of Corollary 7.6.

Lemma 7.5. *Let $\mathcal{A} \to S$ be the universal family of abelian varieties of $(K,-1,n)$-Weil type over the corresponding moduli space, with $n \geq 2$. At a very general point $s \in S$ one has $\mathrm{End}^0(A_s) = \iota(K)$ and $\mathrm{NS}(A_s)\otimes\mathbb{Q} = \mathbb{Q}\eta_s$, of dimension one.*

Proof. The locus in S where End^0 is strictly larger than $\iota(K)$ is a countable union of proper closed subvarieties: each additional endomorphism imposes a nontrivial algebraic condition on the period point, and there are only countably many candidates, the endomorphisms of the lattice $H_1(A_s,\mathbb{Z})$. So $\mathrm{End}^0(A_s) = \iota(K)$ at a very general point s, and there the Hodge group is the full special unitary group $\mathrm{SU}(V,H)$ of the K-hermitian form H attached to the polarisation (Lemma 4.5), by Weil's computation of the Hodge group of a general abelian variety of Weil type [Wei77]. The space $\mathrm{Hdg}^1(A_s) = \mathrm{NS}(A_s)\otimes\mathbb{Q}$ is the space of Hodge-group invariants in $H^2(A_s,\mathbb{Q})$. By Proposition 5.1 in degree two, $H^2\otimes\mathbb{C} = \bigwedge^2 V_+ \oplus (V_+\otimes V_-) \oplus \bigwedge^2 V_-$. The group $\mathrm{SU}(V,H)$ acts on $V_+\otimes V_- \cong V_+\otimes V_+^\vee$ with a one-dimensional space of invariants, spanned by the class of H, and on $\bigwedge^2 V_\pm$ with no invariants, since $2n \geq 4$ and $\bigwedge^2$ of the standard representation of SL_{2n} has no invariant vector unless $2n = 2$. Hence the invariants are one-dimensional, and they are spanned by η_s. ∎

7.2. **Consequences for Weil classes.** Since a Weil class is by construction not in the subring generated by divisor classes, the theorem excludes Weil classes at once.

Corollary 7.6. *With notation as in Theorem 7.3, suppose in addition that $g = 2n$ and that (A,ι,η) is of $(K,-1,n)$-Weil type. Then $\mathrm{ch}_n(\mathcal{E})$ is a Weil class only if $\mathrm{ch}_n(\mathcal{E}) = 0$.*

The first proof passes through Proposition 3.10. The second replaces that proposition by the characters of Section 5 and explains in terms of representations why the first had to succeed: the $K^\times$-isotypic decomposition of $H^{2n}(A,\mathbb{Q})$ separates the two classes in question (Figure 6).

Proof of Corollary 7.6. We work under the hypothesis of Theorem 7.3. Let $g = 2n$ and let (A,ι,η) be of $(K,-1,n)$-Weil type with $\mathrm{NS}(A)\otimes\mathbb{Q} = \mathbb{Q}\eta$. Then θ is a nonzero multiple of η, and Theorem 7.3 gives $\mathrm{ch}_n(\mathcal{E}) \in \mathbb{Q}\cdot\theta^n = \mathbb{Q}\cdot\eta^n \subset D^{2n}(A)$. The proof of Proposition 3.10 uses the hypothesis $\mathrm{End}^0(A) = \iota(K)$ only to obtain $\mathrm{NS}(A)\otimes\mathbb{Q} = \mathbb{Q}\eta$, so it applies here and gives $W(A)\cap D^{2n}(A) = 0$. Hence if $\mathrm{ch}_n(\mathcal{E})$ is a Weil class, it lies in $W(A)\cap D^{2n}(A) = 0$. ∎

Second proof of Corollary 7.6, via characters. By Lemma 5.2 the class η^n is a $K^\times$-eigenclass for the character N^n, and hence so is every nonzero element of $\mathbb{Q}\eta^n$. As in the first proof, $\mathrm{ch}_n(\mathcal{E}) \in \mathbb{Q}\eta^n$ by Theorem 7.3, so $\mathrm{ch}_n(\mathcal{E})$ satisfies $\iota(\tau)^*\,\mathrm{ch}_n(\mathcal{E}) = \mathrm{N}(\tau)^n\,\mathrm{ch}_n(\mathcal{E})$ for all $\tau \in K^\times$. By Corollary 5.7 it is a Weil class only if it vanishes. ∎

The second proof uses Theorem 7.3, and with it the hypothesis on NS, only to know that $\mathrm{ch}_n(\mathcal{E})$ is an eigenclass for N^n; the rest of the argument uses nothing but this equivariance. We record the resulting statement, which does not mention the ambient space at all.

Theorem 7.7 (Equivariance suffices)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type and let $\mathcal{E} \in D^b(A)$ be an object such that for every τ in the order $\iota^{-1}(\mathrm{End}(A)) \subset K$ one has*

$$\iota(\tau)^* \,\mathrm{ch}_n(\mathcal{E}) = \mathrm{N}_{K/\mathbb{Q}}(\tau)^n \;\mathrm{ch}_n(\mathcal{E}).$$

Then $\mathrm{ch}_n(\mathcal{E})$ is a Weil class only if $\mathrm{ch}_n(\mathcal{E}) = 0$.

Proof. The hypothesis says that $\mathrm{ch}_n(\mathcal{E})$ is an eigenclass for N^n under the nonzero elements of the order, and it is then one under all of $K^\times$. Indeed, since the order spans K over $\mathbb{Q}$, every $\tau \in K^\times$ can be written as σ/m with σ in the order and m a nonzero integer. The endomorphism $\iota(m)$ is multiplication by m, which acts on $H^{2n}(A, \mathbb{Q})$ as $m^{2n} = \mathrm{N}(m)^n$. Hence $\iota(\tau)^* \,\mathrm{ch}_n(\mathcal{E}) = \mathrm{N}(\sigma)^n m^{-2n} \,\mathrm{ch}_n(\mathcal{E}) = \mathrm{N}(\tau)^n \,\mathrm{ch}_n(\mathcal{E})$, and Corollary 5.7 applies. ∎

Remark 7.8 (Comparison of the hypotheses). Theorem 7.3 requires the object to be restricted from an ambient projective space and imposes a condition on $\mathrm{NS}(A^\vee)$; Theorem 7.7 requires only that the resulting class be equivariant for the norm character. The second hypothesis is weaker, and it holds for any construction whose input is K-invariant. This includes the secant constructions, since the secant varieties of a K-invariant subvariety are K-invariant. We state both because the first can be checked by inspection of the construction, while the second requires an equivariance computation.

8. Secant numerology

The obstructions of Theorems 5.6 and 7.3 concern where a class can lie in cohomology, and neither of them uses a count of dimensions. A secant construction is also subject to such a count, and this section shows that the count obstructs nothing. It asks for a polarisation on $A^\vee$ whose Euler characteristic lies below the values at which the theorem of Lefschetz guarantees that a linear system is free of base points or very ample. Those values are sufficient conditions and not necessary ones, and already on abelian surfaces there are base-point-free and very ample systems below them.

8.1. The dimension count.

Setup 8.1 (The secant numerology)**.** Let A be of $(K, -1, n)$-Weil type, so that $\dim A = \dim A^\vee = 2n$. Suppose that $A^\vee$ is embedded in $\mathbb{P}^{N-1}$ by a complete linear system $|M|$ with M ample. An ample line bundle on an abelian variety has no higher cohomology, so $N = h^0(M) = \chi(M)$. The n-th secant variety of a nondegenerate $2n$-fold has expected dimension

$$\dim \mathrm{Sec}^n(A^\vee) = n(2n+1) - 1 = 2n^2 + n - 1,$$

and for a construction that intersects $\mathrm{Sec}^n(A^\vee)$ with $A^\vee$ in the expected dimension inside $\mathbb{P}^{N-1}$,

$$\dim\big(\mathrm{Sec}^n(A^\vee) \cap A^\vee\big) = \big(2n^2 + n - 1\big) + 2n - \big(N - 1\big) = 2n^2 + 3n - N.$$

Requiring this dimension to be n, so that the construction produces a class of the required codimension, forces

$$N = \chi(M) = 2n^2 + 2n. \tag{24}$$

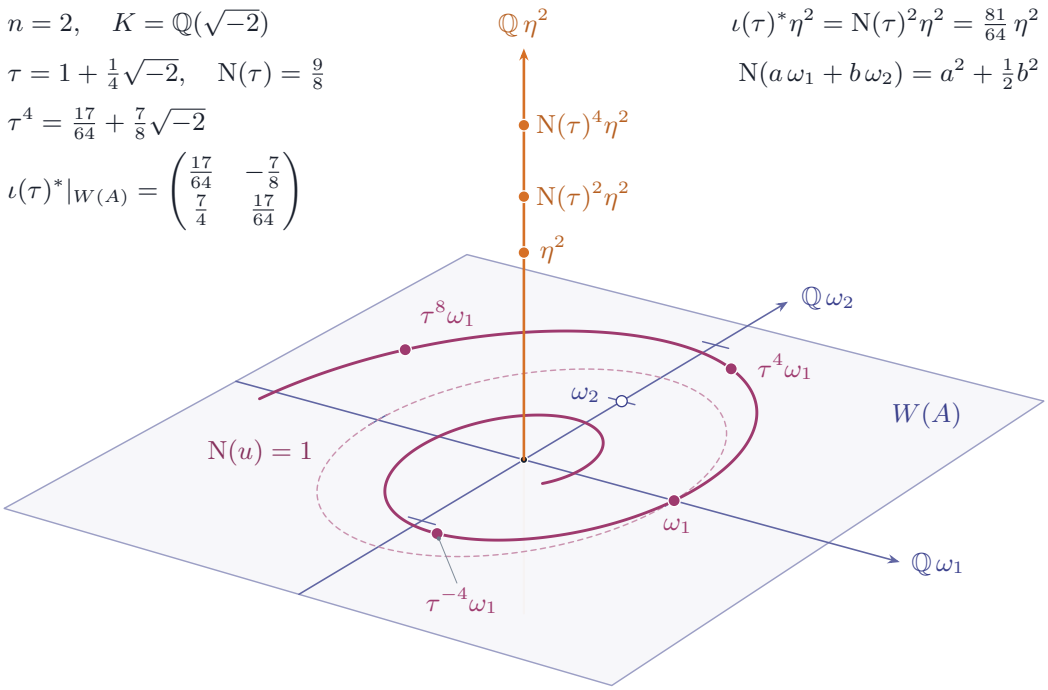


FIGURE 6. The linear algebra behind Theorem 7.3, drawn inside $H^{2n}(A, \mathbb{Q})$. The Weil line $W(A)$ is a plane, free of rank one over K and of rank two over $\mathbb{Q}$, and $K^\times$ acts on it through τ^{2n}, so the orbit of any nonzero class is a spiral that spans the plane and lies on no line in it. The transverse axis carries η^n, on which $K^\times$ acts through the norm. A construction returning a single eigenclass returns a line, and by Proposition 13.1 no line inside $W(A)$ is stable.

Lemma 8.2. *For every $n \geq 2$,*

$$2n^2 + 2n < 2^{2n} < 3^{2n},$$

and the ratio $2^{2n}/(2n^2 + 2n)$ tends to infinity.

Proof. The right inequality is clear. Since $2^{2n} = 4^n$, put $q(n) = 4^n/(2n^2 + 2n)$, so that $q(2) = 16/12 > 1$. The ratio of consecutive values is

$$\frac{q(n+1)}{q(n)} = \frac{4(2n^2 + 2n)}{2(n+1)^2 + 2(n+1)} = \frac{4n}{n+2},$$

which is at least 2 for $n \geq 2$. Hence $q(n) \geq 2^{n-2}q(2) > 1$ for every $n \geq 2$, and $q(n) \to \infty$. ∎

8.2. **The Lefschetz thresholds.** The theorem of Lefschetz says that for M ample on an abelian variety, $M^{\otimes 2}$ is free of base points and $M^{\otimes 3}$ is very ample [BL04, Section 4.5]. If $M = P^{\otimes 2}$ with P ample on an abelian variety of dimension m, then $\chi(M) = 2^m\chi(P) \geq 2^m$, since $\chi(L) = (L^m)/m!$ for every line bundle L by Riemann–Roch, and similarly $\chi \geq 3^m$ in the very ample case. Read as thresholds, these bounds would say that there is no base-point-free system below 2^m and no very ample system below 3^m. Both readings turn a sufficient condition into a necessary one, and both fail already for abelian surfaces.

Proposition 8.3 (The thresholds are not necessary)**.** *Let S be a general complex abelian surface, so $m = 2$.*

(i) *A polarisation of type $(1, 3)$ on S has $\chi = 3 < 4 = 2^2$ and its complete linear system is free of base points; the associated morphism $S \to \mathbb{P}^2$ is a six-sheeted covering.*

(ii) *A polarisation of type $(1, 5)$ on S has $\chi = 5 < 9 = 3^2$ and is very ample; the image is a Horrocks–Mumford abelian surface of degree 10 in $\mathbb{P}^4$* [HM73].

In particular neither 2^m nor 3^m is a lower bound for the Euler characteristic of a base-point-free, respectively very ample, ample line bundle on an abelian variety of dimension m.

Proof. By Riemann–Roch on an abelian surface, a polarisation L of type $(1, d)$ has $\chi(L) = (L^2)/2 = d$. The remaining assertions belong to the classification of $(1, d)$-polarised abelian surfaces; see Gross and Popescu [GP98, §1.2], where the base locus of $|L|$ is determined for each d: it is empty for $d \geq 3$, consists of four points for $d = 2$, and $|L|$ is very ample for $d \geq 5$ on a general surface, the case $d = 5$ being the Horrocks–Mumford embedding of [HM73]. In that case the image has degree $(L^2) = 10$. For $d = 3$ the morphism $S \to \mathbb{P}^2$ defined by $|L|$ contracts no curve, because L is ample, so it is finite and surjective, of degree $(L^2) = 6$. ∎

Corollary 8.4. *The count of Setup 8.1 leads to no contradiction. The requirement $\chi(M) = 2n^2 + 2n$ of (24) is compatible with M being very ample for all n for which a polarisation of that Euler characteristic and suitable type exists on a $2n$-dimensional abelian variety. At $n = 2$ the requirement $\chi = 12$ on an abelian fourfold is met by polarisations of type $(1, 1, 2, 6)$ and $(1, 1, 1, 12)$, and nothing in the numerology excludes very ampleness for either.*

Remark 8.5 (The threshold reading). The supplementary material of the preprint cited on the title page asserts that the polarisation required by (24) admits no base-point-free linear system, on the basis of a table of the values $2n^2 + 2n$, 2^{2n} and 3^{2n}. The table is correct, and Lemma 8.2 reproduces it, but by Proposition 8.3 the inference drawn from it does not hold, and the claim is withdrawn. The obstructions of Theorems 5.6 and 7.3 are of a different nature: they bound where a class can lie in a decomposition of cohomology, and no choice of polarisation, ambient space or object affects them.

9. The semiregularity target

Markman's construction at $n = 2$ deforms a sheaf across a family and controls the obstruction by a semiregularity argument in the sense of Bloch and of Buchweitz and Flenner [Blo72, BF03]. As for the secant construction of Section 8, one may ask whether a dimension count blocks this route for $n \geq 4$. We compute the target of the semiregularity map and show that it grows too fast for the count to impose any such constraint, so that neither route meets a numerical obstruction. The constraint that does exist is structural. In Section 9.2 we show that the annihilator of the Weil line in $H^{0,2}(A)$ has dimension n^2, and that it prevents every object whose Chern character is concentrated on the Weil line from being semiregular as soon as $H^2(\mathcal{O}_A)$ acts on the object faithfully. In Section 9.3 we identify this annihilator with the tangent space to the Weil family.

9.1. The dimension of the target.

Lemma 9.1 (Hodge numbers of an abelian variety)**.** *For an abelian variety A of dimension g one has $h^{p,q}(A) = \binom{g}{p}\binom{g}{q}$.*

Proof. By (1) and the compatibility of the Hodge decomposition with the exterior algebra structure, $H^{p,q}(A) = \bigwedge^p H^{1,0} \otimes \bigwedge^q H^{0,1}$, and $\dim H^{1,0} = \dim H^{0,1} = g$. ∎

Proposition 9.2 (The semiregularity target)**.** *Let A be an abelian variety of dimension $g = 2n$. The target of the semiregularity map, namely $\bigoplus_q H^{q+2}(A, \Omega_A^q)$, has dimension*

$$\sum_{q \geq 0} h^{q,q+2}(A) = \binom{4n}{2n-2}.$$

Proof. By Lemma 9.1 with $g = 2n$,

$$\sum_{q \geq 0} h^{q,q+2} = \sum_{q \geq 0} \binom{2n}{q}\binom{2n}{q+2} = \sum_{q \geq 0} \binom{2n}{q}\binom{2n}{2n-q-2} = \binom{4n}{2n-2},$$

the last step being the Vandermonde convolution. ∎

Corollary 9.3 (The dimension count)**.** *The target dimension $\binom{4n}{2n-2}$ takes the values $28, 495, 8008, 125970, 1961256, 30421755$ for $n = 2, \dots, 7$, growing by a factor exceeding ten at each step. Any source whose dimension grows polynomially in n, for instance the quadratic quantity $\binom{2n}{2} = n(2n-1)$, which takes the values $6, 15, 28, 45, 66, 91$ over the same range, is eventually negligible against it. In particular, for large n a comparison of the two dimensions cannot obstruct the existence of a semiregular sheaf; the inequality it produces points the other way.*

Remark 9.4 (Consistency at $n = 2$). At $n = 2$ the target has dimension 28, and the sheaf produced by Markman has $\dim \mathrm{Ext}^2 = 12$, well below it, which is consistent with the construction succeeding there. Nothing in the numbers changes character between $n = 2$ and $n = 4$, so if the construction cannot be carried out for larger n, the reason is not this comparison.

Remark 9.5 (The remaining requirement). Corollary 9.3 does not show that the semiregularity route works. It removes one apparent obstruction and leaves the actual requirement in place, namely the construction of a sheaf on A whose Chern character has a nonzero component along the Weil line. By Theorem 5.6 such a sheaf cannot be K-equivariant with the norm character, so it cannot be produced by any functorial construction from K-invariant data on $A^\vee$; Markman's sheaf at $n = 2$ is not produced in this way. In this sense the problem for $n \geq 3$ is one of breaking equivariance, and the dimensions play no role in it.

9.2. **The annihilator of the Weil line and objects of rank zero.** By Remark 9.5, what remains is to construct a sheaf whose Chern character has a component along the Weil line. The most economical way to arrange this is to make the Chern character lie *entirely* on the Weil line, that is, to take an object of rank zero with every ch_k vanishing except ch_n, as the construction of Section 14 does. We show that this choice is incompatible with semiregularity whenever $H^2(\mathcal{O}_A)$ acts faithfully on the object, for a reason that depends only on the position of the Chern character and not on how the object is built.

We first recall the semiregularity map, on which the rest of this subsection turns. For a coherent sheaf $\mathcal{E}$ on a smooth projective variety A the semiregularity map is defined through the Atiyah class $\mathrm{At}(\mathcal{E}) \in \mathrm{Ext}^1(\mathcal{E}, \mathcal{E} \otimes \Omega_A^1)$: a class $\xi \in \mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ is sent to the trace of $\xi \cdot \exp(\mathrm{At}(\mathcal{E}))$, and the component of that trace in $H^{q+2}(A, \Omega_A^q) = H^{q,q+2}(A)$ is the q-th coordinate of $\sigma_{\mathcal{E}}(\xi)$. Bloch introduced the map for local complete intersections [Blo72], and Buchweitz and Flenner extended it to arbitrary coherent sheaves and perfect

complexes [BF03]. If A moves in a family $\mathcal{A} \to S$, the obstruction to extending $\mathcal{E}$ to first order along $v \in T_s S$ is a class $\mathrm{ob}_v(\mathcal{E}) \in \mathrm{Ext}^2(\mathcal{E},\mathcal{E})$, and the theorem of Buchweitz and Flenner identifies $\sigma_{\mathcal{E}}(\mathrm{ob}_v(\mathcal{E}))$ with the derivative of $\mathrm{ch}(\mathcal{E})$ in the direction v, projected off the diagonal of the Hodge decomposition. Hence if $\mathrm{ch}(\mathcal{E})$ stays of Hodge type along S, then $\sigma_{\mathcal{E}}(\mathrm{ob}_v(\mathcal{E})) = 0$, and if $\sigma_{\mathcal{E}}$ is also injective, then $\mathrm{ob}_v(\mathcal{E}) = 0$ and $\mathcal{E}$ deforms. Injectivity thus converts a statement about the Chern character into a statement about the sheaf, and it is the one hypothesis in the argument that sees more than $\mathrm{ch}(\mathcal{E})$.

Two special cases are all that is needed below. For a line bundle L one has $\mathrm{Ext}^2(L,L) = H^2(\mathcal{O}_A) = H^{0,2}(A)$, and σ_L is multiplication by $e^{c_1(L)}$. For a direct sum, Ext^2 is the sum of the $\mathrm{Ext}^2(\mathcal{E}_i,\mathcal{E}_j)$, and σ acts on each diagonal block $H^2(\mathcal{O}_A)\cdot \mathrm{id}_{\mathcal{E}_i}$ by multiplication by the Chern character of that summand. Both formulae follow at once from the definition by the trace.

Lemma 9.6 (The annihilator of the Weil line in $H^{0,2}$)**.** *Let A be of $(K,-1,n)$-Weil type and let $\omega \in W(A)\otimes\mathbb{R}$ be any nonzero real element of the Weil line. Then the multiplication map*

$$H^{0,2}(A) \longrightarrow H^{n,n+2}(A), \qquad z \longmapsto z\cdot\omega,$$

has kernel of dimension exactly n^2, spanned by the products $u\wedge v$ with $u \in V_+^{0,1}$ and $v \in V_-^{0,1}$. In particular the kernel is the same subspace for every ω on the line.

Proof. Write $\omega = a\,\alpha_+ + b\,\alpha_-$ with $\alpha_\pm$ generating $\bigwedge^{2n} V_\pm$, which is possible because $W(A)\otimes\mathbb{C} = \bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ (see the proof of Proposition 3.7). Complex conjugation interchanges V_+ and V_-, so $\overline{\alpha}_+ = \mu\,\alpha_-$ for some $\mu \neq 0$, and comparing ω with $\overline{\omega} = \omega$ gives $b = \overline{a}\,\mu$. Since $\omega \neq 0$, the coefficients a and b are therefore both nonzero. As $H^{0,1} = V_+^{0,1} \oplus V_-^{0,1}$ with each summand of dimension n, the space $H^{0,2} = \bigwedge^2 H^{0,1}$ decomposes as

$$H^{0,2} = \bigwedge^2 V_+^{0,1} \;\oplus\; (V_+^{0,1}\otimes V_-^{0,1}) \;\oplus\; \bigwedge^2 V_-^{0,1},$$

of dimensions $\binom{n}{2}$, n^2 and $\binom{n}{2}$. For $u \in V_+^{0,1}$ and $v \in V_-^{0,1}$ the mixed product $u\wedge v$ satisfies $u\wedge v\wedge\alpha_+ = 0$, because $u\wedge\alpha_+$ lies in $\bigwedge^{2n+1} V_+ = 0$, and $u\wedge v\wedge\alpha_- = 0$ for the same reason with v. So the middle summand lies in the kernel.

Conversely, let $z = z_{++} + z_{+-} + z_{--}$ be in the kernel. For the same reason $z_{++}\wedge\alpha_+$, $z_{--}\wedge\alpha_-$ and $z_{+-}\wedge\omega$ vanish, so $z\cdot\omega = a\,z_{--}\wedge\alpha_+ + b\,z_{++}\wedge\alpha_-$. If $n = 1$ the outer summands are zero and there is nothing to prove. If $n \geq 2$ the two terms lie in the distinct summands $\bigwedge^{2n} V_+ \otimes \bigwedge^2 V_-$ and $\bigwedge^2 V_+ \otimes \bigwedge^{2n} V_-$ of $\bigwedge^{2n+2} V_{\mathbb{C}} = \bigwedge^{2n+2}(V_+\oplus V_-)$, so each of them vanishes. Wedging with a generator of $\bigwedge^{2n} V_+$ is injective on $\bigwedge^2 V_-$, and wedging with a generator of $\bigwedge^{2n} V_-$ is injective on $\bigwedge^2 V_+$. Since $a \neq 0$ we get $z_{--} = 0$, and since $b \neq 0$ we get $z_{++} = 0$, so the kernel is the middle summand. ∎

The reality of ω is needed: for the non-real element $\omega = \alpha_+$ of $W(A)\otimes\mathbb{C}$ the same computation gives the kernel $V_+^{0,1}\otimes V_-^{0,1} \oplus \bigwedge^2 V_+^{0,1}$, of dimension $n^2 + \binom{n}{2}$. Figure 8 draws the mixed block that forms the kernel.

Theorem 9.7 (The rank-zero obstruction)**.** *Let A be of $(K,-1,n)$-Weil type. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be coherent sheaves on A such that*

$$\mathrm{ch}(\mathcal{E}_1) - \mathrm{ch}(\mathcal{E}_2) = c\,\omega, \qquad c \neq 0, \quad \omega \in W(A),$$

such that $\mathrm{Ext}^2(\mathcal{E}_1,\mathcal{E}_2) = \mathrm{Ext}^2(\mathcal{E}_2,\mathcal{E}_1) = 0$, and such that the map $H^2(\mathcal{O}_A) \to \mathrm{Ext}^2(\mathcal{E}_1,\mathcal{E}_1)$, $z \mapsto z\cdot\mathrm{id}_{\mathcal{E}_1}$, is injective, as it is whenever $\mathrm{rk}\,\mathcal{E}_1 \neq 0$. Put $\mathcal{E} = \mathcal{E}_1 \oplus \mathcal{E}_2$. Then the

semiregularity map

$$\sigma_{\mathcal{E}}\colon\ \operatorname{Ext}^2(\mathcal{E},\mathcal{E}) \longrightarrow \bigoplus_{q\geq 0} H^{q,q+2}(A)$$

has kernel of dimension at least n^2. In particular $\mathcal{E}$ is not semiregular, for any $n \geq 1$.

Proof. The vanishing of the off-diagonal Ext^2 groups gives $\operatorname{Ext}^2(\mathcal{E},\mathcal{E}) = \operatorname{Ext}^2(\mathcal{E}_1,\mathcal{E}_1) \oplus \operatorname{Ext}^2(\mathcal{E}_2,\mathcal{E}_2)$, and each summand receives $H^2(\mathcal{O}_A)$ through the $\mathcal{O}_A$-module structure, $z \mapsto z \cdot \mathrm{id}_{\mathcal{E}_i}$. For $z \in H^2(\mathcal{O}_A) = H^{0,2}(A)$ consider the element $\xi_z = z \cdot \mathrm{id}_{\mathcal{E}_1} - z \cdot \mathrm{id}_{\mathcal{E}_2}$. Its component in $\operatorname{Ext}^2(\mathcal{E}_1,\mathcal{E}_1)$ is $z \cdot \mathrm{id}_{\mathcal{E}_1}$, so by hypothesis $z \mapsto \xi_z$ is injective. The semiregularity map is compatible with direct sums, and on $H^{0,2} \cdot \mathrm{id}_{\mathcal{E}_i}$ it is multiplication by the Chern character of $\mathcal{E}_i$, so

$$\sigma_{\mathcal{E}}(\xi_z) = z \cdot \operatorname{ch}(\mathcal{E}_1) - z \cdot \operatorname{ch}(\mathcal{E}_2) = c\, z \cdot \omega.$$

If $\omega = 0$ this vanishes for every z, and $\dim H^{0,2} = n(2n-1) \geq n^2$. Otherwise ω, being rational, is a nonzero real element of the Weil line, and by Lemma 9.6 the product vanishes for every z in an n^2-dimensional subspace of $H^{0,2}$. The corresponding ξ_z are linearly independent, and hence $\dim \ker \sigma_{\mathcal{E}} \geq n^2$.

It remains to justify the claim in the statement that the hypothesis on $\mathcal{E}_1$ holds in positive rank. For a coherent sheaf $\mathcal{F}$ the map $H^2(\mathcal{O}_A) \to \operatorname{Ext}^2(\mathcal{F},\mathcal{F})$, $z \mapsto z \cdot \mathrm{id}_{\mathcal{F}}$, is the edge map of the local-to-global spectral sequence, and $\operatorname{tr}(z \cdot \mathrm{id}_{\mathcal{F}}) = \operatorname{rk}(\mathcal{F})\, z$. When $\operatorname{rk}\mathcal{F} \neq 0$ the trace therefore splits the map up to the factor $\operatorname{rk}\mathcal{F}$, and the map is injective; this applies in particular to a direct sum of line bundles. For a sheaf of rank zero the map can vanish: for a skyscraper sheaf it factors through H^2 of the structure sheaf of a point, which is zero. The direct sum $\mathcal{E}_1 \oplus \mathcal{E}_2$ is used only to produce an element of Ext^2 whose image under σ is $z \cdot (\operatorname{ch}\mathcal{E}_1 - \operatorname{ch}\mathcal{E}_2)$ rather than $z \cdot (\operatorname{ch}\mathcal{E}_1 + \operatorname{ch}\mathcal{E}_2)$. ∎

Corollary 9.8 (The construction of Section 14)**.** *Let $L_1, \dots, L_m$ be line bundles on A with $\sum_i c_1(L_i) = 0$ such that the virtual class $\sum_i([L_i] - [L_i^{-1}])$ has Chern character on the Weil line and every difference $\varepsilon\, c_1(L_i) - \varepsilon' c_1(L_j)$ with $\varepsilon, \varepsilon' \in \{\pm 1\}$ and $(\varepsilon, i) \neq (\varepsilon', j)$ is nondegenerate of index different from 2. Then the sheaf $\bigoplus_i (L_i \oplus L_i^{-1})$ has semiregularity kernel of dimension at least n^2. At $n = 3$ the explicit object of Section 14 has kernel of dimension exactly $15 = n^2 + n(n-1)$, of which the $n^2 = 9$ dimensions on the odd side are the ones supplied by Theorem 9.7; the remaining $n(n-1) = 6$ come from the even side and are specific to the line-bundle structure.*

Proof. Take $\mathcal{E}_1 = \bigoplus L_i$ and $\mathcal{E}_2 = \bigoplus L_i^{-1}$. Then $\operatorname{Ext}^2(\mathcal{E}_1,\mathcal{E}_2) = \bigoplus_{i,j} H^2(L_i^{-1} \otimes L_j^{-1})$ and $\operatorname{Ext}^2(\mathcal{E}_2,\mathcal{E}_1) = \bigoplus_{i,j} H^2(L_i \otimes L_j)$, and the first Chern classes $\mp(c_1(L_i) + c_1(L_j))$ occurring here, for all i and j, are differences of the kind in the hypothesis, with $\varepsilon' = -\varepsilon$. A nondegenerate line bundle has cohomology only in the degree equal to its index, so the index hypothesis gives the required Ext^2 vanishing. The sheaf $\mathcal{E}_1$ has rank $m \neq 0$, and the Chern character hypothesis is that of Theorem 9.7. Since e^{λ} and $e^{-\lambda}$ combine into $\cosh\lambda$ and $\sinh\lambda$, which occupy disjoint degrees, the map splits by parity, and the odd part is the annihilator of Lemma 9.6 tensored with the sign vector. At $n = 3$ the dimensions are read off from the proof of [II, Proposition 4.14], where the signs are absorbed in the same way: the kernel consists of the vectors $(u_i^{-k} z)_i$ with z in the mixed block $V_+^{0,1} \otimes V_-^{0,1}$ for $k = 3$, which change sign under $u_i \mapsto -u_i$, and with z in $\bigwedge^2 V_+^{0,1}$ or $\bigwedge^2 V_-^{0,1}$ for $k = 2$ or $k = 4$, which do not. ∎

Remark 9.9 (Uniformity over the configurations). By Corollary 9.8 the kernel has dimension at least n^2 for every configuration of line bundles $L_i^{\pm 1}$ whose virtual Chern character

lies on the Weil line and whose differences are nondegenerate of index different from 2. The deficiency is therefore not special to the six line bundles of [II, Proposition 4.14]: Theorem 9.7 produces it from the vanishing of the rank alone.

Theorem 9.7 excludes the economical choice described at the start of this subsection. An object on which $H^2(\mathcal{O}_A)$ acts faithfully and which realises a Weil class through semiregularity cannot have its Chern character concentrated on the Weil line. It must carry a nonzero rank, or nonzero Chern character in other degrees, so that $z \mapsto z \cdot \mathrm{ch}(\mathcal{E})$ is injective on $H^{0,2}$; the degree-two component of that product is $\mathrm{rk}(\mathcal{E}) \cdot z$, and rank zero is the case in which it vanishes. Markman's sheaves have nonzero rank, and this is essential to his method: the class he shows to be deformation-invariant is the normalised character $\exp(-c_1(\mathcal{E})/\operatorname{rk}\mathcal{E})\,\mathrm{ch}(\mathcal{E})$, which divides by the rank, and the objects at $n = 3$ are reflexive sheaves [Mar25a, Mar25b]. This narrows the fourth ingredient of Remark 14.16: it is a single sheaf of positive rank whose Chern character has a Weil component alongside other components, rather than a difference of two sheaves whose classes cancel outside the Weil line.

Theorem 9.7 is stated for a direct sum because there the element ξ_z is explicit. The mechanism applies to any perfect complex on which $H^2(\mathcal{O}_A)$ acts faithfully, and in particular to the object used in Section 12.

Proposition 9.10 (Complexes with concentrated Chern character)**.** *Let A be of $(K, -1, n)$-Weil type and let E be a nonzero perfect complex on A with $\mathrm{ch}_k(E) = 0$ for $k \neq n$ and $\mathrm{ch}_n(E) = N\omega$ for some $N \neq 0$ and some nonzero $\omega \in W(A)$. If the natural map*

$$H^2(\mathcal{O}_A) \longrightarrow \mathrm{Ext}^2(E, E), \qquad z \longmapsto z \cdot \mathrm{id}_E, \tag{25}$$

is injective, then $\dim \ker \sigma_E \geq n^2$, so E is not semiregular. The map (25) is injective whenever E is represented by a virtual sum of line bundles, in particular at every base point of Corollary 15.4, Theorem 15.25 and Theorem 15.10.

Proof. For any perfect complex the semiregularity map satisfies $\sigma_E(z \cdot \mathrm{id}_E) = z \cdot \mathrm{ch}(E)$, since $\mathrm{tr}\big(z \cdot \mathrm{id}_E \circ \exp \mathrm{At}(E)\big) = z \cdot \mathrm{tr} \exp \mathrm{At}(E) = z \cdot \mathrm{ch}(E)$. Hence for z in the annihilator of ω in $H^{0,2}(A)$,

$$\sigma_E(z \cdot \mathrm{id}_E) = N\, z \cdot \omega = 0,$$

and that annihilator has dimension n^2 by Lemma 9.6. If (25) is injective, the classes $z \cdot \mathrm{id}_E$ for z in a basis of the annihilator are linearly independent, which gives the bound.

For the last sentence, write the class of E in $K_0(A)$ as $\sum_i \varepsilon_i [L_i]$ with $\varepsilon_i = \pm 1$ and L_i line bundles, and realise it by the direct sum of the L_i with $\varepsilon_i = 1$ and of the shifts $L_i[1]$ of those with $\varepsilon_i = -1$. The map (25) is then $z \mapsto (z, \dots, z)$ into $\bigoplus_i \mathrm{Ext}^2(L_i, L_i) = \bigoplus_i H^2(\mathcal{O}_A)$, which is injective. At a split member ω_1 is a polynomial in $\beta, \widehat{\beta}, \ell$ by Theorem 15.2(iii), at a quaternionic member it is a polynomial in divisor classes by Theorem 15.25, and at a product it is one by Theorem 15.10; in each case the Chern character is a ring expression in classes of line bundles, so such a representation exists. ∎

Proposition 9.11 (The class forces the size of the object)**.** *Let A be of $(K, -1, n)$-Weil type, let ω be a nonzero element of the Weil line and let E be a perfect complex on A with*

$$\mathrm{ch}(E) = r + N\omega, \qquad r \in H^0(A, \mathbb{Q}),\ N \neq 0,$$

and no other component. Then

$$\chi(E, E) \;=\; (-1)^n N^2 \int_A \omega^2,$$

independently of r, and consequently

$$\sum_i \dim \operatorname{Ext}^i(E,E) \;\geq\; N^2\Big|\int_A \omega^2\Big| \;>\; 0.$$

Proof. The Todd class of an abelian variety is 1, so Riemann–Roch for the Ext algebra reads $\chi(E,E) = \int_A \operatorname{ch}(E)^\vee \operatorname{ch}(E)$ with $\operatorname{ch}_k^\vee = (-1)^k \operatorname{ch}_k$. Expanding,

$$\operatorname{ch}(E)^\vee \operatorname{ch}(E) = r^2 + (1+(-1)^n) r N\omega + (-1)^n N^2\omega^2,$$

and of the three terms only the last has degree $4n$, so only it survives the integral; the rank enters solely through r^2, which sits in degree zero. For the inequality, $\chi(E,E) = \sum_i (-1)^i \dim \operatorname{Ext}^i(E,E)$, so the total dimension is at least $|\chi(E,E)|$. Finally $\int_A \omega^2 \neq 0$. Write $\omega = \alpha_+ + \alpha_-$ with $\alpha_\pm \in \bigwedge^{2n} V_\pm$, both nonzero by the proof of Lemma 9.6. Each $\alpha_\pm$ is a top form on its own eigenspace and squares to zero, so $\omega^2 = 2\alpha_+\alpha_-$, which is a nonzero multiple of the fundamental class. ∎

Remark 9.12 (The constant). The bound concerns the whole Ext algebra and does not depend on the rank, so increasing the rank does not weaken it. The constant $|\int \omega^2|$ depends on the normalisation of ω; measured against η it picks up factors of d and of the polarisation type, and no dimension-free growth rate can be read off from it. The bound uses only that the Weil class is not isotropic. Together with Proposition 9.10 it constrains an object carrying the class in two ways. If the Chern character is concentrated on the Weil line, the semiregularity map has a kernel of dimension at least n^2 unless $H^2(\mathcal{O}_A)$ acts on the object degenerately; and whatever the rank, the Ext algebra has dimension at least $N^2|\int \omega^2|$.

Remark 9.13 (Consequences for the semiregularity reduction). [II, Theorem 2.3] proves $\Sigma = \mathcal{D}_{n,n}$, that is, the algebraicity of the Weil class on the whole family, under the hypothesis that σ_E is injective for the complex E of [II, Proposition 2.1], whose Chern character is $N\omega_1$ in degree n and zero elsewhere. This complex satisfies the hypotheses of Proposition 9.10 at every base point of Corollary 15.4, Theorem 15.25 and Theorem 15.10, so there the hypothesis of [II, Theorem 2.3] fails for it, by n^2 dimensions.

This does not make [II, Theorem 2.3] vacuous, for two reasons. First, the hypothesis concerns a chosen perfect complex representing a class in $K_0(A)_\mathbb{Q}$, and distinct representatives of one class have distinct Ext^2 groups; what is ruled out is the representative that the construction supplies. Second, Proposition 9.10 needs (25) to be injective, which is a real restriction for an object of rank zero: it fails, for instance, for a sheaf supported at a point. A representative escaping Proposition 9.10 must therefore receive $H^2(\mathcal{O}_A)$ non-injectively, which for a rank-zero object means that its support is degenerate. Of the two reductions of Section 12, the degree bound of Theorem 16.9 is untouched by anything here, and the semiregularity reduction of [II, Theorem 2.3] is unavailable for the complexes that the constructions at these base points supply.

9.3. **The annihilator and the tangent space to the family.** The number n^2 of Lemma 9.6 is also the dimension of the Weil family. We show in this subsection that the polarisation identifies the annihilator with the tangent space to the family, canonically and in every dimension. Two consequences follow. First, the deficiency of the semiregularity map at an object of rank zero cannot be removed by a better choice of object, since it is the family itself, seen inside $H^{0,2}$. Second, the first-order Hodge locus of the Weil line is the Weil family, so the directions that the semiregularity map cannot see are those along which the class stays Hodge and along which it would have to be carried.

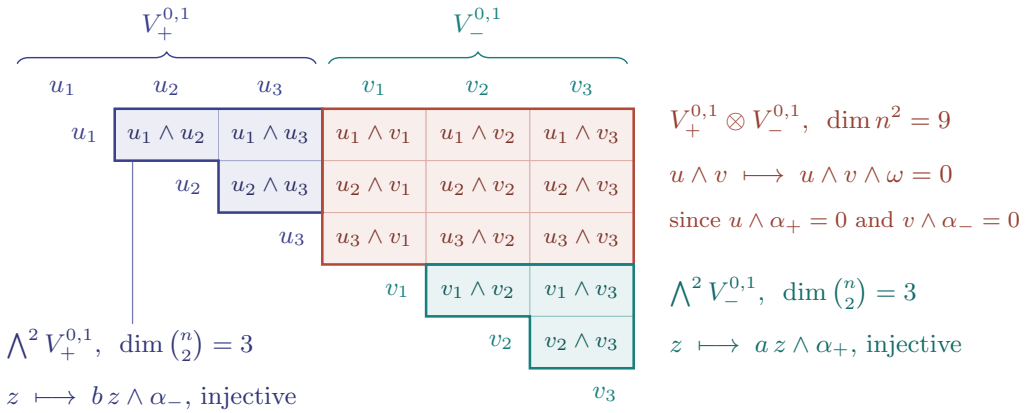


FIGURE 7. The structure behind Lemma 9.6, drawn at $n = 3$. $H^{0,2}(A) = \bigwedge^2 H^{0,1}$ splits into three blocks by how many factors come from $V_+^{0,1}$ and how many from $V_-^{0,1}$. The two outer blocks, of dimension $\binom{n}{2} = 3$ each, are carried injectively by multiplication with the Weil class. The middle block, of dimension $n^2 = 9$ and spanned by the mixed products, goes to zero, because each such product repeats a factor of α_+ and a factor of α_-. Its dimension n^2 is the deficit in Proposition 9.10.

Write $V_\pm$ for the two eigenspaces of K on $H^1(A, \mathbb{C})$, each of dimension $2n$, and

$$V_\pm^{1,0} = V_\pm \cap H^{1,0}, \qquad V_\pm^{0,1} = V_\pm \cap H^{0,1},$$

each of dimension n by Lemma 3.5. Let E denote the polarisation form on $H^1(A, \mathbb{Q})$, extended to $H^1(A, \mathbb{C})$. We use throughout that E vanishes on $V_+ \times V_+$ and on $V_- \times V_-$ and pairs V_+ with V_- perfectly, which is condition (iii) of Definition 3.1 read on the eigenspaces, and that E vanishes on $H^{1,0} \times H^{1,0}$ and on $H^{0,1} \times H^{0,1}$, which is the first Riemann bilinear relation. For the first statement, let $x, y \in V_+$. Then $E(\sqrt{-d}\,x, y) = \sqrt{-d}\,E(x, y)$ from the eigenvalue, and $E(\sqrt{-d}\,x, y) = -E(x, \sqrt{-d}\,y) = -\sqrt{-d}\,E(x, y)$ from Definition 3.1(iii), so $E(x, y) = 0$. The same holds on V_-, and the nondegeneracy of E then forces the pairing between V_+ and V_- to be perfect. Together, the two vanishing statements say that of the four blocks of E on $(V_+^{1,0} \oplus V_+^{0,1}) \times (V_-^{1,0} \oplus V_-^{0,1})$ only $V_+^{1,0} \times V_-^{0,1}$ and $V_+^{0,1} \times V_-^{1,0}$ are nonzero, and each of these two is a perfect pairing of n-dimensional spaces. This block structure is all that the proof of Theorem 9.17 uses of E.

A first-order deformation of A is a class in $H^1(A, T_A)$, which for an abelian variety is canonically $\mathrm{Hom}(H^{1,0}, H^{0,1})$, since the tangent bundle is trivial and $H^{1,0}$ is the dual of its space of sections. Such a v is extended to $H^1(A, \mathbb{C})$ by zero on $H^{0,1}$ and acts on $H^k(A, \mathbb{C}) = \bigwedge^k H^1(A, \mathbb{C})$ as the derivation it generates; we write $v \lrcorner \xi$ for that action. A Hodge class ξ of type (p, p) remains of type (p, p) to first order along v exactly when $v \lrcorner \xi = 0$. We call v *polarised* when $E(vx, y) + E(x, vy) = 0$ for all $x, y \in H^{1,0}$, which is the condition that the deformation preserve the polarisation. The polarised v form a space of dimension $n(2n + 1) = \dim \mathcal{A}_{2n}$.

Proposition 9.14 (The first-order Hodge locus of the Weil line)**.** *Let A be of $(K,-1,n)$-Weil type and let v be a polarised first-order deformation of A. The following are equivalent.*

(i) $v \lrcorner \omega = 0$ *for every ω in the Weil line.*
(ii) v *commutes with the action of K.*
(iii) $v(V_+^{1,0}) \subseteq V_+^{0,1}$ *and* $v(V_-^{1,0}) \subseteq V_-^{0,1}$.

The space of such v has dimension n^2, and under Proposition B.8 it is the tangent space at $[A]$ to the Weil family $\mathcal{D}_{n,n}$. Each of the two rational generators of the Weil line already cuts it out by itself.

Proof. The equivalence of (ii) and (iii) is immediate. Since K acts on $V_\pm$ by the two scalars $\pm\sqrt{-d}$, commuting with K means preserving V_+ and V_-. The extension of v kills $H^{0,1}$ and takes values in $H^{0,1}$, so preserving $V_\pm$ amounts to $v(V_\pm^{1,0}) \subseteq V_\pm \cap H^{0,1} = V_\pm^{0,1}$.

For (i) and (iii), let α_+ generate $\bigwedge^{2n} V_+$ and write $\alpha_+ = e_1 \wedge \cdots \wedge e_{2n}$ for a basis $e_1, \dots, e_{2n}$ of V_+ adapted to $V_+ = V_+^{1,0} \oplus V_+^{0,1}$. As a derivation,

$$v \lrcorner \alpha_+ = \sum_{j=1}^{2n} e_1 \wedge \cdots \wedge v(e_j) \wedge \cdots \wedge e_{2n}.$$

Split $v(e_j)$ along $H^1(A,\mathbb{C}) = V_+ \oplus V_-$. The V_+ components contribute $\operatorname{tr}(\pi_+ \circ v|_{V_+})\,\alpha_+$, and that trace vanishes: the map $\pi_+ \circ v|_{V_+}$ kills $V_+^{0,1}$ and carries $V_+^{1,0}$ into $V_+^{0,1}$, so in the adapted basis its matrix has zero diagonal blocks. Writing $v(e_j)_- = \sum_k c_{jk} f_k$ for a basis $f_1, \dots, f_{2n}$ of V_-, the V_- components contribute

$$\sum_{j,k} \pm\, c_{jk}\; e_1 \wedge \cdots \wedge \widehat{e_j} \wedge \cdots \wedge e_{2n} \wedge f_k,$$

and the $4n^2$ monomials appearing here are distinct members of the standard basis of $\bigwedge^{2n} H^1(A,\mathbb{C})$, hence linearly independent. So $v \lrcorner \alpha_+ = 0$ if and only if every c_{jk} vanishes, which is $v(V_+) \subseteq V_+$. The same argument on α_- gives $v \lrcorner \alpha_- = 0$ if and only if $v(V_-) \subseteq V_-$. The Weil line spans $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ over $\mathbb{C}$ by Proposition 3.7, and each of the two rational generators of Theorem 4.9 has both components nonzero, so a single generator already forces both conditions. This is (i) if and only if (iii).

For the dimension, let v satisfy (iii), and write v_+ and v_- for its restrictions to $V_+^{1,0}$ and $V_-^{1,0}$. If x and y both lie in $V_+^{1,0}$, then v_+x and y both lie in V_+, so both terms of the polarisation condition vanish, and likewise if both lie in $V_-^{1,0}$. For $x \in V_+^{1,0}$ and $y \in V_-^{1,0}$ the condition reads $E(v_+x, y) = -E(x, v_-y)$. The left side is determined by v_+, and the pairing between $V_+^{1,0}$ and $V_-^{0,1}$ is perfect, so for each y there is exactly one $v_-y \in V_-^{0,1}$ satisfying this for all x, and it depends linearly on y. Hence v_- is determined by v_+, which may be chosen freely, and the polarised deformations satisfying (iii) form a space of dimension $\dim \operatorname{Hom}(V_+^{1,0}, V_+^{0,1}) = n^2$. By Proposition B.8 the family $\mathcal{D}_{n,n}$ is an open subset of the Grassmannian of n-planes in V_+, whose tangent space at $V_+^{1,0}$ is $\operatorname{Hom}(V_+^{1,0}, V_+/V_+^{1,0}) = \operatorname{Hom}(V_+^{1,0}, V_+^{0,1})$, and the identification just made is that isomorphism. ∎

Proposition 9.15 (Infinitesimal rigidity of divisor classes)**.** *Keep the notation above, put $P = V_+^{1,0}$ and $P' = V_-^{1,0}$ as in the proof of Proposition B.8, so that $\overline{P} = V_-^{0,1}$ and $\overline{P}' = V_+^{0,1}$, and write a class $\delta \in H^{1,1}$ in the four blocks*

$$a \text{ on } P \otimes \overline{P}, \quad b \text{ on } P' \otimes \overline{P}', \quad c \text{ on } P \otimes \overline{P}', \quad e \text{ on } P' \otimes \overline{P},$$

the first two carrying the norm character of $K^\times$ and the last two the characters τ^2 and $\overline{\tau}^2$; reality makes a and b anti-hermitian and ties e to $-\bar{c}^T$. Identify the tangent space $\mathrm{Hom}(P, \overline{P}')$ *to $\mathcal{D}_{n,n}$ with the $n \times n$ matrices V as in Theorem 9.17, and write* alt *for the alternating part of a matrix, viewed in $\bigwedge^2$. Then contraction with a tangent vector is*

$$\Phi_\delta(V) \;=\; \big(\, Va + b^T V,\quad \mathrm{alt}(Vc),\quad \mathrm{alt}(-V^T e)\,\big), \tag{26}$$

with values in $\overline{P}' \otimes \overline{P} \oplus \bigwedge^2 \overline{P}' \oplus \bigwedge^2 \overline{P} = H^{0,2}$, and $\ker \Phi_\delta$ is the tangent space to the locus where δ stays of type $(1,1)$. Consequently:

(i) *$\Phi_\delta = 0$ if and only if δ is a real multiple of η. The rigid classes form a real line, which is the infinitesimal form of the statement that $\mathrm{NS}(A)_{\mathbb{Q}}$ has rank one at the general member.*

(ii) *For the norm-character class with block a and $b = c = e = 0$,*

$$\dim \ker \Phi_\delta \;=\; n\,(n - \operatorname{rank} a),$$

so such a class cuts a locus of codimension $n \operatorname{rank} a$, at least n.

(iii) *A general class has $\ker \Phi_\delta = 0$: to first order it rigidifies the family to a point.*

Proof. For (26), a polarised deformation commuting with K sends P into $\overline{P}'$ and P' into $\overline{P}$, and the polarisation condition $E(vx, y) + E(x, vy) = 0$ forces the second matrix to be $-V^T$, as in the proof of Proposition 9.14. Applying v to the $(1,0)$ slot of each block and collecting terms in $H^{0,2}$ gives the three displayed components; the sign in the first comes from $\overline{P} \wedge \overline{P}' = -\overline{P}' \wedge \overline{P}$.

(i) If $\Phi_\delta = 0$ then $\mathrm{alt}(Vc) = 0$ for every V, which for $n \geq 2$ forces $c = 0$, and likewise $e = 0$. The first component gives $Va = -b^T V$ for every V; taking V to run over the matrix units shows a is a scalar λ and $b^T = -\lambda$. Anti-hermitian scalars are the purely imaginary ones, so the solutions form a real line. It contains η, whose blocks are $a = i$, $b = -i$, and the two therefore agree.

(ii) With $b = c = e = 0$ the condition is $Va = 0$, that is, the rows of V annihilate the column space of a; the solution space has dimension $n(n - \operatorname{rank} a)$.

(iii) By (ii), the class with a of rank n (for instance $a = i$) and $b = c = e = 0$, which is real since a is anti-hermitian, has $\ker \Phi_\delta = 0$. Since Φ_δ depends linearly on δ, the classes with $\ker \Phi_\delta \neq 0$ are cut out by the vanishing of the maximal minors of Φ_δ and form a proper real algebraic subset of the real classes of type $(1,1)$, which gives (iii). ∎

Remark 9.16 (The divisorial supply). Proposition 9.15(ii) explains why the divisorial supply of Theorem 15.16 cannot be enlarged by a better choice of class. Any norm-character class other than a multiple of η has $\operatorname{rank} a \geq 1$ and so cuts a locus of codimension at least n; that locus is proper, and by (i) only the real multiples of η cut nothing. Each further class therefore yields a subvariety and never an open set, which is Proposition 15.18 with the codimension made explicit.

Theorem 9.17 (The annihilator is the tangent space)**.** *Let A be of $(K, -1, n)$-Weil type and let ω be a rational generator of the Weil line. The polarisation induces an isomorphism*

$$T_{[A]}\mathcal{D}_{n,n} \;\xrightarrow{\;\sim\;}\; \mathrm{Ann}_{H^{0,2}(A)}(\omega),$$

canonical in the sense that it depends on no choice of basis and on no choice of ω on the line. Explicitly, if $p_1, \dots, p_n$ is a basis of $V_+^{1,0}$ and $q_1, \dots, q_n$ the basis of $V_-^{0,1}$ dual to it

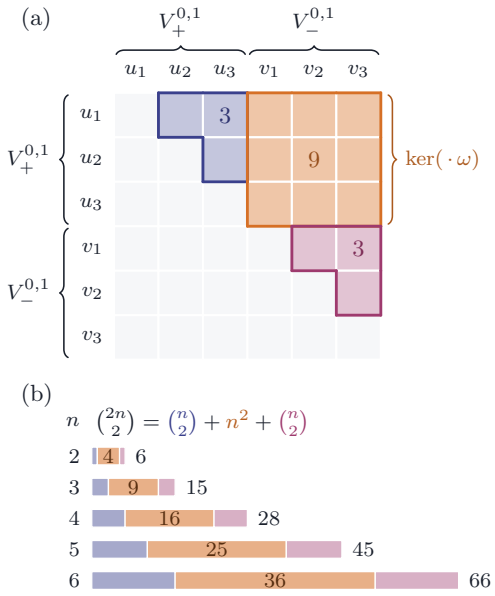


FIGURE 8. The geometry behind Lemma 9.6, drawn schematically for the real shadow of the hermitian form. The ruled surface is the null cone of H; the two transverse planes are the conjugate isotropic subspaces V_+ and V_-, which conjugation interchanges, so that neither is defined over $\mathbb{Q}$ and only their sum is. The segments join a vector of one to its conjugate in the other, and the strip they sweep stands for the mixed block $V_+^{0,1} \otimes V_-^{0,1}$, of dimension n^2, which multiplication by the Weil class sends to zero and which is the deficit in Proposition 9.10.

under E, then $\varphi \in \mathrm{Hom}(V_+^{1,0}, V_+^{0,1})$ corresponds to

$$\sum_{a=1}^{n} q_a \wedge \varphi(p_a) \;\in\; V_-^{0,1} \otimes V_+^{0,1} \;\subset\; \textstyle\bigwedge^2 H^{0,1} = H^{0,2}(A).$$

Proof. The form E pairs V_+ with V_- perfectly and vanishes on $V_+^{1,0} \times V_-^{1,0}$, the latter because both spaces lie in $H^{1,0}$. Hence E restricted to $V_+^{1,0} \times V_-$ kills $V_-^{1,0}$ and descends to a pairing

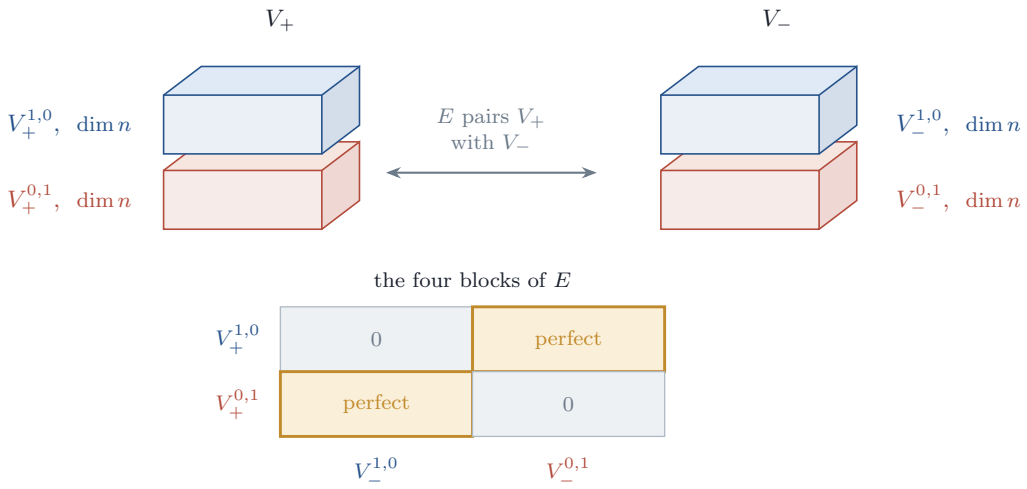
$$V_+^{1,0} \times V_-^{0,1} \longrightarrow \mathbb{C}.$$

This pairing is perfect: both spaces have dimension n, and a vector of $V_+^{1,0}$ orthogonal to $V_-^{0,1}$ is orthogonal to all of V_-, hence zero. It gives a canonical isomorphism $(V_+^{1,0})^* \cong V_-^{0,1}$, and therefore

$$T_{[A]}\mathcal{D}_{n,n} = \mathrm{Hom}(V_+^{1,0}, V_+^{0,1}) = (V_+^{1,0})^* \otimes V_+^{0,1} \;\cong\; V_-^{0,1} \otimes V_+^{0,1},$$

the first equality being the tangent space computation at the end of the proof of Proposition 9.14. Since $H^{0,1} = V_+^{0,1} \oplus V_-^{0,1}$, the map $q \otimes u \mapsto q \wedge u$ identifies $V_-^{0,1} \otimes V_+^{0,1}$ with the middle summand of $\bigwedge^2 H^{0,1}$, and that summand is the annihilator by Lemma 9.6. The displayed formula is the composite of these isomorphisms written in a dual pair of bases, so it is independent of the basis chosen; and Lemma 9.6 gives the same subspace for every ω on the line, so the isomorphism does not depend on ω either. ∎

K splits $H^1(A,\mathbb{C})$ into two halves, and each half splits by Hodge type.



The two zeros are the first Riemann relation, $E = 0$ on $H^{1,0} \times H^{1,0}$ and on $H^{0,1} \times H^{0,1}$. The upper right block is the one that moves the tangent space.

$$T_{[A]}\mathcal{D}_{n,n} = \operatorname{Hom}\big(V_+^{1,0}, V_+^{0,1}\big) \xrightarrow{\;E\;} V_-^{0,1} \otimes V_+^{0,1} \xrightarrow{\;q \otimes u \mapsto q \wedge u\;} \operatorname{Ann}(\omega) \subset H^{0,2}(A)$$

Both spaces have dimension n^2. The composite depends on no basis and on no choice of ω on the line.

$n = 2$: $\dim T = \dim \operatorname{Ann} = 4$ $\quad n = 3$: 9 $\quad n = 4$: 16

checked exactly over $\mathbb{Q}(i)$

FIGURE 9. Theorem 9.17. The polarisation has only two nonzero blocks on $V_+ \times V_-$: it pairs $V_+^{1,0}$ with $V_-^{0,1}$ and $V_+^{0,1}$ with $V_-^{1,0}$, each perfectly, and the other two blocks vanish by the first Riemann relation. The upper right block turns the tangent space $\operatorname{Hom}(V_+^{1,0}, V_+^{0,1})$ to the Weil family into $V_-^{0,1} \otimes V_+^{0,1}$, which is the middle block of $\bigwedge^2 H^{0,1}$ drawn in Figure 7 and the annihilator of the Weil line. The two n^2-dimensional spaces are therefore the same subspace of $H^{0,2}$, and the directions that the semiregularity map cannot see are the directions of the family along which the class stays Hodge.

Corollary 9.18 (The kernel contains the tangent space to the family)**.** *Let A be of $(K,-1,n)$-Weil type and let E be a nonzero perfect complex on A satisfying the hypotheses of Proposition 9.10, that is, with Chern character concentrated on the Weil line and with (25) injective. Then* $\ker\sigma_E$ *contains a subspace canonically isomorphic to $T_{[A]}\mathcal{D}_{n,n}$, namely the image of the composite*

$$T_{[A]}\mathcal{D}_{n,n} \;\cong\; \mathrm{Ann}_{H^{0,2}}(\omega) \;\hookrightarrow\; H^2(\mathcal{O}_A) \xrightarrow{z\mapsto z\cdot\mathrm{id}_E} \mathrm{Ext}^2(E,E).$$

By Proposition 9.14 that subspace is also the tangent space to the first-order Hodge locus of the class $\mathrm{ch}_n(E)$*.*

Proof. Immediate from Theorem 9.17, Proposition 9.10 and Proposition 9.14. ∎

Theorem 9.19 (The weakened criterion is unavailable for concentrated Chern characters)**.** *Let A be of $(K,-1,n)$-Weil type and let E be a nonzero perfect complex on A satisfying the hypotheses of Proposition 9.10. Let*

$$\mathrm{ev}_E\colon HH^2(A) \longrightarrow \mathrm{Ext}^2(E,E)$$

be the evaluation on E of the natural transformations $\mathrm{id}\to\mathrm{id}[2]$ *of $D^b(A)$, as in* [Mar25b, Section 11.4]*. Then*

$$\mathrm{ev}_E(H^2(\mathcal{O}_A)) \cap \ker\sigma_E = \Delta := \{\, z\cdot\mathrm{id}_E \;:\; z\in \mathrm{Ann}_{H^{0,2}}(\omega)\,\},$$

a subspace of dimension n^2. In particular σ_E is not injective on the image of ev_E*, so E satisfies neither the hypothesis of* [Mar25b, Lemma 11.3] *nor the weakened hypothesis of* [Mar25b, Question 11.4]*.*

Proof. The Hochschild cohomology $HH^*(A)=\mathrm{Ext}^*_{A\times A}(\mathcal{O}_\Delta,\mathcal{O}_\Delta)$ is the algebra of natural transformations of the identity functor, and the Hochschild–Kostant–Rosenberg isomorphism [Cal05, BF08] writes $HH^2(A)\cong H^2(\mathcal{O}_A)\oplus H^1(A,T_A)\oplus H^0(A,\bigwedge^2 T_A)$, with no correction from the Todd class because T_A is trivial. The first summand is the image of $H^2(\mathcal{O}_A)=\mathrm{Ext}^2(\mathcal{O}_A,\mathcal{O}_A)$ under pullback to $A\times A$, and the natural transformation it defines sends a complex F to the composite $F=F\otimes\mathcal{O}_A\xrightarrow{\mathrm{id}\otimes z}F\otimes\mathcal{O}_A[2]=F[2]$, that is, to $z\cdot\mathrm{id}_F$. This is the statement that the action of $HH^*(A)$ on $\mathrm{Ext}^*(F,F)$ restricts on $H^*(\mathcal{O}_A)$ to the $\mathcal{O}_A$-module structure. Concretely, a class $z\in H^2(\mathcal{O}_A)$ is represented by a 2-extension of $\mathcal{O}_A$ by itself; its pullback π_1^*z to $A\times A$ is a 2-extension of $\mathcal{O}_{A\times A}$ by itself, and tensoring with $\mathcal{O}_\Delta$ gives a 2-extension of $\mathcal{O}_\Delta$ by itself, which is the element of $HH^2(A)$ in question. The Fourier–Mukai transform with kernel $\mathcal{O}_\Delta$ is the identity, and the transform with kernel this extension is the extension of the identity by itself twisted by z, so its value on F is $z\cdot\mathrm{id}_F$. Markman uses this identification when he writes the left vertical arrow of the diagram in the proof of [Mar25b, Lemma 11.3] as the HKR isomorphism followed by contraction with $\mathrm{ch}(E)$. On the summand $H^2(\mathcal{O}_A)$ contraction is the cup product $z\mapsto z\cdot\mathrm{ch}(E)$, which is $\sigma_E(z\cdot\mathrm{id}_E)$ by the computation in the proof of Theorem 9.7.

Hence $\mathrm{ev}_E(z)=z\cdot\mathrm{id}_E$ for $z\in H^2(\mathcal{O}_A)$, and $\sigma_E(z\cdot\mathrm{id}_E)=z\cdot\mathrm{ch}(E)=N\,z\cdot\omega$ by the proof of Proposition 9.10, which vanishes exactly for $z\in\mathrm{Ann}_{H^{0,2}}(\omega)$. So $\mathrm{ev}_E(H^2(\mathcal{O}_A))\cap\ker\sigma_E=\Delta$, and $\dim\Delta=\dim\mathrm{Ann}_{H^{0,2}}(\omega)=n^2$ by Lemma 9.6 and the injectivity of (25). Since $\Delta\neq 0$, the restriction of σ_E to the image of ev_E has a nonzero kernel. The hypothesis of [Mar25b, Lemma 11.3] is that $\ker\mathrm{ev}_E$ equals the kernel of contraction with $\mathrm{ch}(E)$ and that ev_E is surjective; the two conditions together would make σ_E injective on all of $\mathrm{Ext}^2(E,E)$, contradicting Proposition 9.10. The weakened hypothesis of [Mar25b, Question 11.4] is injectivity of σ_E on the image, which we have just shown to fail. ∎

Theorem 9.20 (A line bundle summand off the polarisation line obstructs)**.** *Let A be of $(K,-1,n)$-Weil type with $n \geq 2$ and let $T = T_{[A]}\mathcal{D}_{n,n}$. Let E be a perfect complex on A with a direct summand $L[p]$, L a line bundle, and suppose that $\mathrm{ch}(E)$ remains of Hodge type to first order along $\mathcal{D}_{n,n}$, that is $v \lrcorner \mathrm{ch}(E) = 0$ for every $v \in T$. If $c_1(L)$ is not a real multiple of η, then there is $v \in T$ with*

$$\mathrm{ev}_E(v) \neq 0 \qquad \text{and} \qquad \sigma_E(\mathrm{ev}_E(v)) = 0.$$

Consequently E is not semiregular, does not satisfy the weakened hypothesis of [Mar25b, Question 11.4]*, and is obstructed to first order along v. The same holds for a twisted complex, with $c_1(L)$ replaced by the twisted first Chern class of the summand, which is $c_1(L)$ shifted by the class common to all summands.*

Proof. The class $c_1(L)$ is a real class of type $(1,1)$ and not a real multiple of η, so by Proposition 9.15(i) the contraction $\Phi_{c_1(L)} \colon T \to H^{0,2}(A)$ is not zero: there is $v \in T$ with $v \lrcorner c_1(L) \neq 0$. On the summand $H^1(A, T_A)$ of $HH^2(A)$ the evaluation map is contraction with the Atiyah class, $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$. This is the top row of the diagram of Buchweitz and Flenner reproduced as [Mar25b, (2.1)]; its identification with the Hochschild evaluation is in [BF08], and Markman uses it in the proof of [Mar25b, Lemma 11.3]. The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of $L[p]$ is $c_1(L)$, so the component of $\mathrm{ev}_E(v)$ in the summand $\mathrm{Ext}^2(L[p], L[p]) = H^{0,2}(A)$ of $\mathrm{Ext}^2(E,E)$ is $v \lrcorner c_1(L) \neq 0$. On the other hand the same diagram gives $\sigma_E(\mathrm{ev}_E(v)) = v \lrcorner \mathrm{ch}(E)$, which vanishes by hypothesis. So $\mathrm{ev}_E(v)$ is a nonzero element of $\ker \sigma_E \cap \mathrm{im}\, \mathrm{ev}_E$; this is the failure of both hypotheses, and $v \lrcorner \mathrm{At}(E) \neq 0$ is the statement that E does not lift to a first-order deformation of the pair along v, as recalled in [Mar25b, Section 2]. Twisting by a class B replaces the Atiyah class of every rank one summand by its shift by B and $\mathrm{ch}(E)$ by the twisted Chern character, and nothing else in the argument changes. ∎

Remark 9.21 (Scope of the kernel computation). The semiregularity criterion converts a statement about the Chern character into a statement about the object by requiring σ_E to be injective. Corollary 9.18 says that for an object whose Chern character is concentrated on the Weil line this requirement fails, and that the failure is measured by the tangent space to the family across which the class is to be transported, in every dimension and with no hypothesis beyond (25). This concerns the size and the position of the kernel. By itself it does not say that the obstruction classes are nonzero, nor that the Weil class fails to deform. For the explicit object of [II, Proposition 4.14] the obstruction classes are computed in [II, Proposition 4.16]: they are nonzero in every direction of the family normal to the split locus, they lie in $\ker \sigma_E$, and they meet the subspace of Corollary 9.18 only in zero. The Weil class deforms along the family, by Proposition 9.14, while that object does not.

The same computation bears on a weaker criterion, which it settles only in part. Markman asks in [Mar25b, Question 11.4] whether the semiregularity theorem survives when σ_E is required to be injective only on the image of the evaluation map ev_E rather than on all of $\mathrm{Ext}^2(E,E)$, and records in his Section 12 that an affirmative answer would carry the method to higher dimensions and to fields of larger degree. By Theorem 9.19 the classes of Corollary 9.18 lie in that image, so for concentrated Chern characters the weakened hypothesis fails as well, in every dimension and by the same n^2 dimensions. By Theorem 9.20 it fails for every object with a line bundle summand off the polarisation line, whatever the rank and at every member of the family, and by [II, Corollary 4.21] this excludes every sum of line bundles. For the explicit split object of [II, Proposition 4.14] the

failure is complete: [II, Proposition 4.18] shows that every one of the fifteen dimensions of $\ker \sigma_E$ is the value of ev_E on a Hochschild class that already preserves $\mathrm{ch}(E)$ to first order.

The next question is whether an object of positive rank can satisfy the weakened hypothesis without satisfying the original one. Markman's own sheaves are of positive rank and are not of the concentrated shape (Remark 9.9); for them he avoids the question by descending to a quotient on which σ_E is injective on an invariant subspace containing the image of ev_E [Mar25b, Proposition 11.5]. In his Section 12 he proposes a rank one complex on a principally polarised abelian fourfold, built from $(d+9)/2$ translates of the theta divisor, as a candidate for the weakened criterion in dimension eight. [II, Theorem 4.61] shows that the candidate fails, for every discriminant, at every isolated double point of its support.

Voisin's account of the subject [Voi26] makes the same point qualitatively, for the variational Hodge conjecture in general: semiregular objects are hard to construct and too restricted to carry a general argument. Corollary 9.18 gives this observation a quantitative form for one family of classes, in which the restriction amounts to n^2 dimensions and those dimensions are the tangent directions of the family. Totaro's account [Tot26] describes Markman's method as the construction of enough semiregular sheaves to prove the conjecture for abelian fourfolds and fivefolds, and records that it needed the semiregularity theorem for twisted sheaves, which Pridham supplied [Pri24].

10. The hyperplane obstruction

This section sets the constructions on abelian varieties aside and turns to the other classical approach: induction on the dimension of the variety through hyperplane sections. One places the variety in a family of hyperplane sections, uses the algebraicity of the Hodge locus to control where a class stays of type (p,p), appeals to the conjecture in one dimension lower, and transports the resulting cycles back up. Each ingredient of such an argument is a theorem: the Lefschetz hyperplane theorem, hard Lefschetz, the algebraicity of the Hodge locus [CDK95], and the properness of relative Hilbert schemes. The question is whether they compose into an induction. They do not, for an elementary reason that applies to every argument of this shape, however the family of hypersurfaces is chosen.

Throughout, X is a smooth complex projective variety of dimension d, $j\colon Y \hookrightarrow X$ is a smooth hypersurface in the linear system of an ample divisor, and $L = c_1(\mathcal{O}_X(Y)) \in H^2(X,\mathbb{Q})$.

10.1. **Two classical inputs.**

Theorem 10.1 (Lefschetz hyperplane theorem)**.** *The restriction $j^*\colon H^k(X,\mathbb{Q}) \to H^k(Y,\mathbb{Q})$ is an isomorphism for $k \le d-2$ and is injective for $k = d-1$.*

This is the topological form of the theorem. The standard proof, due to Andreotti and Frankel, is by Morse theory on the smooth affine variety $X \smallsetminus Y$; see [Voi03, Chapter 1].

Lemma 10.2 (Gysin and restriction)**.** *For every k and every $x \in H^k(X,\mathbb{Q})$,*

$$j_* j^* x = x \cup L.$$

Moreover $j_*\colon H^k(Y,\mathbb{Q}) \to H^{k+2}(X,\mathbb{Q})$ is injective for $k \ge d$.

Proof. The first identity is the projection formula together with $j_*1 = [Y] = L$. For the second, Poincaré duality on X and on Y identifies $j_*\colon H^k(Y) \to H^{k+2}(X)$ with the dual of

$$j^*\colon H^{2d-k-2}(X) \longrightarrow H^{2(d-1)-k}(Y) = H^{2d-k-2}(Y),$$

so j_* is injective on $H^k(Y)$ if and only if this j^* is surjective. By Theorem 10.1 that j^* is surjective whenever $2d-k-2 \le d-2$, that is whenever $k \ge d$. ∎

10.2. **The middle degree.**

Theorem 10.3 (Hyperplane obstruction)**.** *Let $d = 2p$ and let $\alpha \in H^{2p}(X, \mathbb{Q})$ be primitive. Then $j^*\alpha = 0$ for every smooth ample hypersurface $j\colon Y \hookrightarrow X$.*

The primitive middle degree is where the Hodge conjecture is hard, and it is where the restriction map returns nothing. Figure 10 draws the Lefschetz decomposition of an abelian fourfold and the part of it that restriction annihilates. The proof takes three lines.

Proof of Theorem 10.3. Let $d = 2p$ and let $\alpha \in H^{2p}(X, \mathbb{Q})$ be primitive. A class of degree $k \le d$ is primitive when L^{d-k+1} annihilates it, so in the middle degree primitivity means $L\alpha = 0$, that is, $H^d_{\mathrm{prim}}(X, \mathbb{Q}) = \ker\bigl(L\colon H^d(X, \mathbb{Q}) \to H^{d+2}(X, \mathbb{Q})\bigr)$. By Lemma 10.2,

$$j_*\bigl(j^*\alpha\bigr) = \alpha \cup L = 0.$$

The class $j^*\alpha$ lies in $H^d(Y, \mathbb{Q})$, on which j_* is injective by the case $k = d$ of Lemma 10.2. Hence $j^*\alpha = 0$. ∎

Corollary 10.4 (Blindness in the middle degree)**.** *Let $d = 2p$ and let $\alpha, \beta \in H^{2p}(X, \mathbb{Q})$ be primitive Hodge classes. Then $j^*\alpha = j^*\beta = 0$ for every smooth ample hypersurface Y. In particular no property of the restricted classes distinguishes α from β, and no statement about Hodge classes on hypersurfaces of X, however strong, carries any information about the primitive middle cohomology of X.*

Remark 10.5 (The size of what is lost)*.* The primitive middle cohomology is not a small correction term. When X is a smooth hypersurface of degree m in $\mathbb{P}^{d+1}$, the dimension of $H^d_{\mathrm{prim}}(X, \mathbb{Q})$ grows like m^{d+1}; and when $d = 2p$, the Hodge conjecture for X in degree $2p$ is the statement that the Hodge classes in $H^d_{\mathrm{prim}}(X, \mathbb{Q})$ are algebraic, since the remaining Hodge classes of degree $2p$ are the multiples of L^p. By Corollary 10.4, all of this cohomology is invisible to hyperplane restriction.

10.3. **Below the middle degree.** Below the middle degree the restriction map is injective, and in degrees at most $d-2$ it is an isomorphism, so a class can be transported downwards. The difficulty moves to the return trip.

Definition 10.6 (Lefschetz standard conjecture)**.** For a smooth projective variety X of dimension d with ample class L, hard Lefschetz gives isomorphisms $L^{d-k}\colon H^k(X, \mathbb{Q}) \to H^{2d-k}(X, \mathbb{Q})$, and makes $H^*(X, \mathbb{Q})$ a representation of $\mathfrak{sl}_2$ in which L raises the degree by 2 and H acts on H^k by $k - d$. Let Λ denote the unique endomorphism of degree -2 with $[L, \Lambda] = H$. The Lefschetz standard conjecture $B(X)$ asserts that Λ is induced by an algebraic cycle on $X \times X$; this is equivalent to Kleiman's formulations [Kle68, Kle94].

Lemma 10.7 (Λ undoes L on primitive classes)**.** *Let $\alpha \in H^k(X, \mathbb{Q})$ be primitive with $k \le d$. Then $\Lambda L\alpha = (d-k)\,\alpha$.*

Proof. This follows from the relation $[L, \Lambda] = H$, where H acts on H^k by the scalar $k - d$, and from $\Lambda\alpha = 0$, which holds because the primitive classes of degree at most d are the lowest weight vectors of the $\mathfrak{sl}_2$-representation. Explicitly, $\Lambda L\alpha = L\Lambda\alpha - [L, \Lambda]\alpha = 0 - (k-d)\alpha = (d-k)\alpha$. ∎

Theorem 10.8 (The return trip)**.** *Let $2p \le d-2$, so that $j^*\colon H^{2p}(X,\mathbb{Q}) \to H^{2p}(Y,\mathbb{Q})$ is an isomorphism. If $j^*\alpha$ is algebraic on Y, then $L\alpha$ is algebraic on X. The implication "$L\alpha$ algebraic $\Rightarrow$ α algebraic", asked of all smooth projective X and all Hodge classes α, is equivalent to the Lefschetz standard conjecture.*

Proof of Theorem 10.8. Let $2p \le d-2$, so $j^*\colon H^{2p}(X,\mathbb{Q}) \to H^{2p}(Y,\mathbb{Q})$ is an isomorphism by Theorem 10.1. If $j^*\alpha = \mathrm{cl}(W)$ for some $W \in \mathrm{CH}^p(Y)_{\mathbb{Q}}$, then $j_*W \in \mathrm{CH}^{p+1}(X)_{\mathbb{Q}}$ and

$$\mathrm{cl}(j_*W) = j_*j^*\alpha = L\alpha$$

by Lemma 10.2 and the compatibility of the cycle class map with proper push-forward, so $L\alpha$ is algebraic.

For the second statement, suppose $B(X)$ holds and α is primitive of degree $2p < d$ with $L\alpha$ algebraic. Then Λ is induced by an algebraic correspondence, hence carries algebraic classes to algebraic classes, and $\alpha = (d-2p)^{-1}\Lambda(L\alpha)$ by Lemma 10.7, so α is algebraic. The same holds without primitivity. Write P^m for the primitive part of $H^m(X,\mathbb{Q})$. Iterating the relation $[L,\Lambda] = H$ as in the proof of Lemma 10.7 shows that ΛL acts on the summand L^iP^{2p-2i} of the Lefschetz decomposition of $H^{2p}(X,\mathbb{Q})$ by the scalar $(i+1)(d-2p+i)$, which is nonzero because $2p < d$. So ΛL is invertible on $H^{2p}(X,\mathbb{Q})$, its inverse there is a polynomial in ΛL with rational coefficients, and $\alpha = (\Lambda L)^{-1}\Lambda(L\alpha)$ is algebraic.

Conversely, suppose that for every smooth projective variety and every Hodge class α on it the algebraicity of $L\alpha$ implies that of α. Applied repeatedly, this shows that on every smooth projective variety of dimension e the hard Lefschetz isomorphisms $L^{e-k}\colon H^k \to H^{2e-k}$, $k \le e$, carry the algebraic classes of degree k onto those of degree $2e-k$, since the preimage of an algebraic class under L^{e-k} is a Hodge class. This is Kleiman's conjecture A for every smooth projective variety, and A for $X \times X$ implies $B(X)$ [Kle68, Kle94]. ∎

Together the two theorems give the following.

Corollary 10.9 (No dimensional induction through hyperplane sections)**.** *Let X be smooth projective of dimension d and let $\alpha \in \mathrm{Hdg}^p(X)$. An argument that deduces the algebraicity of α from the Hodge conjecture for smooth ample hypersurfaces of X, using only j^* and j_*, returns no information when $2p = d$ and α is primitive, and otherwise establishes the algebraicity of $L\alpha$ rather than of α.*

Proof of Corollary 10.9. Suppose the argument uses only j^* and j_* to relate classes on X to classes on smooth ample hypersurfaces $Y \subset X$. First let $2p = d$ with α primitive. Then $j^*\alpha = 0$ by Theorem 10.3. A cycle on Y of codimension $p-1$ has its class in $H^{d-2}(Y,\mathbb{Q})$, which is $j^*H^{d-2}(X,\mathbb{Q})$ by Theorem 10.1, so by Lemma 10.2 its image under j_* lies in $L \cdot H^{d-2}(X,\mathbb{Q})$. By the Lefschetz decomposition this subspace meets the primitive part of $H^d(X,\mathbb{Q})$ only in 0, so j_* produces α only when $\alpha = 0$, and no conclusion about α is available. In the remaining cases the best that j_*j^* produces is $L\alpha$, and Theorem 10.8 identifies the passage from $L\alpha$ to α with the Lefschetz standard conjecture. ∎

10.4. **Two gaps in inductive arguments.** In practice an induction of this shape breaks at one of two steps, each of which looks harmless.

Remark 10.10 (The dimension bookkeeping). A common formulation places X in a Lefschetz pencil by choosing a generic pencil $\{X_t\}_{t\in\mathbb{P}^1}$ of hyperplane sections *of* X and then setting $X = X_{t_0}$ for some t_0. The two requirements are incompatible: the members of a pencil of hyperplane sections of X have dimension $d-1$, so none of them is X. An induction that reads $\dim X_t = d-1 < d$ in its inductive step, and $X_{t_0} = X$ when it transports the resulting cycle back, is using both readings of the same symbol.

Neither way of resolving the ambiguity rescues the argument. If X is required to be a member of the pencil, then the pencil lives in an ambient variety of dimension $d+1$ and every member has dimension d; the inductive parameter does not decrease and the induction is circular. If instead $\dim X_t = d-1$, then the class α lives on X and not on the members, and the only available transports are j^* and j_*. This is the situation of Theorems 10.3 and 10.8, and Corollary 10.9 applies.

Remark 10.11 (Pointwise existence is not a family). The second gap appears in arguments that produce cycles fibrewise and then assemble them. Suppose an inductive hypothesis furnishes, for each t in the base S of a family, a cycle Z_t with a prescribed class. It does not follow that the Hilbert polynomial $t \mapsto P_{Z_t}$ is constructible, or semicontinuous, or takes finitely many values. Those are theorems about a flat family, and the input here is a bare existence statement at each point with no coherence between points. Without constructibility there is no section of a relative Hilbert scheme over a dense open subset, and therefore nothing for the valuative criterion of properness to extend. The step looks safe because the relative Hilbert scheme of each fixed Hilbert polynomial is proper over S; but properness extends a section that exists over a punctured curve, and supplies no section where none is given.

11. The Kuga–Satake construction

Theorem 10.3 blocks induction on dimension through hyperplane sections in the primitive middle degree. One classical construction does reach a middle degree Hodge class without any such induction, and we set it out in full, both because it shows what a successful route looks like and because its boundary is numerical and can be located. It is the construction of Kuga and Satake [KS67], which builds on Satake's Clifford algebra families and their Riemann forms [Sat66]. We follow the form given by Deligne [Del72], to whom the Hodge-theoretic properties that make it useful are due, in particular the embedding of Proposition 11.7. After building it we show that Theorem 10.3 has no bearing on it. For abelian varieties of Weil type with Hodge group $\mathrm{SU}(V,H)$ we then determine when it applies to a piece of H^2: exactly when $n=1$, where the Weil line consists of divisor classes and there is nothing to prove, or $n=2$ with trivial discriminant, where the Weil line lies in H^4.

11.1. The construction.

Setup 11.1 (K3 type)**.** Let T be a $\mathbb{Q}$-vector space of dimension b with a nondegenerate symmetric form q, and let T carry a Hodge structure of weight two, $T_{\mathbb{C}} = T^{2,0} \oplus T^{1,1} \oplus T^{0,2}$, polarised by q, with

$$\dim_{\mathbb{C}} T^{2,0} = 1.$$

Such a structure is said to be *of K3 type*. Write $T^{2,0} = \mathbb{C}\sigma$. The Hodge–Riemann relations give $q(\sigma,\sigma) = 0$ and $q(\sigma,\overline{\sigma}) > 0$, and make q negative definite on the real points of $T^{1,1}$; so the real plane $P = \langle \Re\sigma, \Im\sigma \rangle \subset T_{\mathbb{R}}$ is positive definite and q has signature $(2, b-2)$ on $T_{\mathbb{R}}$. Since $q(\Re\sigma, \Im\sigma) = 0$ and $q(\Re\sigma) = q(\Im\sigma)$, we may normalise σ so that $e_1 = \Re\sigma$ and $e_2 = \Im\sigma$ are orthonormal.

Definition 11.2 (Clifford algebra)**.** $C(T,q)$ is the quotient of the tensor algebra on T by the ideal generated by $v \otimes v - q(v)$ for $v \in T$. It is $\mathbb{Z}/2$-graded, $C = C^+ \oplus C^-$, by the parity of the word length.

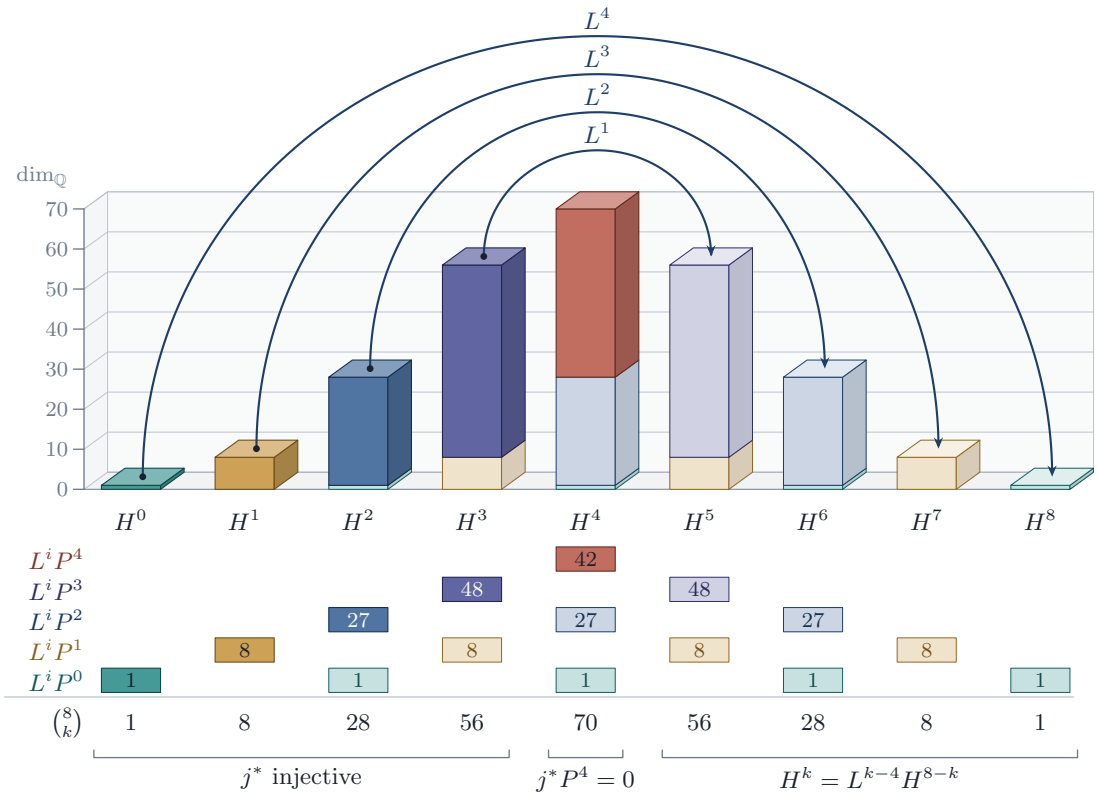


FIGURE 10. The Betti numbers $\binom{8}{k}$ of an abelian fourfold, drawn as prisms and divided according to the Lefschetz decomposition $H^k = \bigoplus_i L^i P^{k-2i}$: the primitive part P^k is drawn dark, and each image $L^i P^{k-2i}$ is drawn light in the colour of P^{k-2i}. The arcs are the hard Lefschetz isomorphisms $L^{4-k}\colon H^k \to H^{8-k}$. Below the middle degree the restriction j^* to a smooth ample hypersurface is injective and a class descends, but the return trip j_*j^* multiplies by L, and recovering the class is the Lefschetz standard conjecture; above the middle degree a class is carried to the lower half by an arc. The primitive middle cohomology P^4, of dimension $70 - 28 = 42$, is annihilated by j^* (Theorem 10.3), and neither an arc nor a restriction reaches it.

Lemma 11.3. *For an orthogonal basis $f_1, \dots, f_b$ of T the products f_S over subsets $S \subseteq \{1, \dots, b\}$, taken in increasing order, form a basis of $C(T, q)$. Hence $\dim_{\mathbb{Q}} C = 2^b$ and $\dim_{\mathbb{Q}} C^{\pm} = 2^{b-1}$, and $C^- \cdot C^+ \cdot C^- \subseteq C^+$.*

Proof. The relations $f_i^2 = q(f_i)$ and $f_i f_j = -f_j f_i$ for $i \neq j$, the latter obtained by expanding $(f_i + f_j)^2 = q(f_i) + q(f_j)$, let one rewrite any word in the f_i as a scalar times an increasing product, so the f_S span; they are independent by the standard filtration argument comparing C with $\bigwedge^* T$. The dimension counts are $\sum_k \binom{b}{k} = 2^b$ and $\sum_{k \text{ even}} \binom{b}{k} = 2^{b-1}$. The last inclusion holds because the grading is multiplicative and odd plus even plus odd is even. ∎

Lemma 11.4 (The complex structure)**.** *With e_1, e_2 as in Setup 11.1, put $J = e_1 e_2 \in C^+ \otimes_{\mathbb{Q}} \mathbb{R}$. Then $J^2 = -1$, and left multiplication by J is an $\mathbb{R}$-linear operator on $C^+ \otimes_{\mathbb{Q}} \mathbb{R}$ whose square is $-\mathrm{id}$.*

Proof. $J^2 = e_1 e_2 e_1 e_2 = -e_1 e_1 e_2 e_2 = -q(e_1) q(e_2) = -1$, since the orthogonal vectors e_1 and e_2 anticommute and $e_i^2 = q(e_i) = 1$. Left multiplication by an element of C^+ preserves C^+, and the square of left multiplication by J is left multiplication by $J^2 = -1$. ∎

Figure 11 draws the positive plane P, the null cone of q and the period domain in the smallest case $b = 3$.

Definition 11.5 (Kuga–Satake variety)**.** Fix a lattice $C^+_{\mathbb{Z}} \subset C^+$ stable under multiplication. The complex torus

$$\mathrm{KS}(T) = \big(C^+ \otimes_{\mathbb{Q}} \mathbb{R},\, J\big) / \, C^+_{\mathbb{Z}}$$

has dimension $\frac{1}{2} \dim_{\mathbb{Q}} C^+ = 2^{b-2}$. Satake's construction of a Riemann form on C^+ compatible with J [Sat66] makes it an abelian variety, and by construction

$$H^1\big(\mathrm{KS}(T), \mathbb{Q}\big) = C^+(T)$$

as a Hodge structure of weight one, the Hodge decomposition being the eigenspace decomposition of left multiplication by J.

Example 11.6. For a projective K3 surface, T is the transcendental lattice inside $H^2(X, \mathbb{Q})$; taking all of H^2 with $b = 22$ gives $\dim \mathrm{KS} = 2^{20} = 1{,}048{,}576$. At the other end, $b = 3$ gives $\dim \mathrm{KS} = 2$, an abelian surface, and $b = 6$ gives $\dim \mathrm{KS} = 16$; these are the two cases that occur below, for $n = 1$ and for $n = 2$.

11.2. **The transport of Hodge classes.**

Proposition 11.7 (The embedding)**.** *Fix $v_0 \in T$ with $q(v_0) \neq 0$. The map*

$$\Psi : T \longrightarrow \mathrm{End}_{\mathbb{Q}}(C^+), \qquad \Psi(v)(x) = v\, x\, v_0,$$

is injective and is a morphism of Hodge structures $T(1) \to \mathrm{End}\big(H^1(\mathrm{KS}(T), \mathbb{Q})\big)$, both sides having weight zero. Consequently a Hodge class in T, that is a rational class of type $(1,1)$, is carried to a Hodge class of type $(0,0)$ in $\mathrm{End}(H^1(\mathrm{KS}(T), \mathbb{Q}))$, that is to an element of $\mathrm{End}^0(\mathrm{KS}(T))$.

Proof. The map lands in $\mathrm{End}(C^+)$ by the last part of Lemma 11.3, since $v, v_0 \in T \subset C^-$. It is injective because $\Psi(v)(1) = v\, v_0$ and v_0 is invertible in $C(T, q)$, with $v_0^{-1} = v_0 / q(v_0)$, so $\Psi(v) = 0$ forces $v = 0$. That it respects the Hodge structures is Deligne's computation [Del72, §4]: the Hodge structure on $\mathrm{End}(C^+)$ is induced by J acting on both sides, and Ψ intertwines the weight-two structure on T, twisted once, with it. The last sentence follows from the definition of a morphism of Hodge structures and the Lefschetz theorem on $(1,1)$

classes, in the form which says that for an abelian variety B a Hodge class of type $(0,0)$ in $\mathrm{End}(H^1(B,\mathbb{Q}))$ is an element of $\mathrm{End}^0(B)$. ∎

Remark 11.8 (Algebraicity of the transport). An element of $\mathrm{End}^0(\mathrm{KS}(T))$ is a rational multiple of an endomorphism, whose graph is an algebraic cycle on $\mathrm{KS}(T)\times\mathrm{KS}(T)$, so the image of a Hodge class under Ψ is algebraic. Carrying that conclusion back to the variety X from which T came requires the correspondence Ψ itself to be induced by an algebraic cycle on $X\times\mathrm{KS}(T)\times\mathrm{KS}(T)$. What is proved in general is that Ψ is absolute Hodge, by Deligne's theorem for abelian varieties together with the analysis of the Kuga–Satake correspondence, and that it is motivated in André's sense [And96a]; van Geemen [vG00] surveys the cases in which algebraicity has been established. The Kuga–Satake construction therefore converts the Hodge conjecture for T into the algebraicity of one specific correspondence.

11.3. **Independence from the hyperplane obstruction.**

Proposition 11.9. *Theorems 10.3 and 10.8 do not constrain the Kuga–Satake route.*

Proof. Both theorems are statements about the pair of maps j^* and j_* attached to the inclusion $j\colon Y\hookrightarrow X$ of a smooth member of an ample linear system, and both proofs use two properties of that pair which a general correspondence does not have: the Lefschetz hyperplane theorem, which makes j^* an isomorphism in degrees at most $d-2$ and, through Lemma 10.2, forces the vanishing of Theorem 10.3 in the primitive middle degree; and the identity $j_*j^* = \cdot\cup L$ of Lemma 10.2, which makes the return trip multiply by the polarisation instead of recovering the class. The Kuga–Satake correspondence is a cohomology class on a triple product of varieties of unrelated dimensions, $\dim\mathrm{KS}(T)=2^{b-2}$ against $\dim X$, and it has neither property; in particular the composite of Ψ with its transpose is a nonzero multiple of the identity of T, not a power of a polarisation of X. So neither of the two properties on which Theorem 10.3 rests holds for Ψ. ∎

Proposition 11.9 shows what an escape from Theorem 10.3 must look like: a correspondence whose composite with its transpose is not the Lefschetz operator. Section 13 makes the analogous identification for Theorems 5.6 and 7.3.

11.4. **Abelian varieties of Weil type.** The limitation of the Kuga–Satake construction is the hypothesis $\dim T^{2,0}=1$. For abelian varieties of Weil type the rational sub-Hodge structures of H^2, and hence the possible values of $\dim T^{2,0}$, can be listed completely.

Proposition 11.10 (The pieces of H^2)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type. In the character grading of Proposition 5.1,*

$$H^2(A,\mathbb{C}) = \textstyle\bigwedge^2 V_+ \;\oplus\; (V_+\otimes V_-) \;\oplus\; \bigwedge^2 V_-, \tag{27}$$

the middle summand is the isotypic component of the norm character and is defined over $\mathbb{Q}$, and the outer two are interchanged by conjugation so their sum is defined over $\mathbb{Q}$. The Hodge numbers are

$$(h^{2,0},h^{1,1},h^{0,2}) = \begin{cases} \left(\binom{n}{2},\, n^2,\, \binom{n}{2}\right) & \text{on } \bigwedge^2 V_\pm,\\ (n^2,\, 2n^2,\, n^2) & \text{on } V_+\otimes V_-, \end{cases}$$

and the total is $2\binom{2n}{2}+4n^2=\binom{4n}{2}=\dim H^2(A,\mathbb{Q})$.

Proof. The splitting is (16) at $k=2$. The characters are τ^2, $\tau\overline{\tau}=\mathrm{N}(\tau)$ and $\overline{\tau}^2$, so the middle summand is the norm-isotypic component, and it is self-conjugate, hence defined over $\mathbb{Q}$; the outer two are exchanged. For the Hodge numbers use (3) and Lemma 3.5: $V_\pm^{1,0}$ and $V_\pm^{0,1}$ have

dimension n each, so $(\bigwedge^2 V_+)^{2,0} = \bigwedge^2 V_+^{1,0}$ has dimension $\binom{n}{2}$ and $(\bigwedge^2 V_+)^{1,1} = V_+^{1,0} \otimes V_+^{0,1}$ has dimension n^2, while $(V_+ \otimes V_-)^{2,0} = V_+^{1,0} \otimes V_-^{1,0}$ has dimension n^2 and the $(1,1)$ part has dimension $2n^2$. The total is $2(2\binom{n}{2} + n^2) + 4n^2 = 2(2n^2 - n) + 4n^2 = 8n^2 - 2n = \binom{4n}{2}$. ∎

Proposition 11.11 (Sub-Hodge structures of K3 type)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$. Then $H^2(A, \mathbb{Q})$ is the direct sum of the three rational sub-Hodge structures*

$$\mathbb{Q}\,\eta, \qquad (V_+ \otimes V_-)_{\mathbb{Q}} \ominus \mathbb{Q}\eta, \qquad R = \Big(\bigwedge^2 V_+ \oplus \bigwedge^2 V_-\Big)_{\mathbb{Q}},$$

with $h^{2,0}$ equal to 0, n^2 and $n(n-1)$ respectively. The first two are irreducible, and so is R when $n \geq 3$. When $n = 2$, the structure R is irreducible if the discriminant of (A, ι, η) in the sense of Definition 15.22 is nontrivial, and it is the direct sum of two isomorphic sub-Hodge structures of K3 type, each of dimension six, if the discriminant is trivial. Consequently $H^2(A, \mathbb{Q})$ contains a rational sub-Hodge structure of K3 type if and only if either $n = 1$, or $n = 2$ and the discriminant is trivial.

Proof. A rational sub-Hodge structure $W \subseteq H^2(A, \mathbb{Q})$ is stable under $\mathrm{Hg}(A)$: the orthogonal projector onto W for the polarisation is a Hodge class in $\mathrm{End}(H^2)$, hence is fixed by $\mathrm{Hg}(A)$, hence W is a $\mathrm{Hg}(A)$-submodule. Conversely a $\mathrm{Hg}(A)$-submodule is a sub-Hodge structure, because the Hodge decomposition is the eigenspace decomposition of an element of $\mathrm{Hg}(A)(\mathbb{R})$. So the rational sub-Hodge structures are the $\mathbb{Q}$-subrepresentations of $\mathrm{SU}(V, H)$. For the same reason the Hodge numbers of a sub-Hodge structure depend only on its complexification as a representation of $\mathrm{Hg}(A)_{\mathbb{C}}$.

After complexification $\mathrm{SU}(V, H)$ becomes $\mathrm{SL}(V_+)$, acting on V_+ tautologically and on $V_- \cong V_+^\vee$ dually. In (27) the outer summands $\bigwedge^2 V_+$ and $\bigwedge^2 V_- \cong \bigwedge^2 V_+^\vee$ are irreducible, and the middle summand is $V_+ \otimes V_+^\vee = \mathrm{End}(V_+) = \mathfrak{sl}(V_+) \oplus \mathbb{C}$, the sum of the adjoint representation and the trivial one. Conjugation fixes the two pieces of the middle summand and exchanges the outer two, so the three listed subspaces are rational sub-Hodge structures, and the first two, having irreducible complexifications, are irreducible. The trivial summand is spanned by η by Lemma 5.2, and has $h^{2,0} = 0$ since η is of type $(1,1)$. The adjoint summand has $h^{2,0} = n^2$, since removing the one-dimensional trivial piece from $V_+ \otimes V_-$ removes a $(1,1)$ class only. The structure R has $h^{2,0} = 2\binom{n}{2} = n(n-1)$ by Proposition 11.10.

The complexification of a rational subrepresentation of R is a subrepresentation of $\bigwedge^2 V_+ \oplus \bigwedge^2 V_-$ stable under conjugation. For $n \geq 3$ the two summands are not isomorphic, because $\bigwedge^2 V_+^\vee \cong \bigwedge^{2n-2} V_+$ and $2n - 2 \neq 2$. The subrepresentations are then 0, the two summands and the whole; conjugation exchanges the summands, so only 0 and the whole are stable under it, and R is irreducible. For $n = 1$ both summands are the trivial representation, and R is the plane of Hodge classes $W(A) \subset H^2(A, \mathbb{Q})$.

Let $n = 2$. Then $\bigwedge^2 V_+^\vee \cong \bigwedge^2 V_+ \otimes (\bigwedge^4 V_+)^\vee \cong \bigwedge^2 V_+$ as representations of $\mathrm{SL}(V_+)$, so $\mathrm{End}_{\mathrm{Hg}(A)}(R) \otimes \mathbb{C} \cong M_2(\mathbb{C})$ and $\mathrm{End}_{\mathrm{Hg}(A)}(R)$ is a quaternion algebra over $\mathbb{Q}$. To identify it, put $V = H^1(A, \mathbb{Q})$, a K-vector space of dimension four by Lemma 3.2. The natural map $\bigwedge^2_{\mathbb{Q}} V \to \bigwedge^2_K V$ is $\mathrm{SU}(V, H)$-equivariant, and its kernel has complexification $V_+ \otimes V_-$, so it restricts to an isomorphism of R onto $\bigwedge^2_K V$; in particular $K \subseteq \mathrm{End}_{\mathrm{Hg}(A)}(R)$. The wedge product $B\colon \bigwedge^2_K V \times \bigwedge^2_K V \to \bigwedge^4_K V \cong K$ is a symmetric K-bilinear form, invariant under $\mathrm{SU}(V, H)$ because $\mathrm{SU}(V, H)$ acts trivially on $\bigwedge^4_K V$. Let H_2 be the hermitian form induced by H on $\bigwedge^2_K V$, and let φ be the K-antilinear map defined by $B(x, y) = H_2(y, \varphi(x))$. Then φ commutes with $\mathrm{SU}(V, H)$, so φ^2 is a K-linear endomorphism commuting with $\mathrm{SU}(V, H)$; it is a scalar by Schur's lemma, and a rational one because it commutes with

φ. Choose a K-basis $e_1, \dots, e_4$ of V orthogonal for H, with $H(e_i, e_i) = h_i \in \mathbb{Q}^\times$, and identify $\bigwedge^4_K V$ with K through $e_1 \wedge e_2 \wedge e_3 \wedge e_4$. Then $\varphi(e_1 \wedge e_2) = (h_3 h_4)^{-1} e_3 \wedge e_4$ and $\varphi(e_3 \wedge e_4) = (h_1 h_2)^{-1} e_1 \wedge e_2$, so $\varphi^2 = (\det H)^{-1}$. Hence $\mathrm{End}_{\mathrm{Hg}(A)}(R) = K \oplus K\varphi$ is the quaternion algebra $(-d, \det H)_{\mathbb{Q}}$. It is split if and only if $\det H \in \mathrm{N}_{K/\mathbb{Q}}(K^\times)$, that is, since $(-1)^2 \det H = \det H$, if and only if the discriminant is trivial. If it is not split it is a division algebra, and R is irreducible. If it is split, $R = R_1 \oplus R_2$ with $R_1 \cong R_2$ of dimension six, and the complexification of each R_i is isomorphic to $\bigwedge^2 V_+$ as a representation of $\mathrm{SL}(V_+)$, so by Proposition 11.10 each R_i has Hodge numbers $(1, 4, 1)$. Each R_i lies in the primitive part of $H^2(A, \mathbb{Q})$, which is the $\mathrm{Hg}(A)$-stable complement of $\mathbb{Q}\eta$, and is polarised by the restriction of the Hodge–Riemann form; so R_i is of K3 type.

Finally, a rational sub-Hodge structure of $H^2(A, \mathbb{Q})$ is a direct sum of irreducible ones, each isomorphic to $\mathbb{Q}\eta$, to the adjoint structure, to R, or, for $n = 2$ and trivial discriminant, to R_1; and $h^{2,0}$ is additive. A structure of K3 type has $h^{2,0} = 1$. For $n = 1$ the adjoint structure has $h^{2,0} = 1$. For $n \geq 3$ the nonzero values n^2 and $n(n-1)$ both exceed 1. For $n = 2$ the values are 0, 4, and 2 for R or 1 for R_1, so a structure of K3 type exists if and only if the discriminant is trivial. ∎

Corollary 11.12 (Where the construction applies)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$. The Kuga–Satake construction applies to a rational sub-Hodge structure of $H^2(A, \mathbb{Q})$ if and only if either $n = 1$, or $n = 2$ and the discriminant is trivial. For $n = 1$ the Weil line lies in $H^2(A, \mathbb{Q})$ and consists of divisor classes, hence of algebraic classes by the Lefschetz theorem on $(1, 1)$ classes, so nothing needs transporting. For $n = 2$ the Weil line lies in $H^4(A, \mathbb{Q})$, outside the structures to which the construction applies.*

Proof. The first clause is Proposition 11.11 with Setup 11.1. For $n = 1$ we have $2n = 2$, so $W(A) \subseteq H^2(A, \mathbb{Q}) \cap H^{1,1}(A)$, which is $\mathrm{NS}(A) \otimes \mathbb{Q}$ by the Lefschetz theorem on $(1, 1)$ classes. For $n = 2$ the Weil line lies in degree $2n = 4$. ∎

Remark 11.13 (Higher weight)*.* For $n \geq 2$ the Weil line sits in $H^{2n}(A, \mathbb{Q})$ with $2n \geq 4$, and the Clifford construction has no known counterpart in weight greater than two. It turns a quadratic form into an algebra; the polarisation of a Hodge structure of weight w is symmetric only when w is even, and even then there is no reason for its Clifford algebra to carry the right complex structure. One can instead apply the construction to a sub-Hodge structure of H^2 and try to recover the degree $2n$ class from the resulting abelian variety. Proposition 11.11 rules this out for $n \geq 3$, and for $n = 2$ at nontrivial discriminant. At $n = 2$ and trivial discriminant the two structures of K3 type exist, but the Hodge conjecture for abelian fourfolds of Weil type is in any case a theorem of Markman [Mar25a, Mar25b], proved at trivial discriminant also by Floccari and Fu [FF26]. What remains conceivable for $n \geq 3$ is a construction of Kuga–Satake shape for the weight $2n$ structure directly, producing an auxiliary variety on which the Weil line becomes a divisor class or an endomorphism. None is known, and Theorem 5.6 does not obstruct one: such a construction is a correspondence and need not produce a $K^\times$-eigenclass. This is the gap of [III, Remark 2.4], seen from the other side.

12. Reduction to the middle degree

The two preceding sections dealt with restriction to hypersurfaces and with the one classical correspondence that avoids it. Here we combine the first of them with the Lefschetz decomposition to locate the difficulty of the Hodge conjecture in general. The result, Theorem 12.2, is an equivalence in which neither side is easier than the other. Its

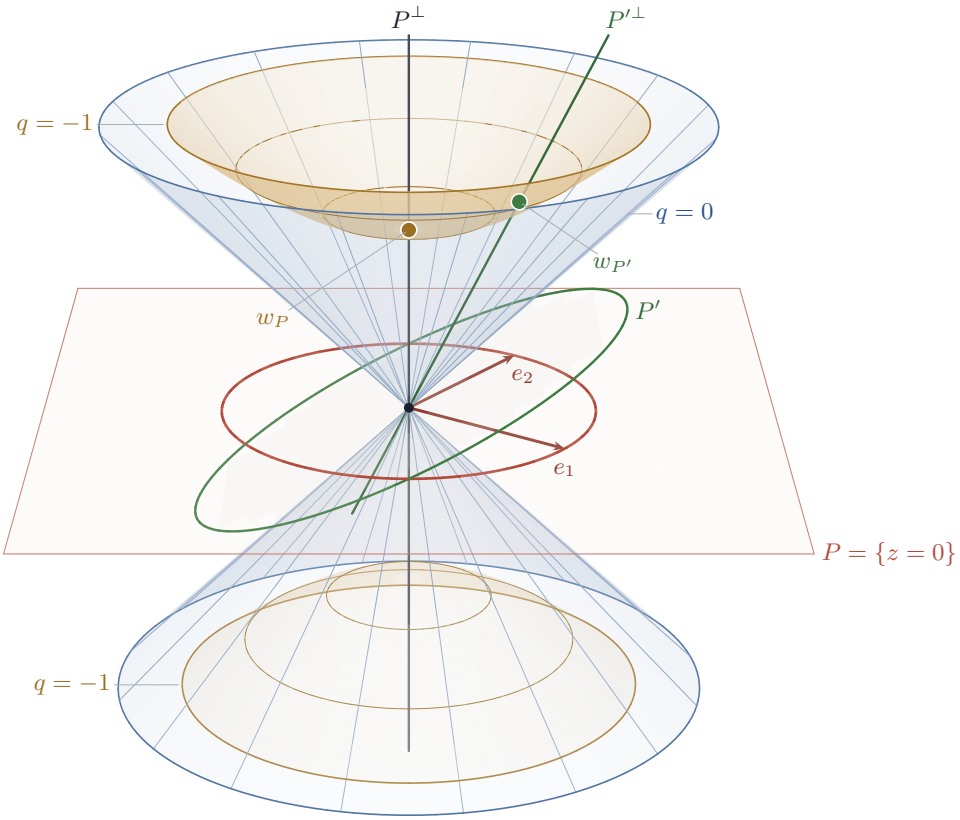


FIGURE 11. The geometry from which the Kuga–Satake construction starts, in the smallest case $b = 3$. The quadratic form is $q = x^2 + y^2 - z^2$, of signature $(2, 1)$, with its null cone $q = 0$; the horizontal plane $P = \{z = 0\}$ is positive definite and meets the cone only at the origin. The construction needs such a plane, and Setup 11.1 supplies one: $P = \langle \Re\sigma, \Im\sigma \rangle$ for the period σ spanning $T^{2,0}$. An orthonormal basis e_1, e_2 of P gives $J = e_1 e_2$ with $J^2 = -1$ by Lemma 11.4, and J is the complex structure of $\mathrm{KS}(T)$. A positive definite plane P is determined by the negative line $P^\perp$, which meets the upper sheet of the hyperboloid $q = -1$ in one point w_P; that sheet is the period domain, and a second plane P' with its point $w_{P'}$ is drawn for comparison. The hypothesis behind all of this is $\dim T^{2,0} = 1$, and by Proposition 11.11 an abelian variety of Weil type with Hodge group $\mathrm{SU}(V, H)$ supplies it in H^2 only when $n = 1$, or $n = 2$ at trivial discriminant.

use is to isolate the primitive classes of middle degree as the one place where a new method is needed, and by Theorem 10.3 restriction to hyperplane sections sees nothing there.

12.1. **Reduction to primitive classes.**

Proposition 12.1 (Lefschetz decomposition of Hodge classes)**.** *Let X be smooth projective of dimension d with ample class L. Every $\alpha \in \mathrm{Hdg}^p(X)$ can be written*

$$\alpha = \sum_{r \geq \max(0,\, 2p-d)} L^r \beta_r, \qquad \beta_r \in \mathrm{Hdg}^{p-r}(X) \cap H^{2p-2r}_{\mathrm{prim}}(X, \mathbb{Q}),$$

and the β_r are uniquely determined. Consequently the Hodge conjecture holds for X if and only if every primitive rational Hodge class on X is algebraic.

Proof. The Lefschetz decomposition $H^k(X,\mathbb{Q}) = \bigoplus_r L^r H^{k-2r}_{\mathrm{prim}}(X,\mathbb{Q})$ is a decomposition of rational Hodge structures, because L is a rational class of type $(1,1)$ and the primitive parts are kernels of morphisms of Hodge structures. The rationality of L is essential here: for a Kähler class that is not a real multiple of a rational one the primitive subspaces are in general not rational, and a rational Hodge class need not decompose into rational primitive pieces. A Hodge class α therefore has Hodge classes $L^r\beta_r$ as its components. By hard Lefschetz, L^r is injective on $H^{2p-2r}_{\mathrm{prim}}(X,\mathbb{Q})$ for $r \geq \max(0, 2p-d)$, and it is a morphism of Hodge structures of bidegree (r,r), so each β_r is uniquely determined and is itself a Hodge class. For the last statement, if each primitive β_r is algebraic then so is $L^r\beta_r$, since L is algebraic and algebraic classes form a subring, and hence so is α; the converse holds because primitive Hodge classes are Hodge classes. ∎

12.2. **The equivalence.**

Theorem 12.2. *The following are equivalent.*

(i) *The Hodge conjecture holds for every smooth complex projective variety.*
(ii) *The Lefschetz standard conjecture $B(X)$ holds for every smooth complex projective variety X, and every primitive rational Hodge class of middle degree on every smooth complex projective variety of even dimension is algebraic.*

Proof. (i) implies (ii). The second half of (ii) is a special case of (i). For $B(X)$, note that Λ is a morphism of Hodge structures of bidegree $(-1,-1)$, hence is given by a Hodge class on $X \times X$ through the standard correspondence between morphisms of Hodge structures and Hodge classes on the product, and (i) applied to $X \times X$ makes that class algebraic. This is Kleiman's argument [Kle68, §2].

(ii) implies (i). We argue by induction on $d = \dim X$. For $d \leq 1$ the Hodge conjecture is classical. Let $d \geq 2$ and assume the conjecture for all smooth projective varieties of dimension less than d. By Bertini there is a smooth hypersurface $j\colon Y \hookrightarrow X$ in $|mH|$ for an ample H and $m \gg 0$; put $L = c_1(\mathcal{O}_X(Y))$, and take primitivity and the operator Λ with respect to this L, which is harmless because $B(X)$ does not depend on the choice of ample class [Kle68, Kle94]. By Proposition 12.1 it suffices to prove that a primitive Hodge class $\alpha \in H^{2p}(X,\mathbb{Q})$ is algebraic and, by hard Lefschetz, primitive classes occur only in degrees at most d, so $2p \leq d$.

If $2p = d$, then α is a primitive Hodge class of middle degree on a variety of even dimension, so it is algebraic by the second half of (ii).

If $2p < d$, the class $j^*\alpha$ is a rational Hodge class on Y because j^* is a morphism of Hodge structures. As $\dim Y = d-1 < d$, the inductive hypothesis makes $j^*\alpha$ algebraic, say $j^*\alpha = \mathrm{cl}(W)$ with $W \in \mathrm{CH}^p(Y)_{\mathbb{Q}}$. Then $\mathrm{cl}(j_*W) = j_*j^*\alpha = L\alpha$ by Lemma 10.2, so

$L\alpha$ is algebraic. By $B(X)$ the operator Λ carries algebraic classes to algebraic classes, and by Lemma 10.7, $\alpha = (d-2p)^{-1}\Lambda(L\alpha)$, which is therefore algebraic; the scalar is defined because $d - 2p \geq 1$. ∎

Remark 12.3 (The content of the equivalence). $B(X)$ is itself a consequence of (i), so Theorem 12.2 does not reduce the problem to a smaller one; it locates the difficulty. The induction in the proof that (ii) implies (i) runs through every degree except the middle one, where it rests entirely on the hypothesis. By Theorem 10.3, the step it uses below the middle degree, restriction to a hypersurface, returns zero on primitive classes of middle degree, so no refinement of that step extends the induction to that degree.

Corollary 12.4 (Primitive classes of middle degree)**.** *Any proof of the Hodge conjecture must, at some point, produce an algebraic cycle representing a primitive Hodge class of middle degree on a variety of even dimension, by a method that does not factor through restriction to hyperplane sections of that variety.*

Proof. By Theorem 12.2, the Hodge conjecture includes the algebraicity of the primitive Hodge classes of middle degree on varieties of even dimension, and by Corollary 10.4 every such class restricts to zero on every smooth ample hypersurface. ∎

Remark 12.5 (Methods that do not factor through hyperplane sections). Corollary 12.4 does not say that the conjecture is out of reach, and the settled cases come from known methods outside its scope. The Kuga–Satake construction (Section 11) replaces a weight-two Hodge structure of K3 type by the H^1 of an abelian variety; the class it produces is of type $(1,1)$ on a product, hence algebraic by Lefschetz, and it is not a restriction to a subvariety. Degenerations and limits of Hodge structures move the variety instead of cutting it. Correspondences from other varieties, of which the Kuga–Satake construction is one instance and Markman's construction [Mar25a, Mar25b] another, are also outside the scope. Arguments through the coniveau filtration, which show that a class supported on a proper closed subvariety is controlled by that subvariety, are compatible with everything here: they lower the dimension legitimately, because the class is supported on the smaller variety, whereas a primitive class of coniveau zero is supported on no proper closed subvariety.

13. Equivariance and Markman's construction

We return to abelian varieties of Weil type. By Theorem 5.6, a $K^\times$-eigenclass for the norm character has no component along the Weil line, and we ask what the contrapositive demands of a construction that does reach the Weil line. The demand does not concern the size of the object, the ambient geometry or any dimension count; it concerns the way the object transforms under $K^\times$. We state it in a concrete form and then show how Markman's construction meets it.

13.1. **The orbit criterion.**

Proposition 13.1 (Weil classes generate a plane)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type and let $\alpha \in W(A)$ be nonzero. Then*

$$\operatorname{span}_{\mathbb{Q}}\{\iota(\tau)^*\alpha \;:\; \tau \in K^\times\} \;=\; W(A),$$

a two-dimensional $\mathbb{Q}$-vector space, and as a representation of $K^\times$ it is irreducible over $\mathbb{Q}$.

Proof. By Proposition 3.7(i) the space $W(A)$ is free of rank one over K, and $\iota(\tau)^*$ acts on it as multiplication by the scalar $\tau^{2n} \in K$. Hence the span in question is $S \cdot \alpha$, where $S = \operatorname{span}_{\mathbb{Q}}\{\tau^{2n} : \tau \in K^\times\} \subseteq K$. Taking $\tau = 1$ shows $1 \in S$. By the argument in the proof

of Lemma 5.4 the map $c(\tau)=\tau/\overline{\tau}$ has infinite image while $\mu(K)$ is finite, so there is τ_0 with $(\tau_0/\overline{\tau_0})^{2n}\neq 1$, that is with $\tau_0^{2n}\neq\overline{\tau_0}^{2n}$, that is with $\tau_0^{2n}\notin\mathbb{Q}$. Then $S\supseteq\mathbb{Q}+\mathbb{Q}\tau_0^{2n}=K$, so $S=K$ and the span is $K\alpha=W(A)$.

For irreducibility, a proper nonzero $K^\times$-stable $\mathbb{Q}$-subspace would be a line $\mathbb{Q}\alpha'$, on which $K^\times$ would act by a $\mathbb{Q}$-valued character. Since $\iota(\tau)^*\alpha'=\tau^{2n}\alpha'$ and $W(A)$ is free over K, this forces $\tau^{2n}\in\mathbb{Q}$ for all τ, which we have just excluded. ∎

Corollary 13.2 (The orbit criterion)**.** *Let $\mathcal{E}$ be an object of $D^b(A)$, or Z a cycle in $\mathrm{CH}^n(A)_\mathbb{Q}$, whose class $\mathrm{ch}_n(\mathcal{E})$, respectively $\mathrm{cl}(Z)$, is a nonzero Weil class. Then there is no function $\chi\colon K^\times\to\mathbb{Q}$ with*

$$\iota(\tau)^*\,\mathrm{ch}_n(\mathcal{E})=\chi(\tau)\,\mathrm{ch}_n(\mathcal{E})\qquad\text{for all }\tau\in K^\times.$$

Equivalently, the $K^\times$-orbit of the class spans a plane and not a line.

Proof. Such a χ would make $\mathbb{Q}\cdot\mathrm{ch}_n(\mathcal{E})$ a $K^\times$-stable line inside $W(A)$, contradicting Proposition 13.1. ∎

Corollary 13.2 says in what sense a construction that reaches the Weil line must break equivariance. Suppose that a construction assigns to the data (A,ι,η) an object $\mathcal{E}(A,\iota,\eta)$ naturally, in the sense that an isomorphism of the data induces an isomorphism of the objects. For $\tau\neq 0$ in the order of K the endomorphism $\iota(\tau)$ is an isogeny of A commuting with ι and multiplying η by $\mathrm{N}(\tau)$, and one expects naturality to give $\iota(\tau)^*\mathcal{E}\cong\mathcal{E}$ up to a normalisation imposed by the change of polarisation, whose effect on ch_n is multiplication by a rational number $\chi(\tau)$, a power of the norm. Whenever that is the case, Corollary 13.2 shows that $\mathrm{ch}_n(\mathcal{E})$ is not a nonzero Weil class. A construction that reaches the Weil line must therefore depend on a choice that $\mathrm{Gal}(K/\mathbb{Q})$ moves. The rest of this section describes such a choice.

13.2. **The two conjugate isotropic subspaces.** The choice that $\mathrm{Gal}(K/\mathbb{Q})$ moves is already visible in Section 5. The splitting $V_\mathbb{C}=V_+\oplus V_-$ of (2) is defined over K and not over $\mathbb{Q}$, since conjugation interchanges the two summands. Hence $\bigwedge^{2n}V_+$ and $\bigwedge^{2n}V_-$ are each defined over K, and only their sum is rational. A class in $W(A)$ is of the form $w+\overline{w}$ with $w\in\bigwedge^{2n}V_+$, and we call $\bigwedge^{2n}V_+$ and $\bigwedge^{2n}V_-$ the two conjugate halves.

Definition 13.3 (Conjugate pair data)**.** By *conjugate pair data* for (A,ι,η) we mean a choice of one of the two summands $V_\pm$, equivalently of one of the two embeddings $K\hookrightarrow\mathbb{C}$, equivalently of one of the two points of a $\mathrm{Gal}(K/\mathbb{Q})$-orbit of size two attached to the K-structure.

Proposition 13.4 (Corestriction produces the plane)**.** *Let $w\in\bigwedge^{2n}V_+$ be nonzero. Then $w+\overline{w}$ and $\sqrt{-d}\,(w-\overline{w})$ lie in $W(A)$ and span it over $\mathbb{Q}$. In particular, the rational classes in the Weil line are the $\mathrm{Gal}(K/\mathbb{Q})$-corestrictions of the classes in one conjugate half, and any construction producing such a class must produce, in effect, a pair of conjugate inputs and combine them.*

Proof. Conjugation interchanges w and $\overline{w}$ and sends $\sqrt{-d}$ to $-\sqrt{-d}$, so both elements are fixed by it and are therefore rational. Both lie in $\bigwedge^{2n}V_+\oplus\bigwedge^{2n}V_-$ by construction, hence in $W(A)$ by Definition 3.6. They are linearly independent over $\mathbb{Q}$ because w and $\overline{w}$ are linearly independent over $\mathbb{C}$: a relation $q_1(w+\overline{w})+q_2\sqrt{-d}\,(w-\overline{w})=0$ with $q_1,q_2\in\mathbb{Q}$ forces $q_1+q_2\sqrt{-d}=0$, that is $q_1=q_2=0$. Since $W(A)$ has dimension two by Proposition 3.7(i), they span it. ∎

A construction reaching the Weil line therefore has to be, in a suitable sense, bilinear in the two conjugate halves: it takes one input attached to V_+ and one attached to V_- and pairs them. A construction fed a single K-invariant input cannot do this, by Corollary 13.2.

13.3. **Markman's construction.** Markman's construction [Mar25a, Mar25b] produces Weil classes, and it meets the criterion in the manner just described. We set out only the structure relevant here; the details are in [Mar25b].

The construction does not begin on A or on $A^\vee$. It begins on an abelian n-fold X, that is, on a variety of half the dimension of A. Inside $H^{\mathrm{ev}}(X, \mathbb{Q})$ one selects a plane P spanned by Hodge classes, and one requires that P meet the spinor variety of even pure spinors in two points. Those two points are conjugate over the imaginary quadratic field K; they correspond to two maximal isotropic subspaces W_1 and W_2 that are interchanged by $\mathrm{Gal}(K/\mathbb{Q})$. This is the conjugate pair data of Definition 13.3, realised concretely.

To the two points one attaches *secant sheaves* on X, coherent sheaves whose Chern characters are rational points of P. In the worked case in which X is the Jacobian of a curve of genus three, the secant sheaf is $\mathcal{I}_{\cup C_i}(\Theta)$, the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi image of the curve. Call the two sheaves used $\mathcal{F}_1$ and $\mathcal{F}_2$.

The abelian variety of Weil type is then $A = X \times \widehat{X}$, of dimension $2n$, with its K-action and polarisation inherited from P. The object that carries the Weil class is

$$E = \Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2), \tag{28}$$

where $\Phi\colon D^b(X \times X) \to D^b(X \times \widehat{X})$ is Orlov's derived equivalence. One then shows that the characteristic class $\kappa(E)$ remains of Hodge type under every deformation of $X \times \widehat{X}$ as a polarised abelian $2n$-fold of Weil type, which is what makes the class algebraic on the whole family. The conclusion in [Mar25b] is the algebraicity of the Weil classes for abelian sixfolds of Weil type of trivial discriminant in the normalisation of Definition 15.22 (discriminant -1 in Markman's normalisation), and, through Schoen's product argument [Sch98, Proposition 10], for all abelian fourfolds of Weil type, whence the Hodge conjecture for abelian fourfolds.

Proposition 13.5 (Which hypotheses are escaped)**.** *The object* (28) *satisfies none of the hypotheses of Theorem 7.3 and does not satisfy the hypothesis of Theorem 7.7. Specifically:*

(i) The source is not a projective space. *The input $\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2$ lives on $X \times X$, an abelian variety. The proof of Theorem 7.3 used that $H^*(\mathbb{P}^{N-1}, \mathbb{Q})$ is generated in degree two, so that* ch *of any object is a polynomial in the hyperplane class. The cohomology of $X \times X$ is not generated in degree two: it contains Künneth classes of every degree, and $\mathrm{ch}(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$ is not confined to any divisor subring.*

(ii) The transform is not $\Phi_{\mathcal{P}}$ on $A \times A^\vee$. *The relevant equivalence is between $D^b(X \times X)$ and $D^b(X \times \widehat{X})$, and Theorem 6.5, which computes the transform with Poincaré kernel on powers of the polarisation of A, says nothing about it.*

(iii) The construction is not K-equivariant. *The two tensor factors of* (28) *carry the two conjugate points of $P \cap$ (spinor variety), one each. Interchanging them is the action of the nontrivial element of* $\mathrm{Gal}(K/\mathbb{Q})$*, and it exchanges E with a different object. Hence $\mathrm{ch}_n(E)$ is not a $K^\times$-eigenclass for the norm, and Corollary 13.2 is satisfied rather than violated.*

Proof. Items (i) and (ii) are a comparison of hypotheses and need no argument beyond reading (28). For (iii), if $\mathrm{ch}_n(E)$ were an eigenclass for the norm, then by Corollary 5.7

it would have zero component along the Weil line, whereas Markman shows that this component is nonzero [Mar25b]. ■

The three hypotheses of Proposition 13.5 are related. A construction that takes its input from projective space is normally equivariant for whatever acts on the input, and an equivariant construction normally returns a single eigenclass, so the three tend to hold or fail together.

Remark 13.6 (Where secant geometry must be placed). Secant geometry can produce Weil classes: Markman's construction is a secant construction. The obstructions bear on its placement. Secant geometry placed on the dual $A^\vee$ of the $2n$-fold inside an ambient projective space, and transported by the Poincaré kernel, has three defects at once, by Proposition 13.5: its input lies where the cohomology is generated by a divisor, it uses the transform to which Theorem 6.5 applies, and it feeds the construction K-invariant data, since a secant variety of a K-invariant subvariety is K-invariant. Placing the secant geometry on an abelian n-fold removes all three, and it does so because halving the dimension makes room for the two conjugate inputs.

13.4. **Beyond the known range.** The algebraicity of the Weil classes is known for $n = 2$, for every discriminant, and for $n = 3$ with trivial discriminant; see [Mar25b] and, for a different route at $n = 2$ and trivial discriminant, [FF26]. We state what a construction of the same shape would have to supply for $n \geq 4$.

Question 13.7. Fix $n \geq 4$ and an imaginary quadratic field K. Does there exist an abelian n-fold X, a plane $P \subset H^{\mathrm{ev}}(X, \mathbb{Q})$ spanned by Hodge classes meeting the spinor variety in two points conjugate over K, secant sheaves $\mathcal{F}_1, \mathcal{F}_2$ on X realising those two points, and a deformation argument showing that $\kappa(\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2))$ remains of Hodge type across the moduli of polarised abelian $2n$-folds of Weil type with multiplication by K?

Each of the four ingredients is a concrete demand. By Corollary 13.2, any answer must be bilinear in a pair of conjugate inputs, since that is the only way a rational class can acquire a $K^\times$-orbit spanning a plane. Section 14 supplies the first three ingredients explicitly, for every $n \geq 2$ and every imaginary quadratic field, with X an arbitrary polarised abelian n-fold. The fourth, the deformation argument, is isolated in Remark 14.16, which says exactly what it requires.

14. An explicit secant plane in every dimension

Question 13.7 asks for four things: an abelian n-fold X, a plane $P \subset H^{\mathrm{ev}}(X, \mathbb{Q})$ of Hodge classes meeting the spinor variety in two points conjugate over K, a pair of secant sheaves realising the plane, and a deformation argument. This section supplies the first three, for every $n \geq 2$ and every imaginary quadratic field K. The first two are written in closed form (Theorems 14.2 and 14.5). The third is first reduced to an explicit equation on a Chern character (Theorem 14.7) and then solved, in every dimension, by sheaves of positive rank whose classes are combinations of the powers of a single line bundle (Theorem 14.13). The fourth ingredient is the deformation argument, and Remark 14.16 says exactly what it requires.

The framework is Markman's [Mar25a, Mar25b]: the Weil structure on $X \times \widehat{X}$ obtained by embedding K into $\mathrm{End}^0(X \times \widehat{X})$, the notion of a K-secant sheaf, whose Chern character lies in a plane meeting the spinor variety at two conjugate points, and the deformation step that completes the argument at $n = 3$ are all his. What we add is the explicit form of the first three ingredients, uniformly in n and K. The closed form rests on two facts: the two

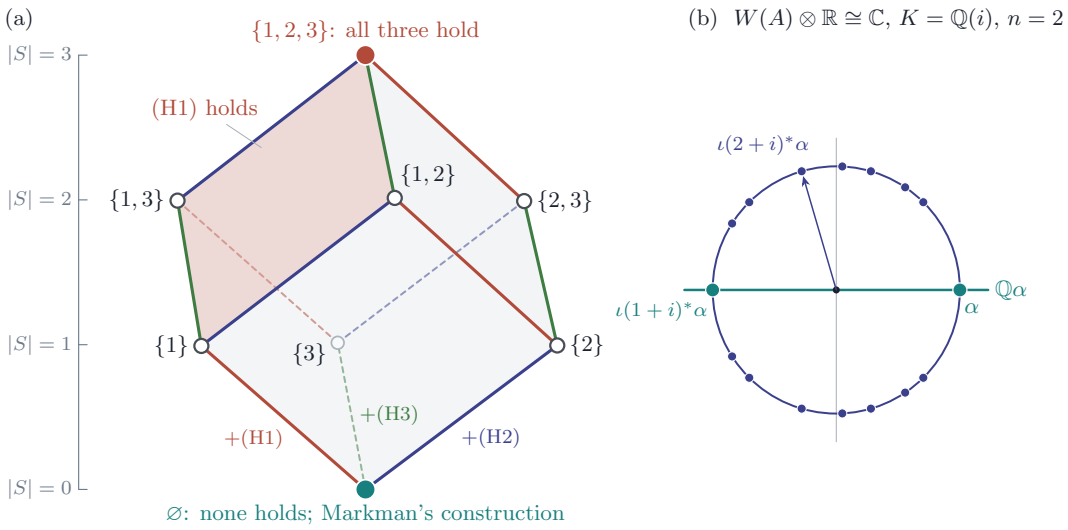


FIGURE 12. (a) The eight positions a construction can occupy, drawn as the Boolean lattice of the subsets S of three hypotheses: (H1) the Chern character of the input lies in the subring generated by divisor classes; (H2) the output is a $K^\times$-eigenclass; (H3) the output spans a single line rather than a plane. (H1) and (H2) are the hypotheses negated in (i) and (iii) of Proposition 13.5, and (H3) is what Corollary 13.2 excludes for a nonzero Weil class. At the top vertex all three hold, and both Theorem 7.3 and Theorem 5.6 apply; on the shaded face (H1) holds, and Theorem 7.3 alone already applies. Markman's construction sits at the bottom vertex, where none of the three holds. (b) The orbit of a Weil class α under $K^\times$ for $K = \mathbb{Q}(i)$ and $n = 2$, in $W(A) \otimes \mathbb{R} \cong \mathbb{C}$ and normalised to the unit circle. Since $\iota(\tau)^*$ acts through τ^4, the class $\iota(1+i)^*\alpha = -4\alpha$ stays on the line $\mathbb{Q}\alpha$, while $\iota(2+i)^*\alpha$, with $(2+i)^4 = -7+24i$, does not; the orbit spans the plane, as Proposition 13.1 asserts.

conjugate isotropic subspaces demanded by Corollary 13.2 are the graphs of multiples of a single element of the Néron–Severi group, and the pure spinor of the graph of a two-form is the exponential of that form.

14.1. **The Weil structure.**

Setup 14.1. Let X be a complex abelian variety of dimension n with $H_1(X, \mathbb{Q}) = U$ of dimension $2n$ and complex structure J on $U \otimes \mathbb{R}$. Let d be a squarefree positive integer, $s = \sqrt{-d}$ and $K = \mathbb{Q}(s)$. Let $\beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$ be nondegenerate, regarded as an alternating form on U, so that the Riemann relation $J^\top \beta J = \beta$ holds, and write $\phi_\beta \colon X \to \widehat{X}$ for the associated isogeny. Put

$$A = X \times \widehat{X}, \qquad \dim A = 2n, \qquad H_1(A, \mathbb{Q}) = U \oplus U^\vee,$$

and let

$$M = \begin{pmatrix} 0 & -\beta^{-1} \\ d\,\beta & 0 \end{pmatrix} \in \mathrm{End}^0(A), \tag{29}$$

acting on $H_1(A, \mathbb{Q}) = U \oplus U^\vee$ by $(u, \phi) \mapsto (-\beta^{-1}\phi,\ d\,\beta u)$.

Theorem 14.2 (An explicit Weil structure)**.** *In Setup 14.1:*

(i) $M^2 = -d$, *so* $\iota(s) = M$ *defines an embedding* $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$*;*
(ii) M *commutes with the complex structure of* A*, and for an alternating form* β *on* U *this is equivalent to the Riemann relation, that is to* $\beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$*;*
(iii) M *acts on* $H^{1,0}(A)$ *with eigenvalues* s *and* $-s$*, each of multiplicity* n*, so* (A, ι, η) *is of* $(K, -1, n)$*-Weil type for a suitable polarisation* η*;*
(iv) *the eigenspaces* $W_\pm \subset H_1(A, \mathbb{C})$ *of* M *for* $\pm s$ *are maximal isotropic for the canonical pairing on* $U \oplus U^\vee$*, are interchanged by complex conjugation, and are the graphs of the two-forms* $\mp s\beta$*.*

Proof. (i) Block multiplication gives $M^2 = \mathrm{diag}(-\beta^{-1}\, d\beta,\ -d\beta\, \beta^{-1}) = -d \cdot \mathrm{id}$.

(ii) The complex structure of $A = X \times \widehat{X}$ is $J_A = \mathrm{diag}(J, -J^\top)$. The relation $MJ_A = J_A M$ amounts to the two identities $d\beta J = -dJ^\top \beta$ and $-\beta^{-1}(-J^\top) = J(-\beta^{-1})$, and each is equivalent to $\beta J = -J^\top \beta$. This is the Riemann relation: from $J^\top \beta J = \beta$ and $J^2 = -1$ one gets $J^\top \beta = \beta J^{-1} = -\beta J$, as required. Conversely $\beta J = -J^\top \beta$ gives $J^\top \beta J = -\beta J J = \beta$.

(iii) M is block anti-diagonal and $H^{1,0}(A) = H^{1,0}(X) \oplus H^{1,0}(\widehat{X})$ with both summands of dimension n, and M exchanges them. If $M(x, y) = \lambda(x, y)$ then $-\beta^{-1} y = \lambda x$ and $d\beta x = \lambda y$, so $y = d\beta x/\lambda$ and, substituting, $-dx/\lambda = \lambda x$, that is $\lambda^2 = -d$. For each of the two values of λ the assignment $x \mapsto (x, d\beta x/\lambda)$ is an isomorphism from $H^{1,0}(X)$ onto the λ-eigenspace, which is therefore of dimension n. That is the balanced signature of Definition 3.1(ii). For (iii) of that definition, let η_0 be any polarisation of A and put $\eta(x, y) = \eta_0(x, y) + d^{-1}\eta_0(Mx, My)$. A positive integer multiple mM of M is an isogeny, so the second term is the positive multiple $(dm^2)^{-1}(mM)^*\eta_0$ of a polarisation, and η is a polarisation. Since $M^2 = -d$, $\eta(Mx, My) = \eta_0(Mx, My) + d\,\eta_0(x, y) = d\,\eta(x, y)$, and as $M^{-1} = -M/d$ this gives $\eta(Mx, y) = -\eta(x, My)$: the Rosati involution of η sends M to $-M$, that is $\iota(\tau)^\dagger = \iota(\overline{\tau})$. When β is a polarisation an explicit choice of η is recorded in Proposition 15.19(i).

(iv) The same computation identifies $W_\lambda = \{(x, d\beta x/\lambda)\}$, the graph of the form $d\beta/\lambda$, and $d\beta/\lambda = d\beta/(\pm s) = \mp s\beta$ since $d/s = -s$. The canonical pairing on $U \oplus U^\vee$ is $\langle (x, \xi), (x', \xi') \rangle = \xi'(x) + \xi(x')$, and on the graph it gives $d\beta(x', x)/\lambda + d\beta(x, x')/\lambda = 0$ because β is alternating; so W_λ is isotropic, and it has dimension $2n$, half of $\dim(U \oplus U^\vee) = 4n$, hence is maximal. Conjugation sends s to $-s$ and so interchanges the two. ∎

The placement of the factor d in (29) is a convention, fixed once and for all. The endomorphism $\left(\begin{smallmatrix} 0 & -d\beta^{-1} \\ \beta & 0 \end{smallmatrix}\right)$ also squares to $-d$, commutes with the complex structure and defines a Weil structure on the same A, but it is a different embedding of K: its eigenspaces are the graphs of $\mp s\beta/d$ rather than of $\mp s\beta$, its polarisation is $\beta + d\widehat{\beta}$ rather than $d\beta + \widehat{\beta}$, and the class in $\bigwedge^2 V_+ \oplus \bigwedge^2 V_-$ it produces is $\beta - d\widehat{\beta}$ rather than $\gamma = d\beta - \widehat{\beta}$. The convention of (29) is the one whose action on cohomology is $x \mapsto -Bx$, $\xi \mapsto dB^{-1}\xi$ in the proof of Theorem 15.2, and every formula below refers to it. For $d = 1$ the two conventions coincide.

Remark 14.3. Theorem 14.2 says that abelian $2n$-folds of Weil type of the split form $X \times \widehat{X}$ exist for every n, every imaginary quadratic K and every polarised abelian n-fold X, and that the Weil structure is written down by one element of $\mathrm{NS}(X)$. These are not general members of the moduli of Weil type. The Weil classes on them are of interest because Markman's deformation argument, where it can be carried out, propagates algebraicity from them to the general member; see Remark 14.16.

14.2. **The secant plane.** Write $t = \beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$, so that by Theorem 14.2(iv) the two conjugate maximal isotropic subspaces are the graphs of $\mp s\, t$; we use a separate letter because everything below holds for any nondegenerate $t \in \mathrm{NS}(X) \otimes \mathbb{Q}$. The pure spinor attached to the graph of a two-form ω is e^{ω}, so the two conjugate points of the even spinor variety determined by the Weil structure are $[e^{-st}]$ and $[e^{st}]$.

Definition 14.4 (The secant plane)**.** With t and d as above, put

$$u_t = \sum_{j \geq 0} \frac{(-d)^j}{(2j)!}\, t^{2j}, \qquad v_t = \sum_{j \geq 0} \frac{(-d)^j}{(2j+1)!}\, t^{2j+1}, \tag{30}$$

both sums being finite because $t^k = 0$ for $k > n$, and let $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t \subset H^{\mathrm{ev}}(X, \mathbb{Q})$.

Theorem 14.5 (The plane is an explicit rational secant)**.** *With notation as above:*

(i) *u_t and v_t are rational Hodge classes, and $e^{\pm st} = u_t \pm s\, v_t$;*

(ii) *P_t is a two-dimensional $\mathbb{Q}$-subspace of $H^{\mathrm{ev}}(X, \mathbb{Q})$ spanned by Hodge classes, and $\mathbb{P}(P_t)$ is a line defined over $\mathbb{Q}$ passing through the two conjugate points $[e^{\pm st}]$ of the even spinor variety;*

(iii) *that line is not contained in the spinor variety, so it meets it in exactly those two points.*

Proof. (i) Each t^k is a rational Hodge class because $t \in \mathrm{NS}(X) \otimes \mathbb{Q}$ is of type $(1,1)$ and the algebraic classes form a subring. Separating $e^{st} = \sum_k s^k t^k / k!$ into even and odd powers of s and using $s^{2j} = (-d)^j$, $s^{2j+1} = (-d)^j s$ gives $e^{st} = u_t + s v_t$, and conjugating gives $e^{-st} = u_t - s v_t$.

(ii) The two elements are linearly independent over $\mathbb{Q}$ because u_t has nonzero component in H^0 and v_t does not, while $v_t \neq 0$ since $t \neq 0$. From (i), both e^{st} and e^{-st} lie in $P_t \otimes K$, so $\mathbb{P}(P_t)$ passes through the two points.

(iii) The even spinor variety is cut out by quadrics, by Cartan's description of pure spinors, so a line either lies inside it or meets it in a subscheme of length at most two. The point $[u_t]$ of $\mathbb{P}(P_t)$ is not a pure spinor: a pure spinor with nonzero component in H^0 is a scalar multiple of e^{ω} for a two-form ω, and u_t has zero component in H^2 while its component in H^4 is $-\frac{d}{2}t^2 \neq 0$, whereas e^{ω} with vanishing H^2 part is 1. So the line is not contained in the spinor variety. The two points of (ii) are distinct, since $u_t + s v_t$ and $u_t - s v_t$ have the same component 1 in H^0 and $v_t \neq 0$; hence they exhaust the intersection. ∎

Figure 15 draws the case $n = 2$, in which the line $\mathbb{P}(P_t)$ meets the spinor quadric in the two conjugate points $[e^{\pm st}]$.

Example 14.6. For $n = 4$ and $d = 1$, $u_t = 1 - \frac{1}{2}t^2 + \frac{1}{24}t^4$ and $v_t = t - \frac{1}{6}t^3$. For $n = 4$ and $d = 3$, $u_t = 1 - \frac{3}{2}t^2 + \frac{3}{8}t^4$ and $v_t = t - \frac{1}{2}t^3$. All four classes are rational and lie in the subring $\mathbb{Q}[t]$.

14.3. **The Chern character equation.** The next theorem computes the invariants that a sheaf with Chern character in the secant plane must have. On the side of sheaves, the demands of Question 13.7 reduce to this one equation.

Theorem 14.7 (The invariants of a secant sheaf)**.** *Let X, t, d be as above and let $\mathcal{F}$ be a coherent sheaf on X with $\mathrm{ch}(\mathcal{F}) \in P_t$, say $\mathrm{ch}(\mathcal{F}) = a\, u_t + b\, v_t$ with $a, b \in \mathbb{Q}$. Then*

$$\mathrm{rk}\, \mathcal{F} = a, \qquad c_1(\mathcal{F}) = b\, t, \qquad \mathrm{ch}_2(\mathcal{F}) = -\tfrac{ad}{2}\, t^2, \qquad c_2(\mathcal{F}) = \tfrac{ad+b^2}{2}\, t^2,$$

and more generally $\mathrm{ch}_k(\mathcal{F})$ *is the degree* k *term of* (30) *scaled by* a *or* b *according to the parity of* k*. The Bogomolov discriminant is*

$$\Delta(\mathcal{F}) = 2a\, c_2(\mathcal{F}) - (a-1)c_1(\mathcal{F})^2 = \mathrm{N}_{K/\mathbb{Q}}(b+as)\cdot t^2. \tag{31}$$

In particular, if t *is ample and* $\mathcal{F}\neq 0$*, then* $\Delta(\mathcal{F})\cdot H^{n-2}>0$ *for every ample* H*, so the Bogomolov inequality places no obstruction on the existence of* $\mathcal{F}$*.*

Proof. The first four identities are read off from (30) degree by degree, using $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ to convert ch_2 into c_2. For (31),

$$2a\cdot\tfrac{ad+b^2}{2}t^2 - (a-1)b^2t^2 = (a^2d + ab^2 - ab^2 + b^2)t^2 = (a^2d+b^2)t^2,$$

and $\mathrm{N}_{K/\mathbb{Q}}(b+as) = (b+as)(b-as) = b^2 + a^2d$. If $\mathcal{F}\neq 0$ then $\mathrm{ch}(\mathcal{F})\neq 0$, so $(a,b)\neq(0,0)$ and the norm is positive; and $t^2H^{n-2}>0$ for t and H ample. This gives the last assertion. ∎

Remark 14.8 (The norm form as the discriminant). Identity (31) says that the discriminant of the sheaf is the norm form of K evaluated at the coordinates (a,b) of its Chern character in the plane. The two conjugate points of the spinor variety are the two isotropic directions of this norm form after extension of scalars to K, and no nonzero rational point of the plane is isotropic. The Chern character of a secant sheaf is a rational point of the line $\mathbb{P}(P_t)$, and so never an isotropic one, while the Weil structure is carried by the two isotropic points of that line, which are defined over K. In this sense the conjugate pair of Corollary 13.2 is built into the invariants of the sheaf.

Corollary 14.9 (The case of an abelian surface)**.** *Let* $n=2$*, so* X *is an abelian surface and* $A = X\times\widehat{X}$ *an abelian fourfold of Weil type, and let* t *be ample. A nonzero sheaf with* $\mathrm{ch}(\mathcal{F})\in P_t$ *has Mukai vector* $v(\mathcal{F}) = (a,\, bt,\, -\frac{ad}{2}t^2)$ *and Mukai self-pairing*

$$\langle v(\mathcal{F}), v(\mathcal{F})\rangle = \mathrm{N}_{K/\mathbb{Q}}(b+as)\cdot t^2 > 0.$$

Proof. On an abelian surface the Mukai pairing of $v=(r,c,e)$ with itself is c^2-2re, which here is $b^2t^2 - 2a(-\frac{ad}{2}t^2) = (b^2+a^2d)t^2$. It is positive because $(a,b)\neq(0,0)$ and $t^2>0$. ∎

Corollary 14.9 is consistent with the case $n=2$ being settled: on an abelian surface, positivity of the self-pairing of a primitive Mukai vector of positive rank is sufficient for the moduli spaces of stable sheaves with that vector to be non-empty.

14.4. **The genus three Jacobian.** At $n=3$ the equation of Theorem 14.7 determines a secant sheaf of rank one almost completely, and it reproduces the shape of the object Markman uses. The computation serves as a check on the framework of this section.

Proposition 14.10 (Rank one at $n=3$)**.** *Let* X *be the Jacobian of a smooth curve of genus three, let* Θ *be the theta divisor and* $C\subset X$ *the Abel–Jacobi curve, so that* $\Theta^2/2 = [C]$*. Take* $t=\Theta$ *and* $a=1$ *in Theorem 14.7. Then a rank one torsion-free sheaf* $\mathcal{F}$ *with* $\mathrm{ch}(\mathcal{F})\in P_\Theta$ *and* $c_1(\mathcal{F}) = b\Theta$ *is of the form* $\mathcal{F} = \mathcal{I}_Z(b\Theta)$ *with*

$$[Z] = \frac{b^2+d}{2}\,\Theta^2 = (b^2+d)\,[C].$$

If Z *is a disjoint union of* m *translates of* C*, then* $b=1$ *and*

$$m = d+1.$$

Proof. A rank one torsion-free sheaf on a smooth projective variety is the tensor product of its double dual, which is a line bundle, with the ideal sheaf of a subscheme of codimension at least two. With $c_1 = b\Theta$ this gives $\mathcal{F} = \mathcal{I}_Z(b\Theta)$ for such a subscheme $Z \subset X$, and $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = e^{b\Theta}(1 - \operatorname{ch}(\mathcal{O}_Z))$. Comparing the components in H^4 with Theorem 14.7 at $a = 1$, which gives $\operatorname{ch}_2 = -\frac{d}{2}\Theta^2$, yields $\frac{b^2}{2}\Theta^2 - [Z] = -\frac{d}{2}\Theta^2$, that is $[Z] = \frac{b^2+d}{2}\Theta^2$. The Poincaré formula $\Theta^{g-1}/(g-1)! = [C]$ at $g = 3$ reads $\Theta^2/2 = [C]$, whence $[Z] = (b^2 + d)[C]$.

Suppose now that Z is a disjoint union of m translates of C. Translates of C are homologous to C, so $m = b^2 + d$. Each translate is a smooth curve of genus three and X has trivial tangent bundle, so Grothendieck–Riemann–Roch gives $\operatorname{ch}(\mathcal{O}_Z) = m[C] - 2m[\mathrm{pt}]$. Using $\Theta^3 = 6[\mathrm{pt}]$ and $\Theta \cdot [C] = 3[\mathrm{pt}]$, the component of $\operatorname{ch}(\mathcal{I}_Z(b\Theta))$ in H^6 is $(b^3 - 3bm + 2m)[\mathrm{pt}]$, and Theorem 14.7 requires it to equal the degree three term $-bd\,\Theta^3/3! = -bd\,[\mathrm{pt}]$. Substituting $m = b^2 + d$ gives $-2(b-1)(b^2+d) = 0$, so $b = 1$ and $m = d + 1$. ∎

d	1	2	3	5	7	11
$m = d+1$	2	3	4	6	8	12

TABLE 2. The number $m = d + 1$ of translates of the Abel–Jacobi curve in Proposition 14.10, as a function of the field $K = \mathbb{Q}(\sqrt{-d})$.

Remark 14.11. The sheaf used in [Mar25b] on a genus three Jacobian is $\mathcal{I}_{\cup C_i}(\Theta)$, the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi curve. This is the shape that Proposition 14.10 forces, including the twist by Θ itself, with the number of translates determined by the field.

14.5. **Existence of secant sheaves.** Theorem 14.7 describes a secant sheaf but does not produce one. We show that such sheaves exist on every polarised abelian variety, in every dimension and for every imaginary quadratic field, and that their classes can be assembled from the powers of a single line bundle.

Two observations do the work. The first is that the Chern character wanted is a rational point of a plane, and a plane has no preferred scale: a positive integer multiple Mp of a rational point p of P_t is again a rational point of P_t, and since the Hodge conjecture concerns classes with rational coefficients, replacing p by Mp costs nothing. The second is that, once denominators may be cleared, the Chern characters of the powers of one line bundle span the subring $\mathbb{Q}[t]$ in which the secant plane lies, because the matrix expressing them in the basis $t^k/k!$ is a Vandermonde matrix on distinct nodes. What remains is the passage from a class in K_0 to a sheaf, and for that the only hypothesis needed is that the rank be positive.

Lemma 14.12 (Positive rank is enough)**.** *Let Y be a smooth projective variety over a field and let $\alpha \in K_0(Y)$ satisfy $\operatorname{rk}\alpha > 0$. Then $\alpha = [\mathcal{F}]$ for some coherent sheaf $\mathcal{F}$ on Y.*

Proof. Write $m = \dim Y$ and $a = \operatorname{rk}\alpha \geq 1$, and let $F^cK_0(Y)$ be the topological filtration, the subgroup generated by the classes $[\mathcal{O}_Z]$ of closed integral subschemes $Z \subseteq Y$ of codimension at least c. Since Y is smooth, every coherent sheaf has a finite locally free resolution, so $K_0(Y) = K_0'(Y)$ and the filtration is exhaustive with $F^{m+1} = 0$. The class of a coherent sheaf supported in codimension at least c lies in F^c and is congruent modulo F^{c+1} to $\sum_i \ell_i[\mathcal{O}_{Z_i}]$, where $\sum_i \ell_i Z_i$ is its cycle in codimension c; and F^1 is the kernel of the

rank, while F^1/F^2 is $\mathrm{Pic}(Y)$ through the first Chern class [Ful98, §15.1 and Ex. 15.1.5]. We construct coherent sheaves $\mathcal{F}_c$ of rank a with $\alpha - [\mathcal{F}_c] \in F^c$, by induction on c.

Start at $c = 2$. Let L be a line bundle with $c_1(L) = c_1(\alpha)$ and set $\mathcal{F}_2 = L \oplus \mathcal{O}_Y^{\oplus(a-1)}$. Then $\alpha - [\mathcal{F}_2]$ has rank zero and first Chern class zero, hence lies in F^2.

For the inductive step let $\delta = \alpha - [\mathcal{F}_c] \in F^c$ with $c \geq 2$. Modulo F^{c+1} the class δ is a finite sum $\sum_j m_j[\mathcal{O}_{Z_j}]$ with $m_j \in \mathbb{Z}$ and $\operatorname{codim} Z_j = c$. The terms with $m_j > 0$ are removed by replacing $\mathcal{F}_c$ with $\mathcal{F}_c \oplus \bigoplus_{m_j>0} \mathcal{O}_{Z_j}^{\oplus m_j}$, which changes neither the rank nor the codimension of the remaining difference. For a term with $m_j < 0$ we must subtract, and here the rank is used. The restriction $\mathcal{F}_c \otimes \mathcal{O}_{Z_j}$ is a quotient of $\mathcal{F}_c$ and has rank at least $a \geq 1$ at the generic point of the integral scheme Z_j, by semicontinuity of the fibre dimension. Dividing $\mathcal{F}_c \otimes \mathcal{O}_{Z_j}$ by a subsheaf of generic corank one gives a surjection $\mathcal{F}_c \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ a coherent $\mathcal{O}_{Z_j}$-module of rank one. Then $[\mathcal{Q}] \in F^c$ with associated cycle $[Z_j]$, so $[\mathcal{Q}] - [\mathcal{O}_{Z_j}] \in F^{c+1}$. The kernel of that surjection is a coherent sheaf, still of rank a because $\mathcal{Q}$ is supported in positive codimension, and its class is $[\mathcal{F}_c] - [\mathcal{Q}]$. Performing this $|m_j|$ times for each j with $m_j < 0$ produces $\mathcal{F}_{c+1}$ with $\alpha - [\mathcal{F}_{c+1}] \in F^{c+1}$ and $\operatorname{rk} \mathcal{F}_{c+1} = a$. At $c = m + 1$ the difference lies in $F^{m+1} = 0$, so $\alpha = [\mathcal{F}_{m+1}]$. ∎

Theorem 14.13 (Secant sheaves exist in every dimension)**.** *Let X be an abelian variety of dimension n, let $t \in \mathrm{NS}(X) \otimes \mathbb{Q}$ be a polarisation and let $d > 0$. Let p be a rational point of the secant plane P_t whose u_t coordinate is positive. Then there are a positive integer M, an integer $D > 0$ with $Dt \in \mathrm{NS}(X)$, integers $n_0, \dots, n_n$ and a coherent sheaf $\mathcal{F}$ on X such that, setting*

$$\alpha = \sum_{j=0}^{n} n_j [\mathcal{O}_X(jDt)] \in K_0(X), \tag{32}$$

one has $\alpha = [\mathcal{F}]$ in $K_0(X)$, $\operatorname{ch}(\mathcal{F}) = M\,p$ and $\operatorname{rk} \mathcal{F} = M\,a > 0$, where a is the u_t coordinate of p. The integers M and n_j depend only on n, d, D and the coordinates of p, and are obtained by inverting one Vandermonde matrix. Moreover Mp is again a rational point of P_t; applying the statement to two linearly independent rational points of P_t produces a pair of coherent sheaves whose Chern characters span P_t.

Proof. If the conclusion holds for Np, where N is a positive integer, with multiple M', then it holds for p with $M = NM'$, and the hypothesis on p is unchanged by the scaling. Clearing the denominators of the coordinates of p, we may therefore assume $p = au_t + bv_t$ with $a, b \in \mathbb{Z}$ and $a > 0$. Choose $D > 0$ with $Dt \in \mathrm{NS}(X)$, so that $\mathcal{O}_X(jDt)$ is a line bundle for every $j \in \mathbb{Z}$.

Since t is a polarisation, $t^n \neq 0$ and $t^{n+1} = 0$ in $H^{\mathrm{ev}}(X, \mathbb{Q})$, so the $n + 1$ classes $t^k/k!$ with $0 \leq k \leq n$ are a basis of the subring $\mathbb{Q}[t]$. In that basis

$$p = \sum_{k=0}^{n} c_k \frac{t^k}{k!}, \qquad c_k = \begin{cases} a(-d)^{k/2}, & k \text{ even}, \\ b(-d)^{(k-1)/2}, & k \text{ odd}, \end{cases}$$

by (30), and all the c_k are integers. On the other side,

$$\operatorname{ch}(\mathcal{O}_X(jDt)) = e^{jDt} = \sum_{k=0}^{n} (jD)^k \frac{t^k}{k!},$$

so in the same basis the $n + 1$ vectors $\operatorname{ch}(\mathcal{O}_X(jDt))$ for $j = 0, \dots, n$ are the columns of the Vandermonde matrix $\left((jD)^k\right)_{0 \leq j,k \leq n}$ on the $n + 1$ distinct nodes $0, D, 2D, \dots, nD$. Its

determinant is $\prod_{0\le i<j\le n}(jD-iD) = D^{n(n+1)/2}\prod_{0\le i<j\le n}(j-i)\neq 0$, so the system

$$\sum_{j=0}^{n} x_j\,(jD)^k = c_k \qquad (0\le k\le n)$$

has a unique solution $x\in\mathbb{Q}^{n+1}$. Let M be the least common multiple of the denominators of the x_j and put $n_j = Mx_j\in\mathbb{Z}$. With α as in (32),

$$\operatorname{ch}(\alpha) = \sum_j n_j e^{jDt} = \sum_{k=0}^{n}\Big(\sum_j Mx_j(jD)^k\Big)\frac{t^k}{k!} = M\sum_{k=0}^{n} c_k\frac{t^k}{k!} = Mp.$$

In particular $\operatorname{rk}\alpha = \operatorname{ch}_0(\alpha) = Mc_0 = Ma > 0$, so Lemma 14.12 gives a coherent sheaf $\mathcal{F}$ with $[\mathcal{F}]=\alpha$. The Chern character is a function of the class in $K_0(X)$, whence $\operatorname{ch}(\mathcal{F}) = Mp$ and $\operatorname{rk}\mathcal{F} = Ma$. Finally $Mp = (Ma)u_t + (Mb)v_t$ is a rational point of P_t, and two linearly independent rational points stay linearly independent after scaling, which gives the last assertion. ∎

Figure 14 draws the theorem at $n=2$ and $d=2$, with the points of the secant plane that integer combinations of line bundles reach.

Remark 14.14 (The factorials and the multiple M). The integer M is not an artefact of the proof. The lattice spanned by the Chern characters of the powers of a single line bundle is spanned equally well by the classes $(e^t-1)^k$ for $0\le k\le n$, because the $(n+1)$st finite difference of a polynomial of degree n vanishes; and $(e^t-1)^k = t^k+\cdots$ has leading term t^k, not $t^k/k!$. The secant plane, by contrast, is spanned by classes whose coordinates in the basis $t^k/k!$ are integers. The index between the two lattices can be computed exactly. Write $j^k = \sum_{i\le k} S(k,i)\,i!\binom{j}{i}$ with the Stirling numbers $S(k,i)$ of the second kind. As j runs over $0,\dots,n$, the vectors $\big(\binom{j}{i}\big)_{0\le i\le n}$ form a unitriangular basis of $\mathbb{Z}^{n+1}$, so the vectors $(j^k)_{0\le k\le n}$, which are the coordinates of $\operatorname{ch}\mathcal{O}_X(jt)$ in the basis $t^k/k!$, span the image of $\mathbb{Z}^{n+1}$ under the lower triangular matrix $T = \big(S(k,i)\,i!\big)_{k,i}$, whose diagonal is $0!,1!,\dots,n!$. The index is therefore $\prod_{k\le n}k!$. The inverse of T is $\big(s(k,i)/k!\big)_{k,i}$, with the Stirling numbers $s(k,i)$ of the first kind, because $\binom{j}{k} = \sum_i s(k,i)j^i/k!$; so $n!\,T^{-1}$ is integral, and for every integral point p of the secant plane the least admissible M divides $n!$. It is often larger than 1. At $n=4$, $d=3$, $a=b=1$ the least multiple is $M=6$, a divisor of $4!$, and the witness is

$$\alpha = -23\,[\mathcal{O}_X] + 66\,[\mathcal{O}_X(t)] - 54\,[\mathcal{O}_X(2t)] + 20\,[\mathcal{O}_X(3t)] - 3\,[\mathcal{O}_X(4t)],$$

of rank $-23+66-54+20-3 = 6 = Ma$; in degree one its coefficient is $66-108+60-12 = 6 = Mb$, and in degrees two to four the sums $\sum_j n_j j^k$ are -18, -18 and 54, which are $6c_2$, $6c_3$ and $6c_4$. The solution of the Vandermonde system is unique and the n_j have no common factor, 23 being prime to 6, so $M=6$ is the least multiple, and the freedom to rescale is needed. Figure 13 shows the least multiple in further cases.

For a principal polarisation θ the classes $\theta^k/k!$ are integral: in a symplectic basis $x_1,\dots,x_g,y_1,\dots,y_g$ of $H^1(X,\mathbb{Z})$, $\theta=\sum_i x_iy_i$, the products x_iy_i commute and square to zero, and so $\theta^k/k! = \sum_{|I|=k}\prod_{i\in I}x_iy_i$. This is Poincaré's formula in a form that does not need a curve, and it is why $t^k/k!$ is the natural integral basis in which to take the coordinates c_k.

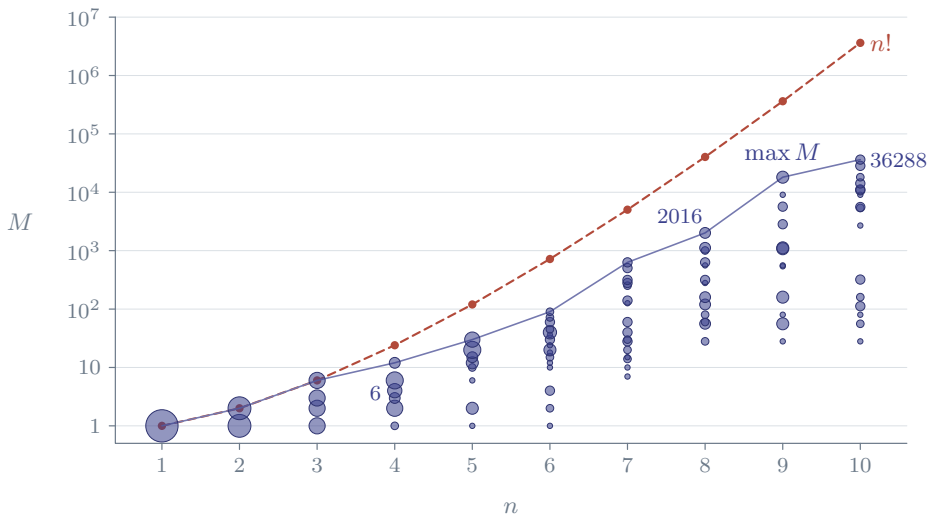


FIGURE 13. Remark 14.14. The least multiple M of Theorem 14.13 for $1 \le n \le 10$, over the seventy-two cases $d \in \{1,2,3,5,7,11\}$, $a \in \{1,2,3\}$, $b \in \{\pm 1, \pm 2\}$ for each n, read off from the solution of the Vandermonde system, on a logarithmic scale; the area of a disc counts the cases with that value. The solid line joins the largest values and the dashed one is $n!$, which every M divides by the remark: the quotient of $\mathbb{Z}^{n+1}$ by the lattice of the characters $(1, j, \dots, j^n)$ of the powers of a line bundle is dual to the integer-valued polynomials of degree at most n, spanned by the $\binom{x}{k}$, whose coefficients have denominators dividing $n!$.

14.6. **The three ingredients.**

Theorem 14.15 (Three of the four ingredients)**.** *For every $n \ge 2$, every squarefree $d > 0$ and every polarised abelian n-fold X carrying a nondegenerate $\beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$, the first three of the four ingredients required by Question 13.7 exist and are explicit:*

(a) *the abelian $2n$-fold of Weil type is $A = X \times \widehat{X}$ with the K-action (29), by Theorem 14.2;*
(b) *the plane is P_t of Definition 14.4 with $t = \beta$, it is spanned by Hodge classes and it meets the spinor variety in exactly the two conjugate points $[e^{\pm st}]$, by Theorem 14.5;*
(c) *a pair of coherent sheaves $\mathcal{F}_1, \mathcal{F}_2$ on X whose Chern characters are linearly independent rational points of P_t exists, by Theorem 14.13, and each may be taken to have the class of an explicit integer combination of powers of one line bundle.*

The invariants of any such sheaf are forced by the explicit system of Theorem 14.7, which imposes $n-1$ linear conditions on a Chern character already constrained to $\mathbb{Q}[t]$, and, when β is ample, are unobstructed numerically by (31).

Proof. Items (a), (b) and (c) are the cited theorems. For (c), the proof of Theorem 14.13 uses of t only that $t^n \neq 0$ and that a positive multiple of t is integral; both hold for $t = \beta$, the first because β is nondegenerate. For the count in the last sentence, a Chern character in the subring $\mathbb{Q}[t] \subset H^{\mathrm{ev}}(X, \mathbb{Q})$ is a polynomial of degree at most n in t, so lies in a space of dimension $n+1$, while P_t has dimension two, and $(n+1) - 2 = n - 1$. ∎

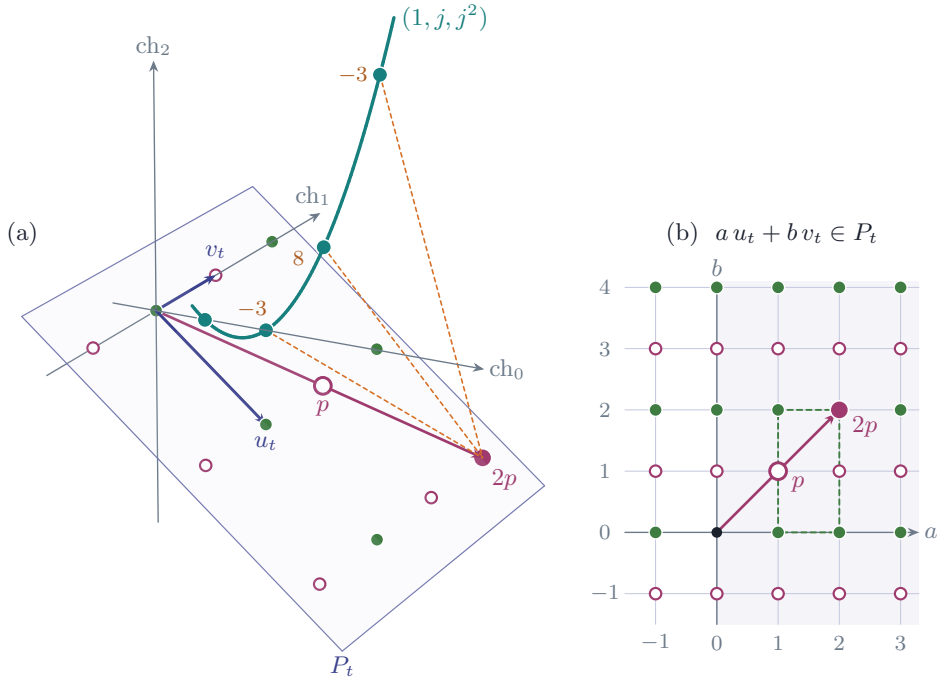


FIGURE 14. Theorem 14.13 at $n = 2$ and $d = 2$. (a) The coordinates $(\mathrm{ch}_0, \mathrm{ch}_1, \mathrm{ch}_2)$ in the basis 1, t, $t^2/2$. The Chern characters of the powers of one line bundle lie on the moment curve $j \mapsto (1, j, j^2)$, and the Vandermonde determinant says that any three of its points are affinely independent. The integral points of $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t$ split by the criterion $\mathrm{ch}_2 \equiv \mathrm{ch}_1 \bmod 2$ into those reached by integer combinations of line bundles, drawn filled, and those missed, drawn open. The point $p = u_t + v_t$ is missed and $2p$ is reached, by the witness $-3[\mathcal{O}_X] + 8[\mathcal{O}_X(t)] - 3[\mathcal{O}_X(2t)]$, whose three terms are joined to $2p$ by dashed segments. (b) The same points in the coordinates (a, b) of $a\,u_t + b\,v_t$. Here $\mathrm{ch}_1 = b$ and $\mathrm{ch}_2 = -2a$, so the criterion reads $b \equiv 0 \bmod 2$; the shaded half-plane $a > 0$ is the range of Theorem 14.13.

Remark 14.16 (The deformation step). The fourth ingredient of Question 13.7 is untouched by this section, and it is the hardest. Given the pair of secant sheaves $\mathcal{F}_1, \mathcal{F}_2$ on X supplied by Theorem 14.13, one must still show that the characteristic class of $\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$ remains of Hodge type under every deformation of $X \times \widehat{X}$ as a polarised abelian $2n$-fold of Weil type, and that the object deforms with the variety, as a twisted sheaf to first order. Markman carries out this step at $n = 3$ [Mar25b], through a semiregularity analysis. Our contribution to it is negative: [II, Theorem 4.56] identifies the Hochschild classes that the weakened form of that analysis must kill for a secant object, and [II, Theorem 4.61] shows that the candidate Markman proposes for $n = 4$ does not kill them. Without the step one obtains algebraicity only on the split locus $X \times \widehat{X}$, where the Weil classes are of much less interest, since the general abelian $2n$-fold of Weil type is simple and is not of that form.

Accordingly, nothing in this section proves the Hodge conjecture for abelian $2n$-folds of Weil type at any $n \geq 4$. What is established is that three of the four ingredients are

written down for every n, the third with an explicit witness in line bundles, and that the fourth is isolated.

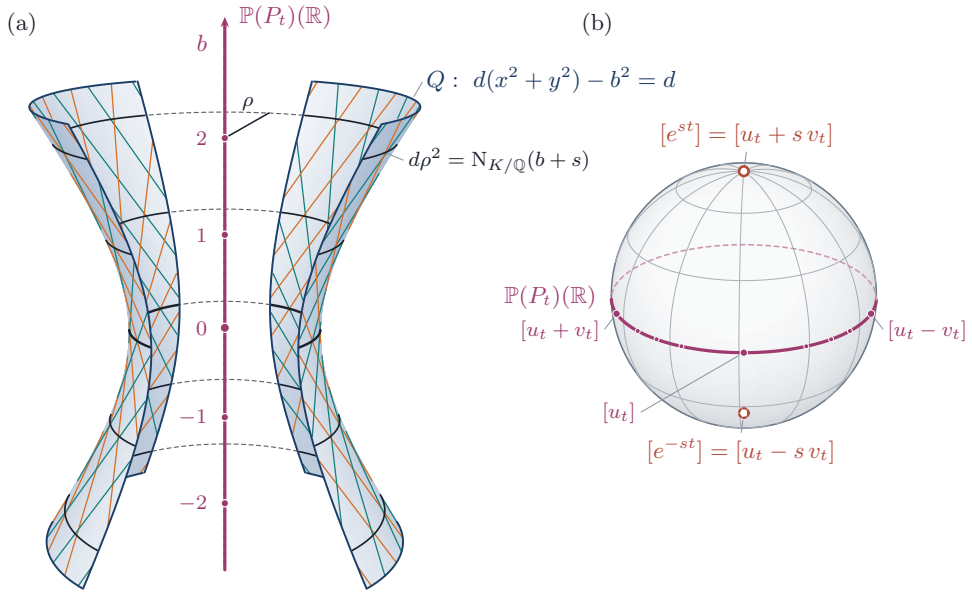


FIGURE 15. The geometry of Theorem 14.5 in the first case, $n=2$. The even spinor variety is then the smooth six-dimensional quadric of classes isotropic for the Mukai pairing in $\mathbb{P}(H^{\mathrm{ev}}(X,\mathbb{C}))\cong\mathbb{P}^7$, and the figure shows a real two-dimensional section of it, a quadric surface with its two rulings. The line $\mathbb{P}(P_t)$ of Definition 14.4 is defined over $\mathbb{Q}$ and meets the quadric only in the two points $[e^{\pm st}]$, which are conjugate over K and not individually rational; they are drawn as real points. This is the configuration the Weil line demands: a class defined over $\mathbb{Q}$ cannot single out one of the two points, so the construction must carry the pair, and the pair spans a plane rather than a line. Proposition 13.1 is the same statement on the cohomological side, and Figure 12 places it among the hypotheses a construction can satisfy.

15. Base points in every Weil family

The secant construction of Section 14 lives on the split variety $X\times\widehat{X}$ and stops short of the deformation step. In this section we study the split member and the other special members of a Weil family through the Weil classes themselves. On the split member the Weil classes are polynomials in three divisor classes (Theorem 15.2); on products of abelian surfaces of Weil type they are polynomials in divisor classes (Theorem 15.10); and on abelian varieties with quaternionic multiplication extending K, and on their products, they are polynomials in two divisor classes, which yields an algebraic base point in every Weil family, of every discriminant (Theorem 15.27). What remains is the propagation of one flat class over one connected period domain, formulated in Sections 15.3 and 15.7.

15.1. **The split member in closed form.** We begin with the split member, where a single formula, valid for every n and every d, writes the Weil classes as polynomials in three divisor classes. The split locus is thin. A general abelian $2n$-fold of Weil type is simple, hence not isogenous to a product, and for $n \geq 2$ the members $X \times \widehat{X}$ form a subvariety of $\mathcal{D}_{n,n}$ of positive codimension (Proposition 15.19(iii)). Corollary 15.4 is therefore a base case, and Question 15.13 is a question about propagation.

Notation 15.1. On $A = X \times \widehat{X}$ write

$$\beta \in H^2(A, \mathbb{Q}), \qquad \widehat{\beta} \in H^2(A, \mathbb{Q}), \qquad \ell = c_1(\mathcal{P}) \in H^2(A, \mathbb{Q})$$

for the pullback of the polarisation of X, the pullback of the dual polarisation of $\widehat{X}$, and the class of the Poincaré bundle. All three are divisor classes, so every polynomial in them is algebraic. Put

$$\gamma = d\,\beta - \widehat{\beta}. \tag{33}$$

Theorem 15.2 (The Weil line of the split member)**.** *Let $A = X \times \widehat{X}$ carry the K-action (29) determined by a principal β, and let $V_{\pm} \subset H^1(A, \mathbb{C})$ be the eigenspaces of $\iota(\sqrt{-d})$. Then*

(i) *$\gamma \mp \sqrt{-d}\,\ell$ lies in $\bigwedge^2 V_{\pm}$, so it is an eigenclass for the character $\tau \mapsto \tau^2$ on $H^2(A, \mathbb{C})$ respectively for its conjugate;*

(ii) *the generators of $\bigwedge^{2n} V_{\pm}$ are*

$$\omega_{\pm} = \frac{(-1)^{n(n-1)/2}}{n!\,d^{\,n}} (\gamma \mp \sqrt{-d}\,\ell)^n; \tag{34}$$

(iii) *consequently the Weil line of A is*

$$W(A) = \Big\langle\ \Re(\gamma - \sqrt{-d}\,\ell)^n,\ \ \Im(\gamma - \sqrt{-d}\,\ell)^n\ \Big\rangle_{\mathbb{Q}},$$

and every Weil class on A is a polynomial with rational coefficients in the three divisor classes $\beta, \widehat{\beta}, \ell$, hence is algebraic.

Proof. Write $U = H^1(X, \mathbb{Q})$ with basis $x_1, \dots, x_{2n}$ chosen so that $\beta = \sum_{j \leq n} x_j \wedge x_{n+j}$, let $\xi_1, \dots, \xi_{2n}$ be the dual basis of $H^1(\widehat{X}, \mathbb{Q}) = U^\vee$, and let $B \colon U \to U^\vee$ be the isomorphism induced by β, so that $Bx_j = \xi_{n+j}$ and $Bx_{n+j} = -\xi_j$. The endomorphism (29) acts on homology, and its transpose M^* acts on $H^1(A, \mathbb{Q}) = U \oplus U^\vee$ by $u \mapsto -Bu$ and $\phi \mapsto d\,B^{-1}\phi$. Indeed, in the basis $u_1, \dots, u_{2n}$ of $H_1(X, \mathbb{Q})$ dual to the x_i, the map $\beta \colon H_1(X) \to H^1(X)$ sends $u_j \mapsto x_{n+j}$ and $u_{n+j} \mapsto -x_j$, so (29) sends $u_j \mapsto dx_{n+j}$, $u_{n+j} \mapsto -dx_j$, $x_{n+j} \mapsto -u_j$ and $x_j \mapsto u_{n+j}$, where x_i is regarded as an element of $H_1(\widehat{X}, \mathbb{Q}) = H^1(X, \mathbb{Q})$ and ξ_i is its dual class; pairing with the dual basis gives the stated action. The square of M^* is $-d$; so, writing $\delta = \sqrt{-d}$,

$$V_{\pm} = \Big\{\, u \pm \tfrac{\delta}{d} Bu \ :\ u \in U_{\mathbb{C}} \Big\}, \tag{35}$$

since $M^*(u \pm \frac{\delta}{d} Bu) = -Bu \pm \delta u = \pm\delta(u \pm \frac{\delta}{d} Bu)$.

(i) Take the sum over $j \leq n$ of the wedges of the two eigenvectors attached to x_j and to x_{n+j}:

$$\sum_{j=1}^{n} \Big(x_j \pm \tfrac{\delta}{d} Bx_j\Big) \wedge \Big(x_{n+j} \pm \tfrac{\delta}{d} Bx_{n+j}\Big) = \beta \pm \tfrac{\delta}{d}\Sigma - \tfrac{1}{d}\widehat{\beta},$$

using $\delta^2 = -d$, where $\Sigma = \sum_j (x_j \wedge Bx_{n+j} + Bx_j \wedge x_{n+j})$ and $\sum_j Bx_j \wedge Bx_{n+j} = \sum_j \xi_{n+j} \wedge (-\xi_j) = \widehat{\beta}$. Now

$$\Sigma = \sum_{j=1}^{n} (x_j \wedge (-\xi_j) + \xi_{n+j} \wedge x_{n+j}) = -\sum_{i=1}^{2n} x_i \wedge \xi_i = -\ell.$$

So the displayed sum is $\frac{1}{d}(d\beta - \widehat{\beta} \mp \delta\ell) = \frac{1}{d}(\gamma \mp \delta\ell)$. It is a sum of wedges of pairs of vectors of $V_\pm$, hence lies in $\bigwedge^2 V_\pm$, which proves (i); the statement about characters is Proposition 5.1 at $(r,s) = (2,0)$ and $(0,2)$.

(ii) By (i) the n-th power $(\gamma \mp \delta\ell)^n$ lies in $\bigwedge^{2n} V_\pm$, which is a line by Proposition 3.7. It is nonzero: expanding $\gamma \mp \delta\ell = d\beta - \widehat{\beta} \mp \delta\ell$, the only term of the n-th power lying in $\bigwedge^{2n} U$ is $d^n\beta^n$, and $\beta^n \neq 0$ because β is nondegenerate; the remaining terms all involve at least one ξ, so no cancellation can occur. Hence $(\gamma \mp \delta\ell)^n$ spans $\bigwedge^{2n} V_\pm$ and is a scalar multiple of $\omega_\pm = \bigwedge_{i=1}^{2n}(x_i \pm \frac{\delta}{d} Bx_i)$. To find the scalar, compare the components in $\bigwedge^{2n} U$: that of $\omega_\pm$ is $x_1 \wedge \cdots \wedge x_{2n}$, while

$$d^n\beta^n = d^n\, n! \bigwedge_{j=1}^{n} (x_j \wedge x_{n+j}) = d^n\, n!\, (-1)^{n(n-1)/2}\, x_1 \wedge \cdots \wedge x_{2n},$$

the sign being that of the permutation carrying $(1, n+1, 2, n+2, \ldots, n, 2n)$ to $(1, 2, \ldots, 2n)$. Inverting gives (34).

(iii) The two displayed classes are the real and imaginary parts of $(\gamma - \delta\ell)^n$, which up to the nonzero rational factor in (34) are $\omega_- + \omega_+$ and $(\omega_- - \omega_+)/\delta$; by Proposition 3.7(i) these span $W(A)$ over $\mathbb{Q}$. Expanding $(\gamma - \delta\ell)^n$ by the binomial theorem and separating even from odd powers of δ writes each of them as a rational polynomial in $\beta, \widehat{\beta}, \ell$, because $\delta^{2m} = (-d)^m$ and $\delta^{2m+1} = (-d)^m\delta$. Products of divisor classes are algebraic, so both are algebraic. ∎

Example 15.3. The $2n$ classes $x_1, \ldots, x_{2n}$ form a K-basis of $H^1(A, \mathbb{Q})$, on which $\sqrt{-d}$ acts by $x \mapsto -Bx$, so (15) applies to them; with (34) it gives the integral generators of Theorem 4.9 for this basis in closed form,

$$\omega_1 + \sqrt{-d}\,\omega_2 = \frac{(-1)^{n(n-1)/2}}{n!}\,(\gamma + \sqrt{-d}\,\ell)^n.$$

At $n = 2$ this reads

$$\omega_1 = \tfrac{1}{2}(d\,\ell^2 - \gamma^2), \qquad \omega_2 = -\gamma\,\ell,$$

and at $n = 3$,

$$\omega_1 = \tfrac{1}{2}\,d\,\gamma\,\ell^2 - \tfrac{1}{6}\gamma^3, \qquad \omega_2 = \tfrac{1}{6}\,d\,\ell^3 - \tfrac{1}{2}\gamma^2\ell,$$

with $\gamma = d\beta - \widehat{\beta}$ throughout.

Corollary 15.4 (Algebraicity on the split locus)**.** *For every $n \geq 1$, every squarefree $d > 0$ and every principally polarised abelian n-fold X, the Weil classes of $A = X \times \widehat{X}$ with the K-action (29) are algebraic, and an algebraic representative is given by (34) as an explicit rational combination of intersections of three divisors.*

Proof. This is Theorem 15.2(iii), together with the Lefschetz theorem on $(1,1)$ classes, which makes $\beta, \widehat{\beta}, \ell$ algebraic, and the fact that the algebraic classes form a subring of $H^*(A, \mathbb{Q})$. ∎

Remark 15.5 (Consistency with the character obstruction). Corollary 15.4 does not contradict Theorem 5.6. At the split point $\mathrm{NS}(A)\otimes\mathbb{Q}$ contains the three independent classes $\beta, \widehat{\beta}, \ell$, so the hypothesis $\mathrm{NS}(A)\otimes\mathbb{Q} = \mathbb{Q}\eta$ of Proposition 3.10 fails, and this is the situation of [III, Remark 2.5]. What Theorem 5.6 forbids is a $K^\times$-eigenclass for the norm character meeting the Weil line, and $\gamma \mp \sqrt{-d}\,\ell$ is an eigenclass for τ^2 and for $\overline{\tau}^2$, not for the norm. The formula (34) is the first place in the paper where a class in the relevant eigenspace is exhibited, and such a class exists because the split member has enough divisors to build one.

15.2. **Products of abelian surfaces.** A second source of members with enough divisors is the product, an operation that the Weil line respects. Throughout this subsection K is fixed and all Weil structures are taken with respect to it.

Lemma 15.6 (Products of Weil type)**.** *Let (A_1, ι_1, η_1) be of $(K, -1, n_1)$-Weil type and (A_2, ι_2, η_2) of $(K, -1, n_2)$-Weil type. Then $A = A_1 \times A_2$, with the diagonal action $\iota = \iota_1 \times \iota_2$ and the product polarisation $\eta = p_1^*\eta_1 + p_2^*\eta_2$, is of $(K, -1, n_1 + n_2)$-Weil type.*

Proof. Write $V_i = H^1(A_i, \mathbb{Q})$, so $V = H^1(A, \mathbb{Q}) = V_1 \oplus V_2$ as K-vector spaces and $\dim_K V = 2n_1 + 2n_2$. The Rosati involution of the product polarisation is the product of the two Rosati involutions, so it restricts on K to conjugation, which is the sign -1 of Definition 3.1. For the multiplicities, the Hodge decomposition of V is the direct sum of those of V_1 and V_2, and the eigenspace decomposition is likewise $V_\pm = V_{1,\pm} \oplus V_{2,\pm}$. Hence

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_{1,+}^{1,0} + \dim_{\mathbb{C}} V_{2,+}^{1,0} = n_1 + n_2,$$

and the same for the other three, which is Lemma 3.5 for $n = n_1 + n_2$. ∎

For an abelian variety A of $(K, -1, n)$-Weil type and a K-basis of $H^1(A, \mathbb{Q})$, let $\omega_1(A), \omega_2(A)$ be the generators of the Weil line given by (11) in that basis, and write

$$\omega(A) = \omega_1(A) + \sqrt{-d}\,\omega_2(A) \;\in\; K \otimes_{\mathbb{Q}} H^{2n}(A, \mathbb{Q})$$

for the K-valued Weil class; by (15) its image in $H^{2n}(A, \mathbb{C})$ is $d^{-n}\prod_j (d\,x_j + \delta\,y_j)$. On a product we use the K-basis obtained by concatenating bases of the factors.

The Weil line is the top exterior power over K, and a top exterior power of a direct sum is the tensor product of the top exterior powers. This is the content of the next theorem; what has to be checked is its effect on the rational generators, which are defined through the real structure rather than over K.

Theorem 15.7 (The Weil class is multiplicative)**.** *With the notation of Lemma 15.6, and cup product on $H^*(A, \mathbb{Q})$ taken after pullback along the two projections,*

$$\begin{aligned} \omega_1(A) &= \omega_1(A_1)\,\omega_1(A_2) - d\,\omega_2(A_1)\,\omega_2(A_2), \\ \omega_2(A) &= \omega_1(A_1)\,\omega_2(A_2) + \omega_2(A_1)\,\omega_1(A_2), \end{aligned} \tag{36}$$

that is $\omega(A_1)\,\omega(A_2) = \omega(A_1 \times A_2)$ for the K-valued Weil class. For k factors, $\prod_j \omega(A_j) = \omega(A_1 \times \cdots \times A_k)$.

Proof. Let $u_1, \ldots, u_{2n_1}$ and $u_{2n_1+1}, \ldots, u_{2n}$ be the K-bases of $H^1(A_1, \mathbb{Q})$ and $H^1(A_2, \mathbb{Q})$, so that together they form the K-basis of $H^1(A, \mathbb{Q})$, and put $x_j = u_j$ and $y_j = \sqrt{-d}\,u_j$ as

in (15). By (15),

$$\omega(A) = d^{-n}\prod_{j=1}^{2n}(d\,x_j + \delta\,y_j)$$
$$= \Big(d^{-n_1}\prod_{j\le 2n_1}(d\,x_j+\delta\,y_j)\Big)\Big(d^{-n_2}\prod_{j>2n_1}(d\,x_j+\delta\,y_j)\Big) = \omega(A_1)\,\omega(A_2)$$

in $H^{2n}(A,\mathbb{C})$, the products being taken in the order of the basis, so that no sign arises. Expanding $\big(\omega_1(A_1)+\delta\omega_2(A_1)\big)\big(\omega_1(A_2)+\delta\omega_2(A_2)\big)$ with $\delta^2=-d$, and comparing rational parts and δ-parts, which is legitimate because the classes $\omega_j(A_i)$ are rational and $1,\delta$ are linearly independent over $\mathbb{Q}$, gives (36). The statement for k factors follows by induction. ■

Figure 16 shows the multiplicativity and the loci it reaches.

Corollary 15.8 (Algebraicity propagates along products)**.** *If the Weil classes of A_1 and of A_2 are algebraic, then so are those of $A_1\times A_2$, and an algebraic representative is the rational combination of external products given by* (36)*.*

Proof. Algebraic classes are closed under pullback along the projections and under cup product, and (36) expresses $\omega_1(A)$ and $\omega_2(A)$ in those operations applied to the $\omega_j(A_i)$. ■

In dimension two there is nothing to prove.

Lemma 15.9 (Dimension two)**.** *Let A_0 be of $(K,-1,1)$-Weil type, so $\dim A_0=2$. Then $W(A_0)\subset H^2(A_0,\mathbb{Q})$ and both generators are algebraic.*

Proof. The Weil classes lie in $H^{2n}=H^2$ and are of type $(1,1)$ by Proposition 3.7(ii), so they are algebraic by the Lefschetz theorem on $(1,1)$ classes. No hypothesis on A_0 beyond the Weil structure is used. ■

Theorem 15.10 (An algebraic Weil class in every dimension)**.** *Let $n\ge 1$, let $d>0$ be squarefree, and let $A_1,\dots,A_n$ be abelian surfaces of $(K,-1,1)$-Weil type. Then $A=A_1\times\cdots\times A_n$ is of $(K,-1,n)$-Weil type and its Weil classes are algebraic; explicitly, by Theorem 15.7 and Lemma 15.9 they are the rational part and the coefficient of $\sqrt{-d}$ in $\prod_{j=1}^{n}(\omega_1(A_j)+\sqrt{-d}\,\omega_2(A_j))$, a polynomial with integer coefficients in $2n$ divisor classes.*

Proof. Lemma 15.6 gives the Weil structure, Lemma 15.9 gives algebraicity of each factor's classes, and Corollary 15.8 applied $n-1$ times gives the conclusion; the displayed form is Theorem 15.7 for $k=n$ factors. Each $\omega_j(A_i)$ is a divisor class by Lemma 15.9, so the product is a polynomial in $2n$ of them. ■

Remark 15.11 (The product locus). Theorem 15.10 is unconditional and holds for every n; like Corollary 15.4 and Theorem 15.27, it produces an algebraic Weil class in every dimension using nothing beyond the Lefschetz theorem on $(1,1)$ classes. The product construction supplies a second family of base points, defined for every n and every d, at which the class is written down explicitly. It also shows that the algebraicity locus is closed under products, so that a settled case n_0 propagates to every multiple of n_0, and settled cases propagate to their sums. Combined with Markman's theorem at $n=2$ [Mar25a, Mar25b], this gives, for every n, algebraicity on the locus of products of abelian fourfolds and surfaces of Weil type, which has dimension $2n$ when n is even.

The reach of these results is limited. The family of polarised abelian $2n$-folds of $(K,-1,n)$-Weil type has dimension n^2, and the locus of products of n surfaces has dimension n, so for

$n \geq 2$ it is a proper closed subvariety, smaller than the quaternionic locus of Theorem 15.25, whose dimension is $n(n+1)/2$. A general member of the family is simple, hence not isogenous to a product, and neither Theorem 15.10 nor Corollary 15.4 applies to it. The general member lies outside all of these loci, and by Theorem 15.14 what joins them to it is the propagation statement.

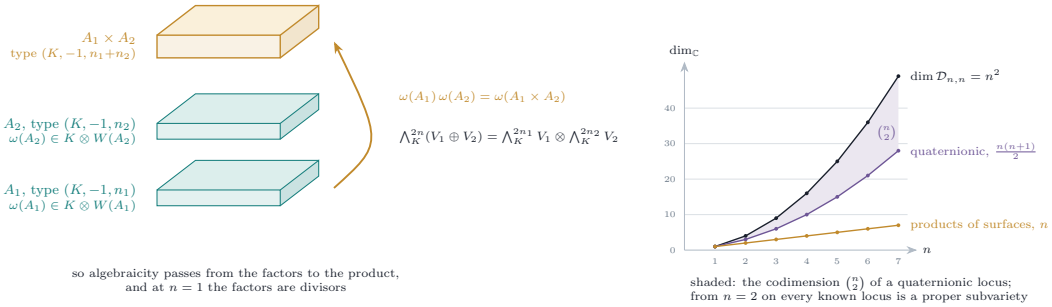


FIGURE 16. Theorem 15.7 and its reach. Left: the Weil line of each factor is a K-line, the Weil line of the product is the tensor product of the two over K, and the K-valued Weil class is multiplicative. Algebraicity therefore passes from the factors to the product, and at $n = 1$ the factors are divisor classes, so a product of n abelian surfaces of Weil type carries an explicit algebraic Weil class in every dimension. Right: the dimensions of the loci on which the class is known to be algebraic, against the dimension n^2 of the family. Both the quaternionic locus and the product locus are proper subvarieties once $n \geq 2$, and a general member lies on neither.

15.3. **The residual statement.** The results above allow the remaining question to be stated in one sentence, without auxiliary hypotheses.

Setup 15.12 (The family)**.** Fix $n \geq 2$ and a squarefree $d > 0$. Let $\mathcal{D}_{n,n}$ be the period domain of Proposition B.8 and let

$$\pi \colon \mathcal{A} \longrightarrow \mathcal{D}_{n,n}$$

be the family of polarised abelian $2n$-folds of $(K, -1, n)$-Weil type it parametrises, with ω the flat section of $R^{2n}\pi_*\mathbb{Q}$ given fibrewise by the class ω_1 of (11). Let $\mathcal{D}^{\mathrm{spl}} \subset \mathcal{D}_{n,n}$ be the locus of split members $X \times \widehat{X}$ carrying the action (29).

Question 15.13 (The residual statement)**.** Is ω_s algebraic for every $s \in \mathcal{D}_{n,n}$?

Theorem 15.14 (The reduction)**.** *Fix $n \geq 2$ and a squarefree $d > 0$, and keep the notation of Setup 15.12.*

(a) *If Question 15.13 has a positive answer, then the Hodge conjecture holds, in every degree, for every abelian $2n$-fold of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.*
(b) *Conversely, if the Hodge conjecture holds for every abelian $2n$-fold of $(K, -1, n)$-Weil type, then Question 15.13 has a positive answer.*
(c) *$\mathcal{D}^{\mathrm{spl}}$ is nonempty, ω_s is algebraic at every $s \in \mathcal{D}^{\mathrm{spl}}$ with the explicit representative (34), and $\mathcal{D}_{n,n}$ is connected.*

Thus Question 15.13 is the variational Hodge conjecture for one flat class on one connected family; it is necessary as well as sufficient, and its base case is supplied. Theorem 15.27 supplies a base case in every family, for every discriminant, so nonemptiness of the base locus is never the obstacle.

Proof. (a) Let A be of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A)=\mathrm{SU}(V,H)$. By Proposition B.8 its Hodge structure is a point s of $\mathcal{D}_{n,n}$, so A is a fibre of π, and the Weil line of A is generated over K by ω_s. A positive answer makes ω_s algebraic, hence makes the whole Weil line algebraic by the argument of Corollary 4.15. By Theorem 4.14 the Weil line together with the powers of η exhausts $\mathrm{Hdg}^*(A)$, and the powers of η are algebraic, so every Hodge class on A is algebraic.

(b) For every s the class ω_s is a Hodge class on the fibre A_s, by Proposition 3.7(ii), and A_s is of $(K,-1,n)$-Weil type by construction. So the Hodge conjecture for A_s gives its algebraicity.

(c) $\mathcal{D}^{\mathrm{spl}}$ is nonempty because the construction (29) applies to any principally polarised abelian n-fold and produces a point of $\mathcal{D}_{n,n}$ by Theorem 14.2; algebraicity there is Corollary 15.4; and $\mathcal{D}_{n,n}$ is connected because it is a bounded symmetric domain, by Proposition B.8. ∎

Remark 15.15 (The target). The preceding sections remove three parts of the problem. Theorem 4.14 settles how many classes are at issue: one, in one degree, on each fibre. Theorem 7.3 and Theorem 5.6 exclude two families of candidate constructions. Theorem 15.2 supplies the base case: the class is algebraic somewhere in the family, with a representative that can be written on a single line.

The next step is propagation, which is Grothendieck's variational Hodge conjecture for the pair (π,ω) of Setup 15.12. Theorem 15.14 shows that for the Hodge conjecture on abelian varieties of Weil type it is necessary as well as sufficient, so every route to the conjecture meets an equivalent difficulty. Section 16.1 analyses the propagation; there the passage from a formal deformation to an algebraic one, usually the harder half, is supplied by properness of the Hilbert scheme. What survives is either an infinitesimal statement at a single point or a bound on the complexity of representing cycles along one orbit, and the second is equivalent to its conclusion (Theorem 16.17).

15.4. **The divisorial criterion.** Theorem 15.2 is a computation on one member of the family, but the mechanism behind it is not special to that member. Isolating it gives a criterion that can be tested at any point; we then determine where it holds.

Theorem 15.16 (The divisorial criterion)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type and suppose there exists $\Theta\in\bigwedge^2 V_-$ with*

(i) *Θ nondegenerate as an alternating form on $V_-^\vee$, equivalently $\Theta^n\neq 0$;*
(ii) *Θ of Hodge type $(1,1)$;*
(iii) *$\Theta+\overline{\Theta}$ rational, that is in $H^2(A,\mathbb{Q})$.*

Then the Weil classes of A are algebraic. Explicitly, writing $g=\Theta+\overline{\Theta}$ and $\lambda=(\Theta-\overline{\Theta})/\sqrt{-d}$, both g and λ are divisor classes and

$$W(A)=\left\langle\, \Re(g+\sqrt{-d}\,\lambda)^n,\ \ \Im(g+\sqrt{-d}\,\lambda)^n \,\right\rangle_{\mathbb{Q}}, \tag{37}$$

so every Weil class is a rational polynomial in two divisor classes.

Proof. Conjugation interchanges $\bigwedge^2 V_-$ and $\bigwedge^2 V_+$, so $\overline{\Theta}\in\bigwedge^2 V_+$, and both g and λ are fixed by conjugation, hence real. Rationality of g is (iii); that of λ comes from the action

of $K^\times$. Fix $\tau = a + b\sqrt{-d} \in K^\times$ with $ab \neq 0$. By Proposition 5.1 the operator $\iota(\tau)^*$ acts on $\bigwedge^2 V_+$ by $\tau^2 = a^2 - db^2 + 2ab\sqrt{-d}$ and on $\bigwedge^2 V_-$ by $\overline{\tau}^2$, so

$$J := \frac{\iota(\tau)^* - (a^2 - db^2)}{2ab}$$

is a $\mathbb{Q}$-rational endomorphism of $H^2(A, \mathbb{Q})$ preserving the space R of Proposition 11.11 and acting on $\bigwedge^2 V_\pm$ by $\pm\sqrt{-d}$. Applying it to $g = \Theta + \overline{\Theta}$ gives $J(g) = -\sqrt{-d}\,\Theta + \sqrt{-d}\,\overline{\Theta} = -\sqrt{-d}\cdot\sqrt{-d}\,\lambda = d\,\lambda$, so $\lambda = J(g)/d$ is rational. Both g and λ are of type $(1,1)$ by (ii) and conjugation symmetry, so both lie in $\mathrm{NS}(A)\otimes\mathbb{Q}$ by the Lefschetz theorem on $(1,1)$ classes, and both are algebraic.

By (i) the form Θ on the $2n$-dimensional space V_- has nonzero Pfaffian, so Θ^n is a nonzero element of the line $\bigwedge^{2n} V_-$, and it therefore spans it; conjugating, $\overline{\Theta}^n$ spans $\bigwedge^{2n} V_+$. Hence $\Theta^n + \overline{\Theta}^n$ and $(\Theta^n - \overline{\Theta}^n)/\sqrt{-d}$ span $W(A)$ over $\mathbb{Q}$, by Proposition 3.7(i). Finally $2^n\Theta^n = (g + \sqrt{-d}\lambda)^n$, and expanding by the binomial theorem with $(\sqrt{-d})^{2m} = (-d)^m$ writes the real and imaginary parts as rational polynomials in g and λ. Products of divisor classes are algebraic. ∎

Corollary 15.17. *The split member of Theorem 15.2 satisfies the criterion, with* $\Theta = \gamma - \sqrt{-d}\,\ell$ *and* $\gamma = d\beta - \widehat{\beta}$*; the two divisor classes of* (37) *are* 2γ *and* -2ℓ*.*

Proof. Theorem 15.2(i) places $\Theta = \gamma - \sqrt{-d}\,\ell$ in $\bigwedge^2 V_+$ and its conjugate in $\bigwedge^2 V_-$. The hypotheses and the conclusion of Theorem 15.16 are symmetric under replacing Θ by $\overline{\Theta}$, which interchanges V_+ and V_-, fixes g and changes the sign of λ, so the criterion may be applied to Θ. The classes γ and ℓ are divisor classes, hence of type $(1,1)$, so Θ is of type $(1,1)$ and $\Theta + \overline{\Theta} = 2\gamma$ is rational. Nondegeneracy is the nonvanishing of Θ^n, proved in Theorem 15.2(ii). ∎

The base case is therefore one instance of a general criterion. The next two results determine how far the criterion reaches, and both limit it.

Proposition 15.18 (The criterion fails at a general point)**.** *Let* (A, ι, η) *be of* $(K, -1, n)$*-Weil type with* $n \geq 2$ *and* $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$*. Put*

$$R = \left(\bigwedge^2 V_+ \oplus \bigwedge^2 V_-\right) \cap H^2(A, \mathbb{Q}).$$

Then $R \cap \mathrm{NS}(A)_\mathbb{Q} = 0$*, and no* Θ *as in Theorem 15.16 exists.*

Proof. By Proposition 11.11 the space R is a rational sub-Hodge structure of $H^2(A, \mathbb{Q})$, and the intersection $R \cap \mathrm{NS}(A)_\mathbb{Q}$ is a rational sub-Hodge structure of R of pure type $(1,1)$. The complexification of a nonzero rational sub-Hodge structure of R is a nonzero $\mathrm{Hg}(A)_\mathbb{C}$-subrepresentation of $\bigwedge^2 V_+ \oplus \bigwedge^2 V_-$, so it contains a constituent isomorphic to $\bigwedge^2 V_+$ or to $\bigwedge^2 V_-$; by the proof of Proposition 11.11, Hodge numbers depend only on the representation, and by Proposition 11.10 each of $\bigwedge^2 V_\pm$ has $h^{2,0} + h^{0,2} = 2\binom{n}{2} = n(n-1)$, which is positive for $n \geq 2$. So no nonzero rational sub-Hodge structure of R is of pure type $(1,1)$, and $R \cap \mathrm{NS}(A)_\mathbb{Q} = 0$. The argument covers the case $n = 2$ with trivial discriminant, where R is reducible. A Θ as in Theorem 15.16 would give the nonzero element $\Theta + \overline{\Theta}$ of $R \cap \mathrm{NS}(A)_\mathbb{Q}$. ∎

Proposition 15.19 (The split locus)**.** *Fix* $n \geq 1$ *and a squarefree* $d > 0$*, and let* $A = X \times \widehat{X}$ *carry the* K*-action* (29) *determined by a nondegenerate* $\beta \in \mathrm{NS}(X)\otimes\mathbb{Q}$ *of elementary divisors* $(d_1, \ldots, d_n)$*.*

(i) *Let B be the Gram matrix of β in a basis of U, and let $\widehat{\beta}$ be the form on $U^\vee$ with Gram matrix $-B^{-1}$ in the dual basis, which for principal β is the class of Notation 15.1. The polarisation of A is $\eta = \beta + \widehat{\beta}/d$, a positive rational multiple of $d\beta+\widehat{\beta}$; its Gram matrix on $H_1(A,\mathbb{Q}) = U \oplus U^\vee$ is the block form $Q = B \oplus (-B^{-1}/d)$, and M is anti-self-adjoint for it.*

(ii) *In the K-basis $u_1, \dots, u_{2n}$ of $H_1(A,\mathbb{Q})$ the hermitian form of Lemma 4.5 has Gram matrix $\sqrt{-d}\,B$, so*

$$\det H = (-d)^n \det B = (-d)^n (d_1 \cdots d_n)^2,$$

and $(-1)^n \det H = d^n (d_1 \cdots d_n)^2$ is a norm from K. Every split member therefore lies in the Weil family of trivial discriminant, whatever β is chosen.

(iii) *The split members form a locus of dimension $n(n+1)/2$ inside $\mathcal{D}_{n,n}$, of codimension $n(n-1)/2$, which is zero only for $n = 1$ and is a divisor for $n = 2$.*

(iv) *For X with $\mathrm{Hg}(X) = \mathrm{Sp}(H^1(X,\mathbb{Q}))$ one has $\mathrm{NS}(A)_\mathbb{Q} = \langle \beta, \widehat{\beta}, \ell \rangle$, of rank three, and in degree $2n$ the monomials in those three classes span the whole space of Hodge classes of A.*

Proof. (i) In the chosen basis of U and the dual basis of $U^\vee$ the map $\beta\colon U \to U^\vee$ has matrix $B^\top = -B$ and its inverse has matrix $-B^{-1}$, so (29) becomes $M = \left(\begin{smallmatrix} 0 & B^{-1} \\ -dB & 0 \end{smallmatrix}\right)$, with $M^2 = -d$ and $M^\top = \left(\begin{smallmatrix} 0 & dB \\ -B^{-1} & 0 \end{smallmatrix}\right)$, using $B^\top = -B$ and $(B^{-1})^\top = -B^{-1}$. For a block form $Q = \left(\begin{smallmatrix} P & \Lambda \\ -\Lambda^\top & S \end{smallmatrix}\right)$ the Rosati relation $M^\top Q + QM = 0$ reads $B\Lambda^\top + \Lambda B = 0$, $dBS + PB^{-1} = 0$, $B^{-1}P + dSB = 0$ and $B^{-1}\Lambda + \Lambda^\top B^{-1} = 0$. Taking $\Lambda = 0$ and $P = B$ forces $S = -B^{-1}/d$ from the second relation, and the third then reads $-I + I = 0$. So $Q = B \oplus (-B^{-1}/d)$, which is the Gram matrix of $\beta + \widehat{\beta}/d$. For principal β one may take $B = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, so that $-B^{-1} = B$ and $\widehat{\beta} = \sum_j \xi_j \wedge \xi_{n+j}$ as in Notation 15.1.

(ii) The set $u_1, \dots, u_{2n}$ is a K-basis because $MU = d\beta(U) = U^\vee$, and $H(u_i,u_j) = Q(Mu_i,u_j) + \sqrt{-d}\,Q(u_i,u_j)$ by (10); the first term vanishes since $Mu_i \in U^\vee$, $u_j \in U$ and Q has no cross block by (i), and the second is $\sqrt{-d}\,\beta_{ij}$. Hence the Gram matrix is $\sqrt{-d}\,B$ and $\det H = (\sqrt{-d})^{2n} \det B = (-d)^n \det B$. The determinant of an alternating matrix is the square of its Pfaffian, here $(d_1 \cdots d_n)^2$. Now $d = \mathrm{N}_{K/\mathbb{Q}}(\sqrt{-d})$ is a norm and every square is a norm, and the norms form a subgroup of $\mathbb{Q}^\times$, so $d^n (d_1 \cdots d_n)^2$ is a norm.

(iii) A split member is determined by X together with β; the moduli space of polarised abelian n-folds has dimension $\dim \mathcal{A}_n = n(n+1)/2$, and the type of β is discrete. By Proposition B.8 the ambient domain has dimension n^2, and $n^2 - n(n+1)/2 = n(n-1)/2$.

(iv) The Hodge classes of $A = X \times \widehat{X}$ are the invariants of $\mathrm{Hg}(A) = \mathrm{Hg}(X) = \mathrm{Sp}(U)$ acting on $\bigwedge^*(U \oplus U^\vee)$. In degree two the invariants are the three lines spanned by β, $\widehat{\beta}$ and ℓ, one from each Künneth summand $\bigwedge^2 U$, $U \otimes U^\vee$ and $\bigwedge^2 U^\vee$. In higher degree, identify $U^\vee$ with U through β, so that $\bigwedge^*(U \oplus U^\vee) \cong \bigwedge^*(U \otimes \mathbb{Q}^2)$ as representations of $\mathrm{Sp}(U)$. By the first fundamental theorem of invariant theory for the symplectic group, the invariants of $\mathrm{Sp}(U)$ in $U^{\otimes 2m}$ are spanned by the tensors attached to the pairings of the $2m$ factors, each a product of copies of the invariant element of $U \otimes U$ dual to β. The projection of $(U \otimes \mathbb{Q}^2)^{\otimes 2m}$ onto $\bigwedge^{2m}(U \otimes \mathbb{Q}^2)$ is equivariant and surjective, hence surjective on invariants since $\mathrm{Sp}(U)$ is reductive, and it carries such a tensor, tensored with vectors of $\mathbb{Q}^2$, to a product of invariants of degree two. So the invariants are generated by β, $\widehat{\beta}$ and ℓ, and in degree $2n$ the monomials $\beta^i \widehat{\beta}^j \ell^k$ with $i + j + k = n$ span the Hodge classes. ∎

Remark 15.20 (Two residual questions). The Weil families for a fixed K and a fixed n are classified by the discriminant of the hermitian form, an element of the infinite group $\mathbb{Q}^\times_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^\times)$. By Proposition 15.19(ii) the split construction of Section 15.1 produces a base point in only one of them, the family of trivial discriminant; products of abelian surfaces of Weil type (Theorem 15.10, Lemma 15.26) reach every discriminant, but only on a locus of dimension n. Read together with Theorem 15.14, this divides Question 15.13 into two questions of different kinds.

(R1) *Trivial discriminant.* A base point exists, with the explicit cycle (34), and what is missing is propagation off a locus of codimension $n(n-1)/2$. At $n=2$ that locus is a divisor and the propagation is known, by Markman [Mar25a, Mar25b] and by Floccari and Fu [FF26]; combined with Theorem 15.14(a) this recovers the Hodge conjecture for abelian fourfolds of Weil type of trivial discriminant with full Hodge group. At $n=3$ it is settled as well, by Markman's theorem on the sixfolds of trivial discriminant [Mar25a, Mar25b], for every K. For $n\geq 4$ it is the propagation statement, and by [III, Proposition 2.13] it is then the only question in the imaginary quadratic case.

(R2) *Nontrivial discriminant.* The split construction never lands there, by Proposition 15.19(ii), so it supplies no base point; and by Proposition 15.18 no divisorial argument can supply one at a general member. A base point must come from a special member at which R acquires Hodge classes.

Question (R2) is settled in Section 15.5: the special members required are built from abelian varieties with quaternionic multiplication extending K and their products, they exist in every family, and on them Theorem 15.16 applies. What then remains is the propagation question of (R1), posed in every family, each of which has a base point.

Remark 15.21 (Codimension of the split locus). A component of the Noether–Lefschetz locus of a weight two variation has codimension at most $h^{2,0}$, here $n(n-1)$. The split locus has codimension $n(n-1)/2$ by Proposition 15.19(iii), half of that, so it is a component of half the expected codimension. This reflects Proposition 15.19(iv): at a split point R acquires two independent Hodge classes at once, namely γ and ℓ, and each of them costs $h^{2,0}/2$ conditions rather than $h^{2,0}$. Whether there are components of the expected codimension carrying only one class, and whether the criterion of Theorem 15.16 can be met on them, is not needed below: question (R2) is answered in Section 15.5 through quaternionic multiplication.

15.5. **A base point in every family.** By Proposition 15.19(ii) the split construction reaches a single discriminant class. Theorem 15.16 does not depend on the origin of Θ, and in this subsection Θ comes from an endomorphism rather than from a pair of divisors chosen by hand. This source reaches every class, and it settles question (R2) of Remark 15.20.

Definition 15.22 (Discriminant)**.** For (A,ι,η) of $(K,-1,n)$-Weil type with hermitian form H as in Lemma 4.5, the *discriminant* is the class of $(-1)^n \det H$ in $\mathbb{Q}^\times_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^\times)$. The sign makes the representative positive, because H has signature (n,n); and the class is well defined because changing the K-basis multiplies $\det H$ by a norm. Hermitian forms of rank $2n$ over K with signature (n,n) are classified by this class, so it labels the Weil families.

The classes that enter Theorem 15.16 have a description by endomorphisms: a class in $R\cap \mathrm{NS}(A)_\mathbb{Q}$ is the same thing as a symmetric endomorphism that conjugates K.

Lemma 15.23 (Classes in R and endomorphisms)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type. The map $\mu\mapsto g_\mu$, $g_\mu(x,y)=\eta(\mu x,y)$, is a bijection between*

(a) *the elements $\mu\in\mathrm{End}^0(A)$ with $\mu^\dagger=\mu$ and $\mu\,\iota(\tau)=\iota(\overline{\tau})\,\mu$ for all $\tau\in K$, and*

(b) *the elements of $R\cap\mathrm{NS}(A)_{\mathbb{Q}}$.*

Under it, μ invertible corresponds to g_μ nondegenerate, hence to a nondegenerate Θ. If such a $\mu\neq 0$ exists and the centraliser of $\iota(K)$ in $\mathrm{End}^0(A)$ is $\iota(K)$, then $\mu^2\in\mathbb{Q}^\times$ and $\iota(K)+\iota(K)\mu$ is a quaternion algebra $(-d,\mu^2)_{\mathbb{Q}}$ inside $\mathrm{End}^0(A)$.

Proof. A class in $\mathrm{NS}(A)_{\mathbb{Q}}$ is a symmetric homomorphism $A\to A^\vee$ up to isogeny, that is $\eta\mu$ with $\mu^\dagger=\mu$. The condition $g_\mu(\iota(\tau)x,y)=g_\mu(x,\iota(\tau)y)$, which says that g_μ is K-bilinear and hence lies in $R=\bigwedge^2_K V$, reads $\eta(\mu\iota(\tau)x,y)=\eta(\mu x,\iota(\tau)y)=\eta(\iota(\tau)^\dagger\mu x,y)=\eta(\iota(\overline{\tau})\mu x,y)$, using Definition 3.1(iii), so it is the condition $\mu\iota(\tau)=\iota(\overline{\tau})\mu$. For the last sentence, μ^2 commutes with $\iota(K)$, hence lies in $\iota(K)$ by hypothesis, and $\mu\mu^2\mu^{-1}=\mu^2$ forces it to be fixed by conjugation, so $\mu^2\in\mathbb{Q}^\times$. Then $\iota(\sqrt{-d})$ and μ generate the quaternion algebra with the stated invariants. ∎

The criterion therefore asks for quaternionic multiplication extending K. Abelian varieties with this structure are easy to construct, and their invariants can be computed.

Setup 15.24 (The quaternionic model)**.** Let $b\in\mathbb{Q}_{>0}$ and let $B=(-d,b)_{\mathbb{Q}}$ be the quaternion algebra with basis $1,i,j,ij$ and $i^2=-d$, $j^2=b$, $ij=-ji$. Let $V=B^n$ as a left B-module, of dimension $4n$ over $\mathbb{Q}$, let $K=\mathbb{Q}(i)\subset B$ act through i, and fix $a_1,\dots,a_n\in\mathbb{Q}^\times$ and

$$E(x,y)=\sum_{k=1}^{n} a_k\,\mathrm{trd}(\overline{x_k}\,i\,y_k), \tag{38}$$

where trd is the reduced trace and $\bar{}$ the canonical involution of B.

Theorem 15.25 (Weil type from quaternionic multiplication)**.** *With the data of Setup 15.24:*

(i) *E is alternating and nondegenerate, and its Rosati involution is $\beta\mapsto i^{-1}\overline{\beta}i$, which restricts to complex conjugation on K and fixes j; it is positive, the form $\mathrm{trd}(\beta\beta^\dagger)$ being diagonal with entries $2,2d,2b,2db$ on the basis;*

(ii) *in the K-basis $e_1,e_1j,\dots,e_n,e_nj$ of V the hermitian form of Lemma 4.5 is*

$$H=\mathrm{diag}(2a_1d,\ -2a_1bd,\ \dots,\ 2a_nd,\ -2a_nbd),$$

of signature (n,n);

(iii) *there is a complex structure on $V_{\mathbb{R}}$ commuting with B and polarised by E, and for every such the resulting abelian variety A is of $(K,-1,n)$-Weil type, the balanced signature being forced rather than imposed;*

(iv) *the discriminant of A is the class of b^n;*

(v) *$\mu=$ left multiplication by j satisfies Lemma 15.23(a), so $g=g_\mu$ lies in $R\cap\mathrm{NS}(A)_{\mathbb{Q}}$ and is nondegenerate, the divisorial criterion Theorem 15.16 applies, and the Weil classes of A are algebraic.*

Proof. (i) $\mathrm{trd}(z)=\mathrm{trd}(\overline{z})$ and $\overline{\overline{x}i\overline{y}}=\overline{y}\,\overline{i}\,x=-\overline{y}ix$ give $E(y,x)=-E(x,y)$. Nondegeneracy is the nondegeneracy of the reduced trace pairing. For the Rosati, $E(\beta x,y)=E(x,\beta^\dagger y)$ for all x,y reads $\overline{\beta}i=i\beta^\dagger$, that is $\beta^\dagger=i^{-1}\overline{\beta}i$. Then $i^\dagger=-i$ and, since $(-j)i=ij$, also $j^\dagger=i^{-1}(ij)=j$. Positivity is the displayed diagonal, which is positive because $d>0$ and $b>0$.

(ii) $V = Ke_k \oplus Ke_k j$ over all k, since $B = K \oplus Kj$. By (10), $H(x,y) = E(ix,y) + \sqrt{-d}\,E(x,y)$. On one factor, $E(ix,y) = a\,\mathrm{trd}(\overline{ix}\,i\,y) = -a\,\mathrm{trd}(\overline{x}i^2y) = ad\,\mathrm{trd}(\overline{x}y)$, so

$$H(x,y) = a\big[d\,\mathrm{trd}(\overline{x}y) + \sqrt{-d}\,\mathrm{trd}(\overline{x}iy)\big].$$

Now $\mathrm{trd}(1) = 2$, $\mathrm{trd}(i) = \mathrm{trd}(j) = \mathrm{trd}(ij) = 0$, $\overline{j}j = -b$ and $\overline{j}ij = ib$, which give $H(1,1) = 2ad$, $H(j,j) = -2abd$ and $H(1,j) = 0$. Since $b > 0$ the two entries of each block have opposite signs, so the signature is (n,n).

(iii) A complex structure commuting with B and compatible with E is the same as a subspace $V^{1,0} \subset V_{\mathbb{C}}$, stable under B, with $V_{\mathbb{C}} = V^{1,0} \oplus \overline{V^{1,0}}$ and the two Riemann relations. Being K-stable it splits as $P \oplus P'$ with $P \subset V_+$ and $P' \subset V_-$; left multiplication by j is σ-semilinear, so it carries V_+ to V_-, and B-stability forces $P' = jP$. Conversely, for any n-dimensional H-negative definite $P \subset V_+$ the subspace $P \oplus jP$ is B-stable and satisfies the Riemann relations, by the computation in the proof of Theorem 4.4 applied blockwise; such P exists because H has signature (n,n) by (ii). Since j is invertible and carries P isomorphically to P', one gets $\dim P = \dim P' = n$, which is Definition 3.1(ii). The general theory of abelian varieties with prescribed endomorphism structure is in [BL04, Ch. 9].

(iv) By (ii), $\det H = \prod_k (2a_k d)(-2a_k bd) = (-1)^n b^n \big(2^n d^n \prod_k a_k\big)^2$, so $(-1)^n \det H$ is b^n times a square, and squares are norms.

(v) Left multiplication by j satisfies $\mu^\dagger = \mu$ by (i) and $\mu\,\iota(\tau) = \iota(\overline{\tau})\,\mu$ because $ji = -ij$; it is invertible because $\mu^2 = b \neq 0$. Lemma 15.23 puts g_μ in $R \cap \mathrm{NS}(A)_{\mathbb{Q}}$ and makes it nondegenerate, and Theorem 15.16 applies to the component Θ of g_μ in $\bigwedge^2 V_-$. ∎

Lemma 15.26 (Products and discriminants)**.** *Let (A_1, ι_1, η_1) and (A_2, ι_2, η_2) be of $(K, -1, n_1)$- and $(K, -1, n_2)$-Weil type. Then $A_1 \times A_2$ with the diagonal K-action and the product polarisation is of $(K, -1, n_1 + n_2)$-Weil type, its hermitian form is the orthogonal sum, its discriminant is the product of the two discriminants, and if both factors satisfy the divisorial criterion then so does the product.*

Proof. The first assertion is Lemma 15.6. Signatures and hermitian forms add over an orthogonal sum, so the signature is $(n_1 + n_2, n_1 + n_2)$ and $\det H = \det H_1 \det H_2$, whence $(-1)^{n_1+n_2} \det H$ is the product of the two normalised determinants. For the criterion, $V_- = V_{1,-} \oplus V_{2,-}$, and $\Theta = \Theta_1 + \Theta_2$ lies in $\bigwedge^2 V_-$, is of type $(1,1)$, has $\Theta + \overline{\Theta}$ rational, and is nondegenerate on the direct sum because each summand is nondegenerate on its factor. ∎

Theorem 15.27 (A base point in every Weil family)**.** *Let $K = \mathbb{Q}(\sqrt{-d})$, let $n \geq 1$, and let δ be any class in $\mathbb{Q}^\times_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^\times)$. Then there is an abelian $2n$-fold A of $(K, -1, n)$-Weil type with discriminant δ whose Weil classes are algebraic, with an explicit representative as a rational polynomial in two divisor classes.*

Proof. Choose $b \in \mathbb{Q}_{>0}$ representing δ.

If n is odd, take the quaternionic model of Setup 15.24 with this b. Its discriminant is $b^n = b \cdot (b^{(n-1)/2})^2$, which is δ, and its Weil classes are algebraic by Theorem 15.25(v).

If n is even, then b^n is a square and the model alone reaches only the trivial class. Take instead $A = A_1 \times A_2$ with A_1 the quaternionic model at $n_1 = 1$ and parameter b, and A_2 the quaternionic model at $n_2 = n - 1$ and parameter d. Both factors satisfy the criterion, so the product does by Lemma 15.26; and the discriminant is $b^1 \cdot d^{n-1}$, which is δ because $d = \mathrm{N}_{K/\mathbb{Q}}(\sqrt{-d})$ is a norm. The dimension is $2(1 + (n-1)) = 2n$.

In both cases the representative is (37) for the class g_μ of Theorem 15.25(v), or, in the second case, for the sum of the two such classes. ∎

Corollary 15.28 (Reduction to propagation)**.** *Every Weil family contains a member at which the Weil classes are algebraic. Consequently Question 15.13 reduces to propagation alone: the Hodge conjecture for abelian $2n$-folds of Weil type, for every n and every discriminant, holds if and only if the class ω of Setup 15.12 propagates from the nonempty locus where it is known to be algebraic to the rest of its connected family.*

Proof. The first sentence is Theorem 15.27 applied to each class, together with the fact that the families are indexed by those classes (Definition 15.22). The second is Theorem 15.14, the base locus being nonempty in every family by the first. ∎

Remark 15.29 (Two features of the residual statement)*.* By Theorem 15.14 and Corollary 15.28, the residual statement is the variational Hodge conjecture for one flat class on one connected family, with a base point in every family, and it is necessary as well as sufficient. Two of its features show which tools cannot settle it.

The base locus is thin. By Proposition 15.18 the divisorial criterion fails at every point with full Hodge group, so the locus where it holds is contained in the Noether–Lefschetz locus of R, a countable union of proper closed algebraic subvarieties of $\mathcal{D}_{n,n}$ by Theorem B.6. The divisorial method cannot enlarge the base locus beyond special points: every point it produces is again special.

The propagation is not formal. By Remark B.11 its infinitesimal form would follow from semiregularity, in the sense of Bloch, of the cycle at a base point, and that cycle is explicit in every family, being a rational polynomial in two divisor classes. Its semiregularity is a computation in the normal bundle that we do not carry out; for the perfect complexes of [II, Proposition 2.1], whose Chern character is concentrated in the degree of the Weil class, semiregularity fails (Proposition 9.10, [II, Theorem 3.5]). The other half of the difficulty, that semiregularity gives deformations only in the formal category, is removed in Section 16.1: once the algebraic locus is known to be a countable union of closed subvarieties, a formal deformation at one point is enough.

The secant route of Section 14 needs, at $n \geq 4$, an object on X whose Chern character lies on the secant plane and which also supplies the deformation step of Remark 14.16. No such object is known, but its numerical invariants raise no obstruction.

Proposition 15.30 (Numerical admissibility of the secant plane)**.** *Let t be a principal polarisation on X and let $\mathrm{ch} = au+bv$ be a rational point of the secant plane of Theorem 14.5, with $a, b \in \mathbb{Z}$. Then*

(i) *every ch_k is an integral multiple of the integral class $t^k/k!$, the multiple being $\pm ad^j$ in even degree $k = 2j$ and $\pm bd^j$ in odd degree $k = 2j+1$;*

(ii) *for all integers a, b and every positive integer d, the Chern classes obtained from ch by Newton's identities are integral in every degree: $c_k = \gamma_k\, t^k/k!$ with $\gamma_0 = 1$, $\gamma_1 = b$ and*

$$\gamma_{k+1} \;=\; b\,\gamma_k + d\,k\,(a-k+1)\,\gamma_{k-1} \qquad (k \geq 1);$$

(iii) *the discriminant is $\Delta = 2rc_2 - (r-1)c_1^2 = (a^2d + b^2)t^2 = \mathrm{N}_{K/\mathbb{Q}}(b + a\sqrt{-d})\, t^2$, which is positive, so Bogomolov's inequality never binds.*

Proof. (i) is the definition of u and v: the coefficient of t^k is $a(-d)^j/(2j)!$ or $b(-d)^j/(2j+1)!$, and multiplying by $k!$ clears the factorial and leaves $\pm ad^j$ or $\pm bd^j$. For (ii), write the total Chern class as a power series $c(t) = \sum_k \gamma_k t^k/k!$ in the variable t. Newton's identities in generating-function form say $\log c = \sum_{k\geq 1}(-1)^{k-1}(k-1)!\,\mathrm{ch}_k$, and with the coefficients of

(i) the two halves of this sum are recognised series:

$$\log c(t) \;=\; -a\sum_{j\ge 1}\frac{(-d)^j t^{2j}}{2j} + b\sum_{j\ge 0}\frac{(-d)^j t^{2j+1}}{2j+1} \;=\; \frac{a}{2}\log(1+dt^2) + \frac{b}{\sqrt{d}}\arctan(\sqrt{d}\,t).$$

Differentiating, $(1+dt^2)\,c'(t) = (b+adt)\,c(t)$. Comparing the coefficients of $t^k/k!$ on the two sides gives $\gamma_{k+1} + d\,k(k-1)\gamma_{k-1} = b\gamma_k + ad\,k\,\gamma_{k-1}$, which is the stated recursion. Its coefficients are integers and it starts from $\gamma_0 = 1$, $\gamma_1 = b$, so every γ_k is an integer, and $c_k = \gamma_k\,t^k/k!$ is an integral class because $t^k/k!$ is. (iii) is the computation recorded in Theorem 14.7, and the norm form of an imaginary quadratic field is positive definite. ∎

Remark 15.31 (Numerical consistency of the secant object). Proposition 15.30 excludes a purely arithmetic obstruction. If the object required by the construction at $n \ge 4$ were forbidden by its numerical invariants, the secant route would fail on arithmetic grounds alone. It is not: the Chern character is integral, the Chern classes are integral in every degree by Proposition 15.30(ii), the discriminant is positive, and neither Riemann–Roch nor Bogomolov's inequality stands in the way. Together with Proposition 9.11, which bounds the size of the object from below without excluding it, and the parameter count of Remark 14.11, which fixes the rank one shape at $n = 3$, every count available to us is consistent with the existence of the object, and none of them produces it. What is missing at $n \ge 4$ is the object itself.

15.6. **The quaternionic locus as a Lagrangian Grassmannian.** Theorem 15.25 produces points of the family carrying an admissible endomorphism ψ. The locus of all such points has dimension $n(n+1)/2$, by a Morita argument recalled in Remark 16.26. Here we identify the locus itself, and the identification shows where the residual difficulty lies.

Proposition 15.32 (The quaternionic locus)**.** *Keep Setup 15.24 and let ψ be K-semilinear, invertible, Rosati-symmetric, with $\psi^2 = b$. For $x, y \in V_+$ put*

$$\omega_\psi(x,y) \;=\; E(x,\psi y).$$

(i) *ω_ψ is a nondegenerate alternating form on V_+, and it is alternating precisely because ψ is Rosati-symmetric.*

(ii) *A point $P = V_+^{1,0}$ of $\mathcal{D}_{n,n}$ lies on the locus where ψ is a morphism of Hodge structures if and only if P is isotropic for ω_ψ.*

(iii) *Since $\dim_{\mathbb{C}} V_+ = 2n$ and $\dim_{\mathbb{C}} P = n$, such a P is Lagrangian, so the locus is the open subset of $\mathrm{LG}(n, V_+, \omega_\psi)$ cut out by negative definiteness of H. Its dimension is $n(n+1)/2$ and its codimension in $\mathcal{D}_{n,n}$ is $n(n-1)/2$.*

Proof. (i) ψ carries V_+ to V_-, since it is semilinear, and E pairs V_+ with V_- perfectly, so ω_ψ is defined and nondegenerate. Rosati symmetry is the identity $E(\psi x, y) = E(x, \psi y)$; with E alternating it gives $\omega_\psi(y,x) = E(y,\psi x) = -E(\psi x, y) = -E(x,\psi y) = -\omega_\psi(x,y)$.

(ii) Suppose ψ preserves the Hodge structure. Being semilinear it carries $V_+^{1,0}$ into $V_-^{1,0}$, so $\psi P \subseteq V^{1,0}$, and E vanishes on $V^{1,0}$ by the first Riemann relation; hence $E(P,\psi P) = 0$, which is the isotropy of P. Conversely, if P is isotropic then ψP lies in the E-annihilator of P inside V_-. That annihilator has dimension n, because E pairs V_+ with V_- perfectly, and it contains $V_-^{1,0}$, which also has dimension n and is E-orthogonal to P since both lie in $V^{1,0}$. The two therefore agree, so $\psi P = V_-^{1,0}$ and ψ preserves $V^{1,0}$.

(iii) A Lagrangian Grassmannian $\mathrm{LG}(n, 2n)$ has dimension $n(n+1)/2$, its tangent space at a Lagrangian P being the symmetric forms on P; negative definiteness is an open

condition and does not change the dimension. Subtracting from $\dim \mathcal{D}_{n,n} = n^2$ gives $n(n-1)/2$. ∎

Figure 17 draws these loci as a pencil of leaves.

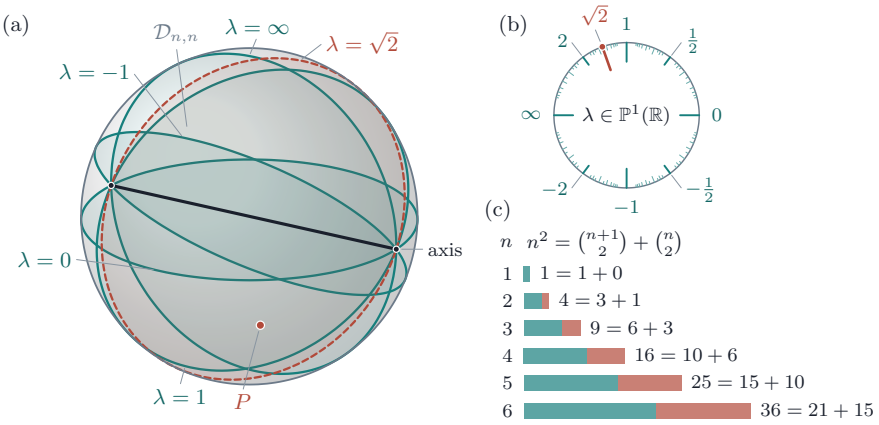


FIGURE 17. The quaternionic loci of Proposition 15.32, drawn as a pencil. Each admissible ψ gives one leaf, an open piece of $\mathrm{LG}(n, V_+, \omega_\psi)$ of dimension $n(n+1)/2$. As ψ varies over the reals the leaves through a common axis form a continuous family and sweep out $\mathcal{D}_{n,n}$. The two heavy leaves are those with ψ rational, and only such leaves are quaternionic loci of abelian varieties. The drawing is faithful to the continuity of the family and to the discreteness of its rational members, but not to dimensions: in the family the leaves have dimension $n(n+1)/2$ inside n^2, hence codimension $n(n-1)/2$, which vanishes only at $n = 1$ (at $n = 2$ the leaves are divisors, Remark 16.26), and ψ ranges over a real variety rather than an interval.

Remark 15.33 (Real and rational quaternionic loci). Fix a point P of $\mathcal{D}_{n,n}$, and let ψ_0 be left multiplication by j at a point P_0 given by Theorem 15.25(iii). The group $\mathrm{SU}(V, H)(\mathbb{R})$ acts transitively on the bounded symmetric domain $\mathcal{D}_{n,n}$ of Proposition B.8, so $P = gP_0$ for some real g; since g is K-linear and preserves H, hence E, the map $\psi = g\psi_0 g^{-1}$ is K-semilinear, invertible and Rosati-symmetric, satisfies $\psi^2 = b$, and preserves the Hodge structure at P. It satisfies every condition of Proposition 15.32 except rationality, and P lies on its locus. So the quaternionic loci, taken over the *real* ψ, cover $\mathcal{D}_{n,n}$; they cover it with room to spare, since each has dimension $n(n+1)/2$ and they are parametrised by a positive-dimensional family. What prevents the quaternionic loci of abelian varieties from covering it is that ψ must lie in $\mathrm{End}^0(A)$, hence be rational, and the rational ψ form a countable set.

This is the dichotomy of Corollary 16.5 in its simplest form. The obstruction is the countability of a set of parameters inside an uncountable one, and it involves neither deformation theory nor positivity. What is needed is a mechanism that produces, from one rational ψ, a family of cycles representing ω over an open subset of $\mathcal{D}_{n,n}$ rather than over the locus of ψ alone; by Corollary 16.5, such a family would answer Question 15.13 in the affirmative.

15.7. **Absolutely Hodge classes and the exact form.** The algebraicity of a Hodge class is usually approached through an intermediate property. A rational class of type (p,p) on a smooth projective variety X over $\mathbb{C}$ is *absolutely Hodge* if, for every automorphism σ of $\mathbb{C}$, the conjugate under σ of its de Rham component is the de Rham component of a rational class of type (p,p) on the conjugate variety X^{σ}. The implications

$$\text{algebraic} \implies \text{absolutely Hodge} \implies \text{Hodge} \tag{39}$$

are both elementary. The Hodge conjecture asserts the converse of their composite. The converse of the second is the weaker statement, and Deligne proved it for abelian varieties, so on abelian varieties the Hodge conjecture is the converse of the first.

Theorem 15.34 (Deligne [Del82])**.** *Every Hodge class on a complex abelian variety is absolutely Hodge. Moreover,* Principle B*: if $\pi\colon \mathcal{X} \to S$ is smooth projective over a connected smooth base, and α is a flat section of $R^{2p}\pi_*\mathbb{Q}(p)$ that is Hodge at every point of S and absolutely Hodge at one point, then it is absolutely Hodge at every point.*

Together with Theorem 15.14 and Theorem 15.27, this locates the remaining problem.

Corollary 15.35 (Reduction to an absolutely Hodge class)**.** *Let $\pi\colon \mathcal{A} \to \mathcal{D}_{n,n}$ and ω be as in Setup 15.12. Then ω_s is absolutely Hodge for every s, and the following are equivalent:*

(i) *the Hodge conjecture holds, in every degree, for every abelian $2n$-fold of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$;*

(ii) *the absolutely Hodge class ω_s is algebraic for every s.*

This holds for every K, every n and every discriminant.

Proof. The first assertion is Theorem 15.34. It also follows from Principle B together with Theorem 15.27, which supplies a point of $\mathcal{D}_{n,n}$ at which ω is algebraic and hence absolutely Hodge, and with the connectedness of $\mathcal{D}_{n,n}$ from Proposition B.8.

That (ii) implies (i) is Theorem 15.14(a). Conversely, the Hodge group of A_s is $\mathrm{SU}(V,H)$ for s outside a countable union of proper closed analytic subsets of $\mathcal{D}_{n,n}$, by Weil's theorem [Wei77] recalled before Theorem 4.14. Under (i) the class ω_s is algebraic at all such s, so the set Σ of points at which it is algebraic is not meagre, and Corollary 16.5 gives $\Sigma = \mathcal{D}_{n,n}$, which is (ii). ∎

16. The algebraic locus and the propagation criteria

The results of Section 15 are pointwise: a class, a criterion and a cycle at a point, and in Corollary 15.35 an implication to be proved at every point. This section studies the set of points at which the Weil class is algebraic. Properness of the Hilbert scheme makes it a countable union of closed subvarieties (Theorem 16.4), and rational isogenies make it dense (Proposition 16.7); together these turn the propagation question into a dichotomy (Corollary 16.5). Two criteria for propagation follow, a bound along one orbit (Theorem 16.9) and a rank at one point ([II, Theorem 2.3]). We treat the bound first, together with the transport of cycles along isogenies, which decides one of its two readings, and with the limits of representatives built from products of divisors; the rest of the section is devoted to the rank.

16.1. **The algebraic locus.** Question 15.13 is a question about the set

$$\Sigma \;=\; \{\, s \in \mathcal{D}_{n,n} \;:\; \omega_s \text{ is algebraic on } A_s \,\}.$$

This set is restricted in shape, and the restriction changes the form of the remaining problem: instead of a class to construct at every point, one is left with a bound to prove

along one orbit, or a rank to compute at one point. The reason is that the Hilbert scheme is proper, so each piece of Σ cut out by bounded data is Zariski closed. The mechanism is standard: the locus where a fixed Hodge class becomes algebraic is a countable union of closed subvarieties, the companion of the statement that the Hodge locus itself is one (see [Voi03, Chapter 5] and, in the form used in Theorem B.6, [CDK95]). What is particular to the present situation is that the class is determined by Theorem 4.14 and written out by Theorem 4.9, that its algebraic locus is nonempty in every family by Theorem 15.27, and that the group acting on the family is known.

Setup 16.1 (The quotient and the universal family)**.** Keep Setup 15.12. Fix an integer $N \geq 3$ and let Γ be the subgroup of $\mathrm{SU}(V,H)(\mathbb{Q})$ preserving the lattice $L \subset V$ used to build the family and acting trivially on L/NL. Write

$$p\colon \mathcal{D}_{n,n} \longrightarrow \mathcal{S} = \Gamma\backslash\mathcal{D}_{n,n}.$$

Since $N \geq 3$ the group Γ is torsion-free, so it acts freely on $\mathcal{D}_{n,n}$, and by Baily and Borel [BB66] the quotient $\mathcal{S}$ is a smooth quasi-projective complex variety; it is irreducible because $\mathcal{D}_{n,n}$ is connected. The level structure rigidifies, so $\mathcal{S}$ carries a universal family

$$f\colon \mathcal{A} \longrightarrow \mathcal{S},$$

and the relative polarisation 3η is relatively very ample, which fixes a relative projective embedding and hence a notion of degree in the fibres. Write $G = \mathrm{SU}(V,H)$, an algebraic group over $\mathbb{Q}$, so that $G(\mathbb{R})$ acts transitively on $\mathcal{D}_{n,n}$ and $\Gamma \subset G(\mathbb{Q})$.

We record two facts that are used repeatedly below.

Lemma 16.2 (The section descends)**.** *The section ω of Setup 15.12 is Γ-invariant, hence descends to a flat section of $R^{2n}f_*\mathbb{Q}$ over $\mathcal{S}$ that is of type (n,n) at every point.*

Proof. The Weil line is $W = \bigwedge_K^{2n} V$, and an element of $\mathrm{SU}(V,H)$ is by definition a K-linear automorphism of V of K-determinant 1. Its action on the top K-exterior power is multiplication by that determinant, so Γ acts trivially on W and in particular fixes ω. Flatness is then immediate, and the Hodge type at every point is Proposition 3.7(ii). ∎

Lemma 16.3 (One component is enough)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type with $\iota(\mathcal{O}_K) \subseteq \mathrm{End}(A)$, and let $W \subset H^{2n}(A,\mathbb{Q})$ be its Weil line. If one nonzero element of W is algebraic, every element of W is algebraic.*

Proof. Among the elements $\tau_m = m + \sqrt{-d}$ of $\mathcal{O}_K$ with $1 \leq m \leq 7$ the ratios $\tau_m/\overline{\tau}_m$ are pairwise distinct, since $\tau_m\overline{\tau}_{m'}$ is rational only when $m = m'$. The group of roots of unity in an imaginary quadratic field has order 2, 4 or 6, so some m in that range has $\tau = \tau_m$ with $\tau/\overline{\tau}$ not a root of unity, and then $\tau^{2n} \neq \overline{\tau}^{2n}$, that is $\tau^{2n} \notin \mathbb{Q}$, for every $n \geq 1$. A range is needed, since the witness $m = 1$ fails for two fields: for $d = 1$ the ratio attached to $1 + \sqrt{-1}$ is a primitive fourth root of unity, and for $d = 3$ the ratio attached to $1 + \sqrt{-3}$ is a primitive cube root of unity, so $(1 + \sqrt{-d})^{2n}$ is rational when $d = 1$ and n is even, and when $d = 3$ and $3 \mid n$.

Since $\tau \in \mathcal{O}_K$, the element $\iota(\tau)$ is an endomorphism of A, so $\iota(\tau)^*$ on $H^{2n}(A,\mathbb{Q})$ is induced by the graph correspondence and therefore carries algebraic classes to algebraic classes. On H^1 it is multiplication by τ, so on $\bigwedge_K^{2n} V$ it is multiplication by τ^{2n}. As $\tau^{2n} \notin \mathbb{Q}$ and W is a one-dimensional K-vector space, the pair w, $\tau^{2n}w$ is a $\mathbb{Q}$-basis of W for every $w \neq 0$. Explicitly, fix a K-basis w_0 of W and write $\tau^{2n} = a + b\sqrt{-d}$, so that $b \neq 0$, and $w = (w_1 + w_2\sqrt{-d})\,w_0$; the determinant of the pair $w, \tau^{2n}w$ in the $\mathbb{Q}$-basis $w_0, \sqrt{-d}\,w_0$ is $b(w_1^2 + dw_2^2)$, which is nonzero when $w \neq 0$. If w is algebraic then so is $\iota(\tau)^*w = \tau^{2n}w$, and the two together span. ∎

Lemma 16.3 writes out the last step of the proof of Corollary 4.15, where the witness τ has to be chosen.

Theorem 16.4 (The algebraic locus is a countable union of closed subvarieties)**.** *Keep Setup 16.1. For a finite tuple $\underline{P} = (P_1, \dots, P_m)$ of Hilbert polynomials of degree n and a tuple $\underline{q} \in \mathbb{Q}^m$ put*

$$\mathcal{H}_{\underline{P}} \;=\; \mathrm{Hilb}^{P_1}(\mathcal{A}/\mathcal{S}) \times_{\mathcal{S}} \cdots \times_{\mathcal{S}} \mathrm{Hilb}^{P_m}(\mathcal{A}/\mathcal{S}),$$

$$\Sigma_{\underline{P},\underline{q}} \;=\; \mathrm{pr}\Big(\{\, \underline{Z} \in \mathcal{H}_{\underline{P}} \;:\; \textstyle\sum_i q_i [Z_i] = \omega \,\} \Big) \;\subseteq\; \mathcal{S}.$$

Then each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed in $\mathcal{S}$, and

$$p^{-1}\Big(\bigcup_{\underline{P},\underline{q}} \Sigma_{\underline{P},\underline{q}} \Big) \;=\; \Sigma,$$

the union running over a countable index set.

Proof. Each $\mathrm{Hilb}^{P_i}(\mathcal{A}/\mathcal{S})$ is projective over $\mathcal{S}$ by Grothendieck's construction of the Hilbert scheme [Gro61], so the fibre product $\mathcal{H}_{\underline{P}}$ is projective over $\mathcal{S}$, and being of finite type it has finitely many connected components.

Fix a connected component T, replace it by its reduction, which changes neither the underlying space nor any cohomology group, and let $\mathcal{Z}_i \subset \mathcal{A} \times_{\mathcal{S}} T$ be the pullback of the i-th universal subscheme. It is flat over T and its fibres have Hilbert polynomial P_i of degree n, so the associated n-cycles, the top-dimensional components counted with their multiplicities, are the fibres of a cycle on $\mathcal{A} \times_{\mathcal{S}} T$ whose class restricts on the fibre over t to $[Z_{i,t}]$. Restriction of a global class to the fibres of a topologically locally trivial family is a flat section of the local system, so $t \mapsto \sum_i q_i [Z_{i,t}]$ is flat on T; and ω is flat on $\mathcal{S}$ by Lemma 16.2, hence flat after pullback. Two flat sections of a local system over a connected base agree everywhere as soon as they agree at one point, so the locus where they agree is a union of connected components of $\mathcal{H}_{\underline{P}}$. That locus is therefore itself projective over $\mathcal{S}$, and the image of a proper morphism is Zariski closed.

For the equality of sets, let $s \in \Sigma$. Then $\omega_s = \sum_i q_i [Z_i]$ for finitely many closed integral subschemes $Z_i \subset A_s$ of dimension n and rationals q_i; taking P_i to be the Hilbert polynomial of Z_i puts $p(s)$ in $\Sigma_{\underline{P},\underline{q}}$. The converse holds because the class of the n-cycle of a closed subscheme of dimension n is algebraic. The index set is countable, since a Hilbert polynomial is determined by its finitely many integer coordinates in the binomial basis, and $\mathbb{Q}$ is countable. ∎

The bound on the Hilbert polynomials cannot be dropped: without it one takes the image of an infinite disjoint union, and an infinite union of closed sets need be neither closed nor constructible, and the dichotomy below is what can be said of Σ itself. Tuples are needed because a rational cycle is a rational combination of several integral ones, which may have different Hilbert polynomials. The relative Chow variety of effective n-cycles of bounded degree would serve equally well, with the same proof, since only properness over the base and local constancy of the cycle class are used.

Corollary 16.5 (The dichotomy)**.** *Exactly one of the following holds.*

(a) *$\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$-fold of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ in the family.*

(b) *Σ has empty interior in the analytic topology. Its complement is then a dense G_δ, so ω_s fails to be algebraic for every s outside a meagre set, and the Hodge conjecture fails.*

In particular, if ω_s is algebraic for all s in some nonempty open subset of $\mathcal{D}_{n,n}$, then it is algebraic for all s.

Proof. Suppose Σ contains a nonempty open set; then so does its image in $\mathcal{S}$, and we may pick a nonempty connected open $U \subseteq \mathcal{S}$ contained in that image. By Theorem 16.4 the countably many sets $\Sigma_{\underline{P},\underline{q}} \cap U$ are closed in U and cover it. A connected complex manifold is a Baire space, so one of them has nonempty interior in U. A Zariski closed subset of the irreducible variety $\mathcal{S}$ either is all of $\mathcal{S}$ or is a proper closed subvariety, and a proper closed subvariety of a connected complex manifold has empty interior. Hence that $\Sigma_{\underline{P},\underline{q}}$ is all of $\mathcal{S}$, so $\Sigma = \mathcal{D}_{n,n}$ and (a) holds by Theorem 15.14(a). If Σ has empty interior, its complement is the intersection of the countably many dense open sets $\mathcal{D}_{n,n} \setminus p^{-1}(\Sigma_{\underline{P},\underline{q}})$, and Baire again gives that this intersection is dense. At each of its points ω_s is a Hodge class by Lemma 16.2 and is not algebraic, so the Hodge conjecture fails there. ∎

Figure 18 draws the two cases.

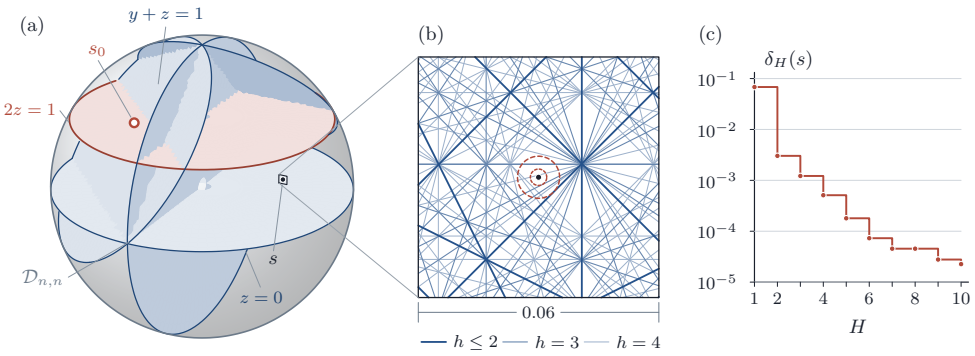


FIGURE 18. The dichotomy of Corollary 16.5. Each stratum $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed by Theorem 16.4, and there are countably many of them; the base point s_0 of Theorem 15.27 lies on one. By Proposition 16.7 the union of the strata is dense, so every neighbourhood of a point s lying off all of them meets infinitely many strata. The conjecture for the family is the assertion that no such s exists, and by Corollary 16.5 it is enough that one stratum has nonempty interior.

Lemma 16.6 (Invariance under rational isogenies)**.** *Σ is stable under the action of $G(\mathbb{Q})$ on $\mathcal{D}_{n,n}$.*

Proof. A point of $\mathcal{D}_{n,n}$ is a complex structure J on $V_{\mathbb{R}}$ compatible with H, and $g \in G(\mathbb{Q})$ carries J to gJg^{-1}. The map g is then $\mathbb{Q}$-rational and $\mathbb{C}$-linear from $(V_{\mathbb{R}}, J)$ to $(V_{\mathbb{R}}, gJg^{-1})$, so after clearing denominators it induces an isogeny $A_J \to A_{gJ}$, whose graph is an algebraic correspondence. Algebraicity of a rational cohomology class is preserved by such a correspondence and by its transpose. Finally g lies in $\mathrm{SU}(V, H)$, so by the computation in Lemma 16.2 it fixes the Weil line pointwise; the correspondence therefore carries ω_J to a nonzero rational multiple of ω_{gJ}, and ω_{gJ} is algebraic when ω_J is. ∎

Two density statements should be distinguished. For a class that does not remain of Hodge type over the whole family, the locus where it is of Hodge type is a countable union of proper closed subvarieties, each algebraic by Theorem B.6, and density there concerns

the distribution of special points. For the class ω of Setup 15.12 the Hodge locus is the whole family by Proposition 3.7(ii), so Theorem B.6 gives no information, and the set that carries information is the algebraic locus. The next proposition concerns that set, and its proof uses no Hodge theory: the algebraic locus is a union of orbits of a group whose rational points are dense in its real points.

Proposition 16.7 (The algebraic locus is dense)**.** *Let $n \geq 2$. Then Σ is dense in $\mathcal{D}_{n,n}$ in the analytic topology, and its image in $\mathcal{S}$ is dense.*

Proof. The hermitian form H has rank $2n \geq 4$ over K and signature (n, n), so it is isotropic over $\mathbb{R}$; over every $\mathbb{Q}_p$ a hermitian form of rank at least three over a quadratic extension is isotropic; and hermitian forms over an imaginary quadratic field satisfy the Hasse principle [Sch85, Ch. 10, §6]. So H is isotropic over $\mathbb{Q}$ and G is $\mathbb{Q}$-isotropic. The group $G = \mathrm{SU}(V, H)$ is simply connected and almost $\mathbb{Q}$-simple, so by Platonov's theorem $G(\mathbb{Q})$ is generated by the rational points of the unipotent radicals of its proper parabolic $\mathbb{Q}$-subgroups [PR94, Thm. 7.6 and §7.2]. For a unipotent group U over $\mathbb{Q}$ the exponential identifies $U(\mathbb{Q})$ with a $\mathbb{Q}$-vector space inside the corresponding $\mathbb{R}$-vector space $U(\mathbb{R})$, so $U(\mathbb{Q})$ is dense in $U(\mathbb{R})$. The closure of $G(\mathbb{Q})$ in $G(\mathbb{R})$ therefore contains $U(\mathbb{R})$ for every such U, and those generate the identity component $G(\mathbb{R})^+$, which acts transitively on $\mathcal{D}_{n,n}$. Hence every $G(\mathbb{Q})$-orbit in $\mathcal{D}_{n,n}$ is dense.

By Theorem 15.27 the set Σ is nonempty, and by Lemma 16.6 it is $G(\mathbb{Q})$-stable, so it contains a dense orbit. The projection p is continuous and surjective, so the image is dense. ∎

In particular, if case (b) of Corollary 16.5 occurs, then Σ and its complement are both dense, and the two sets are distinguished by Baire category and not by position.

Corollary 16.8 (A Baire category criterion)**.** *Let $n \geq 2$. The Hodge conjecture holds for every abelian $2n$-fold of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ in the family of Setup 15.12 if and only if Σ is non-meagre in $\mathcal{D}_{n,n}$, and if and only if Σ has nonempty interior.*

Proof. If $\Sigma = \mathcal{D}_{n,n}$, both conditions hold, and so does the conjecture, by Theorem 15.14(a). Otherwise case (b) of Corollary 16.5 holds, so Σ has empty interior; it is then the countable union of the closed sets $p^{-1}(\Sigma_{\underline{P},\underline{q}})$, each with empty interior, and so it is meagre. The points s with $\mathrm{Hg}(A_s) = \mathrm{SU}(V, H)$ form the complement of a countable union of proper closed analytic subsets of $\mathcal{D}_{n,n}$, by Weil's theorem [Wei77] recalled before Theorem 4.14, so by Baire they are not all contained in the meagre set Σ. At a point of this kind outside Σ the Hodge conjecture fails for A_s. ∎

16.2. **The first criterion: a bound along one orbit.** The proof of Proposition 16.7 produces the dense algebraic locus by transporting one cycle along rational isogenies, and an isogeny multiplies degrees. The dense set so obtained is therefore stratified by unbounded data, and this is what prevents Corollary 16.5 from applying to it. A bound on the data along one orbit removes the difficulty.

Theorem 16.9 (A bound along the orbit closes the family)**.** *Let $n \geq 2$ and let $s_0 \in \Sigma$ be one of the base points supplied by Theorem 15.27. Suppose that there is a finite set $\mathcal{T}$ of pairs $(\underline{P}, \underline{q})$ such that for every $g \in G(\mathbb{Q})$ the class $\omega_{g \cdot s_0}$ is represented by some cycle whose Hilbert data lies in $\mathcal{T}$. Then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$-fold of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ in that family.*

Proof. The orbit $G(\mathbb{Q})\cdot s_0$ is dense by the proof of Proposition 16.7, and by hypothesis it is the union of the finitely many sets $G(\mathbb{Q})\cdot s_0 \cap p^{-1}(\Sigma_{\underline{P},\underline{q}})$ with $(\underline{P},\underline{q})\in\mathcal{T}$. The union of the finitely many closed sets $p^{-1}(\Sigma_{\underline{P},\underline{q}})$, $(\underline{P},\underline{q})\in\mathcal{T}$, therefore contains a dense set, and so it is all of $\mathcal{D}_{n,n}$; in particular $\Sigma=\mathcal{D}_{n,n}$. In fact one of these sets is already all of $\mathcal{S}$: each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed in the irreducible variety $\mathcal{S}$, hence either equal to $\mathcal{S}$ or nowhere dense, and a finite union of nowhere dense sets is nowhere dense. The Hodge conjecture for the family follows by Theorem 15.14(a). ∎

Remark 16.10 (Choice of the representing cycle). We call the hypothesis of Theorem 16.9 the bounded criterion (P1); [II, Remark 3.7] lists it beside the rank criterion (P2) of [II, Theorem 2.3]. It quantifies over representing cycles at each point of the orbit and asks only that, at each point, one of them have bounded complexity. A stronger hypothesis would fix the cycle, taking at $g\cdot s_0$ the transport of the cycle at s_0 along the isogeny of Lemma 16.6. Theorem 16.15 below shows that this stronger form is never satisfied by a base cycle with a component of finite stabiliser, since the transported data is then unbounded. The proof above uses only the weaker hypothesis, which is what the covering argument needs.

The hypothesis of Theorem 16.9 can be sharpened by locating the part of the Hilbert data that is at issue. The leading coefficient is not.

Proposition 16.11 (The leading coefficient is constant on the orbit)**.** *Let $s_0\in\Sigma$ and $g\in G(\mathbb{Q})$, and let ω_s denote the Weil class at s and η the polarisation class. Then*

$$\int_{A_{g\cdot s_0}} \omega_{g\cdot s_0}\cdot\frac{\eta^n}{n!} \;=\; \int_{A_{s_0}} \omega_{s_0}\cdot\frac{\eta^n}{n!},$$

so the degree of the transported class with respect to the polarisation, which is the leading coefficient of the Hilbert polynomial of any cycle representing it, is the same at every point of the orbit.

Proof. By Lemma 16.6 the correspondence transporting the class is induced by a $\mathbb{Q}$-rational map $g\in\mathrm{SU}(V,H)$. It preserves the hermitian form and hence the polarisation class, it fixes the Weil line pointwise by Lemma 16.2, and its determinant over $\mathbb{Q}$ is $\mathrm{N}_{K/\mathbb{Q}}(\det_K g)=1$, so it acts trivially on the top exterior power of $V^\vee$, in which both integrals are evaluated. The two integrals are therefore equal. The identification of the number with the leading coefficient is Hirzebruch–Riemann–Roch on A, whose Todd class is 1 (for the embedding by 3η of Setup 16.1 the leading coefficient is 3^n times the number); the remaining coefficients involve the structure sheaf of the cycle and not only its class. ∎

Remark 16.12 (The source of unboundedness). Write a representing cycle at a point of the orbit, the transported cycle or any other, as a difference $\xi^+-\xi^-$ of effective cycles. By Proposition 16.11 the difference of their degrees is constant along the orbit, and by the boundedness of effective cycles of bounded degree in the fibres of the polarised family $f\colon\mathcal{A}\to\mathcal{S}$, if $\deg\xi^+$ were bounded along the orbit then only finitely many tuples $(\underline{P},\underline{q})$ could occur and Theorem 16.9 would apply. So the hypothesis of that theorem is equivalent to a bound on the effective part alone.

This bound is not automatic, and the mechanism that threatens it can be seen directly. The transport at g is by the isogeny obtained from g by clearing denominators; if g has denominator m on the reference lattice, which has rank $4n$, the isogeny has degree m^{4n} (Lemma 16.13 below), and pushing a cycle forward along it multiplies its multiplicities and degrees by powers of m (Lemma 16.14). Elements of $G(\mathbb{Q})$ of bounded denominator form a discrete subset of $G(\mathbb{R})$, so along a dense orbit the denominators, and with them the

isogeny degrees, are unbounded. What Theorem 16.9 asks is that the cycles nevertheless simplify: that at each point of the orbit the class ω, whose degree has not changed, admits an effective representation no more complicated than before. For the transported cycle itself the answer is negative as soon as a component of the base cycle has finite stabiliser (Theorem 16.15 below), and for a cycle chosen at each point the bound is equivalent to the conclusion (Theorem 16.17). The present remark shows only that the question reduces from the whole Hilbert datum to the degree of the effective part, and that the part of the datum determined by cohomology is constant.

16.3. **Transport along isogenies.** The mechanism of Remark 16.12 can be made explicit, and doing so settles one of the two readings of (P1): the reading in which the cycle at s_0 is transported along the orbit rather than chosen afresh at each point. Write Λ for the reference lattice, of rank $4n$ over $\mathbb{Z}$, and E for the alternating form of the polarisation, so that a point s of $\mathcal{D}_{n,n}$ is a complex structure on $\Lambda\otimes\mathbb{R}$ and $A_s=(\Lambda\otimes\mathbb{R})/\Lambda$. Throughout, $\dim A_s=2n$, and a component of a cycle representing ω_s has dimension n.

Lemma 16.13 (The transport in coordinates)**.** *Let $g\in G(\mathbb{Q})$ and let $c=c(g)$ be the least positive integer with $cg\Lambda\subseteq\Lambda$. Then $\varphi_g:=cg$ induces an isogeny $A_{s_0}\to A_{g\cdot s_0}$ with*

(i) $\deg\varphi_g=c^{4n}$;

(ii) $\varphi_g^*\eta_{g\cdot s_0}=c^2\eta_{s_0}$ *and* $\varphi_g^*\omega_{g\cdot s_0}=c^{2n}\omega_{s_0}$;

(iii) *the transported cycle representing $\omega_{g\cdot s_0}$ is $c^{-2n}\,\varphi_{g*}(Z)$, where Z represents ω_{s_0}.*

Moreover $\sup_{g\in G(\mathbb{Q})}c(g)=\infty$.

Proof. (i) The index of $cg\Lambda$ in Λ is $|\det_{\mathbb{Q}}(cg)|=c^{4n}|\det_{\mathbb{Q}}g|$, and $\det_{\mathbb{Q}}g=\mathrm{N}_{K/\mathbb{Q}}(\det_K g)=1$ for $g\in\mathrm{SU}(V,H)$.

(ii) g preserves E, so $(cg)^*E=c^2E$, which is the first identity. On $H^{2n}=\bigwedge^{2n}(\Lambda^\vee\otimes\mathbb{Q})$ the map φ_g^* is $c^{2n}\bigwedge^{2n}g^\vee$, and $g\in\mathrm{SU}(V,H)$ fixes the Weil line pointwise by Lemma 16.2.

(iii) $\varphi_{g*}\varphi_g^*=\deg\varphi_g$, so $\omega_{g\cdot s_0}=(\deg\varphi_g)^{-1}\varphi_{g*}\varphi_g^*\omega_{g\cdot s_0}=c^{-4n}\,c^{2n}\varphi_{g*}\omega_{s_0}=c^{-2n}\varphi_{g*}\omega_{s_0}$ by (i) and (ii).

For the last sentence, suppose that $c(g)\le C$ for every $g\in G(\mathbb{Q})$. Then $(C!)\,g\Lambda\subseteq\Lambda$ for every g, so $G(\mathbb{Q})$ lies in the discrete subset $(C!)^{-1}\,\mathrm{End}_{\mathbb{Z}}(\Lambda)$ of $\mathrm{End}(\Lambda\otimes\mathbb{R})$. But by the proof of Proposition 16.7 the closure of $G(\mathbb{Q})$ in $G(\mathbb{R})$ contains $G(\mathbb{R})^+$, a Lie group of positive dimension, which is a contradiction. ∎

Lemma 16.14 (Push-forward along an isogeny)**.** *Let $\varphi\colon A\to A'$ be an isogeny of abelian varieties and $Z\subseteq A$ an integral closed subvariety. Put $G(Z)=\{x\in A:Z+x=Z\}$, a closed algebraic subgroup. Then*

$$\varphi_*[Z]=m\,[\varphi(Z)],\qquad m=\bigl|\ker\varphi\cap G(Z)\bigr|.$$

If moreover $\dim Z=n$ *and* $\varphi^*\eta'=c^2\eta$, *then*

$$\deg_{\eta'}\varphi(Z)=\frac{c^{2n}}{m}\,\deg_{\eta}Z.$$

Proof. The map $\varphi|_Z\colon Z\to\varphi(Z)$ is finite and surjective, and its degree m is the number of points of a general fibre. For $z\in Z$ the fibre through z is $Z\cap(z+\ker\varphi)$, which as a set of translations is $\{t\in\ker\varphi:z+t\in Z\}$. For a fixed t the locus $\{z\in Z:z+t\in Z\}=Z\cap(Z-t)$ is all of Z when $t\in G(Z)$ and a proper closed subset otherwise; since $\ker\varphi$ is finite, a general z avoids the finitely many proper subsets, and there $\{t\in\ker\varphi:z+t\in Z\}=\ker\varphi\cap G(Z)$. This is the first identity, and $\varphi_*[Z]=m[\varphi(Z)]$ by the definition of the push-forward of cycles.

For the degree, the projection formula gives

$$m \deg_{\eta'} \varphi(Z) = \int \varphi_*[Z] \cdot \frac{\eta'^n}{n!} = \int [Z] \cdot \frac{(\varphi^*\eta')^n}{n!} = c^{2n} \int [Z] \cdot \frac{\eta^n}{n!} = c^{2n} \deg_\eta Z.$$

■

For example, for multiplication by c on an abelian variety of dimension $2n$, an abelian subvariety Z of dimension n has $G(Z) = Z$, so $m = |Z[c]| = c^{2n}$ and $\deg \varphi(Z) = \deg Z$, while a Z whose stabiliser meets $A[c]$ trivially has $m = 1$ and $\deg \varphi(Z) = c^{2n} \deg Z$.

Theorem 16.15 (The transported cycle)**.** *Let $s_0 \in \Sigma$ and let $Z = \sum_i q_i Z_i$, with $q_i \neq 0$ and the Z_i integral of dimension n, represent ω_{s_0}.*

(i) *For $g \in G(\mathbb{Q})$ the transported cycle is $\sum_i c(g)^{-2n} m_i(g)\, q_i\, [\varphi_g(Z_i)]$ with $m_i(g) = |\ker \varphi_g \cap G(Z_i)|$, and the i-th component has degree $c(g)^{2n} \deg_\eta(Z_i)/m_i(g)$.*

(ii) *If some Z_i has finite stabiliser $G(Z_i)$, then $\deg \varphi_g(Z_i) \geq c(g)^{2n} \deg_\eta(Z_i)/|G(Z_i)|$, which is unbounded on $G(\mathbb{Q})$. The Hilbert data of the transported cycle then takes infinitely many values, and the hypothesis of Theorem 16.9 in the form that fixes the transported cycle is not satisfied by Z.*

(iii) *Consequently, a base cycle for which the transported data is bounded must have every component of positive-dimensional stabiliser, that is, every Z_i must be stable under translation by a positive-dimensional abelian subvariety of A_{s_0}.*

Proof. (i) is Lemma 16.13(iii) and Lemma 16.14.

(ii) $\ker \varphi_g \cap G(Z_i) \subseteq G(Z_i)$, so $m_i(g) \leq |G(Z_i)|$, a constant; the displayed bound is (i), and $c(g)$ is unbounded by Lemma 16.13. Distinct degrees of the components give distinct Hilbert polynomials, so the data takes infinitely many values.

(iii) is the contrapositive of (ii), together with the fact that an algebraic subgroup of positive dimension in an abelian variety has an abelian subvariety of positive dimension as its identity component. ■

Remark 16.16 (The two readings of (P1)). Three points should be kept apart.

First, (P1) has two readings, and Theorem 16.15 settles one of them. Read with the cycle fixed at s_0 and transported, it fails for every base cycle written down in this paper: those cycles are intersections of divisors, and an intersection of n divisors on an abelian $2n$-fold has finite stabiliser unless the divisors are pulled back from a quotient, which the ones here are not. Read with the cycle chosen afresh at each point of the orbit, which is the form of the hypothesis of Theorem 16.9, it is not affected by Theorem 16.15.

Second, the difference between the two readings is the content of Lemma 16.14. The class $\omega_{g \cdot s_0}$ has constant degree along the orbit, by Proposition 16.11; what grows is the complexity of the particular representative that the transport produces, because the isogeny spreads each component out by the factor c^{2n}/m_i. When a component of the base cycle has finite stabiliser, the transport does not produce a representative of bounded complexity, whether or not one exists.

Third, an approach to (P1) must either produce a base cycle supported on subvarieties with positive-dimensional stabiliser, a strong requirement for a Weil class, or abandon the transport and bound instead the minimal complexity of a representative, which is a statement about the Chow variety of the family and not about any one cycle. Neither is achieved here.

The bound on the minimal complexity of a representative is the hypothesis of Theorem 16.9, and it is the weakest hypothesis under which the covering argument runs.

The next theorem shows that it is equivalent to the conclusion. Write $\mathbf{W}(K, n, \delta)$ for the statement that every abelian $2n$-fold of $(K, -1, n)$-Weil type with discriminant δ (Definition 15.22) has algebraic Weil classes, δ being the discriminant of the family of Setup 15.12.

Theorem 16.17 (Equivalence of the bounded criterion)**.** *Let $n \geq 2$, let $s_0 \in \Sigma$ and keep Setup 15.12. The following are equivalent.*

(a) *There is a finite set $\mathcal{T}$ of pairs $(\underline{P}, \underline{q})$ such that at every point of the orbit $G(\mathbb{Q}) \cdot s_0$ the class ω is represented by a cycle whose Hilbert data lies in $\mathcal{T}$; that is,* (P1) *holds at s_0.*

(b) *There is a single pair $(\underline{P}, \underline{q})$ such that at every point of $\mathcal{D}_{n,n}$, not merely of the orbit, the class ω is represented by a cycle with that datum.*

(c) $\Sigma = \mathcal{D}_{n,n}$.

(d) $\mathbf{W}(K, n, \delta)$ *holds, that is, the Weil classes are algebraic on every member of the family.*

In particular Theorem 16.9 is an equivalence, and (P1) *is a reformulation of the conjecture for the family and not a reduction of it. The same holds for every CM field, with Theorem 17.7(iv) in place of Theorem 16.9.*

Proof. (a) implies (c) is Theorem 16.9. (b) implies (a) is immediate, the orbit being contained in $\mathcal{D}_{n,n}$ and $\mathcal{T}$ being the one pair. (d) implies (c) trivially, and (c) implies (d) because the algebraicity of ω_s makes the whole Weil line of A_s algebraic, by the argument of Corollary 4.15, written out in Lemma 16.3.

It remains to deduce (b) from (c). If $\Sigma = \mathcal{D}_{n,n}$ then Σ has nonempty interior, and the proof of Corollary 16.5 shows that some $\Sigma_{\underline{P},\underline{q}}$ is then all of $\mathcal{S}$: the countably many sets $\Sigma_{\underline{P},\underline{q}} \cap U$ are closed in a nonempty connected open $U \subseteq \mathcal{S}$ and cover it, so by Baire one of them has nonempty interior in U, and a proper Zariski closed subvariety of the irreducible variety $\mathcal{S}$ has empty interior. By the definition of $\Sigma_{\underline{P},\underline{q}}$ in Theorem 16.4 as the image of a proper morphism, every $s \in \mathcal{S}$ then carries a tuple $\underline{Z}$ of subschemes with Hilbert polynomials $\underline{P}$ and $\sum_i q_i[Z_i] = \omega$, which is (b).

For a CM field the same argument applies without change, using Theorem 17.7(iii) in place of Theorem 16.4 and Corollary 16.5. ∎

Remark 16.18 (Status of the first criterion). By Theorem 16.17, (P1) is a restatement of the conjecture for the family and not a route to it. Theorem 16.9 is a valid implication, and its covering argument is the only place in the paper where density of the rational orbit is converted into a global statement; but the implication has a converse, so verifying its hypothesis for a family is as hard as proving the conclusion for that family. The boundedness is nevertheless a concrete finiteness statement about the Chow variety of the universal family, and a reformulation of $\mathbf{W}(K, n, \delta)$ that mentions neither Hodge theory nor deformation theory.

The two propagation criteria are therefore not on the same footing. The rank criterion (P2) of [II, Theorem 2.3] is a sufficient condition with no evident converse: injectivity of one semiregularity map at one point implies the conjecture for the whole family, while $\Sigma = \mathcal{D}_{n,n}$ carries no information about the deformation theory of any particular perfect complex. In its pure form, with Chern character a Weil class, it is never satisfied ([II, Theorem 3.5]); the polarised criterion (P2′) of [II, Remark 3.6], in which the Chern character may also carry powers of the polarisation, is the form studied in [II, Section 3].

16.4. **Products of divisors.** The intersection numbers needed below are Pfaffians. Let E be a nondegenerate alternating form on a real vector space V of dimension $2g$, let η be its class, with the orientation for which $e=\int\eta^g>0$, and for an E-self-adjoint endomorphism ϕ write $x_\phi(u,v)=E(\phi u,v)$, an alternating form.

Lemma 16.19 (Pfaffians of self-adjoint classes)**.** *For real s, $\int(s\eta+x_\phi)^g=e\,p_\phi(s)$, where p_ϕ is the monic polynomial of degree g with $p_\phi^2=\det(s+\phi)$. In particular:*

(i) *$2g\int x_\phi\eta^{g-1}=e\,\mathrm{tr}\,\phi$, and if $\mathrm{tr}\,\phi=0$ then $2g(g-1)\int x_\phi^2\eta^{g-2}=-e\,\mathrm{tr}(\phi^2)$;*

(ii) *if $\mathrm{tr}\,\phi=0$ and $\phi^2=\beta$ is a nonzero real scalar, then g is even and $\int x_\phi^g=e\,(-\beta)^{g/2}$.*

Proof. In a basis with Gram matrix G for E, the class $s\eta+x_\phi$ has Gram matrix $(s+\phi)^TG$, of determinant $\det G\det(s+\phi)$. Its Pfaffian divided by $\mathrm{Pf}(G)$ is therefore a polynomial in s, monic of degree g, whose square is $\det(s+\phi)$; and $\int\omega^g/e$ is $\mathrm{Pf}(\omega)/\mathrm{Pf}(G)$ for every alternating form ω. This is the identity. Write $p_\phi(s)=s^g+p_1s^{g-1}+p_2s^{g-2}+\cdots$ and compare with $\det(s+\phi)=s^{2g}+\mathrm{tr}(\phi)\,s^{2g-1}+\frac12((\mathrm{tr}\,\phi)^2-\mathrm{tr}(\phi^2))s^{2g-2}+\cdots$: then $2p_1=\mathrm{tr}\,\phi$ and $2p_2+p_1^2=\frac12((\mathrm{tr}\,\phi)^2-\mathrm{tr}(\phi^2))$, while the binomial expansion of $\int(s\eta+x_\phi)^g$ gives $g\int x_\phi\eta^{g-1}=e\,p_1$ and $\binom g2\int x_\phi^2\eta^{g-2}=e\,p_2$; this is (i). Under the hypothesis of (ii) the eigenvalues of ϕ are $\pm\sqrt\beta$, each with multiplicity g because the trace vanishes, so $\det(s+\phi)=(s^2-\beta)^g$; this is the square of a polynomial only for g even, and then $p_\phi(s)=(s^2-\beta)^{g/2}$ and $\int x_\phi^g=e\,p_\phi(0)$. ∎

At the quaternionic points of Setup 15.24 the mechanism of Remark 16.12 can be seen explicitly. Take $n=2$ and the integral lattice $\Lambda_b=\mathcal O_b^2$, $\mathcal O_b=\mathbb Z\langle1,i,j,ij\rangle$, for $b\in\mathbb Z_{>0}$. Put

$$g_b(x,y)=E_b(jx,y),\qquad \lambda_b(x,y)=-\tfrac1d\,g_b(ix,y),\qquad Z_b=g_b^2-d\lambda_b^2,\qquad Z_b'=2g_b\lambda_b,$$

so that $Z_b+\sqrt{-d}\,Z_b'=(g_b+\sqrt{-d}\,\lambda_b)^2$ and Z_b,Z_b' span the Weil plane. Write $x\in\mathcal O_b$ as $x=u+vj$ with $u,v\in\mathbb Z[i]$, where $i^2=-d$ as in Setup 15.24, so that $\mathbb Z[i]=\mathbb Z[\sqrt{-d}]$; the pair (Λ_b,i) is then the same $\mathbb Z[i]$-module $\mathbb Z[i]^4$ for every b, and we use this identification throughout.

Proposition 16.20 (Divisibility of the quaternionic Weil cycle)**.** *With these data, for every $b\in\mathbb Z_{>0}$:*

(i) *$E_b=E_u+b\,E_v$, $g_b=b\,g_1$ and $\lambda_b=b\,\lambda_1$, where E_u,E_v,g_1,λ_1 are integral on $\mathbb Z[i]^4$ and do not depend on b; in particular $g_1,\lambda_1\in\mathrm{NS}(A_b)$;*

(ii) *$Z_b=b^2Z_1$ and $Z_b'=b^2Z_1'$ in $\mathrm{CH}^2(A_b)$ modulo algebraically trivial cycles, for every choice of line bundles representing g_b,λ_b; in cohomology the equalities hold in $\wedge^4\Lambda^*$;*

(iii) *$\int\eta_b^4=b^2\int\eta_1^4$, $\int g_b^2\eta_b^2=-\frac b3\int\eta_b^4$ and $\int Z_b^2=\frac{4b^2}3\int\eta_b^4$, so that $(\int g_b^2\eta_b^2)^2=\frac{1}{12}\int Z_b^2\int\eta_b^4$;*

(iv) *for $\alpha\in K^\times$ the element αj generates the same algebra at the same point, and replacing j by αj multiplies the two ratios in* (iii) *by $N(\alpha)$ and $N(\alpha)^2$.*

Proof. For $x=u+vj$ one has $\overline x=\overline u-vj$, and the relations $ju=\overline uj$, $j^2=b$ give $\overline x\,i\,y\equiv\overline u\,i\,u'+b\,v\,i\,\overline{v'}$ modulo Kj. Since trd vanishes on Kj and restricts to $\mathrm{tr}_{K/\mathbb Q}$ on K,

$$E_b(x,y)=\textstyle\sum_k a_k\big(\mathrm{tr}(i\,\overline{u_k}u_k')+b\,\mathrm{tr}(i\,v_k\overline{v_k'})\big).$$

From $jx=b\overline v+\overline uj$ one reads off $g_b(x,y)=b\sum_k a_k\big(\mathrm{tr}(i\,v_ku_k')+\mathrm{tr}(i\,\overline{u_k}\overline{v_k'})\big)$, which is $b\,g_1$, and $\lambda_b=b\lambda_1$ follows. The traces take integral values on $\mathbb Z[i]$, and $g_1=b^{-1}g_b$ and $\lambda_1=b^{-1}\lambda_b$ are rational multiples of Hodge classes, so they are integral Hodge classes and lie in $\mathrm{NS}(A_b)$ by the Lefschetz theorem on $(1,1)$-classes; this proves (i). Then $g_b+\sqrt{-d}\,\lambda_b=b(g_1+\sqrt{-d}\,\lambda_1)$

and (ii) holds in cohomology. Choose L, M with $c_1(L) = g_1$ and $c_1(M) = \lambda_1$. The cycle $c_1(L^b)^2 - d\,c_1(M^b)^2$ equals $b^2(c_1(L)^2 - d\,c_1(M)^2)$ in $\mathrm{CH}^2(A_b)$, and another choice changes both sides by algebraically trivial cycles.

For (iii), E_u and E_v live on complementary halves U, V' of $\Lambda \otimes \mathbb{Q}$ of rank four, so $\eta_u^3 = \eta_v^3 = 0$ and $\int \eta_b^4 = 6b^2 \int \eta_u^2 \eta_v^2$. The form g_b is x_ϕ for $\phi = j$, which is E_b-self-adjoint by Theorem 15.25(i), anticommutes with i, so has trace 0, and has $j^2 = b$; hence Lemma 16.19(i), with $g = 4$, gives $\int g_b^2 \eta_b^2 = -\frac{1}{24}\operatorname{tr}(j^2) \int \eta_b^4 = -\frac{8b}{24} \int \eta_b^4$. For real a, c the class $a g_b + c\lambda_b$ is x_ϕ for $\phi = j(a - \frac{c}{d}\, i)$, again self-adjoint of trace 0, and $\phi^2 = (a + \frac{c}{d} i)\, j^2\, (a - \frac{c}{d} i) = b\,(a^2 + c^2/d)$. So Lemma 16.19(ii) gives $\int (a g_b + c\lambda_b)^4 = b^2(a^2 + c^2/d)^2 \int \eta_b^4$, and comparing coefficients, $\int g_b^4 = b^2 \int \eta_b^4$, $\int g_b^2 \lambda_b^2 = \frac{b^2}{3d} \int \eta_b^4$ and $\int \lambda_b^4 = \frac{b^2}{d^2} \int \eta_b^4$. Hence $\int Z_b^2 = \int g_b^4 - 2d \int g_b^2 \lambda_b^2 + d^2 \int \lambda_b^4 = \frac{4b^2}{3} \int \eta_b^4$.

For (iv), $(\alpha j)^2 = N(\alpha) b$, and $g_{\alpha j} + \sqrt{-d}\, \lambda_{\alpha j}$ is α or $\overline{\alpha}$ times $g_j + \sqrt{-d}\, \lambda_j$. ∎

Remark 16.21 (Growth of the polarisation). At these points no multiple growing with b enters when the Weil class is written in integral terms: by Proposition 16.20(ii) the cycle Z_b is $b^2 Z_1$, with Z_1 one integral algebraic cycle built from the two fixed divisor classes g_1, λ_1. The factor that the orbit argument would otherwise have to absorb into the coefficients $\underline{q}$ of Theorem 16.9 is therefore absent. What grows is the polarisation: $\int \eta_b^4 = b^2 \int \eta_1^4$, so the points A_b carry polarisations of unbounded degree, and an isogeny carrying η_b to the polarisation of one fixed family has degree $\int \eta_b^4 / \int \eta^4$, which grows like b^2. This is the unbounded isogeny degree of Remark 16.12, located at an explicit family of points, and the integrality of Z_1 does not affect it.

The base points of Theorem 15.27, and every algebraicity statement for Weil classes proved in this paper, represent ω through products of divisor classes. The next result shows that a representation of that kind can hold with bounded degrees only on a nowhere dense set, so the bound asked for in Theorem 16.9 has to come from cycles that are not products of divisors. For a class $x \in H^2(A_s, \mathbb{R})$ put

$$t(x) = \frac{\int x\, \eta^{2n-1}}{\int \eta^{2n}}, \qquad I(x) = t(x)^2 \int \eta^{2n} - \int \bigl(x - t(x)\eta\bigr)^2 \eta^{2n-2}.$$

Since $\mathcal{D}_{n,n}$ is contractible, the local system $R^2\pi_*\mathbb{Z}$ is trivial; write $H^2_{\mathbb{Z}}$ for its fibre, identified with $H^2(A_s, \mathbb{Z})$ at every s. Both t and I are polynomial functions on $H^2_{\mathbb{Z}}$ and do not depend on s.

Lemma 16.22 (The Hodge norm of a divisor)**.** *Let h_s be the Hodge norm on $H^2(A_s, \mathbb{R})$ defined by the polarisation, normalised by $h_s(\eta) = \int \eta^{2n}$. Then*

(i) *$h_s(x) = I(x)$ for every real class x of type $(1,1)$ at s;*

(ii) *if D is an effective divisor on A_s then $I([D]) \le 2\bigl(\int [D]\eta^{2n-1}\bigr)^2 / \int \eta^{2n}$.*

Proof. Write $x = t\eta + x_0$ with $t = t(x)$, so that x_0 is primitive. The Lefschetz decomposition is orthogonal for h_s, and on primitive classes $h_s(x_0) = -\int x_0 \wedge C_s x_0 \wedge \eta^{2n-2}$, where C_s is the Weil operator, positive by the Hodge–Riemann relations. For x_0 of type $(1,1)$, $C_s x_0 = x_0$, which gives (i). An effective divisor on an abelian variety is nef, since a translate of it meets any given curve properly, so $\int [D]^2 \eta^{2n-2} \ge 0$. Expanding, $\int [D]^2 \eta^{2n-2} = t^2 \int \eta^{2n} + \int x_0^2 \eta^{2n-2}$, hence $-\int x_0^2 \eta^{2n-2} \le t^2 \int \eta^{2n}$ and $I([D]) \le 2t^2 \int \eta^{2n}$, which is (ii). ∎

Proposition 16.23 (Divisor products of bounded degree)**.** *In Setup 15.12, fix $C > 0$ and let $\Sigma_C \subset \mathcal{D}_{n,n}$ be the set of points s at which ω_s is a $\mathbb{Q}$-linear combination of classes $[D_1] \cdots [D_n]$ with every D_i an effective divisor on A_s of degree $\int [D_i] \eta^{2n-1} \le C$. Then Σ_C*

is contained in a locally finite union of proper closed analytic subsets of $\mathcal{D}_{n,n}$. In particular Σ_C is nowhere dense, and on any dense subset of $\mathcal{D}_{n,n}$ at which ω is a combination of products of effective divisors the degrees of those divisors are unbounded on every nonempty open set.

Proof. For $x \in H^2_{\mathbb{Z}}$ let $\mathrm{Hdg}(x) \subset \mathcal{D}_{n,n}$ be the closed analytic set where x is of type $(1,1)$. If $x \notin \mathbb{Q}\eta$ it is proper, because $\mathrm{NS}(A_s) \otimes \mathbb{Q} = \mathbb{Q}\eta$ at a very general point by Lemma 7.5. Let $s \in \Sigma_C$. Not every $[D_i]$ in the representation lies in $\mathbb{Q}\eta$, since $\mathbb{Q}\eta^n$ and the Weil line carry different characters by Lemma 5.4. So $s \in \mathrm{Hdg}(x)$ for some x in

$$X_C = \{x \in H^2_{\mathbb{Z}} \setminus \mathbb{Q}\eta \ : \ I(x) \le 2C^2 / \textstyle\int \eta^{2n}\},$$

by Lemma 16.22(ii). Now let $U \subset \mathcal{D}_{n,n}$ be open with compact closure and $s_0 \in U$. The Hodge norm depends continuously on the point, so $h_{s_0} \le c_U\, h_s$ for some c_U and all $s \in U$. If $\mathrm{Hdg}(x)$ meets U at s and $x \in X_C$, then by Lemma 16.22(i) $h_{s_0}(x) \le c_U h_s(x) = c_U I(x) \le 2c_U C^2 / \int \eta^{2n}$, and only finitely many integral classes satisfy this. So the sets $\mathrm{Hdg}(x)$, $x \in X_C$, form a locally finite family of proper closed analytic subsets. Their union is closed with empty interior in the connected manifold $\mathcal{D}_{n,n}$, and it contains Σ_C. ∎

Remark 16.24 (Divisorial representatives). The loci on which this paper proves ω algebraic are the split, quaternionic and product loci and their translates under $G(\mathbb{Q})$, and on each of them ω is a rational polynomial in divisor classes, since an isogeny carries $\mathrm{NS} \otimes \mathbb{Q}$ isomorphically. Proposition 16.23 shows that no bounded family of such cycles covers a dense set, so the finiteness in Theorem 16.9 cannot be obtained by bounding the divisors that produce ω at the points where it is known to be algebraic. It would have to come from representatives of another kind, such as the transported cycles of Lemma 16.6, which are push-forwards along isogenies rather than products of divisors on the target; and by Theorem 16.15 their data is unbounded as soon as a component of the base cycle has finite stabiliser. The unbounded isogeny degree of Remark 16.12 and Remark 16.21 is thus unavoidable for divisorial constructions. The argument uses nothing about ω beyond the fact that it is not a multiple of η^n, and nothing about the family beyond the generic Picard number, so it applies equally to any variational argument that rests on products of divisors.

At $n = 2$ the phenomenon can be exhibited on one fixed polarised lattice. Take $\Lambda = \mathcal{O}^2$ with $\mathcal{O} = \mathbb{Z}\langle 1, i, j, ij\rangle$, $i^2 = -d$, $j^2 = 1$, and E as in (38) with $a_1 = a_2 = 1$. The lattice and the polarisation are fixed, and all the loci below lie in the one family $\mathcal{D}_{2,2}$ that they define. Let $R = \{x : \ x(iu, v) = x(u, iv)\} \subset H^2(\Lambda, \mathbb{Q})$ be the space of K-bilinear classes, where the divisor classes of the quaternionic points live, and for $x \in R$ let $\phi_x = E^{-1}x$, so that $x(u, v) = E(\phi_x u, v)$. Put $e = \int \eta^4$ and, for $k \in \mathbb{Z}$,

$$C_k = \begin{pmatrix} 1 & k \\ k & -1 \end{pmatrix}, \qquad \phi_k(u_1, u_2) = \tfrac{1}{2d}\big(j(u_1 + ku_2),\, j(ku_1 - u_2)\big), \qquad x_k(u, v) = E(\phi_k u, v),$$

so that ϕ_k is left multiplication by the matrix $(2d)^{-1} C_k\, j$ and $x_k = x_0 + k(x_1 - x_0)$ is a pencil.

Proposition 16.25 (A pencil of quaternionic loci)**.** *In this setting, for every d and every $k \in \mathbb{Z}$:*

(i) *$e = 384\, d^4$; R has dimension 12, every $x \in R$ has $t(x) = 0$ and $I(x) = \frac{1}{24}\operatorname{tr}(\phi_x^2)\, e$, and the form I has signature $(8, 4)$ on R;*

(ii) *x_k is a primitive integral class in R, and $\phi_{x_k}^2 = \beta(k)$ with $\beta(k) = (1 + k^2)/(4d^2)$;*

(iii) *the set* $\mathrm{Hdg}(x_k)$ *is a quaternionic locus with algebra* $(-d,\beta(k)) \cong (-d,1+k^2)$*, nonempty of dimension* 3*, and loci with different values of* β *are not related by any automorphism of* (Λ,E,i)*;*

(iv) *at a very general point of* $\mathrm{Hdg}(x_k)$ *one has* $\mathrm{NS} = \mathbb{Z}\,\frac{1}{2d}\eta \oplus \mathbb{Z}x_k \oplus \mathbb{Z}\,\iota x_k$*, where* $\iota x = x(i\cdot,\cdot)$*, and the minimum of* I *on* $\mathrm{NS}\setminus\mathbb{Q}\eta$ *is* $\mu_k = 32\,d^2(1+k^2)$*.*

Consequently, at a very general point of $\mathrm{Hdg}(x_k)$ *every effective divisor* D *whose class is not a multiple of* η *has* $\int[D]\eta^3 \geq 32\sqrt{6}\,d^3\sqrt{1+k^2}$*, which grows linearly in* $|k|$*.*

Proof. Two facts about $\mathcal{O}$ are used throughout. The reduced trace form $\mathrm{trd}(wz)$ is diagonal on $1,i,j,ij$ with entries $2,-2d,2,2d$, so the dual lattice $\mathcal{O}^\sharp = \{w\in B : \mathrm{trd}(w\mathcal{O})\subseteq\mathbb{Z}\}$ is $\mathbb{Z}\frac{1}{2}\oplus\mathbb{Z}\frac{i}{2d}\oplus\mathbb{Z}\frac{j}{2}\oplus\mathbb{Z}\frac{ij}{2d}$. And a form $y(u,v)=\sum_{k,l}\mathrm{trd}(\overline{u_l}\,w_{lk}\,v_k)$ on $\mathcal{O}^2$ with $w_{lk}\in B$ is integral if and only if every w_{lk} lies in $\mathcal{O}^\sharp$, because $\mathrm{trd}(\overline{u}wv)=\mathrm{trd}(wv\overline{u})$ and the products $v\overline{u}$ with $u,v\in\mathcal{O}$ span $\mathcal{O}$.

(i) The Gram matrix of $\mathrm{trd}(\overline{u}iv)$ on $1,i,j,ij$ pairs 1 with i and j with ij, with entries $\mp 2d$ and $\pm 2d$ and zeros elsewhere, so E has elementary divisors $2d$ four times on Λ and $e = 4!\cdot(2d)^4 = 384\,d^4$. The involution $\sigma(x) = d^{-1}x(i\cdot,i\cdot)$ of $\bigwedge^2 V^*$ has R as its (-1)-eigenspace, since $x(iu,v)=x(u,iv)$ is equivalent to $x(iu,iv)=-d\,x(u,v)$. Over $\mathbb{C}$, $V_\mathbb{C} = W\oplus\overline{W}$ with $W,\overline{W}$ the eigenspaces of i for $\pm\sqrt{-d}$, each of dimension 4, and σ is -1 on $\bigwedge^2 W^*\oplus\bigwedge^2\overline{W}^*$ and $+1$ on $W^*\otimes\overline{W}^*$; so $\dim R = 6+6=12$. For $x\in R$ the map ϕ_x is E-self-adjoint, as x and E are alternating, and anticommutes with i, as i is E-skew-adjoint by Theorem 15.25(i): $E(\phi_x iu,v)=x(u,iv)=E(\phi_x u,iv)=-E(i\phi_x u,v)$. Hence $\mathrm{tr}\,\phi_x = -\mathrm{tr}(i\phi_x i^{-1}) = -\mathrm{tr}\,\phi_x = 0$, and Lemma 16.19(i), with $g=4$, gives $t(x)=0$ and $I(x) = -\int x^2\eta^2 = \frac{1}{24}\mathrm{tr}(\phi_x^2)e$.

For the signature regard $V_\mathbb{R}$ as a complex vector space through $i/\sqrt{d}$, with the hermitian form H of Lemma 4.5, which has signature $(2,2)$ by Theorem 15.25(ii), and an H-orthogonal basis in which $H=\mathrm{diag}(h_1,\dots,h_4)$. The maps ϕ_x, $x\in R\otimes\mathbb{R}$, are the E-self-adjoint antilinear maps. Writing $\phi(v)=A\overline{v}$, self-adjointness for $E = d^{-1/2}\,\mathrm{Im}\,H$ says exactly that $S=h\overline{A}$ is antisymmetric, and then $\mathrm{tr}_\mathbb{R}(\phi^2) = 2\,\mathrm{Re}\,\mathrm{tr}(A\overline{A}) = -4\sum_{k<l}|S_{kl}|^2/(h_k h_l)$. Four of the six pairs $k<l$ have $h_kh_l<0$ and two have $h_kh_l>0$, and each S_{kl} is a complex coordinate, so I has signature $(8,4)$.

(ii) ϕ_k anticommutes with i because $ji=-ij$ and the entries of C_k are rational. It is E-self-adjoint because C_k is symmetric and left multiplication by j is self-adjoint for $\mathrm{trd}(\overline{u}iv)$ on each factor, by Theorem 15.25(i); so $x_k\in R$. Since the entries of C_k commute with j, $\phi_k^2 = (2d)^{-2}C_k^2 j^2 = (1+k^2)/(4d^2)$. In the form of the first paragraph x_k has $w_{lk} = -(2d)^{-1}c_{kl}\,j\,i = (2d)^{-1}c_{kl}\,ij$, with c_{kl} the entries of C_k, which lie in $\mathcal{O}^\sharp$; so x_k is integral, and $x_k((1,0),(ij,0)) = (2d)^{-1}\mathrm{trd}(ij\cdot ij) = 1$, so it is primitive.

(iii) Write $\phi=\phi_k$ and $\beta=\beta(k)$. The map ϕ is E-self-adjoint, anticommutes with i, and $\phi^2=\beta>0$, so $V_\mathbb{R} = V_+\oplus V_-$ with $V_\pm$ the $\pm\sqrt{\beta}$ eigenspaces, E-orthogonal to each other, and i exchanges them. A complex structure J on V_+ compatible with $E|_{V_+}$, extended to $V_- = iV_+$ by $J(iw)=iJw$, commutes with i and ϕ and is compatible with E, and i has eigenvalues $\pm\sqrt{-d}$ each with multiplicity 2 on $V^{1,0}$; so these J are points of $\mathcal{D}_{2,2}$ at which x_k is of type $(1,1)$, and they form a copy of the Siegel space of V_+, of dimension 3, in agreement with Proposition 15.32. The algebra generated by i and ϕ is $(-d,\beta)$, and $(-d,\beta)\cong(-d,1+k^2)$ because the two differ by the square $(2d)^2$. An automorphism γ of (Λ,E,i) sends ϕ_x to $\gamma\phi_x\gamma^{-1}$ and so preserves $\mathrm{tr}(\phi_x^2)=8\beta$.

For the first half of (iv), let $S_k\subset\mathcal{D}_{2,2}$ be the Siegel space just described and $\mathcal{C}_k\subset\mathrm{Sp}(V_\mathbb{R},E)$ the centraliser of i and ϕ. An element of $\mathcal{C}_k$ preserves V_+ and is determined by its restriction there, because it commutes with i, so $\mathcal{C}_k\cong\mathrm{Sp}(V_+)$, acting on $V_\mathbb{R}=V_+\oplus iV_+$

as $g \oplus igi^{-1}$. A real class is of type $(1,1)$ at J exactly when it is invariant under J. The points of S_k form one orbit $\{gJ_0g^{-1} : g \in \mathcal{C}_k\}$ of a fixed point J_0. Let z be a real 2-form invariant under every point of S_k, and write $\mathfrak{c} = \mathrm{Lie}(\mathcal{C}_k) \cong \mathfrak{sp}(V_+)$, $\mathfrak{k}$ for the centraliser of J_0 in $\mathfrak{c}$ and $\mathfrak{p} = [\mathfrak{c}, J_0]$. For $X \in \mathfrak{c}$ the curve $g_t = \exp(tX)$ gives $(g_tJ_0g_t^{-1})^*z = z$ for all t, and the derivative at $t = 0$ says that $[X, J_0]$ annihilates z, so $\mathfrak{p}$ annihilates z. The annihilator of z in $\mathfrak{c}$ is a Lie subalgebra, so it contains $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$. Now $\mathfrak{c} = \mathfrak{k} \oplus \mathfrak{p}$ with $[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$ and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, so $\mathfrak{p} + [\mathfrak{p}, \mathfrak{p}]$ is an ideal of $\mathfrak{c}$; it is nonzero, and it is therefore all of $\mathfrak{c}$, because $\mathfrak{c} \cong \mathfrak{sp}_4(\mathbb{R})$ is simple: its complexification $\mathfrak{sp}_4(\mathbb{C})$ is simple, with irreducible root system C_2, and an ideal of a real form complexifies to an ideal. Hence z is annihilated by $\mathfrak{c}$ and, since $\mathrm{Sp}(V_+)$ is connected, invariant under all of $\mathcal{C}_k$. So the classes of type $(1,1)$ at every point of S_k are the invariants of $\mathrm{Sp}(V_+)$ in $\bigwedge^2(V_+ \otimes \mathbb{R}^2) = (\bigwedge^2 V_+ \otimes S^2\mathbb{R}^2) \oplus (S^2V_+ \otimes \bigwedge^2 \mathbb{R}^2)$, a space of dimension 3 spanned by η, x_k and ιx_k. Every other rational class z is of type $(1,1)$ only on a proper closed analytic subset of the connected manifold S_k, and a very general point avoids these countably many subsets.

For the lattice, ιx_k is the form of $\phi_k i = -i\phi_k$, so every class in $\mathbb{Q}\eta + \mathbb{Q}x_k + \mathbb{Q}\iota x_k$ is $y = s\eta + E(\phi_k\lambda\cdot, \cdot)$ with $s \in \mathbb{Q}$ and $\lambda \in K$ acting by left multiplication. In the form of the first paragraph, y has $w_{lk} = \delta_{lk}\, s\, i + (2d)^{-1}c_{kl}\overline{\lambda}\, ij$. The first summand lies in K and the second in Kj, and $\mathcal{O}^\sharp$ is the direct sum of its parts in K and in Kj, so y is integral if and only if $s \in (2d)^{-1}\mathbb{Z}$ and $(2d)^{-1}c_{kl}\overline{\lambda}\, ij \in \mathcal{O}^\sharp$ for all k, l. Writing $\overline{\lambda} = p - q\, i$, one has $\overline{\lambda}\, ij = p\, ij + qd\, j$, and since $c_{11} = 1$ and all c_{kl} are integers this holds exactly when $p, q \in \mathbb{Z}$, that is $\lambda \in \mathbb{Z}[i]$. This is the stated description of NS. Now $t(\eta) = 1$ and t vanishes on R, so $I(s\eta + z) = s^2e + I(z)$ for $z \in R$, and $(\phi_k\lambda)^2 = \phi_k^2\overline{\lambda}\lambda = N(\lambda)\beta(k)$, so by (i)

$$I\big(s\eta + E(\phi_k\lambda\cdot, \cdot)\big) = s^2e + \tfrac{8}{24}N(\lambda)\beta(k)e = s^2e + 32\, d^2(1 + k^2)N(\lambda).$$

A class off $\mathbb{Q}\eta$ has $\lambda \neq 0$, hence $N(\lambda) \geq 1$, and $\lambda = 1$, $s = 0$ gives x_k; so $\mu_k = 32\, d^2(1 + k^2)$. The final assertion follows from Lemma 16.22(ii): an effective D with $[D] \notin \mathbb{Q}\eta$ has $(\int [D]\eta^3)^2 \geq e\, \mu_k/2 = 6144\, d^6(1 + k^2)$. ∎

Each locus of the pencil carries algebraic Weil classes, by the divisorial criterion, since x_k is nondegenerate, and on the k-th of them the divisors that produce those classes have degree at least of order $|k|$. This is Proposition 16.23 seen on explicit quaternionic loci in one fixed polarised lattice, and Figure 19 illustrates it.

Remark 16.26 (The case $n = 2$). Let $n = 2$, so $\dim \mathcal{S} = n^2 = 4$. For every n the quaternionic locus of Setup 15.24 has dimension $n(n + 1)/2$, by a Morita argument. The quaternion algebra $B = (-d, b)$ contains K and acts on V, and $B \otimes \mathbb{R} \cong M_2(\mathbb{R})$ because $b > 0$, so Morita equivalence writes $V_\mathbb{R}$ as $W \otimes_\mathbb{R} \mathbb{R}^2$ with W symplectic of dimension $2n$. The complex structures commuting with B and compatible with E are then the complex structures on W compatible with its symplectic form, that is the points of the Siegel space $\mathfrak{H}_n$, of dimension $n(n + 1)/2$, and the codimension in $\mathcal{D}_{n,n}$ is $n^2 - n(n + 1)/2 = n(n - 1)/2$. At $n = 2$ the quaternionic locus is therefore the symmetric space of $\mathrm{Sp}(4, \mathbb{R})$ inside $\mathcal{D}_{2,2}$, of dimension 3, so it is a divisor. Its $G(\mathbb{Q})$-translates are infinitely many pairwise distinct irreducible closed subvarieties of $\mathcal{S}$ of dimension 3, since a finite union of proper closed analytic subsets cannot be dense. If infinitely many of them carry cycles with one and the same data $(\underline{P}, \underline{q})$, then the closed set $\Sigma_{\underline{P},\underline{q}}$ has an irreducible component containing infinitely many distinct irreducible threefolds, so that component has dimension at least 4 and therefore equals $\mathcal{S}$. No density is needed at $n = 2$, only infinitude. Combined with the theorem of Moonen and Zarhin [MZ99] that every Hodge class on an abelian fourfold lies in the subalgebra generated by divisor classes and Weil classes, the conclusion would be

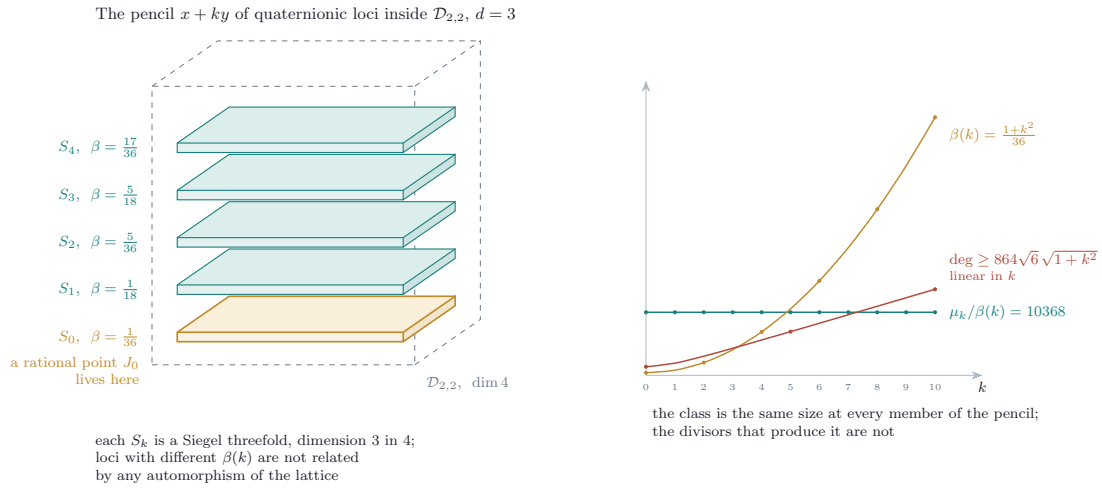


FIGURE 19. Proposition 16.25 at $d = 3$. Left: the members S_k of the pencil, each a Siegel threefold inside the four-dimensional family, with the member $k = 0$, where $\beta = \frac{1}{36}$ is a square and a rational point J_0 can be written down, drawn apart. Right: along the pencil the ratio $\mu_k/\beta(k)$ is the constant $10368 = \int \eta^4/3$, so μ_k grows like k^2 and the least degree of a divisor not proportional to η grows linearly in k. The Weil class is a flat class of the one fixed lattice and does not change; the divisors that produce it must.

the Hodge conjecture for all abelian fourfolds of Weil type, hence for all abelian fourfolds. That conclusion is a theorem of Markman [Mar25a, Corollary 1.6.1], proved by a different route, so at $n = 2$ the argument above would give a second proof; its interest lies in $n \geq 3$.

17. Weil classes of every CM field

The Weil classes of an imaginary quadratic field are the case $m = 1$ of a construction that makes sense for every CM field F of degree $2m$, and the methods of this paper apply to the question whether those classes are algebraic. Every theorem of Section 14 that turns the conjecture for a Weil family into a propagation statement is a theorem about a connected period domain, a flat class on it, a group whose rational points act with dense orbits, and a base point where the class is a polynomial in divisor classes, and none of these four ingredients depends on the field being quadratic. This section carries the four over to every CM field; the only one that needs a new argument is the base point, which Theorem 17.5 supplies. Nothing here is conditional on a question about semiregularity, because the base point is a CM point: the Weil classes there are written out as polynomials with rational coefficients in divisor classes, each divisor class is algebraic by the Lefschetz theorem on $(1,1)$ classes, and products of algebraic classes are algebraic. Nothing is claimed about the whole Hodge ring at a CM point, which can contain classes that are not polynomials in divisor classes.

17.1. **The families.**

Setup 17.1 (Weil type for a CM field)**.** Let F be a CM field of degree $2m$ with maximal totally real subfield F_0, and write $\alpha \mapsto \overline{\alpha}$ for the nontrivial automorphism of F/F_0, which

is complex conjugation under every embedding. The real places of F_0 are $\tau_1,\dots,\tau_m$, and each extends to a conjugate pair $\sigma_\tau,\overline{\sigma}_\tau$ of embeddings of F into $\mathbb{C}$. Let V be an F-vector space of dimension $2n$, let $H\colon V\times V\to F$ be a nondegenerate hermitian form, F-linear in the first variable, and fix a totally imaginary $\xi\in F^\times$, that is $\overline{\xi}=-\xi$. Put

$$E(x,y)=\operatorname{Tr}_{F/\mathbb{Q}}(\xi\,H(x,y)),$$

a nondegenerate alternating $\mathbb{Q}$-bilinear form with $E(\alpha x,y)=E(x,\overline{\alpha}y)$. For each τ let $V_\tau=V\otimes_{F_0,\tau}\mathbb{R}$, a vector space of dimension $2n$ over $F\otimes_{F_0,\tau}\mathbb{R}\cong\mathbb{C}$, and let (p_τ,q_τ) be the signature of the hermitian form H_τ induced by H on it. We say that (V,H) is of (F,n)-*Weil type* if $p_\tau=q_\tau=n$ for every τ, and we call the class of $(-1)^n\det H$ in $F_0^\times/\mathrm{N}_{F/F_0}(F^\times)$ its discriminant δ; it is totally positive, and for $F=K$ it is the discriminant of Definition 15.22. A polarised abelian variety (A,ι,E) of dimension $2mn$ with $\iota\colon F\to\operatorname{End}^0(A)$ and $H^1(A,\mathbb{Q})\cong V$ as F-modules with polarisation E is of (F,n,δ)-Weil type when (V,H) is. As in Lemma 3.2, we work throughout with $V=H^1$; a complex structure on $V_\mathbb{R}$ is the same datum as one on the tangent space. The Weil classes of A are

$$W_F(A)=\textstyle\bigwedge_F^{2n}V\ \subseteq\ \bigwedge_\mathbb{Q}^{2n}V=H^{2n}(A,\mathbb{Q}),$$

the $\mathbb{Q}$-subspace whose complexification is $\bigoplus_\sigma\bigwedge^{2n}V_\sigma$, where $V_\sigma=V\otimes_{F,\sigma}\mathbb{C}$ runs over the $2m$ eigenspaces of F on $V_\mathbb{C}$; it is a one-dimensional F-vector space, of dimension $2m$ over $\mathbb{Q}$. Write

> $\mathbf{W}(F,n,\delta)$: every abelian variety of (F,n,δ)-Weil type has algebraic Weil classes.

The period domain $\mathcal{D}_F$ is the set of complex structures J on $V_\mathbb{R}$ commuting with F and polarised by E, and the group $G_F=\operatorname{Res}_{F_0/\mathbb{Q}}\mathrm{SU}(V,H)$ acts on it. We fix an arithmetic subgroup $\Gamma\subset G_F(\mathbb{Q})$ and write $\mathcal{S}_F=\Gamma\backslash\mathcal{D}_F$ for the quotient, $p\colon\mathcal{D}_F\to\mathcal{S}_F$ for the projection, $\pi\colon\mathcal{A}\to\mathcal{S}_F$ for the universal family and $\Sigma_F\subseteq\mathcal{D}_F$ for the set of s at which $W_F(A_s)$ is algebraic. For $m=1$ this is Setup 15.12 and Setup 16.1, with ω_s a generator of the line.

Remark 17.2 (Every such abelian variety is a member). If A is any abelian variety with a CM field $F\subseteq\operatorname{End}^0(A)$ such that $\dim_F H^1(A,\mathbb{Q})=2n$ and $\dim V_\sigma^{1,0}=n$ for every embedding σ, then A is of (F,n,δ)-Weil type for some δ: the Rosati involution of any polarisation restricts to a positive involution of F, hence to conjugation, and the argument of Lemma 4.5 writes the polarisation as $\operatorname{Tr}_{F/\mathbb{Q}}(\xi H)$ for a unique hermitian H, whose signatures are then forced by the Hodge numbers as in Proposition 17.3(i). These are exactly the abelian varieties whose Weil classes for F are Hodge classes, in the sense of [MZ99], so $\mathbf{W}(F,n,\delta)$ over all (F,n,δ) is the statement that all Weil classes of all CM fields are algebraic.

Proposition 17.3 (The Weil classes of a CM field)**.** *Keep Setup 17.1, with (V,H) of (F,n)-Weil type.*

(i) *A polarised complex structure J is the same as an H_τ-orthogonal decomposition $V_\tau=V_\tau^+\oplus V_\tau^-$ for every τ, with H_τ definite of the sign of $\operatorname{Im}\sigma_\tau(\xi)$ on V_τ^+ and of the opposite sign on V_τ^-; then $\dim V_{\sigma_\tau}^{1,0}=\dim V_\tau^+=n$ and $\dim V_{\overline{\sigma}_\tau}^{1,0}=\dim V_\tau^-=n$, so every class in $W_F(A_J)$ is of type (n,n). The spaces $W_F(A_J)$ form a flat subsystem W_F of $R^{2n}\pi_*\mathbb{Q}$ of rank $2m$ on which G_F acts trivially, and $\mathcal{D}_F\cong\prod_\tau\mathcal{D}_{n,n}$ is connected of dimension mn^2.*

(ii) *If one nonzero element of $W_F(A)$ is algebraic, every element is.*

(iii) *If* $\mathrm{Hg}(A_s) = G_F$*, the Hodge ring of* A_s *is generated by* $\mathrm{NS}(A_s)_{\mathbb{Q}}$ *and by* W_F*, so the Hodge conjecture for* A_s *in every degree is equivalent to the algebraicity of* W_F*. For* $n \geq 2$ *the space* $\mathrm{NS}(A_s)_{\mathbb{Q}}$ *is the* m*-dimensional space of classes* $\eta_t = [E(t\,\cdot, \cdot)]$*,* $t \in F_0$*.*

(iv) *Two members of* (F, n, δ)*-Weil type have isomorphic* (V, H)*, so every one of them is a fibre of* π*.*

Proof. (i) A complex structure commuting with F preserves each V_τ, since $F_0 \otimes \mathbb{R}$ acts diagonally, and is $\mathbb{C}$-linear on it for the structure $F \otimes_{F_0,\tau} \mathbb{R} \cong \mathbb{C}$, so it is $\pm\sqrt{-1}$ on two complementary $\mathbb{C}$-subspaces $V_\tau^{\pm}$. Write $\sigma_\tau(\xi) = \sqrt{-1}\, r_\tau$ with $r_\tau \in \mathbb{R}^\times$. For $x \in V_\tau^{\epsilon}$ and $y \in V_\tau^{\epsilon'}$ one has $H_\tau(Jx, Jy) = \epsilon\epsilon' H_\tau(x, y)$, so $E(Jx, Jy) = E(x, y)$ forces $E(x, y) = 0$ for $\epsilon \neq \epsilon'$, and replacing x by $\sqrt{-1}\, x$ then forces $H_\tau(x, y) = 0$: the decomposition is orthogonal. For $x \in V_\tau^{\epsilon}$, $E(x, Jx) = 2\,\mathrm{Re}(\sqrt{-1}\, r_\tau\, \overline{\epsilon\sqrt{-1}}\, H_\tau(x, x)) = 2\epsilon r_\tau H_\tau(x, x)$, so positivity is the sign condition stated. Conversely such a decomposition defines a polarised J. Under $V_\tau \otimes_{\mathbb{R}} \mathbb{C} = V_{\sigma_\tau} \oplus V_{\overline{\sigma}_\tau}$ the space $V^{1,0}$, the $\sqrt{-1}$-eigenspace of J on $V_{\mathbb{C}}$, meets V_{σ_τ} in the image of V_τ^+ and $V_{\overline{\sigma}_\tau}$ in the conjugate of the image of V_τ^-, which gives the two equalities of dimensions. Since H_τ has signature (n, n) and is definite, of opposite signs, on the complementary orthogonal subspaces $V_\tau^{\pm}$, both have dimension n. So each V_σ is the sum of its intersections with $V^{1,0}$ and $V^{0,1}$, both of dimension n, and a generator α_σ of $\bigwedge^{2n} V_\sigma$ is of type (n, n). The subspace $\bigwedge_F^{2n} V$ of $H^{2n}(A_s, \mathbb{Q})$ is defined by the action of F on V alone and does not depend on s, so it is flat; $\mathrm{SU}(V, H)$ acts on $\bigwedge_F^{2n} V$ by the F-determinant, which is 1. Finally, the choice of V_τ^+ is a point of the bounded symmetric domain of $\mathrm{SU}(V_\tau, H_\tau) \cong \mathrm{SU}(n, n)$, of dimension n^2, and $\mathcal{D}_F$ is the product over τ.

(ii) For $\alpha \in \mathcal{O}_F$ the endomorphism $\iota(\alpha)$ acts on $\bigwedge_F^{2n} V$ by α^{2n}, through an algebraic correspondence, so the algebraic classes in W_F form a $\mathbb{Q}$-subspace stable under multiplication by every α^{2n}. By the polarisation identity, in characteristic zero every product $x_1 \cdots x_{2n}$ of elements of a field is a rational combination of $2n$-th powers of sums of the x_i; taking $x_2 = \cdots = x_{2n} = 1$ shows that the $2n$-th powers span F over $\mathbb{Q}$, and since every element of F is a rational multiple of an element of $\mathcal{O}_F$, so do the α^{2n} with $\alpha \in \mathcal{O}_F$. Hence the $\mathbb{Q}$-span of $\{\alpha^{2n} w\}$ is $Fw = W_F$ for every $w \neq 0$.

(iii) Hodge classes are the invariants of the Hodge group, which here is G_F. Over $\mathbb{C}$ it is $\prod_\tau \mathrm{SL}(U_\tau)$ with $U_\tau = V_{\sigma_\tau}$, and H identifies $V_{\overline{\sigma}_\tau}$ with $U_\tau^\vee$, so $\bigwedge^* V_{\mathbb{C}} = \bigotimes_\tau \bigwedge^*(U_\tau \oplus U_\tau^\vee)$ with the τ-th factor of the group acting on the τ-th tensor factor alone. The invariants of a product group in a tensor product of representations are the tensor product of the invariants, and the invariants of $\mathrm{SL}(U)$ in $\bigwedge^*(U \oplus U^\vee)$ are generated by the pairing $\psi \in U \otimes U^\vee$ and the two determinants, by the computation of Theorem 4.14. The classes η_t are Hodge classes: J commutes with t and preserves E, hence preserves the F_0-twisted form $E_t = E(t\,\cdot, \cdot) = \mathrm{Tr}(\xi t H)$, which for totally positive t is again a polarisation of A_s. The complexified span of the η_t is $\bigoplus_\tau \mathbb{C}\psi_\tau$, and the complexification of W_F contains all the determinants. So the $\mathbb{Q}$-algebra generated by $\mathrm{NS}_{\mathbb{Q}}$ and W_F has complexification equal to the ring of invariants, hence equals $\mathrm{Hdg}^*(A_s)$. Since divisor classes are algebraic, the conjecture reduces to W_F, and by (ii) to one class. For the last sentence, the invariants in $\bigwedge^2 V_{\mathbb{C}}$ are $\bigoplus_\tau \mathbb{C}\psi_\tau$ when $n \geq 2$: a summand $\bigwedge^2 U_\tau$ or $\bigwedge^2 U_\tau^\vee$ has an $\mathrm{SL}(U_\tau)$-invariant only if $\dim U_\tau = 2n$ equals 2, and a summand mixing two different τ has none. So $\mathrm{NS}(A_s)_{\mathbb{Q}}$ has dimension m, and it contains the classes η_t, which form an m-dimensional space because $t \mapsto \eta_t$ is injective on F_0, E being nondegenerate.

(iv) Two hermitian forms over F/F_0 of the same rank, the same discriminant and the same signature at every real place of F_0 are isomorphic [Sch85, Ch. 10, §6]. So the datum

(F, n, δ) determines (V, H) up to isomorphism, and the Hodge structure of any member is, by (i), a polarised complex structure on that $V_{\mathbb{R}}$, that is a point of $\mathcal{D}_F$. ∎

Lemma 17.4 (The Hodge group of a very general member)**.** *Keep Setup 17.1. For every $s \in \mathcal{D}_F$ one has $\mathrm{Hg}(A_s) \subseteq G_F$, and there is a countable union Z of proper closed analytic subsets of $\mathcal{D}_F$ such that $\mathrm{Hg}(A_s) = G_F$ for every $s \notin Z$.*

Proof. The Hodge group of A_s commutes with $\mathrm{End}^0(A_s) \supseteq F$ and preserves E, so it lies in $\mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{U}(V, H)$: an F-linear map preserving $E = \mathrm{Tr}_{F/\mathbb{Q}}(\xi H)$ preserves H, because H is recovered from E and the action of F as in Lemma 4.5. It fixes every Hodge class, in particular the Weil classes, which are Hodge classes by Proposition 17.3(i); an element g of $\mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{U}(V, H)$ acts on $\bigwedge_F^{2n} V$ by $\det_F(g)$, so $\det_F(g) = 1$ and $g \in G_F$.

For the second statement let T be the set of rational tensors $t \in \bigoplus_{a,b} V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ that are not fixed by G_F, a countable set, and for $t \in T$ let Z_t be the set of s at which t is a Hodge class, of type (p, p) with $2p = a - b$, for the Hodge structure induced by A_s. The condition is that t lies in the p-th step of the Hodge filtration, which varies holomorphically with s, so Z_t is a closed analytic subset of $\mathcal{D}_F$. It is proper. Otherwise t is fixed by $h_s(\mathrm{U}(1))$ for every $s \in \mathcal{D}_F$, where h_s is the Hodge representation, which acts on the (p, q) part through z^{q-p}, and hence by the subgroup N of $G_F(\mathbb{R})$ that these tori generate. Since $\mathcal{D}_F$ is one orbit of $G_F(\mathbb{R})^+$ and $h_{gs} = g h_s g^{-1}$, the group N is a connected normal subgroup of $G_F(\mathbb{R})^+ = \prod_\tau \mathrm{SU}(V_\tau, H_\tau)$, so its Lie algebra is an ideal of $\bigoplus_\tau \mathfrak{su}(V_\tau, H_\tau)$, a sum of some of these simple summands. It contains every summand, because $h_s(\sqrt{-1})$ acts on V_τ as the complex structure J, which is $\sqrt{-1}$ on V_τ^+ and $-\sqrt{-1}$ on V_τ^- by Proposition 17.3(i), and is therefore not central in the factor $\mathrm{SU}(V_\tau, H_\tau)$. So $N = G_F(\mathbb{R})^+$, which is Zariski dense in the connected group G_F, and t would be fixed by G_F, contrary to $t \in T$.

Put $Z = \bigcup_{t \in T} Z_t$. The Hodge group of A_s is the subgroup of $\mathrm{GL}(V)$ fixing every rational Hodge class in the spaces $V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ [Del82, §3]. For $s \notin Z$ each such class is fixed by G_F, so $G_F \subseteq \mathrm{Hg}(A_s)$, and equality follows from the first statement. ∎

17.2. **A base point in every family.**

Theorem 17.5 (A base point in every family, for every CM field)**.** *Let (V, H) be of (F, n)-Weil type with discriminant δ. Then $\mathcal{D}_F$ contains a point s_0 with the following properties.*

(i) *A_{s_0} is isogenous to $B_\Phi^n \times B_{\overline{\Phi}}^n$, where B_Φ is an abelian m-fold with complex multiplication by F of type $\Phi = \{\sigma : \mathrm{Im}\, \sigma(\xi) > 0\}$ and $B_{\overline{\Phi}}$ its conjugate; there is an H-orthogonal basis $e_1, \dots, e_{2n}$ of V with $H(e_i, e_i)$ totally positive for $i \le n$ and totally negative for $i > n$, and $Fe_i = H^1(B_\Phi, \mathbb{Q})$ for $i \le n$, $Fe_i = H^1(B_{\overline{\Phi}}, \mathbb{Q})$ for $i > n$.*

(ii) *For $i \le n$ put $i' = i + n$ and let $D_{ii'} \subseteq H^2(A_{s_0}, \mathbb{Q})$ be the F-balanced part of $Fe_i \wedge Fe_{i'}$, spanned by the classes*

$$\delta_i(f) = \sum_k (\omega_k f\, e_i) \wedge (\omega_k^*\, e_{i'}), \qquad f \in F,$$

for dual $\mathbb{Q}$-bases (ω_k), (ω_k^*) of F with respect to the trace form. Every $\delta_i(f)$ is a Hodge class of type $(1, 1)$, hence a divisor class, and $D_{ii'}$ is a one-dimensional F-vector space with $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$.

(iii) *The Weil classes are the products*

$$w(f) = \sum_{k_1, \dots, k_{n-1}} \delta_1(\omega_{k_1} f) \cdot \delta_2(\omega_{k_1}^* \omega_{k_2}) \cdots \delta_{n-1}(\omega_{k_{n-2}}^* \omega_{k_{n-1}}) \cdot \delta_n(\omega_{k_{n-1}}^*), \qquad f \in F,$$

which lie in $W_F(A_{s_0})$, span it as f runs over F, and are polynomials with rational coefficients in divisor classes. In particular $W_F(A_{s_0})$ is algebraic, and $\Sigma_F \neq \emptyset$ for every F, every $n \geq 1$ and every δ.

Proof. (i) Since H_τ has signature (n,n) at every τ, the determinant $\det H$ has sign $(-1)^n$ at every real place, so $b = (-1)^n \det H$, a representative of δ, is totally positive, and the diagonal form $H' = \operatorname{diag}(b, 1, \dots, 1, -1, \dots, -1)$ with n positive and n negative entries has the same dimension, the same discriminant and the same signatures as H. Hermitian forms over F/F_0 are classified by these invariants [Sch85, Ch. 10, §6], so $(V,H) \cong (F^{2n}, H')$ and the standard basis is the required orthogonal basis e_i, with $a_i = H(e_i,e_i)$. Write $\sigma_\tau(\xi) = \sqrt{-1}\, r_\tau$ and define J on $V_\tau = \bigoplus_i \mathbb{C}e_i$ to be $\epsilon_{i,\tau}\sqrt{-1}$ on $\mathbb{C}e_i$, with $\epsilon_{i,\tau} = \operatorname{sign}(r_\tau)\operatorname{sign}(\tau(a_i))$. The decomposition $V_\tau^+ = \bigoplus_{\epsilon_{i,\tau}=+} \mathbb{C}e_i$, $V_\tau^- = \bigoplus_{\epsilon_{i,\tau}=-} \mathbb{C}e_i$ is H_τ-orthogonal, with H_τ of sign $\operatorname{sign}(r_\tau)$ on V_τ^+ and of the opposite sign on V_τ^-, so Proposition 17.3(i) makes J a point s_0 of $\mathcal{D}_F$. On $Fe_i \otimes \mathbb{R} = \prod_\tau \mathbb{C}e_i$ the structure J is multiplication by the element $(\epsilon_{i,\tau}\sqrt{-1})_\tau$ of $F \otimes_{\mathbb{Q}} \mathbb{R}$, so each Fe_i is a rational sub-Hodge structure of weight one with F-action, polarised by the restriction of E; that is, it is the H^1 of an abelian m-fold with complex multiplication by F, of CM type $\Phi_i = \{\sigma : (Fe_i)_\sigma \subseteq V^{1,0}\}$, which by the computation in Proposition 17.3(i) is $\{\sigma_\tau^{\epsilon_{i,\tau}}\}$, where $\sigma^+ = \sigma$ and $\sigma^- = \overline{\sigma}$. For $i \leq n$ the a_i are totally positive, so $\epsilon_{i,\tau} = \operatorname{sign}(r_\tau)$ and $\Phi_i = \{\sigma : \operatorname{Im}\sigma(\xi) > 0\} = \Phi$; for $i > n$ the signs are reversed and $\Phi_i = \overline{\Phi}$. So $V = \bigoplus_i Fe_i$ splits A_{s_0} up to isogeny as stated.

(ii) Write $L_{i,\sigma} = (Fe_i)_\sigma$, a line, so that $V_{\mathbb{C}} = \bigoplus_{i,\sigma} L_{i,\sigma}$ and $L_{i,\sigma}$ has type $(1,0)$ exactly when $\sigma \in \Phi_i$. For dual bases, $\sum_k \sigma(\omega_k)\sigma'(\omega_k^*) = 1$ if $\sigma = \sigma'$ and 0 otherwise, because the matrices $(\sigma(\omega_k))$ and $(\sigma(\omega_k^*))$ are inverse transposes by the definition of the trace form. Let $\ell_{i,\sigma}$ be the component of e_i in $L_{i,\sigma}$, a generator of that line, so that $\omega e_i = \sum_\sigma \sigma(\omega)\,\ell_{i,\sigma}$ for $\omega \in F$. Hence in $Fe_i \otimes Fe_{i'} \otimes \mathbb{C} = \bigoplus_{\sigma,\sigma'} L_{i,\sigma} \otimes L_{i',\sigma'}$ the class $\delta_i(f)$ has component $\sigma(f)\,\ell_{i,\sigma} \wedge \ell_{i',\sigma}$ on the diagonal $\sigma = \sigma'$ and zero off it. In particular it is F-balanced: $\alpha \in F$ acting through the first factor and through the second agree on it, both multiplying the component at σ by $\sigma(\alpha)$, and this is the identity $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$. Since $\Phi_{i'} = \overline{\Phi}_i$, exactly one of $L_{i,\sigma}$, $L_{i',\sigma}$ has type $(1,0)$, so every component, hence $\delta_i(f)$, has type $(1,1)$; and $\delta_i(f)$ is rational, being a sum of wedges of rational vectors. By the Lefschetz theorem on $(1,1)$ classes it is a divisor class. As f varies the components $\sigma(f)$ vary independently, so $D_{ii'} \otimes \mathbb{C} = \bigoplus_\sigma L_{i,\sigma} \otimes L_{i',\sigma}$ and $D_{ii'} \cong F$.

(iii) The complexification of W_F is spanned by $\alpha_\sigma = \pm \prod_{i \leq n} \ell_{i,\sigma} \wedge \ell_{i',\sigma}$, a generator of $\bigwedge^{2n} V_\sigma = \bigwedge^{2n}(\bigoplus_i L_{i,\sigma})$. In the product $w(f)$, the component indexed by $(\sigma_1, \dots, \sigma_n)$, with σ_i the embedding chosen in the i-th factor, carries the coefficient $\sigma_1(f) \prod_{j=1}^{n-1} \sum_{k_j} \sigma_j(\omega_{k_j})\sigma_{j+1}(\omega_{k_j}^*)$, which vanishes unless $\sigma_1 = \cdots = \sigma_n$ and then equals $\sigma_1(f)$. So $w(f) = \sum_\sigma \sigma(f)\,\alpha_\sigma$ up to the sign of the reordering, an element of $W_F \otimes \mathbb{C}$ which is rational because it is a rational polynomial in rational classes; as f runs over F the vectors $(\sigma(f))_\sigma$ span $\mathbb{C}^{2m}$, so the $w(f)$ span W_F. Each $w(f)$ is a rational combination of n-fold products of the divisor classes $\delta_i(\cdot)$, and a product of divisor classes is algebraic. ∎

Remark 17.6 (The base point for a quadratic field). For $m = 1$ the point s_0 is a member isogenous to E^{2n} for an elliptic curve E with complex multiplication by K, at which the whole Hodge ring is generated by divisor classes, since the Hodge group is a one-dimensional torus. It is a second base point in every family of Section 14, beside the quaternionic one of Theorem 15.27, and it lies in the product locus of Theorem 15.10. The formula of Theorem 17.5(iii) then gives the class at that point explicitly.

17.3. **The reduction and the propagation.**

Theorem 17.7 (The reduction and the propagation, for every CM field)**.** *Fix (F, n, δ) with $n \geq 2$, keep Setup 17.1, and let ω_s denote a generator of $W_F(A_s)$ chosen flatly.*

(i) $\mathbf{W}(F, n, \delta)$ *holds if and only if ω_s is algebraic for every $s \in \mathcal{D}_F$, and then the Hodge conjecture holds in every degree for every member of the family with* $\mathrm{Hg}(A_s) = G_F$.

(ii) Σ_F *is stable under $G_F(\mathbb{Q})$, every $G_F(\mathbb{Q})$-orbit in $\mathcal{D}_F$ is dense, and Σ_F is dense.*

(iii) $\Sigma_F = p^{-1}\big(\bigcup_{\underline{P},\underline{q}} \Sigma_{\underline{P},\underline{q}}\big)$ *with each $\Sigma_{\underline{P},\underline{q}}$ Zariski closed in $\mathcal{S}_F$, the union countable; either $\Sigma_F = \mathcal{D}_F$, or $\bar{\Sigma}_F$ is meagre and the conjecture fails on a dense G_δ.*

(iv) *If at every point s of the orbit $G_F(\mathbb{Q}) \cdot s_0$ the class ω_s is represented by a cycle whose Hilbert data comes from one finite set, then $\Sigma_F = \mathcal{D}_F$.*

(v) *If some perfect complex E on A_{s_0} with $\mathrm{ch}(E) = N\omega_{s_0} + P$, where $N \neq 0$ and P is a polynomial with rational coefficients in the classes η_t of Proposition 17.3(iii), has injective semiregularity map, then $\Sigma_F = \mathcal{D}_F$. Complexes with $\mathrm{ch}(E) = N\omega_{s_0}$ exist.*

Proof. (i) By Proposition 17.3(i) and (iv), every member has its Hodge structure at some point of $\mathcal{D}_F$, so $\mathbf{W}(F, n, \delta)$ is the algebraicity of W_F at every point. By Proposition 17.3(ii) this is the algebraicity of one generator at every point, and by Proposition 17.3(iii) it gives the conjecture at the members with full Hodge group.

(ii) The stability is the argument of Lemma 16.6: $g \in G_F(\mathbb{Q})$ carries J to gJg^{-1} and induces, after clearing denominators, an isogeny $A_J \to A_{gJ}$ whose graph transports algebraic classes to algebraic classes, and g fixes W_F pointwise by Proposition 17.3(i).

For the density we show that the closure L of $G_F(\mathbb{Q})$ in $G_F(\mathbb{R}) = \prod_\tau \mathrm{SU}(V_\tau, H_\tau)$ contains the identity component, which acts transitively on $\mathcal{D}_F$. First, the form H is isotropic over F_0. It is indefinite at every real place, its signature being (n, n). At every finite place v its trace form is a quadratic form in $4n \geq 8$ variables over the local field $F_{0,v}$, hence isotropic, and a hermitian form is isotropic as soon as its trace form is [Sch85, Ch. 10, §1]. Since hermitian forms over the quadratic extension F/F_0 satisfy the Hasse principle [Sch85, Ch. 10, §6], H is isotropic. Let $X = \{v \in V : H(v, v) = 0\}$ be the cone of isotropic vectors, a quadric cone in the $4n$-dimensional F_0-space V with a nonzero F_0-point. Projection from that point shows that the projective quadric of its lines is F_0-rational, that is, F_0-birational to a projective space, so it satisfies weak approximation at the archimedean places, F_0 being dense in $F_0 \otimes \mathbb{R} = \prod_\tau \mathbb{R}$. Hence the F_0-points of X are dense in its points over $F_0 \otimes \mathbb{R}$, which are the products $\prod_\tau X_\tau$ of the isotropic cones of the H_τ.

For an isotropic vector $v \in V$ let $U_v \subseteq \mathrm{SU}(V, H)$ be the unipotent radical of the stabiliser of the line Fv, a unipotent F_0-group. The exponential identifies $U_v(F_0)$ with an F_0-vector space and $U_v(F_0 \otimes \mathbb{R}) = \prod_\tau U_{v,\tau}(\mathbb{R})$ with the corresponding real space, so $U_v(\mathbb{Q}) = U_v(F_0)$ is dense in $\prod_\tau U_{v,\tau}(\mathbb{R})$. Hence L contains $\prod_\tau U_{v,\tau}(\mathbb{R})$ for every rational isotropic v, in particular the factor $U_{v,\tau}(\mathbb{R})$ for each τ separately, and its Lie algebra $\mathfrak{l}$ contains $\mathfrak{u}_{v,\tau}$ for every τ and every rational isotropic v. The map $v \mapsto \mathfrak{u}_{v,\tau}$ is continuous on $X_\tau \setminus 0$ and rational v are dense in $\prod_\tau X_\tau$, so the closed subspace $\mathfrak{l} \cap \mathfrak{su}(V_\tau, H_\tau)$ contains $\mathfrak{u}_{v,\tau}$ for every isotropic $v \in V_\tau$. The span of these is stable under the adjoint action of $\mathrm{SU}(V_\tau, H_\tau)$, which permutes the isotropic lines, so it is a nonzero ideal of the simple Lie algebra $\mathfrak{su}(n, n)$, hence everything. Thus $\mathfrak{l} \supseteq \bigoplus_\tau \mathfrak{su}(V_\tau, H_\tau)$ and $L \supseteq G_F(\mathbb{R})^+$, so every $G_F(\mathbb{Q})$-orbit in $\mathcal{D}_F$ is dense. Since Σ_F is nonempty by Theorem 17.5 and $G_F(\mathbb{Q})$-stable, it contains a dense orbit.

(iii) The proofs of Theorem 16.4 and Corollary 16.5 use only three facts: that $\mathcal{S}_F$ is a quasi-projective variety, by the theorem of Baily and Borel [BB66]; that the relative Hilbert schemes of $\mathcal{A}/\mathcal{S}_F$ are projective over it [Gro61]; and that W_F is a flat subsystem consisting of Hodge classes, which is Proposition 17.3(i). They apply verbatim.

(iv) The argument is the proof of Theorem 16.9, which asks for a cycle chosen at each point (Remark 16.10). The orbit is dense by (ii), and a finite union of the closed sets of (iii) covering it forces one of them to be dense and closed, hence everything; then (i) applies. As in the quadratic case, the hypothesis is equivalent to the conclusion (Theorem 16.17), and the transported cycle does not satisfy it (Theorem 16.15).

(v) The existence of complexes with $\mathrm{ch}(E) = N\omega_{s_0}$ is the proof of [II, Proposition 2.1], applied to the algebraic cycle of Theorem 17.5 with the isomorphism $K_0(A)_{\mathbb{Q}} \cong \mathrm{CH}^*(A)_{\mathbb{Q}}$ [Ful98]. The deformation argument is the proof of [II, Theorem 2.3] for a polarised Chern character, as in [II, Remark 3.6]. Its only inputs are the theorem of Buchweitz and Flenner, (iii), and the flatness of $\mathrm{ch}(E)$ as a section of Hodge classes over $\mathcal{D}_F$; the last holds because W_F is flat by Proposition 17.3(i) and the classes η_t remain of Hodge type on the whole family, as in the proof of Proposition 17.3(iii). At a member A_s reached by the deformation the component of degree $2n$ of the character is $N\omega_s + P_{2n}$, with P_{2n} a polynomial in divisor classes, so ω_s is algebraic. ∎

Corollary 17.8 (The Hodge conjecture for the very general member)**.** *Fix (F, n, δ) with $n \geq 2$. The following are equivalent.*

(a) *Σ_F is closed in $\mathcal{D}_F$.*

(b) *Σ_F has an interior point.*

(c) *$\Sigma_F = \mathcal{D}_F$, that is, the Weil classes are algebraic on every member of the family.*

(d) *The Hodge conjecture holds for the very general member of the family: there is a countable union of proper closed analytic subsets of $\mathcal{D}_F$ outside which the Hodge conjecture holds for A_s in every degree.*

Proof. If (a) holds, Σ_F is closed and dense by Theorem 17.7(ii), so (c) holds; and (c) implies (a) and (b). If (b) holds, Σ_F is not meagre, since a nonempty open subset of the manifold $\mathcal{D}_F$ is not meagre by Baire's theorem, and Theorem 17.7(iii) gives (c). If (c) holds, the Hodge conjecture holds in every degree for every member with $\mathrm{Hg}(A_s) = G_F$ by Theorem 17.7(i), and these are the members outside the set Z of Lemma 17.4, so (d) holds. Finally, if (d) holds with an exceptional set Z', then at every $s \notin Z'$ the Weil classes, which are Hodge classes at every point by Proposition 17.3(i), are algebraic. So Σ_F contains the complement of Z', which is not meagre by Baire's theorem, since each proper closed analytic subset of the connected manifold $\mathcal{D}_F$ is nowhere dense; by Theorem 17.7(iii), $\Sigma_F = \mathcal{D}_F$. ∎

Proof of Theorem 1.1. Part (i) is Theorem 17.5, parts (ii) and (iii) are Theorem 17.7(ii) and (iii), and part (iv) is Corollary 17.8. ∎

Remark 17.9 (Scope of the reduction). The propagation statements (iv) and (v) are exactly the two forms (P1) and (P2) of [II, Remark 3.7] with F in place of K, (v) being (P2) in the polarised form of [II, Remark 3.6]. This section proves that they are the whole of the problem for every CM field, and that their hypothesis is never vacuous: the base point exists, its cycle is written down, and the orbit through it is dense. The theorem of [II, Theorem 2.5] on what (v) forces transfers as well, with the annihilator of a class in W_F computed over the $2m$ eigenspaces in place of two, and we do not repeat it. Nothing above depends on Question 11.4 of [Mar25b] or on any secant construction. The route through Weil restriction and CM points is independent of the route through secant sheaves and reaches every field at once; Markman has extended the route through secant sheaves to these fields [Mar25c] (see (C3) in [III, Section 2.5], and [II, Section 5.2] for the smallest case).

Proposition 17.10 (Composite fields)**.** *Let $F = KF_0$ with K imaginary quadratic, and let A be of (F, n, δ)-Weil type. Then A is of $(K, -1, mn)$-Weil type for the restricted action, its Weil classes $W_K(A) \subseteq H^{2mn}(A, \mathbb{Q})$ lie in the $\mathbb{Q}$-span of the m-fold products of elements of $W_F(A)$, and $W_K(A)$ is algebraic whenever $W_F(A)$ is. In particular $\mathbf{W}(F, n, \delta)$ implies the algebraicity of the K-Weil classes on every member of the F-family, a closed subvariety of dimension mn^2 of the K-family of dimension m^2n^2; the converse implication is not available, and Theorem 17.5 is not a consequence of Theorem 15.27.*

Proof. The eigenspace V_{τ_K} of K on $V_{\mathbb{C}}$ for the embedding τ_K is $\bigoplus_{\sigma|\tau_K} V_\sigma$, of dimension $2mn$ with $\dim V_{\tau_K}^{1,0} = mn$ by Proposition 17.3(i), so A is of K-Weil type with balanced signature, and $\bigwedge^{2mn} V_{\tau_K} = \bigotimes_{\sigma|\tau_K} \bigwedge^{2n} V_\sigma$ is spanned by $\prod_{\sigma|\tau_K} \alpha_\sigma$. So $W_K \otimes \mathbb{C}$ is contained in the complexification of the $\mathbb{Q}$-span P of the m-fold products of elements of W_F, and a rational subspace whose complexification lies in that of P lies in P. Since products of algebraic classes are algebraic, $W_K(A)$ is algebraic whenever $W_F(A)$ is. The last sentence of the statement is a dimension count: for $m \geq 2$ the F-family is a proper subvariety of the K-family, and the K-Weil classes on it are products that carry no information about their factors. ∎

17.4. **The special locus and an escaping sequence.** The properties of Σ_F proved so far are shared by a set that is meagre, the locus of members whose Hodge group is smaller than G_F, and every construction of algebraic Weil classes available for $n \geq 4$ produces points of that locus only. In the other direction the bounded criterion (iv) of Theorem 17.7 needs far less than an orbit: by the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture, a bound on one sequence of CM points that leaves every proper special subvariety is enough (Figure 20).

Definition 17.11 (Special subvarieties and the special locus)**.** Keep Setup 17.1, with Γ a congruence subgroup. A *special subvariety* of $\mathcal{S}_F$ is an irreducible component of the image in $\mathcal{S}_F$ of a connected component of the locus of Hodge classes $\mathrm{Hdg}^p(\mathbb{V})$, where $\mathbb{V}$ runs over the variations $(R^1\pi_*\mathbb{Z})^{\otimes a} \otimes (R^1\pi_*\mathbb{Z})^{\vee \otimes b}$ of even weight, or $\mathcal{S}_F$ itself; these are the special subvarieties of $\mathcal{S}_F$ for $R^1\pi_*\mathbb{Z}$ in the sense of [KOU23].[10] A *CM point* is a point $p(s)$ at which A_s is of CM type, that is, at which $\mathrm{Hg}(A_s)$ is a torus; the CM points are the special points of $\mathcal{S}_F$. The *special locus* is

$$\Sigma_F^{\mathrm{sp}} = \{\, s \in \mathcal{D}_F \;:\; \mathrm{Hg}(A_s) \neq G_F \,\}.$$

Proposition 17.12 (The special locus)**.** *Fix (F, n, δ) with $n \geq 2$ and keep Definition 17.11.*

(i) *Special subvarieties are closed algebraic subvarieties of $\mathcal{S}_F$, and there are countably many of them. A special subvariety other than $\mathcal{S}_F$ is proper and nowhere dense.*

(ii) *Σ_F^{sp} is the set Z of Lemma 17.4, and it is the preimage under p of the union of the proper special subvarieties. In particular it is the preimage of a countable union of proper Zariski closed subsets of $\mathcal{S}_F$, and it is meagre.*

(iii) *If $\mathrm{End}^0(A_s) \neq F$, then $s \in \Sigma_F^{\mathrm{sp}}$. In particular Σ_F^{sp} contains every point at which A_s is of CM type, the base point s_0 of Theorem 17.5, and, for F imaginary quadratic, the split members of Setup 15.12, the quaternionic members of Theorem 15.25 and the products of Lemma 15.26.*

[10] By a theorem of Chevalley every algebraic subgroup of $\mathrm{GL}(V)$ is the stabiliser of finitely many tensors [Del82, §3], so the locus where the Hodge group lies in a given subgroup is a locus of Hodge tensors, and these subvarieties are the special subvarieties of $\mathcal{S}_F$ as a connected Shimura variety, the images of the domains of the reductive subgroups of G_F that contain the Hodge group of one of their points.

(iv) Σ_F^{sp} *is stable under* $G_F(\mathbb{Q})$ *and dense in* $\mathcal{D}_F$.

Proof. (i) Each image of a component of $\mathrm{Hdg}^p(\mathbb{V})$ is closed and algebraic by Theorem B.6, applied to the polarised variation $\mathbb{V}$, after a Tate twist when $a < b$, so its irreducible components are irreducible closed subvarieties. There are countably many pairs (a, b), each $\mathrm{Hdg}^p(\mathbb{V})$ has countably many components by Lemma B.4, and an algebraic set has finitely many irreducible components. A proper closed subvariety of the irreducible variety $\mathcal{S}_F$ has empty interior.

(ii) Let $s \in Z$, so that some rational tensor t not fixed by G_F is a Hodge class of A_s, and let C be the component of $\mathrm{Hdg}^p(\mathbb{V})$ through the point $(p(s), Nt)$, for N with Nt integral. Along C the class is carried by parallel transport, so over $\mathcal{D}_F$ it is one of the flat sections γNt with $\gamma \in \Gamma$ on each sheet. If the image of C were $\mathcal{S}_F$, then by Lemma B.4 and Baire's theorem one such section γNt would be a Hodge class on a nonempty open subset of $\mathcal{D}_F$, hence on all of it, since the locus $Z_{\gamma t}$ of Lemma 17.4 is closed analytic in the connected manifold $\mathcal{D}_F$; the proof of Lemma 17.4 then shows that γt, and with it t, is fixed by G_F, a contradiction. So $p(s)$ lies in a proper special subvariety. Conversely, if $p(s)$ lies in a proper special subvariety, an irreducible component of the image of a component C through a point $(p(s), v)$, then v is a Hodge class of A_s. If v were fixed by G_F it would be fixed by Γ, hence a flat global section that is a Hodge class at every point by Lemma 17.4; its graph is open and closed in $\mathrm{Hdg}^p(\mathbb{V})$, so it would be C, with image $\mathcal{S}_F$, and the special subvariety would not be proper. Hence $s \in Z$. The rest follows from (i) and Lemma 17.4.

(iii) The endomorphism algebra $\mathrm{End}^0(A_s)$ is the algebra of Hodge classes in $\mathrm{End}(V)$, that is, the commutant of $\mathrm{Hg}(A_s)$ [Del82, §3]. Over $\mathbb{C}$ the group G_F is $\prod_\tau \mathrm{SL}(V_{\sigma_\tau})$, acting on V_{σ_τ} by the standard representation and on $V_{\overline{\sigma}_\tau}$, which H pairs perfectly with V_{σ_τ}, by its dual. These $2m$ representations are irreducible and pairwise non-isomorphic, the standard representation of SL_{2n} and its dual being non-isomorphic for $2n \geq 3$, so the commutant of G_F in $\mathrm{End}(V_{\mathbb{C}})$ is $\mathbb{C}^{2m}$, and its commutant in $\mathrm{End}_{\mathbb{Q}}(V)$ is a commutative algebra of dimension $2m$ containing F, hence equal to F. So $\mathrm{Hg}(A_s) = G_F$ forces $\mathrm{End}^0(A_s) = F$. A member of CM type has a commutative Hodge group, and G_F is not commutative; at s_0 the Hodge group lies in that of $B_\Phi^n \times B_{\overline{\Phi}}^n$, a torus. The split members contain the projections to X and to $\widehat{X}$ in $\mathrm{End}^0(A)$, the quaternionic members contain a quaternion algebra, and the products contain the projections to their factors; none of these lies in K.

(iv) An element $g \in G_F(\mathbb{Q})$ induces an isogeny $A_s \to A_{gs}$ acting on V by g, so $\mathrm{Hg}(A_{gs}) = g\,\mathrm{Hg}(A_s)g^{-1}$, and $gG_Fg^{-1} = G_F$. By (iii) the set contains s_0, hence its orbit, which is dense by Theorem 17.7(ii). ∎

Corollary 17.13 (What closing the locus requires)**.** *Fix* (F, n, δ) *with* $n \geq 2$. *The set* Σ_F^{sp} *has every property of* Σ_F *used in Theorem 1.1: it contains the explicit CM point* s_0*, it is dense and stable under* $G_F(\mathbb{Q})$*, and it is the preimage of a countable union of Zariski closed subsets of* $\mathcal{S}_F$*; and it is meagre. Consequently* $\Sigma_F = \mathcal{D}_F$ *does not follow from these properties of* Σ_F*, and it holds only if the Weil classes are algebraic at some* s *with* $\mathrm{Hg}(A_s) = G_F$*, where they span, together with the products of divisor classes, every Hodge class of* A_s.

Proof. The properties are Proposition 17.12(ii) to (iv), and meagreness is (ii). If $\Sigma_F = \mathcal{D}_F$, then Σ_F contains the complement of Σ_F^{sp}, which is nonempty by Baire's theorem. The last assertion is Proposition 17.3(iii). ∎

The base points of Theorem 17.5, Theorem 15.27 and Theorem 15.10, the split members of Theorem 15.14(c), their orbits under $G_F(\mathbb{Q})$, and the members on which the divisorial

method of Remark 15.29 works all lie in Σ_F^{sp} by Proposition 17.12(iii). These are all the members at which this paper exhibits algebraic Weil classes, so the part of Σ_F exhibited here fits inside the meagre alternative, and the two criteria of Theorem 17.7, a bound or a semiregular object, are what would leave it. The first can be weakened as follows.

Theorem 17.14 (One escaping sequence suffices)**.** *Fix (F, n, δ) with $n \geq 2$ and keep Definition 17.11.*

(i) *There is a sequence $y_1, y_2, \dots$ of distinct CM points in $p(G_F(\mathbb{Q}) \cdot s_0)$ such that every proper special subvariety of $\mathcal{S}_F$ contains only finitely many of them.*

(ii) *Let $y_1, y_2, \dots$ be any sequence of distinct CM points of $\mathcal{S}_F$ with this property. Then $\Sigma_F = \mathcal{D}_F$ if and only if there is a finite set $\mathcal{T}$ of pairs $(\underline{P}, \underline{q})$ such that for every k the class ω is represented at y_k by a cycle whose Hilbert data lies in $\mathcal{T}$.*

Proof. (i) By Proposition 17.12(i) the proper special subvarieties form a countable family $Y_1, Y_2, \dots$ of nowhere dense closed subsets. The image $p(G_F(\mathbb{Q}) \cdot s_0)$ is dense in $\mathcal{S}_F$, since the orbit is dense in $\mathcal{D}_F$ by Theorem 17.7(ii) and p is continuous and surjective, and it consists of CM points by Proposition 17.12(iii), (iv). For each k the complement of $Y_1 \cup \dots \cup Y_k \cup \{y_1, \dots, y_{k-1}\}$ is a nonempty open set, so it contains a point y_k of the image. Then $y_k \in Y_j$ forces $k < j$, and Y_j contains at most $j - 1$ of the points.

(ii) If $\Sigma_F = \mathcal{D}_F$, then the countably many closed sets $\Sigma_{\underline{P},\underline{q}}$ of Theorem 17.7(iii) cover $\mathcal{S}_F$, so by Baire's theorem one of them has an interior point, and being Zariski closed in the irreducible $\mathcal{S}_F$ it is all of $\mathcal{S}_F$; take $\mathcal{T}$ to be that one pair. Conversely, let W be the union of the finitely many $\Sigma_{\underline{P},\underline{q}}$ with $(\underline{P}, \underline{q}) \in \mathcal{T}$, a Zariski closed set that contains every y_k, and let $Y \subseteq W$ be the Zariski closure of the set of the y_k. The variety $\mathcal{S}_F$ is a connected Shimura variety, and by the theorem of Pila, Shankar and Tsimerman [PST21, Theorem 1.1] a subvariety of a connected Shimura variety contains only finitely many maximal special subvarieties.[11] Each y_k, being a special point of Y, lies in a maximal special subvariety contained in Y, so one of these finitely many, say M, contains infinitely many y_k. By hypothesis M is not proper, so $M = \mathcal{S}_F$, and therefore $W = \mathcal{S}_F$. As $\mathcal{S}_F$ is irreducible, one of the finitely many closed sets making up W is all of $\mathcal{S}_F$, and $\Sigma_F = \mathcal{D}_F$ by Theorem 17.7(iii). ∎

Remark 17.15 (What the sequence changes). Theorem 17.14 replaces the dense orbit of Theorem 17.7(iv) by a discrete sequence, which may be chosen inside the orbit or among CM points with complex multiplication by different fields; what it must do is leave each proper special subvariety. Without the theorem of Pila, Shankar and Tsimerman the Zariski closure of such a sequence could be a proper subvariety that is not special, and a bound on it would give nothing. Like Theorem 16.17, the statement is an equivalence, so it shrinks the set on which a bound is needed and not the problem. Read on the transported cycle the bound remains subject to Theorem 16.15; read on a representative chosen afresh at each y_k, it is not.

Appendix A. The exterior algebra model in coordinates

We record in coordinates the computation behind Theorem 6.5, both to pin down the sign in (20) and because the programs of the archive [BMB26] use this model.

Let $g \geq 1$ and let W be the $\mathbb{Q}$-vector space with basis $e_1, \dots, e_{2g}, f_1, \dots, f_{2g}$. We identify $V = \langle e_1, \dots, e_{2g} \rangle$ with $H^1(A, \mathbb{Q})$ and $V^\vee = \langle f_1, \dots, f_{2g} \rangle$ with $H^1(A^\vee, \mathbb{Q})$, so that $\bigwedge^* W = H^*(A \times A^\vee, \mathbb{Q})$.

[11] This is the André–Oort conjecture, proved for $\mathcal{A}_g$ by Tsimerman [Tsi18] and in general in [PST21].

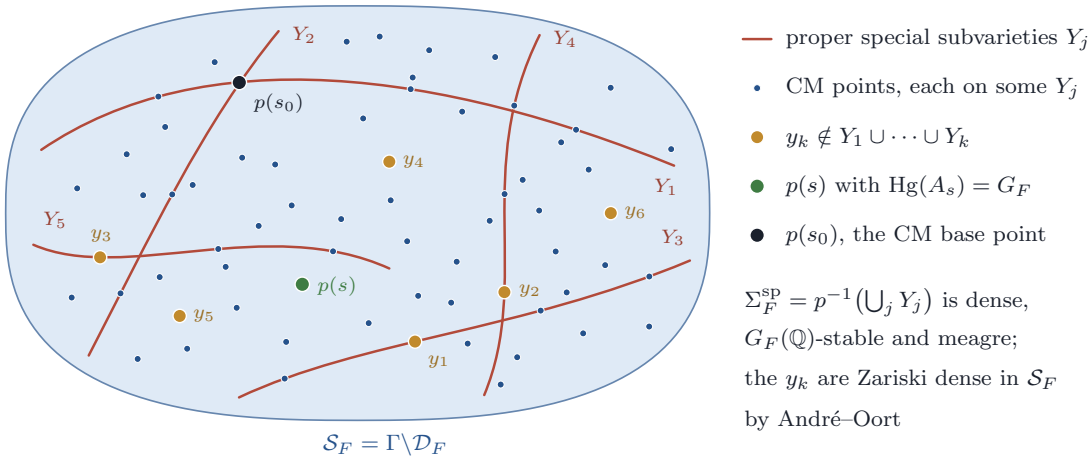


FIGURE 20. Proposition 17.12 and Theorem 17.14 on the quotient $\mathcal{S}_F$. The curves are five of the countably many proper special subvarieties Y_j, whose union is the image of the special locus; it contains $p(s_0)$ and is dense and meagre. The small dots are CM points, each lying on some Y_j, drawn or not. The points $y_1, \dots, y_6$ begin an escaping sequence chosen as in the proof of Theorem 17.14(i): y_k lies on no Y_j with $j \le k$, so y_1 may lie on Y_3 and y_2 on Y_4. A point $p(s)$ at which the Hodge group is all of G_F lies on none of them.

Number the generators so that the generator numbered i is e_i for $i \le 2g$ and f_{i-2g} for $i > 2g$. A basis of $\bigwedge^* W$ is then indexed by the subsets S of $\{1, \dots, 4g\}$, the basis element w_S being the wedge of the generators in S in increasing order. The product is $w_S w_T = \varepsilon(S,T)\, w_{S\cup T}$ when $S \cap T = \emptyset$ and zero otherwise, where

$$\varepsilon(S,T) = (-1)^{\#\{(s,t)\in S\times T\,:\, s>t\}}. \tag{40}$$

The three distinguished elements are

$$\ell = \sum_{i=1}^{2g} e_i \wedge f_i, \qquad \theta = \sum_{j=1}^{g} e_j \wedge e_{g+j}, \qquad \theta' = \sum_{j=1}^{g} f_j \wedge f_{g+j},$$

and p_* is the linear map $\bigwedge^* W \to \bigwedge^* V$ that sends w_S to $\varepsilon(S \cap [1,2g],\, S \setminus [1,2g])\, w_{S\cap[1,2g]}$ when $S \supseteq \{2g+1, \dots, 4g\}$ and to zero otherwise.

Lemma A.1. *Let $0 \le k \le g$, let $T \subseteq \{1, \dots, g\}$ with $|T| = k$, let $U = \{1, \dots, g\} \setminus T$ and $S = U \cup (g + U)$. In the product $\theta'^{\,k} \cdot e^{\ell}$ the unique term contributing w_S after p_* is*

$$k!\,\Big(\prod_{j\in T} f_j f_{g+j}\Big) \cdot \Big(\prod_{i\in S} e_i f_i\Big),$$

and its image under p_* is $k!\,\sigma(g,k)\,\prod_{j\in U} e_j \wedge e_{g+j}$ with $\sigma(g,k) = (-1)^{g(g+1)/2+k}$, independent of T.

Proof. The uniqueness of the term and the identification of S are established in the proof of Theorem 6.5. The sign is the product of the two shuffle signs (40) relating the displayed word to the target word $\big(\prod_{j\in U} e_j e_{g+j}\big)(f_1 \cdots f_{2g})$. Relabelling the symplectic pairs by a permutation π of $\{1, \dots, g\}$ carries the word for T to the word for $\pi(T)$ and multiplies both

the displayed and the target word by the sign of the induced permutation of generators, so the quotient is unchanged. Since the permutations of $\{1,\dots,g\}$ act transitively on its k-subsets, σ depends only on (g,k). Evaluating at $T=\{1,\dots,k\}$, as in the proof of Theorem 6.5, gives the stated value. ∎

Remark A.2. At $g=2$ the computation can be done by hand. There $\ell = e_1f_1+e_2f_2+e_3f_3+e_4f_4$, $\theta = e_1e_3+e_2e_4$ and $\theta' = f_1f_3+f_2f_4$, and one finds $\Phi_{\mathcal{P}}(1) = -\frac{1}{2}\theta^2$, $\Phi_{\mathcal{P}}(\theta') = \theta$ and $\Phi_{\mathcal{P}}(\theta'^2) = -2$, in agreement with (20) for $g=2$, where $(-1)^{g(g+1)/2} = (-1)^3 = -1$.

Appendix B. The locus of Hodge classes

Deformation arguments about Hodge classes rest on one theorem, that of Cattani, Deligne and Kaplan [CDK95]: the locus in a family where a given class stays a Hodge class is an algebraic subvariety, not merely an analytic one. This appendix states the theorem, proves its elementary part, isolates the two analytic inputs that the rest requires, computes the example that matters here, and explains why the theorem is not a route to the Hodge conjecture. The last point is the one that bears on Section 14, and it is why the material is set out here rather than cited.

B.1. **Variations of Hodge structure.**

Definition B.1 (Variation of Hodge structure)**.** Let S be a smooth connected complex quasi-projective variety. A *polarised variation of $\mathbb{Q}$-Hodge structure* of weight w on S consists of

(a) a local system $\mathbb{V}$ of finite dimensional $\mathbb{Q}$-vector spaces on S, with associated holomorphic bundle $\mathcal{V} = \mathbb{V}\otimes_{\mathbb{Q}}\mathcal{O}_S$ carrying the flat connection ∇ whose flat sections are $\mathbb{V}$;
(b) a decreasing filtration $\mathcal{V} = F^0 \supseteq F^1 \supseteq \cdots$ by holomorphic subbundles, inducing on each fibre $\mathbb{V}_s$ a $\mathbb{Q}$-Hodge structure of weight w;
(c) a flat nondegenerate $(-1)^w$-symmetric pairing $Q\colon \mathbb{V}\otimes\mathbb{V}\to\mathbb{Q}_S$ polarising each fibre;

subject to Griffiths transversality

$$\nabla F^p \subseteq F^{p-1}\otimes\Omega^1_S. \tag{41}$$

The example to keep in mind is the geometric one: for $f\colon\mathcal{X}\to S$ smooth projective, $\mathbb{V} = R^w f_*\mathbb{Q}$ with its Gauss–Manin connection is such a variation, and (41) is Griffiths' theorem [Voi02, Ch. 10]. The other example we use is the Weil family of Proposition B.8 below, which is defined in the first instance by a period domain rather than by a family of varieties, and which carries a variation all the same.

Notation B.2. Write $\mathcal{V}^{\mathrm{an}}$ for the total space of $\mathcal{V}$, a complex manifold fibred over S with fibre $\mathbb{V}_s\otimes\mathbb{C}$, and $\mathbb{V}^{\mathrm{tot}}\subset\mathcal{V}^{\mathrm{an}}$ for the subset of pairs (s,α) with $\alpha\in\mathbb{V}_s$, a covering space of S with countable fibres.

B.2. **Hodge loci.**

Definition B.3. Let $\mathbb{V}$ be a polarised variation of weight $2p$ on S. Its *locus of Hodge classes* is

$$\mathrm{Hdg}^p(\mathbb{V}) = \{(s,\alpha)\in\mathbb{V}^{\mathrm{tot}} \;:\; \alpha\in F^p\mathcal{V}_s\}.$$

For $N\in\mathbb{Q}$ write Hdg^p_N for the part with $Q(\alpha,\alpha)=N$.

By conjugation symmetry, a rational α of weight $2p$ lies in F^p exactly when it is of type (p,p). The following two lemmas are elementary, and they give everything about Hdg^p except algebraicity.

Lemma B.4 (Analyticity)**.** $\mathrm{Hdg}^p(\mathbb{V})$ *is a closed analytic subset of* $\mathbb{V}^{\mathrm{tot}}$*, and the projection* $\pi\colon \mathrm{Hdg}^p(\mathbb{V}) \to S$ *is locally biholomorphic onto its image on each connected component. In particular each component is a closed analytic subvariety of an open subset of* S*, and there are countably many components.*

Proof. Work on a simply connected open $U \subseteq S$ over which $\mathbb{V}$ is trivialised, $\mathbb{V}|_U \cong V_0 \times U$ for a fixed $\mathbb{Q}$-vector space V_0. Then $\mathbb{V}^{\mathrm{tot}}|_U$ is the disjoint union over $\alpha \in V_0$ of copies of U. Fix α. The quotient $\mathcal{V}/F^p$ is a holomorphic vector bundle on U, and the image $\bar{\alpha}(s)$ of the flat section α in $(\mathcal{V}/F^p)_s$ is a holomorphic section of it. The set of $s \in U$ with $\alpha \in F^p\mathcal{V}_s$ is the zero locus of $\bar{\alpha}$, a closed analytic subset of U. Taking the union over the countable set V_0 gives the first statement, and shows that on the sheet indexed by α the projection π is the inclusion of that zero locus into U, hence injective. There are countably many components because S is covered by countably many such U, the set V_0 is countable, and each zero locus has countably many components. ∎

Lemma B.5 (Finiteness at bounded norm)**.** *Fix* $s \in S$ *and* $N \in \mathbb{Q}$*. The set of* $\alpha \in \mathbb{V}_s$ *with* α *of type* (p,p)*,* $Q(\alpha,\alpha) = N$*, and* α *in a fixed lattice* $\mathbb{V}_{s,\mathbb{Z}}$*, is finite. The number of such* α *is locally constant on each component of* Hdg^p_N*.*

Proof. Decompose α into its primitive parts for the Lefschetz decomposition attached to the polarisation. On the primitive part of type (p,p) in weight $2p$ the Hodge–Riemann bilinear relations read $(-1)^p\, Q(\alpha,\overline{\alpha}) > 0$ for $\alpha \neq 0$, so the form $(-1)^p Q$ is positive definite on the real span of the primitive real (p,p)-classes; the same holds on each Lefschetz summand with an alternating sign, and the resulting form $\tilde{Q}$ obtained by flipping signs summand by summand is positive definite on the whole space of real (p,p)-classes. The conditions $Q(\alpha,\alpha) = N$ and α of type (p,p) therefore confine α to a compact set for $\tilde{Q}$, and a compact set meets a lattice in finitely many points. The count is locally constant because the lattice and Q are flat and the Hodge filtration varies continuously, so no Hodge class can appear or disappear without its Q-norm changing. ∎

Theorem B.6 (Cattani, Deligne and Kaplan [CDK95])**.** [12] *Let* S *be a smooth complex quasi-projective variety and* $\mathbb{V}$ *a polarised variation of* $\mathbb{Z}$*-Hodge structure of weight* $2p$ *on* S*. Then each connected component of* $\mathrm{Hdg}^p(\mathbb{V})$ *is a closed algebraic subvariety of* $\mathbb{V}^{\mathrm{tot}}$*, and its image in* S *is a closed algebraic subvariety of* S*.*

We do not reproduce the proof, which draws on the whole asymptotic theory of degenerating Hodge structures developed in the two preceding decades. For families of abelian varieties the algebraicity of these loci also follows, by a different route, from Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge; Theorem B.6 is what carries it beyond abelian varieties. We record where the difficulty lies and which two theorems remove it, since the shape of the argument is what one needs in order to judge whether a deformation argument in this area is legitimate.

By Lemma B.4 a component Z of Hdg^p is already a closed analytic subvariety of an open subset of S, and by Lemma B.5 it carries a locally constant Q-norm and is one sheet of a finite cover of its image. Choose a smooth projective compactification $\bar{S} \supseteq S$ with

[12] The paper is short and the proof is not: the two inputs isolated below, the nilpotent orbit theorem and the SL_2-orbit theorem, carry the weight.

$\bar{S} \setminus S$ a normal crossing divisor. By Chow's theorem it is enough to show that Z extends to an analytic subvariety of $\bar{S}$, that is, that Z does not oscillate wildly as one approaches the boundary. This is an asymptotic question about the Hodge filtration, and it is answered by two theorems.

(A) *The nilpotent orbit theorem* of Schmid [Sch73]: near a boundary point the period map is approximated, to exponentially small error in the natural coordinates, by the orbit of a nilpotent Lie algebra element built from the local monodromy logarithms.

(B) *The* SL_2*-orbit theorem* in several variables, of Cattani, Kaplan and Schmid [CKS86]: that nilpotent orbit is in turn approximated by an orbit of a fixed real reductive group, with uniform estimates in the several-variable setting.

Together these give a two-sided estimate for the Hodge norm of a flat section near the boundary, of the shape $\|\alpha\|_s \sim \prod_j (\log |t_j|^{-1})^{m_j}$ with integers m_j determined by the weight filtration. A flat section that is Hodge on Z has its norm for the positive definite form $\tilde{Q}$ of the proof of Lemma B.5 bounded by that lemma, and the estimate converts that bound into a statement that the possible boundary behaviours form a finite list. Algebraicity then follows from the analytic extension across the boundary together with Chow's theorem in $\bar{S}$.

Remark B.7. Both (A) and (B) are theorems about degenerating Hodge structures, and neither mentions algebraic cycles. The algebraicity in Theorem B.6 is algebraicity of a locus of periods, obtained by analysis, and it carries no information about algebraic cycles; Appendix B.4 develops the point.

B.3. **The Weil family.** The example relevant to this paper is the family of complex structures on the model of Setup 4.1.

Proposition B.8 (The Weil period domain)**.** *Fix K, n, the K-space $V = K^{2n}$, and the hermitian form H of signature (n,n) from Lemma 4.5. The set of Hodge structures of weight one on V that are polarised by Q, commute with the K-action, and are of $(K,-1,n)$-Weil type is in bijection with*

$$\mathcal{D}_{n,n} = \big\{ P \subset V_+ \ :\ \dim_{\mathbb{C}} P = n,\ \ H|_P \text{ negative definite} \big\},$$

an open subset of the Grassmannian $\mathrm{Gr}(n,2n)$ and a bounded symmetric domain isomorphic to $\mathrm{U}(n,n)/(\mathrm{U}(n)\times\mathrm{U}(n))$. In particular

$$\dim_{\mathbb{C}} \mathcal{D}_{n,n} = n^2,$$

compared with $\dim \mathcal{A}_{2n} = n(2n+1)$ for the moduli of principally polarised abelian $2n$-folds.

Proof. A Hodge structure of weight one on V is a subspace $V^{1,0} \subset V_{\mathbb{C}}$ with $V_{\mathbb{C}} = V^{1,0} \oplus \overline{V^{1,0}}$. Commuting with the K-action means that $V^{1,0}$ is stable under the operator S of Setup 4.1, so that it splits as $V^{1,0} = P \oplus P'$ with $P = V^{1,0} \cap V_+$ and $P' = V^{1,0} \cap V_-$. Conjugation interchanges V_+ and V_-, so $\overline{V^{1,0}} = \overline{P} \oplus \overline{P'}$ with $\overline{P} \subset V_-$, and the direct sum condition becomes $V_- = P' \oplus \overline{P}$, which determines P' up to the choice of a complement; the Weil condition $\dim P = \dim P' = n$ of Lemma 3.5 together with the polarisation condition pins it down as $P' = \overline{P}^{\perp_H}$. The structure is therefore determined by P.

For the positivity, extend H to the $\mathbb{C}$-bilinear pairing between V_+ and V_-. By the computation (9) in the proof of Theorem 4.4, the second Riemann relation holds exactly when iQ is positive on $V^{1,0}$, and on the V_+ part this is the condition that H be negative definite on P, the sign being fixed by the convention of Lemma 4.5. The set of n-dimensional negative definite subspaces of a hermitian space of signature (n,n) is the standard realisation

of the symmetric domain of $\mathrm{U}(n,n)$: the unitary group acts transitively on it and the stabiliser of a point is $\mathrm{U}(n)\times\mathrm{U}(n)$, the unitary groups of P and of $P^{\perp}$. It is open in $\mathrm{Gr}(n,2n)$ because negative definiteness is an open condition, and $\dim \mathrm{Gr}(n,2n) = n(2n-n) = n^2$.

For the comparison, $\dim \mathcal{A}_g = g(g+1)/2$, which is $n(2n+1)$ for $g = 2n$. ■

Corollary B.9 (The Weil locus is algebraic)**.** *Inside $\mathcal{A}_{2n}$ the image of $\mathcal{D}_{n,n}$, that is the locus of principally polarised abelian $2n$-folds admitting an action of an order of K of $(K,-1,n)$-Weil type, is a closed algebraic subvariety of dimension n^2.*

Proof. The condition of admitting such an action is the condition that a certain class in $\mathrm{End}(H^1)$, namely the graph of $\iota(\sqrt{-d})$, be a Hodge class, together with the balanced signature, which is locally constant. The locus is therefore a component of a locus of Hodge classes for the variation $\mathrm{End}(R^1 f_*\mathbb{Q})$ over $\mathcal{A}_{2n}$ with its level structure, and Theorem B.6 applies. The dimension is Proposition B.8. ■

B.4. **Limits of the theorem.**

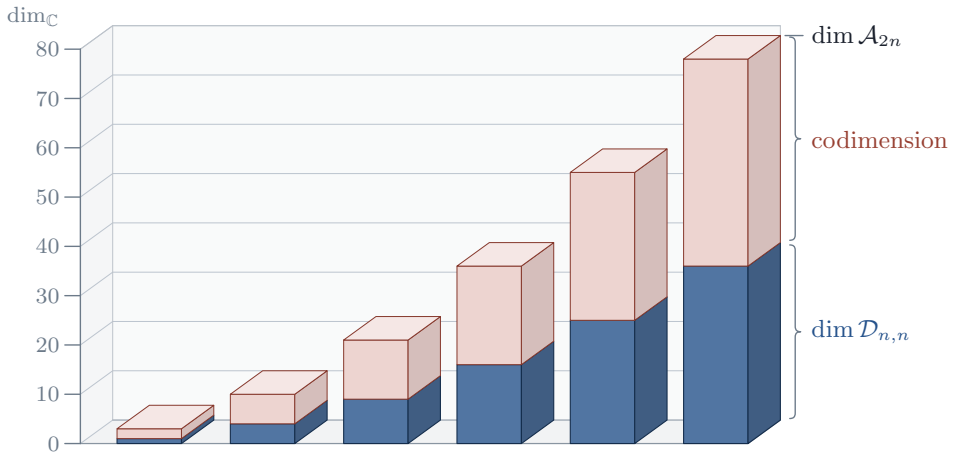
Remark B.10 (Algebraic locus, not algebraic class). Theorem B.6 says that the set of parameters where α is Hodge is cut out by polynomial equations. It says nothing about whether α is the class of a cycle at any one of those parameters. The two statements are independent in both directions: the Hodge locus of a class that happens to be algebraic everywhere is still known to be algebraic only through Theorem B.6, and conversely the algebraicity of the locus supplies no cycle. The distinction is easily blurred, and Corollary B.9 is exactly the kind of statement that invites it: the Weil locus is an algebraic subvariety, even a Shimura subvariety, and none of that produces the class ω_1 of (11). Figure 21 compares the dimension of the Weil locus with that of $\mathcal{A}_{2n}$.

Remark B.11 (The variational Hodge conjecture). The property that would turn Theorem B.6 into a tool is the variational Hodge conjecture of Grothendieck: if $\mathcal{X} \to S$ is smooth projective over a connected base, α is a flat section of $R^{2p}f_*\mathbb{Q}$ that is Hodge at every point, and α_{s_0} is algebraic at one point s_0, then α_s is algebraic at every point. Its infinitesimal form is a theorem under a semiregularity hypothesis: if the cycle at s_0 is semiregular in the sense of Bloch [Blo72], then it deforms to first order, and hence formally, along the locus where its class stays Hodge. That theorem is the subject of Section 9, and Corollary 9.3 shows that its numerical hypothesis is not what obstructs the present problem.

Remark B.12 (The construction of Section 14). The construction of Section 14 produces, for every n, every d and every nondegenerate β, an abelian variety of Weil type together with secant sheaves whose Chern classes are computed in closed form, and its output is algebraic on the split locus. The missing ingredient, described in Remark 14.16, is a statement along the algebraic locus of Corollary B.9: the Hodge type of the deformed object has to be controlled as the parameter moves off the split locus. Theorem B.6 guarantees that the locus along which one deforms is an algebraic variety, which is what makes the question well posed, but it does not answer the question. Question 13.7 states it precisely.

Appendix C. Endomorphism algebras and Weil type

Definition 3.1 asks for three things: an even dimension, an imaginary quadratic field acting with balanced eigenvalue multiplicities, and compatibility with the polarisation. This appendix places these conditions inside Albert's classification of endomorphism algebras. Two statements used in the body come out of it. The first is that the compatibility



n	1	2	3	4	5	6
n^2	1	4	9	16	25	36
$n(n+1)$	2	6	12	20	30	42
$n(2n+1)$	3	10	21	36	55	78

FIGURE 21. The Weil locus inside the moduli of principally polarised abelian $2n$-folds, for $n = 1, \dots, 6$. Each prism has the height $n(2n+1)$ of the moduli space $\mathcal{A}_{2n}$; its lower part, of height n^2, is the dimension of the period domain $\mathcal{D}_{n,n}$ of Proposition B.8, and its upper part is the codimension $n(n+1)$. The codimension grows quadratically, so the Weil locus is thin, and by Corollary B.9 it is an algebraic subvariety. Both facts matter for propagation: the locus along which one would have to deform is small and is a variety, but neither fact produces a cycle on it.

condition is automatic once the field is stable under the Rosati involution, so that it reduces to that stability (Lemma C.4). The second is that the balanced condition is not a normalisation but exactly the condition under which the Weil line carries a Hodge class (Proposition C.7); this is why the subject concerns balanced signatures alone.

C.1. **Positive involutions.**

Definition C.1. Let D be a finite dimensional semisimple $\mathbb{Q}$-algebra. An involution $\dagger$ of D, that is a $\mathbb{Q}$-linear anti-automorphism with $\dagger^2 = \mathrm{id}$, is *positive* if

$$\mathrm{Tr}_{D/\mathbb{Q}}(x\, x^{\dagger}) > 0 \qquad \text{for every } x \in D,\ x \neq 0,$$

where $\mathrm{Tr}_{D/\mathbb{Q}}$ is the trace of left multiplication on D as a $\mathbb{Q}$-vector space.

For an abelian variety A with a polarisation η, the algebra $D = \mathrm{End}^0(A) = \mathrm{End}(A) \otimes \mathbb{Q}$ is semisimple and the Rosati involution $\dagger$ attached to η is positive [BL04, Ch. 5]. Everything below is a statement about the pair $(D, \dagger)$.

Lemma C.2 (Totally real centre)**.** *Let $(D, \dagger)$ be as above with D a division algebra, let F be the centre of D and $F_0 = \{z \in F : z^{\dagger} = z\}$. Then F_0 is a totally real number field.*

Proof. For $z \in F_0$ the positivity condition reads $\mathrm{Tr}_{D/\mathbb{Q}}(z^2) > 0$, and $\mathrm{Tr}_{D/\mathbb{Q}}$ restricted to F_0 is $[D : F_0] \cdot \mathrm{Tr}_{F_0/\mathbb{Q}}$, so $\mathrm{Tr}_{F_0/\mathbb{Q}}(z^2) > 0$ for every nonzero $z \in F_0$. Suppose F_0 had a complex place σ. The embedding of F_0 into the product of its archimedean completions

has dense image, so there is $z \in F_0$ with $\sigma(z)$ within $\frac{1}{10}$ of i and $|\rho(z)| < \varepsilon$ for every other archimedean place ρ. Then $\mathrm{Tr}_{F_0/\mathbb{Q}}(z^2) = \sum_\rho \rho(z)^2$, where a complex place contributes twice the real part of $\rho(z)^2$. The real part of $\sigma(z)^2$ is less than $-1 + \frac{2}{10} + \frac{1}{100} < -\frac{1}{2}$, so the term from σ is less than -1, and the remaining terms are bounded in absolute value by $2[F_0 : \mathbb{Q}]\varepsilon^2$, so the total is negative for ε small. This contradicts positivity, so every place of F_0 is real. ∎

Theorem C.3 (Albert [Alb34, Alb35a, Alb35b])**.** [13] *Let $(D, \dagger)$ be a division algebra over $\mathbb{Q}$ with a positive involution, F its centre, F_0 the subfield fixed by $\dagger$, $e_0 = [F_0 : \mathbb{Q}]$ and $[D : F] = d^2$. Then F_0 is totally real, $[F : F_0] \in \{1, 2\}$, and exactly one of the following holds.*

Type I. *$D = F = F_0$ is a totally real field and $\dagger$ is the identity.*

Type II. *$F = F_0$ is totally real and D is a quaternion algebra over F with $D \otimes_{F,\rho} \mathbb{R} \cong M_2(\mathbb{R})$ at every real place ρ; $\dagger$ is of orthogonal type.*

Type III. *$F = F_0$ is totally real and D is a quaternion algebra over F with $D \otimes_{F,\rho} \mathbb{R} \cong \mathbb{H}$ at every real place ρ; $\dagger$ is the canonical involution.*

Type IV. *F is a CM field with maximal totally real subfield F_0, and $\dagger$ induces complex conjugation on F.*

If moreover $D = \mathrm{End}^0(A)$ for an abelian variety A of dimension g with $\dagger$ the Rosati involution of a polarisation, then

$$\text{I: } e_0 \mid g, \qquad \text{II: } 2e_0 \mid g, \qquad \text{III: } 2e_0 \mid g, \qquad \text{IV: } e_0 d^2 \mid g.$$

We quote Theorem C.3 as a classification. Lemma C.2 proves the part of it that we use directly, and the four types with their divisibility conditions are in [BL04, Ch. 9]. Figure 22 draws the signatures of an imaginary quadratic action and the dimensions of their families.

C.2. **Compatibility with the polarisation.**

Lemma C.4 (Rosati on an imaginary quadratic subfield)**.** *Let A be an abelian variety with a polarisation η and Rosati involution $\dagger$, and let $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$ be an embedding of an imaginary quadratic field such that $\iota(K)$ is stable under $\dagger$. Then*

$$\iota(\tau)^\dagger = \iota(\overline{\tau}) \qquad \text{for every } \tau \in K.$$

That is, condition (iii) *of Definition 3.1 holds automatically, and Weil type is the conjunction of* (i), (ii) *and the $\dagger$-stability of $\iota(K)$.*

Proof. Put $D = \mathrm{End}^0(A)$. By stability, $\dagger$ restricts to an involutive anti-automorphism of the field $\iota(K) \cong K$. An anti-automorphism of a commutative ring is an automorphism, and $\mathrm{Gal}(K/\mathbb{Q})$ has order two, so the restriction is either the identity or complex conjugation. Suppose it were the identity, and put $x = \iota(\sqrt{-d})$. Then $x\,x^\dagger = x^2 = \iota(-d) = -d \cdot 1$, so

$$\mathrm{Tr}_{D/\mathbb{Q}}(x\,x^\dagger) = -d \cdot \dim_{\mathbb{Q}} D < 0,$$

contradicting the positivity of $\dagger$. Hence $\dagger$ restricts to complex conjugation. ∎

[13] The numbering of the four types is Albert's and has stuck, although it follows no natural order: types I to III have a totally real centre and type IV a CM one, so the natural division is two against two rather than three against one. Shimura's normalisation of the same classification, in terms of the associated symmetric domains, is the one that makes the moduli side transparent.

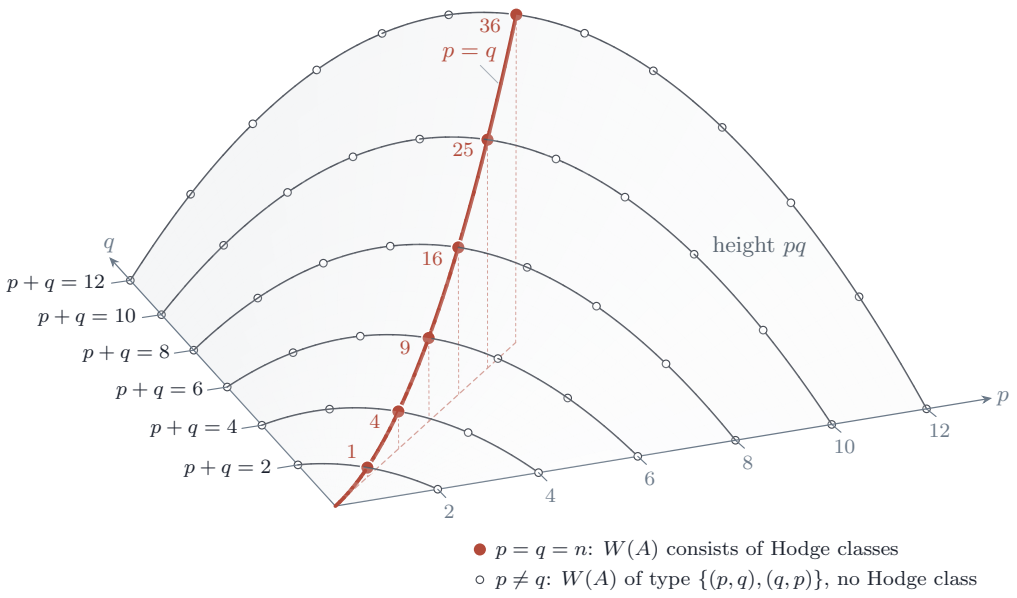


FIGURE 22. The signatures (p, q) of an imaginary quadratic action, with $p + q = \dim A$, drawn on the surface of height pq, which is the dimension of the family of abelian varieties carrying that action, the unitary Shimura variety of signature (p, q). Each curve on the surface collects the signatures available in one dimension, and on it the diagonal point is the highest. The divisibility conditions of Theorem C.3 put no constraint on which of them occur when $\mathrm{End}^0(A) = K$, by Proposition C.9(i). By Proposition C.7 the Weil line consists of Hodge classes exactly at the filled points on the ridge $p = q$, and at every open point it is a rational sub-Hodge structure of type $\{(p, 2n-p), (2n-p, p)\}$ carrying no Hodge class at all. The diagonal is the locus that Corollary B.9 shows to be an algebraic subvariety of the moduli.

Remark C.5. Stability is a real hypothesis: a subfield of $\mathrm{End}^0(A)$ need not be †-stable. What Lemma C.4 removes is the second half of the condition, the identification of $\dagger|_K$ with a specific automorphism. In particular, in the model of Setup 4.1 the polarisation need not be chosen to match the field, since any †-stable embedding does the same work.

C.3. **The balanced signature.**

Definition C.6 (Signature)**.** Let $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$ be as above, $g = \dim A$. The operator $\iota(\sqrt{-d})$ acts on the tangent space $T_0A \cong \mathbb{C}^g$ with the two eigenvalues $\pm i\sqrt{d}$; write p and $q = g - p$ for their multiplicities. The pair (p, q) is the *signature* of (A, ι).

The signature is an isogeny invariant, since an isogeny induces an isomorphism on tangent spaces commuting with the action. It is locally constant in families: the two eigenspaces are the kernels of $\iota(\sqrt{-d}) \mp i\sqrt{d}$ on the relative tangent bundle, so their dimensions are upper semicontinuous, and they sum to g. By duality $\iota(\tau)^*$ acts on $H^{1,0}(A)$ with the same multiplicities, so $\dim V_+^{1,0} = p$ and $\dim V_-^{1,0} = q$ in the notation of Section 3.

Proposition C.7 (The balanced signature is forced)**.** *Let A be an abelian variety of dimension $2n$ with a †-stable embedding $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$ of signature (p, q), $p + q = 2n$,*

and let

$$W(A) = \left(\textstyle\bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_-\right) \cap H^{2n}(A, \mathbb{Q})$$

as in Definition 3.6. Then $W(A)$ is a rational sub-Hodge structure of $H^{2n}(A,\mathbb{Q})$ of dimension two and of Hodge type $\{(p, 2n-p),\ (2n-p, p)\}$, and the following are equivalent:

(i) *$p = q = n$, that is (A, ι, η) is of $(K, -1, n)$-Weil type;*
(ii) *$W(A)$ contains a nonzero Hodge class;*
(iii) *$W(A)$ consists of Hodge classes.*

Proof. The space $\bigwedge^{2n} V_+$ is a line, and by (3) it equals $\bigwedge^{p} V_+^{1,0} \otimes \bigwedge^{2n-p} V_+^{0,1}$, which is of pure Hodge bidegree $(p, 2n-p)$; the conjugate line $\bigwedge^{2n} V_-$ is of bidegree $(2n-p, p)$. Their sum is stable under conjugation and defined over $\mathbb{Q}$ by the argument of Proposition 3.7(i), which does not use the balanced condition, so $W(A)$ is a two-dimensional rational sub-Hodge structure of the stated type.

A Hodge class is an element of bidegree (n, n). A nonzero element of $W(A) \otimes \mathbb{C}$ has a nonzero component in one of the two lines, so it is of bidegree (n, n) only if $(p, 2n-p) = (n, n)$ or $(2n-p, p) = (n, n)$, and either equality says $p = n$. This proves that (ii) implies (i). If $p = n$ then both lines are of bidegree (n, n) and every element of $W(A)$ is a Hodge class, so (i) implies (iii); and (iii) implies (ii) because $\dim W(A) = 2 > 0$. ∎

Remark C.8. Proposition C.7 is the reason the literature speaks of abelian varieties of Weil type rather than of imaginary quadratic multiplication in general. For an unbalanced signature the space $W(A)$ is still a canonical two-dimensional rational sub-Hodge structure, and it is still not in the divisor subring, but it contains no Hodge class and therefore predicts nothing. The exceptional Hodge classes appear exactly at balance, and the locus of balance is the algebraic subvariety of Corollary B.9.

C.4. **Endomorphism algebras of Weil type.**

Proposition C.9 (Weil type in the classification)**.** *Let A be an abelian variety of dimension $2n$ admitting a †-stable embedding of an imaginary quadratic field K with balanced signature, and suppose $\mathrm{End}^0(A)$ is a division algebra D with centre F as in Theorem C.3. Then:*

(i) *if $D = K$ then D is of type IV with $e_0 = 1$ and $d = 1$, and the divisibility condition of Theorem C.3 is vacuous, so every even dimension occurs;*
(ii) *the classification does not exclude type III with $F_0 = \mathbb{Q}$: a totally definite quaternion algebra over $\mathbb{Q}$ contains imaginary quadratic subfields on which the canonical involution is conjugation;*
(iii) *D is not of type I, because a totally real field contains no imaginary quadratic subfield.*

Proof. (i) If $D = K$ then $F = K$ is a CM field with $F_0 = \mathbb{Q}$, so $e_0 = 1$; and $[D : F] = 1$ gives $d = 1$. The involution is conjugation by Lemma C.4, which is type IV. The condition $e_0 d^2 \mid g$ reads $1 \mid 2n$.

(ii) Let D be a quaternion algebra over $\mathbb{Q}$ ramified at the infinite place, with standard generators u, v satisfying $u^2 = a < 0$, $v^2 = b < 0$, $uv = -vu$, and canonical involution $x \mapsto \overline{x}$ fixing $\mathbb{Q}$ and negating u, v, uv. Then $\mathbb{Q}(u) \cong \mathbb{Q}(\sqrt{a})$ is imaginary quadratic, stable under the involution, and the involution restricts to conjugation on it. Whether such a D occurs as $\mathrm{End}^0(A)$ together with a balanced signature is a separate question about the moduli, but nothing in the classification excludes it.

(iii) is immediate, since a totally real field has no element of negative square. ∎

Remark C.10 (The generic case). Case (ii) of Proposition C.9 is a special position: there $\mathrm{End}^0(A)$ is larger than K, the Hodge group is correspondingly smaller, and $\mathrm{Hdg}^n(A)$ can be larger than the three dimensions of Theorem 4.14. It is also a case in which the Weil classes can become accessible: [III, Remark 2.5] records that at such points the Weil line can meet the divisor subring, which is the split situation. The hard case, and the one addressed throughout Sections 5, 7 and 10, is case (i) with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.

Remark C.11 (Dimension four). For $n = 1$ the Weil line lies in $H^2(A, \mathbb{Q})$ and consists of divisor classes, algebraic by the Lefschetz theorem on $(1,1)$ classes, so it predicts nothing new; this is the same boundary as in Proposition 11.11. Moonen and Zarhin determined the Hodge classes on abelian varieties of dimension at most five [MZ99], and on simple abelian fourfolds in [MZ95]: through dimension three the Hodge ring is generated by divisor classes, and the first Hodge classes not of that form occur in dimension four, on abelian fourfolds of Weil type. Thus $n = 2$ is at once the first case of Definition 3.1 with exceptional Hodge classes, the first case where Theorem 4.14 gives three rather than one, and the first case of the Hodge conjecture for abelian varieties that needed more than divisor classes; Markman settled it [Mar25a, Corollary 1.6.1]. The next case is $n = 3$ with nontrivial discriminant, by [III, Proposition 2.12(ii)].

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Ethics approval and consent. This article reports no research involving human participants, human data or animals, so ethics approval and consent to participate or to publish were not required. The results are original and have not been published elsewhere except in the earlier paper named on the title page, which this paper and its companions replace, and the work of others is credited by citation throughout.

Data availability. The archive [BMB26] is deposited on Zenodo under the DOI `10.5281/zenodo.23225483`. It holds programs that repeat much of the finite arithmetic of this paper and its companions [II, III] as an independent check, a Lean 4 certificate of part of that arithmetic, the Macaulay2 computations of local Ext algebras with their output, and the generators and TikZ sources of the figures; its programs perform 2218 checks, all of which pass, and the Lean kernel checks four hundred and nine theorems, each depending on no axiom or on propositional extensionality alone. Deep Bhattacharjee computed all the calculations, in C, Python, Julia, Macaulay2, Lean 4 and shell scripts, with the assistance of Claude Code, an AI model developed by Anthropic. The archive is immutable at that DOI; the command `python3 code/verify_all.py` in it runs every check and ends with their count, and `lean HodgeObstruction.lean` in its folder `lean` checks the certificate and prints the axioms of each theorem. The Lean theorems certify

finite arithmetic only: rank certificates, enumerations, lattice computations and polynomial identities; no geometric argument of the three papers is formalised, and every proof in the text was written and checked by the authors. The authors are responsible for the mathematical statements and their proofs. No other data were generated or analysed.

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